

NEET **CHEMISTRY** MODULE - 4

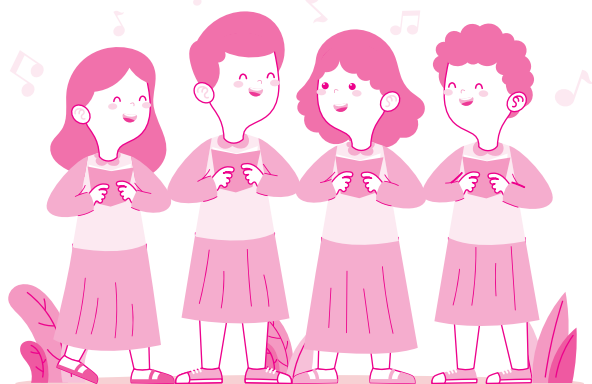
- **Classification & Nomenclature**
- **Isomerism**
- **General Organic Chemistry**
- **Hydrocarbons**
- **Purification of Organic Compound**

सर्वम प्रार्थना

हमको मन की शक्ति, देना मन विजय करें
दूसरों की जय से पहले खुद की जय करें
हमको मन की शक्ति, देना मन विजय करें
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भेद—भाव अपने दिल से साफ कर सकें
दूसरों से भूल हो तो माफ कर सकें
झूठ से बचे रहें, सच का दम भरें
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मुश्किलें पड़े तो हम पे, इतना कर्म करें
साथ दे तो धर्म का, चलें तो धर्म पर
खुदपे होसला रहे, बदी से ना डरें
दूसरों की जय से पहले, खुद को जय करें



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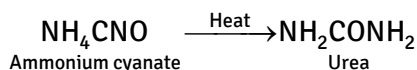
Chapter 1 CLASSIFICATION & NOMENCLATURE

Chapter Summary

- Introduction
- Classification of organic compound
- Homologous Series
- Common Name
- Derived System
- IUPAC Name
- Nomenclature of open chain compounds
- Nomenclature of Alicyclic compounds
- Nomenclature of Aromatic Compounds

Introduction

Around the year 1780, chemists began to distinguish between organic compounds obtained from plants and animals and inorganic compounds prepared from mineral sources. Berzilius, a Swedish chemist proposed that a 'vital force' was responsible for the formation of organic compounds. However, this notion was rejected in 1828 when F. Wohler synthesised an organic compound, urea from an inorganic compound, ammonium cyanate.



The pioneering synthesis of acetic acid by Kolbe (1845) and that of methane by Berthelot (1856) showed conclusively that organic compounds could be synthesised from inorganic sources in a laboratory,

The development of electronic theory of covalent bonding ushered organic chemistry into its modern shape.

(a) Tetravalence of carbon : Shapes of organic compounds

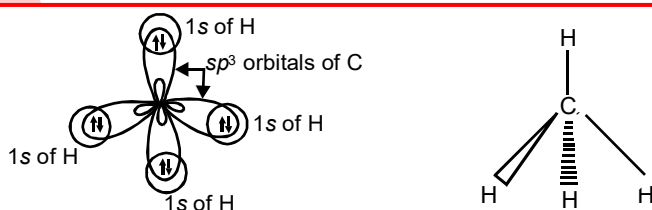
Carbon contains four electrons in outermost shell, so need to make 4 bonds to get nearest noble gas configuration.

Spectroscopic studies and properties of methane have revealed that the four C-H bonds in methane are identical and symmetrically disposed at an angle of $109^\circ 28'$ to each other. This may be accounted for on the basis of Hybridization.

i.e. Carbon (ground state) $1s^2 \quad 2s^2 \quad 2p_x^1 \quad 2p_y^1 \quad 2p_z^0$

Carbon (excited state) $\begin{array}{|c|c|c|c|} \hline 1s & 2s & 2p & \\ \hline \boxed{\uparrow\downarrow} & \boxed{\uparrow} & \boxed{\uparrow} & \boxed{\uparrow} \\ \hline \end{array}$
 sp^3 hybridization

The sp^3 hybrid orbitals of carbon overlap with four $1s$ atomic orbitals of four H atoms.



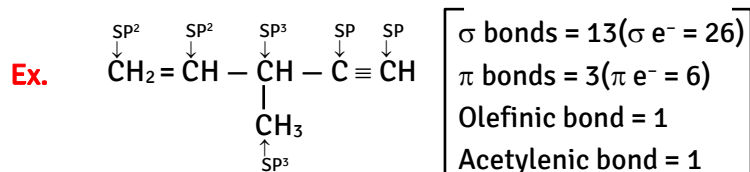
The four valencies of carbon atom can be represented by following way

Structure	σ bonds	π bonds	Hybridisation [No. of σ bond + No. of l.p.]	Shape	Bond Angle	No. of Bond angles
	4	0	sp^3	Tetrahedral (Non planar)	$109^\circ 28'$	6
	3	1	sp^2	Planar (Trigonal)	120°	3
	2	2	sp	Linear	180°	1
	2	2	sp	Linear	180°	1

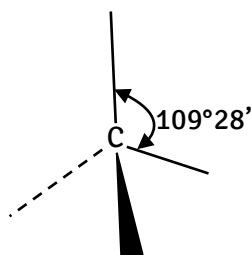
Note: Double bond between carbon atoms is known as olefinic bond and triple bond between carbon atoms is known as Acetylenic bond.

- (b) **Catenation:** Self linking property of C-atom is known as catenation. It is responsible for the variety and large number of organic compounds. It may also give rise to open chain and closed chain nature of compounds. Bond energy for catenation of C is maximum.

Bond energy (in Kcal:) C—C Si—Si N—N P—P
85 54 39 50



- (c) **Tetrahedral shape:** The four covalent bonds are directed towards the four corners of a regular tetrahedron

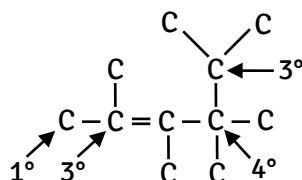


Note: Hybridisation influences the bond length and bond enthalpy (strength) in compounds. The sp hybrid orbital contains more s character and hence it is closer to its nucleus and forms shorter and stronger bonds than the sp^3 hybrid orbital. The sp^2 hybrid orbital is intermediate in s character between sp and sp^3 and, hence, the length and enthalpy of the bonds it forms, are also intermediate between them. The change in hybridisation affects the electronegativity of carbon. The greater the s character of the hybrid orbitals, the greater is the electronegativity. Thus, a carbon atom having an sp hybrid orbital with 50% s character is more electronegative than that possessing sp^2 or sp^3 hybridised orbitals.

Classification of Carbon

There are four types of carbon present in organic compounds. The carbon which is attached to no other carbon atom or directly attached with one, two, three and four carbon atoms are known as primary, secondary, tertiary and quaternary carbon atom respectively.

On the basis of carbon atom, hydrogen atoms bonded with 1° , 2° and 3° C-atoms are named as primary, secondary and tertiary hydrogen atom respectively.



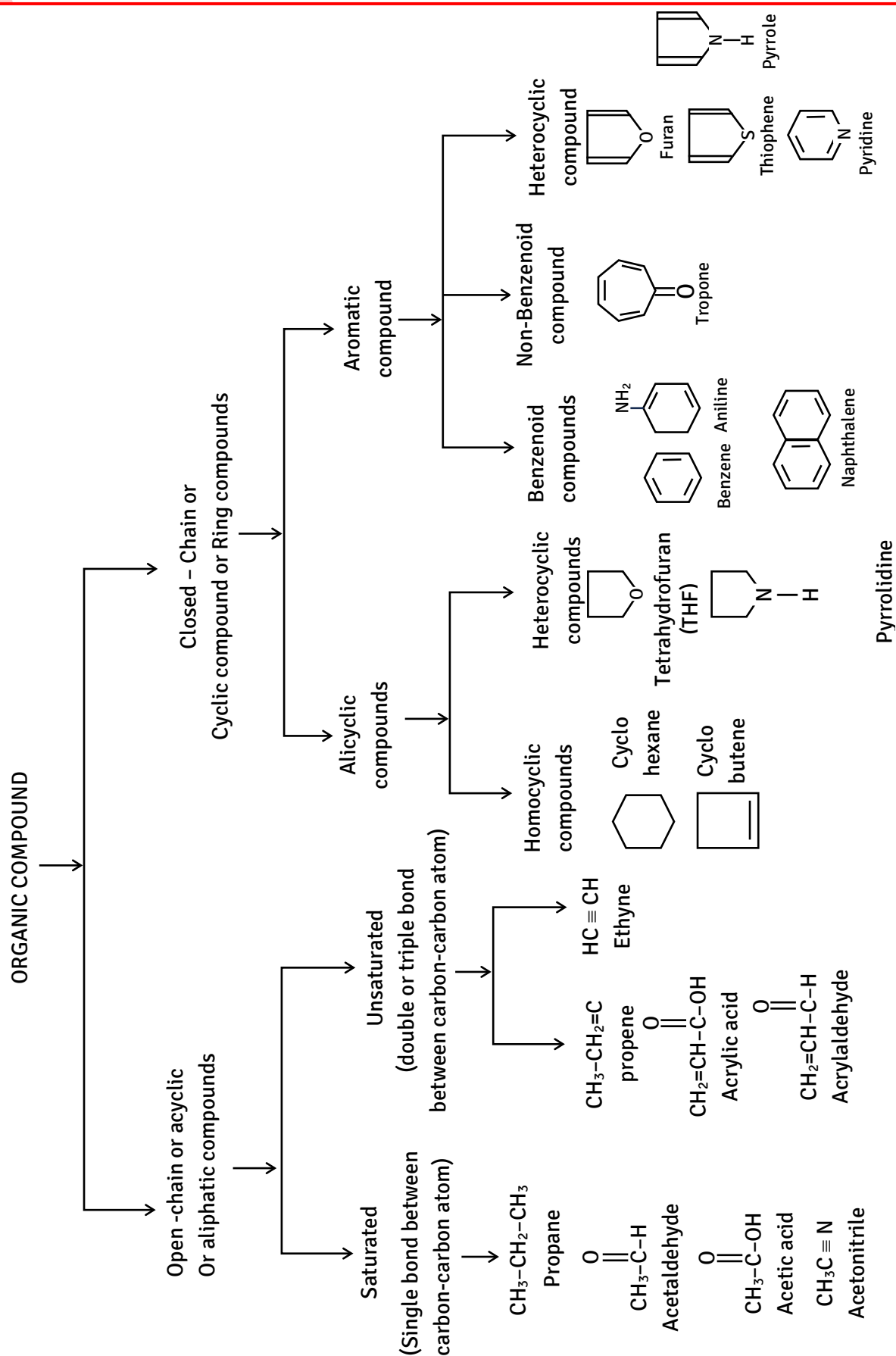
[**Note:** 4°H is not possible because 4°C is attached with four carbons]

Ex.

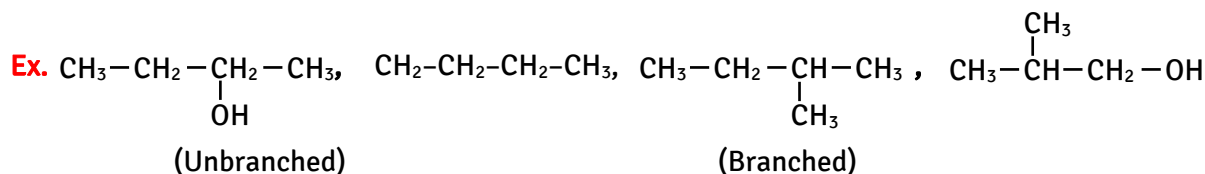
	1°C	2°C	3°C	4°C	1°H	2°H	3°H	Csp^3	Csp^2	Csp
$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{C} - \text{CH}_3 \\ \quad \quad \\ \text{CH}_3 \quad \quad \text{CH}_3 \end{array}$	5	1	1	1	15	2	1	8	0	0
	1	7	3	0	3	14	1	9	2	0

Classification of organic compound

The existing large number of organic compounds and their ever-increasing number has made it necessary to classify them on the basis of their structures. organic compounds are broadly classified as follow:



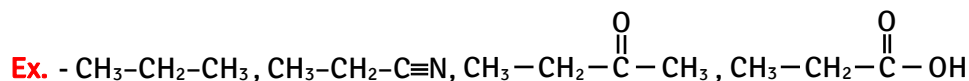
Aliphatic or Open chain compounds: Those compounds which consists of unbranched (straight) or branched chains.



These compounds are further categorised as :

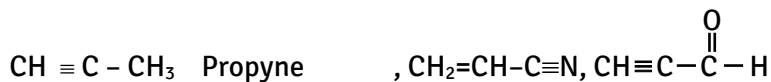
Saturated compounds

- (a) In such type, carbon are bonded with single bonds and do not have any carbon-carbon multiple bond.



Unsaturated Compounds

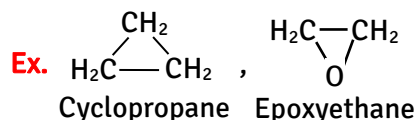
- (a) These compounds have at least one carbon-carbon multiple bond (may be double bond or triple bond)
 $\text{CH}_2 = \text{CH} - \text{CH}_3$ Propene



- (b) Due to presence of π bonds these are more reactive.

Closed chain compounds

In these compounds first & last carbon are attached with each other to form a ring.



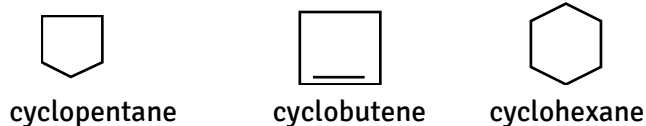
Alicyclic compounds: These are the compounds having the properties like aliphatic compounds.

These may be saturated or unsaturated like aliphatic compounds.

These are further classified as :

- (a) **Homocyclic compounds**

Those compounds in which the ring is formed by one type of atoms only. When ring is composed of C atoms then compound is called carbocyclic.



- (b) **Heterocyclic Compounds**

These are cyclic compounds having ring composed of more than one type of atoms.



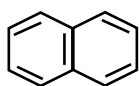
Aromatic compounds Compounds which follow Huck's rule are called aromatic compound.

Huck's rule :

- (i) Compound should be cyclic.
- (ii) Compound should have planer ring.
- (iii) There should be cyclic resonance in ring.
- (iv) Number of electrons involved in cyclic resonance should be $= (4n+2)\pi e^{\ominus}$. Where $n = 0, 1, 2, 3, \dots$

$n = 0$	2π electrons	or 1 pair
$n = 1$	6π electrons	or 3 pairs
$n = 2$	10π electrons	or 5 pairs
$n = 3$	14π electrons	or 7 pairs

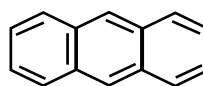
- (a) **Benzoid compounds:** These compounds are composed of Benzene ring.



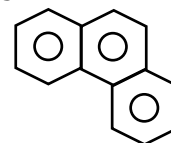
Naphthalene



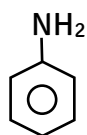
Benzene



Anthracene

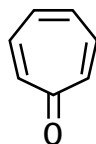


Phenanthrene



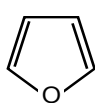
Aniline

- (b) **Non benzoid compounds:** These compounds are composed of other than benzene ring.

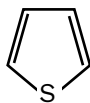


Tropone

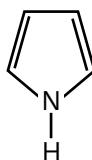
- (c) **Heterocyclic Compounds:** These are cyclic compounds having ring composed of more than one type of atoms.



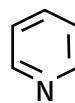
Furan



Thiophene



Pyrrole



Pyridine

Note: Atoms other than carbon in the ring are known as hetero atoms. Like O, S & N in above compound.

Homologous Series

A series of compounds in which compounds having same functional group are arranged in their increasing molecular weight as called homologous series.

Characterstics of Homologous series

- (i) Member of a homologous series are called homologous of each other.
- (ii) All the members of a series have same general formula, general methods of preparation and similar chemical properties due to same functional group.

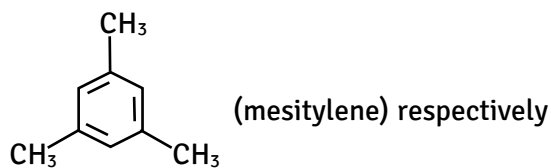
- (iii) The homologues show difference in physical properties due to change in molecular mass and structural arrangement of molecule.
- (iv) Members of a homologous series differ by $n(-CH_2-)$ in their molecular formula and by $n(14)$ in their molecular weight.
- (v) More than one different homologous series can have same general formula.
- (vi) Homologues can never be isomers and isomers can never be homologues since isomers always have same molecular formula while homologues have different molecular formula.

Some Standard Homologous Series Are

S.No.	Name of Series	General Formula	I-homologue	II-homologue
1.	Alkane	C_nH_{2n+2}	CH_4	CH_3-CH_3
2.	Alkene	C_nH_{2n}	$CH_2=CH_2$	$CH_2=CH-CH_3$
3.	Alkyne	C_nH_{2n-2}	$HC\equiv CH$	$HC\equiv C-CH_3$
4.	Halo alkane	$C_nH_{2n+1}X$	CH_3-X	CH_3-CH_2-X
5.	Alcohol	$C_nH_{2n+2}O$	CH_3-OH	CH_3-CH_2-OH
6.	Ether	$C_nH_{2n+2}O$	CH_3-O-CH_3	$CH_3-O-CH_2-CH_3$
7.	Aldehyde	$C_nH_{2n}O$	$H-CHO$	CH_3-CHO
8.	Ketone	$C_nH_{2n}O$	$CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3$	$CH_3-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_3$
9.	Carboxylic acid	$C_nH_{2n}O_2$	$H-COOH$	CH_3-COOH
10.	Ester	$C_nH_{2n}O_2$	$H-\overset{\overset{O}{\parallel}}{C}-O-CH_3$	$H-\overset{\overset{O}{\parallel}}{C}-O-CH_2CH_3$ or $CH_3-\overset{\overset{O}{\parallel}}{C}-O-CH_3$
11.	Anhydride	$C_nH_{2n-2}O_3$	$\begin{array}{c} O \quad O \\ \parallel \quad \parallel \\ H-C-C-H \\ \text{Unstable} \end{array}$	$\begin{array}{c} O \quad O \\ \parallel \quad \parallel \\ CH_3-C-O-C-H \end{array}$
12.	Amide	$C_nH_{2n+1}NO$	$H-CONH_2$	CH_3-CONH_2
13.	Acid halide	$C_nH_{2n-1}XO(X=Cl, -Br, -I, -F)$	$\begin{array}{c} O \\ \parallel \\ H-C-Cl \\ \text{(Unstable)} \end{array}$	$\begin{array}{c} O \\ \parallel \\ CH_3-C-Cl \end{array}$
14.	Nitro alkane	$C_nH_{2n+1}NO_2$	$CH_3-\overset{\nearrow O}{\underset{\parallel O}{N}}$	$CH_3-CH_2-\overset{\nearrow O}{\underset{\parallel O}{N}}$
15.	Amine	$C_nH_{2n+3}N$	CH_3-NH_2	$CH_3-CH_2-NH_2$
16.	Alkenyne	C_nH_{2n-4}	$CH_2=CH-C\equiv CH$	$CH_3-CH=CH-C\equiv CH$
17.	Alkadiene	C_nH_{2n-2}	$CH_2=C=CH_2$	$CH_3-CH=C=CH_2$

Exercise 1.1

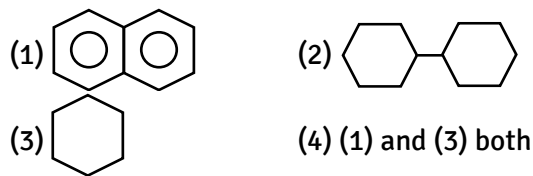
1. Ratio of 1°, 2° & 3° C atoms in



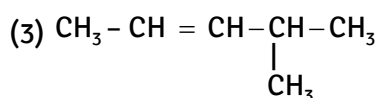
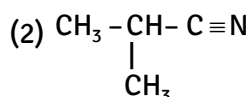
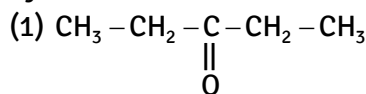
is:-

- (1) 3: 2: 1 (2) 1: 2: 3
(3) 1: 1: 1 (4) 3: 1: 1

2. Which of the following have only one type of H-atoms?



3. Which of the following is/are unsaturated hydrocarbon?

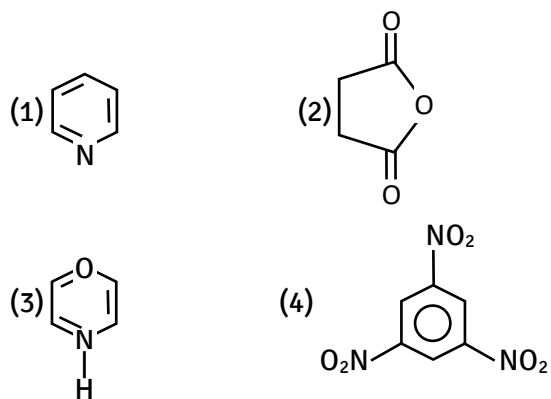


(4) All

4. Which of the following is Homocyclic compound:-

- (1) Pyrrole (2) Naphthaline
(3) Thiophene (4) Pyridine

5. Maximum number of hetero atoms are present in the ring.



6. Which one is not correct for a homologous series -

- (1) All members have same general formula
(2) All members have same chemical properties
(3) All members have same physical properties
(4) All members have same functional group

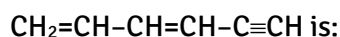
7. Number of C-atoms in 3rd member of ketone homologous series are -

- (1) 5 (2) 3
(3) 4 (4) 6

8. Molecular formula of 2nd member of homologous series of 1° amine is:

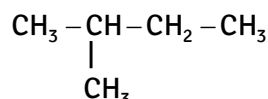
- (1) $\text{C}_2\text{H}_5\text{N}$ (2) $\text{C}_2\text{H}_7\text{N}$
(3) $\text{C}_2\text{H}_6\text{N}$ (4) $\text{C}_3\text{H}_9\text{N}$

9. Number of π bonds in



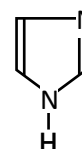
- (1) 2 (2) 3
(3) 4 (4) 5

10. The total number of secondary H-atoms in the structure given below are:



- (1) 1 (2) 4
(3) 3 (4) 2

11. No. of hetero atoms present in the following heterocyclic compound is:



- (1) 3 (2) 2
(3) 1 (4) zero

12. Select the molecule which has only one π -bond.

- (1) $\text{CH} \equiv \text{CH}$ (2) $\text{CH}_2=\text{CHCHO}$
(3) $\text{CH}_3\text{CH}=\text{CH}_2$ (4) CH_3-CN

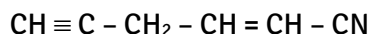
13. Which of the following statement is incorrect regarding alkenes:

- (1) There will be a double bond between any two carbon atoms.
- (2) General formula is C_nH_{2n-2}
- (3) These are called olefins because they reacts with halogens to form oily substances Olefins
- (4) Due to presence of π bonds these are more reactive than alkanes.

14. Number of C-atoms in 2nd member of ester homologous series are:

- (1) 4
- (2) 3
- (3) 5
- (4) 6

15. How many sp C-atoms are present in given compound:

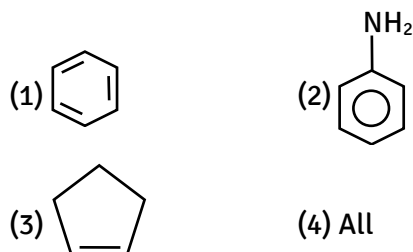


- (1) 2
- (2) 4
- (3) 3
- (4) 5

16. Minimum C required for a open chain compound to have two 3°C and one 2°C:

- (1) 7
- (2) 5
- (3) 8
- (4) 6

17. Which of the following is homocyclic compound



18. Which of the following pair represents homologues

- (1) CH_3-CH_2-OH , $CH_3-CH_2-CH_2-CH_2-OH$
- (2) CH_3-CN , $CH_3-CH_2-NH_2$
- (3) CH_3-NH_2 , $CH_3-NH-CH_3$
- (4) All

19. Incorrect statement is:

- (1) Homologues differ by 'CH₃' in their molecular formula
- (2) Homologues can be isomers
- (3) Different homologous series always have different general formula
- (4) All

20. Molecular formula of third member of ether family (Alkoxy alkane) is:

- (1) C_3H_6O
- (2) C_3H_8O
- (3) $C_4H_{10}O$
- (4) C_4H_8O

Nomenclature Of Organic Compounds

Organic compounds are named mainly in three systems :

- (a) Common Name or Trivial Name System
- (b) Derived Name System
- (c) IUPAC Name or Geneva Name System

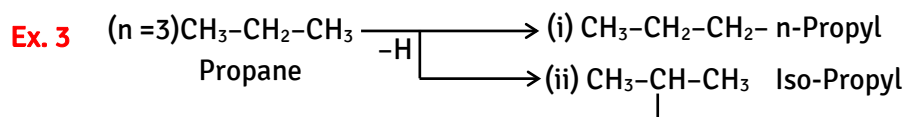
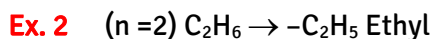
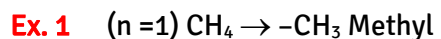
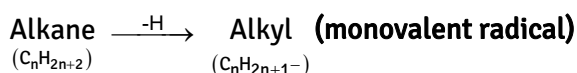
Common Name

Some Common Names Based on Source :

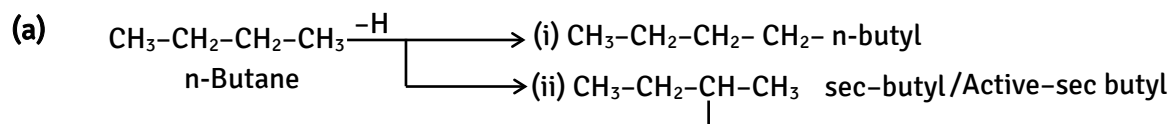
S.No.	Compound	Common Name	Source
1.	CH ₄	Marsh gas (Fire damp)	Marshy Area
2.	CH ₃ OH	Wood spirit (Carbinol)	Destructive distillation of wood
3.	CH ₃ CH ₂ OH	Grain alcohol	Grain
4.	$\begin{array}{c} \text{NH}_2 - \text{C} - \text{NH}_2 \\ \parallel \\ \text{O} \end{array}$	Urea (Carbamide)	Urine
5.	HCOOH	Formic acid	Formica (Red ants)
6.	CH ₃ COOH	Acetic acid	Acetum (Vinegar)
7.	HOOC-COOH	Oxalic acid	Oxalis plant
8.	$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{COOH} \\ \\ \text{OH} \end{array}$	Lactic acid	Lactum (Milk)
9.	CH ₃ CH ₂ CH ₂ COOH	Butyric acid	Butter
10.	$\begin{array}{c} \text{HO} - \text{CH} - \text{COOH} \\ \\ \text{HO} - \text{CH} - \text{COOH} \end{array}$	Tartaric acid	Tamarind
11.	$\begin{array}{c} \text{HO} - \text{CH} - \text{COOH} \\ \\ \text{CH}_2 - \text{COOH} \end{array}$	Malic acid	Malum (Apple)
12.	$\begin{array}{c} \text{CH}_2 - \text{COOH} \\ \\ \text{HO} - \text{C} - \text{COOH} \\ \\ \text{CH}_2 - \text{COOH} \end{array}$	Citric acid	Citrus (Lemon) Fruits

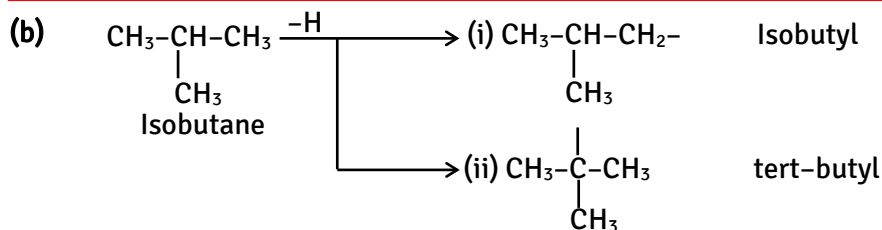
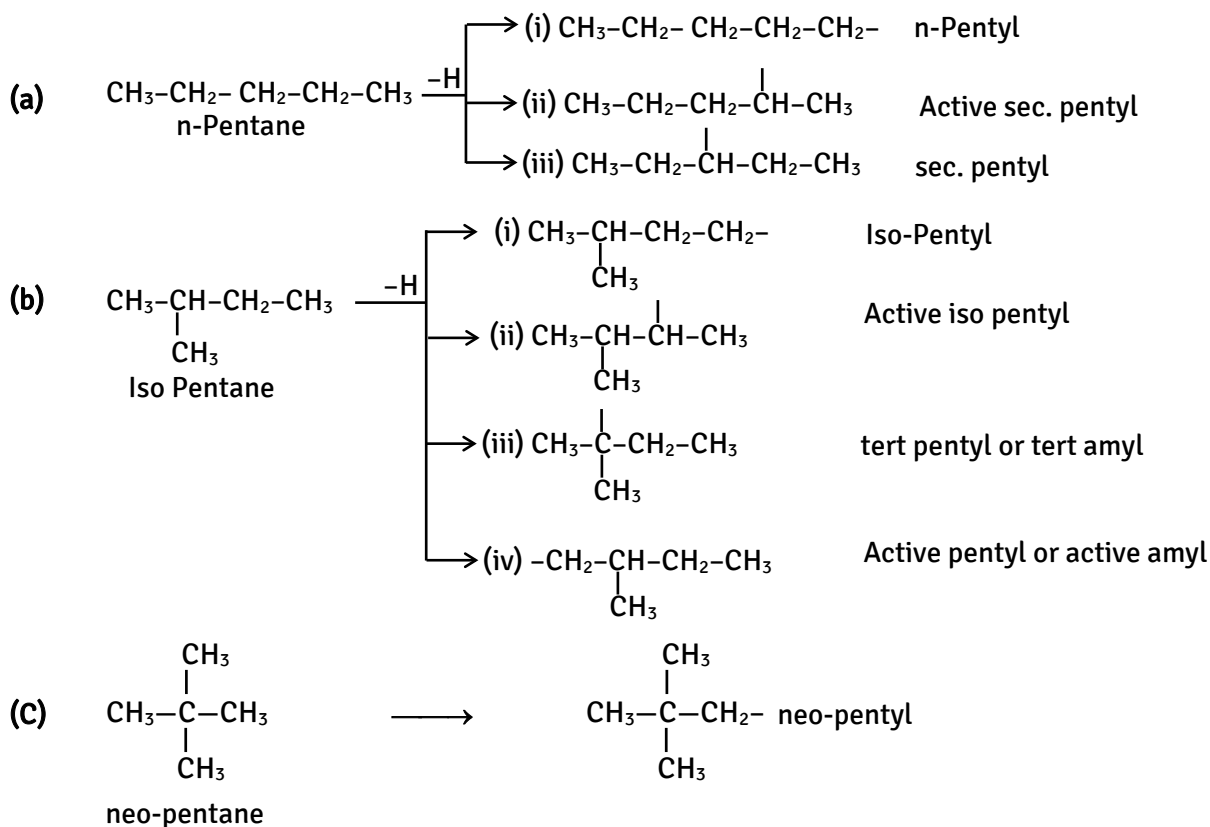
Common Names Of Hydrocarbon Radicals

(A) For saturated hydrocarbons:

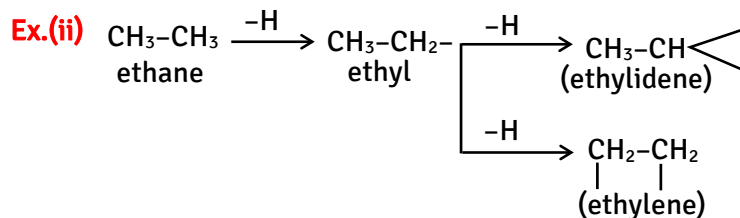
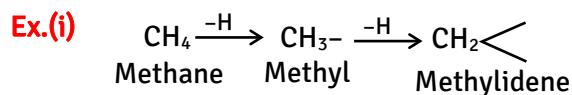
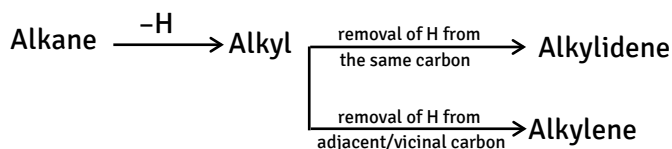


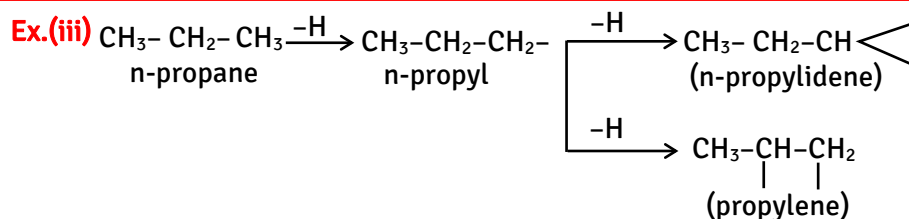
Note: Radicals having same molecular formula but different structures are differentiated in their common names by using prefix as n, iso, neo, secondary, tertiary, active etc.



**Ex.5** (n = 5)

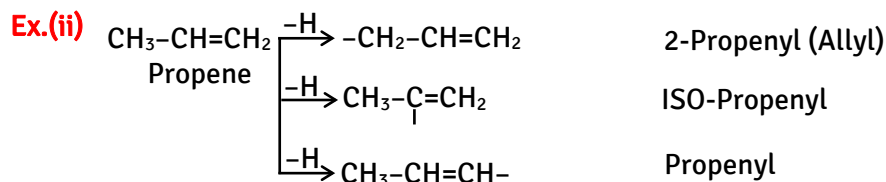
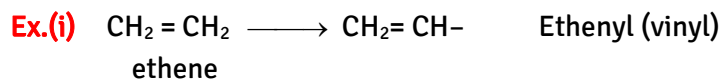
Note: Pentyl is also called amyl group.

Bivalent Radical

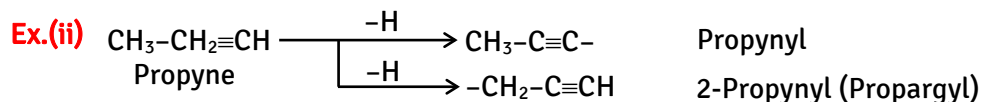
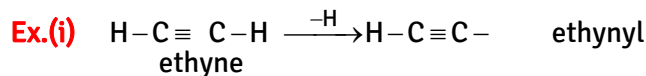


(B) For unsaturated hydrocarbons:

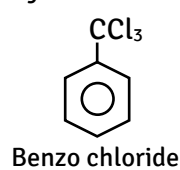
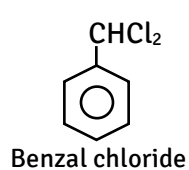
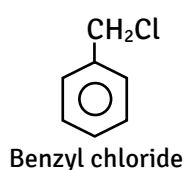
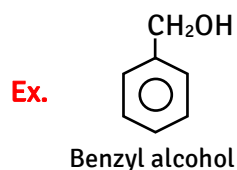
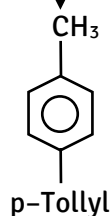
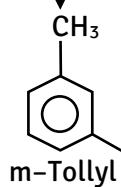
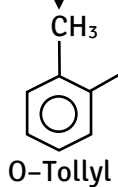
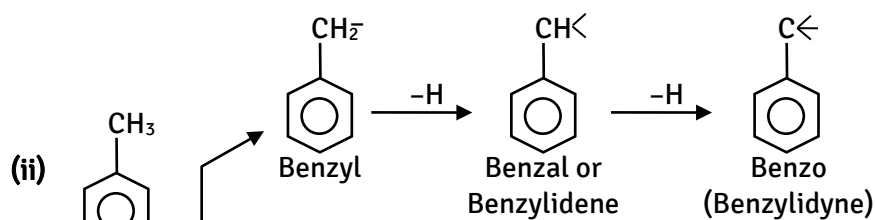
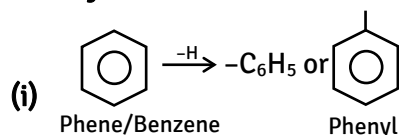
(a) Alkene $\xrightarrow{-\text{H}}$ Alkenyl



(b) Alkyne $\xrightarrow{-\text{H}}$ Alkynyl



Aryl Radical



Note: Aromatic hydrocarbons are called Arenes and respective radicals are called aryl radicals.

1. Common Names of Hydrocarbon Derivatives:

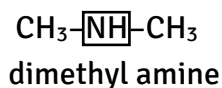
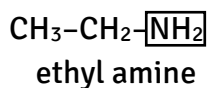
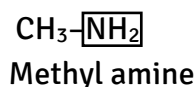
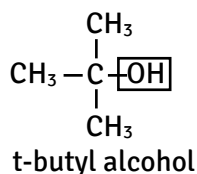
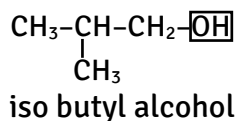
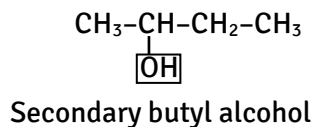
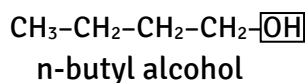
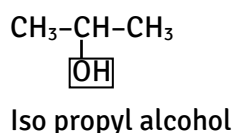
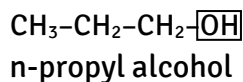
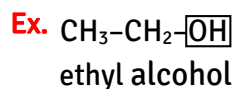
Common names of hydrocarbons derivatives are given in two different systems.

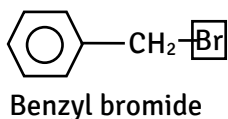
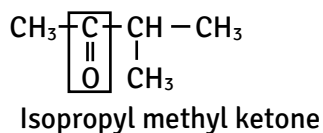
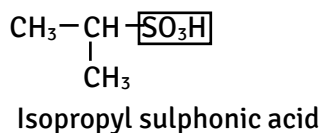
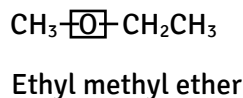
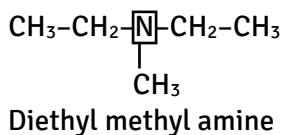
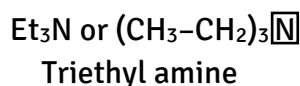
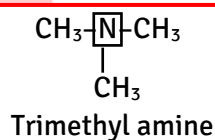
System – I [Used for naming of those compounds which have functional group without carbon]

Format : Name of hydrocarbon radical (R-) + Name of functional group (-Z)

R-Z

S.NO	Functional Group	Name of Functional Group
1	$-\text{SO}_3\text{H}$	Sulphonic acid
2	$-\text{OH}$	Alcohol
3	$-\text{SH}$	Thioalcohol
4	$-\text{NH}_2/-\text{NH}-/-\underset{\text{ }}{\text{N}}-$	Amine
5	$-\text{O}-$	Ether
6	$-\text{S}-$	Thioether
7	$-\text{X}$	Halide
8	$-\underset{\text{ }}{\text{C}}-\text{O}$	Ketone
9	$-\text{C}\equiv\text{N}$	Cyanide
10	$-\text{N}\equiv\text{C}$	isocyanide





Exception : although ketone, cyanide, isocyanide are functional group with carbon can be given name as system-I.

System II:

Format :

Prefix

+

Suffix

On the basis of total no. of carbon in compound including C of functional group

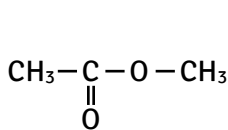
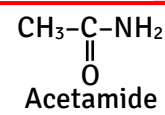
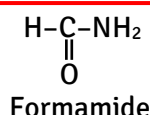
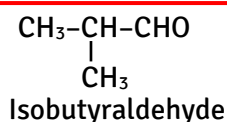
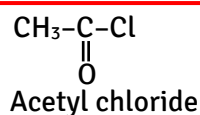
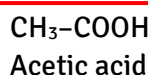
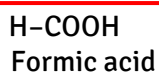
It is suggested for functional group present in the compound

If total carbon $\Rightarrow$	One	Two	Three	Four	Five
Prefix $\Rightarrow$	Form	Acet	Propion	Butyr	Valer

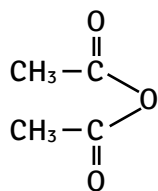
And suffix is given by following table

S.No.	Functional Group	Suffix
1.	$-\text{COOH}$	ic acid
2.	$\begin{array}{c} \text{O} \quad \text{O} \\ \quad \\ -\text{C}-\text{O}-\text{C}- \end{array}$	ic anhydride
3.	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\boxed{\text{OR}} \end{array}$	$\boxed{\text{Alkyl}}$ ate
4.	$-\text{COX}$	yl halide
5.	$-\text{CONH}_2$	amide
6.	$-\text{CHO}$	aldehyde
7.	$-\text{C}\equiv\text{N}$	-o nitrile
8.	$-\text{N}\equiv\text{C}$	-o isonitrile

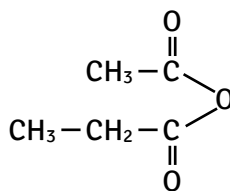
Ex. $\text{H}_2\text{C}=\text{O}$ CH_3CN CH_3-NC CH_3-CHO
 Formaldehyde, Acetonitrile, Acetoisonitrile, Acetaldehyde



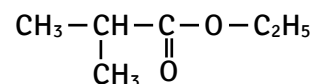
Methyl acetate



Acetic anhydride

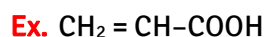


Acetic propionic anhydride

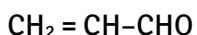


Ethyl isobutyrate

- Prefix **“Acryl”** is used for the compounds which have total three carbon atoms and double bond is on 2nd carbon.

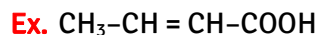


Acrylic acid

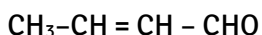


Acryl aldehyde [Acrolein]

- Prefix **“Croton”** is used for the compounds which have total four carbon atoms and double bond is on 2nd carbon



Crotonic acid



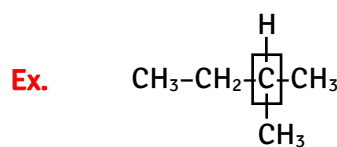
Croton aldehyde

Derived System

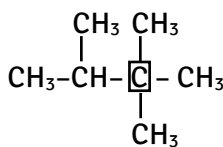
According to this system name of any compound is given by using name of representative compound of the homologous series.

Series	Name of Homologous series	Name of Representative compound	Structure of group
1.	Alkane	Methane	$\begin{array}{c} \\ -\text{C}- \\ \end{array}$
2.	Alkene	Ethylene	$>\text{C}=\text{C}<$
3.	Alkyne	Acetylene	$-\text{C}\equiv\text{C}-$
4.	Alcohol	Carbinol	$\begin{array}{c} \\ -\text{C}-\text{OH} \\ \end{array}$
5.	Aldehyde	Acetaldehyde	$\begin{array}{c} \\ -\text{C}-\text{CHO} \\ \end{array}$
6.	Ketone	Acetone	$\begin{array}{c} \quad \quad \\ -\text{C}-\text{C}-\text{C}- \\ \quad \quad \\ \quad \text{O} \end{array}$
7.	Carboxylic acid	Acetic acid	$\begin{array}{c} \\ -\text{C}-\text{COOH} \\ \end{array}$

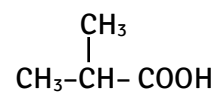
Note: Representative member of a homologous series is that simplest member of the series which represents all the characteristics of series.



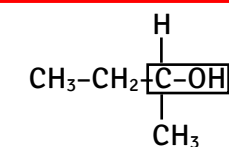
Ethyl dimethyl methane



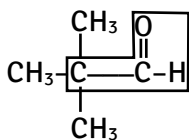
Isopropyltrimethylmethane



Dimethylacetic acid



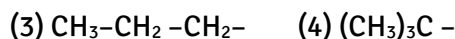
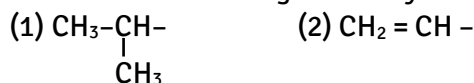
Ethylmethyl carbinol



Trimethyl acetaldehyde

Exercise 1.2

1. Which of the following is tertiary radical:-

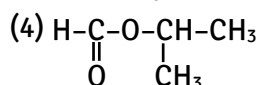
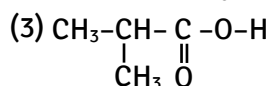
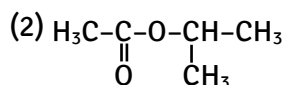
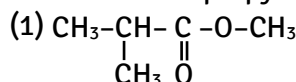
2. Common name of $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{Cl}$ is:-

- (1) Neobutyl chloride
 (2) Neopentyl chloride
 (3) Tert. butyl chloride
 (4) Isobutyl chloride

3. Common name of $\text{CH}_2=\text{CH}-\text{CN}$

- (1) Vinyl Cyanide (2) Acrylonitrile
 (3) Ethyl cyanide (4) Both (1) & (2)

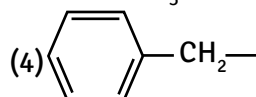
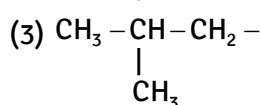
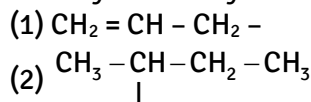
4. Structure of Isopropyl acetate is:-



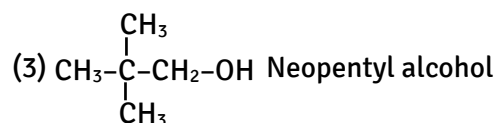
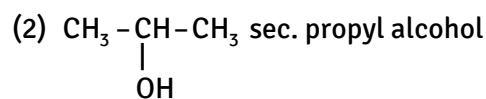
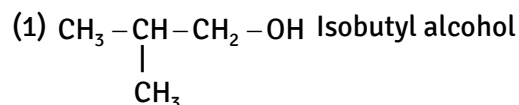
5. Identify primary amine:

- (1) Trimethyl amine
 (2) sec. Butyl methyl amine
 (3) Ethyl methyl amine
 (4) Tert. butyl amine

6. Identify secondary radical:



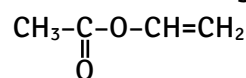
7. Which of the following has wrong common name:



8. The structure of Acetonitrile is:

- (1) $\text{H}-\text{CN}$
 (2) $\text{CH}_3-\text{CH}_2-\text{CN}$
 (3) CH_3-NC
 (4) CH_3-CN

9. Common name of given compound:



- (1) Ethenyl ethanoate
 (2) Vinyl ethanoate
 (3) Vinyl acetate
 (4) Ethenyl acetate

10. Incorrect common name is

- (1) $(\text{CH}_3)_3\text{C}-\text{OH}$ tert. butyl alcohol
 (2) $(\text{CH}_3)_2\text{CH}-\text{OH}$ Isopropyl alcohol
 (3) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$ Acryl alcohol
 (4) CH_3COOH Acetic acid

11. Which is true for Isobutyl alcohol

- (1) It is 1° alcohol
 (2) It has three 1° carbon atoms
 (3) It has one 3° carbon atom
 (4) All

12. sp^2 carbon atoms present in Acryl aldehyde is-
- (1) 0 (2) 3
(3) 2 (4) 1

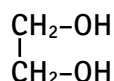
13. Which is correctly named
- (1) $CH_2 = CH - CH = O$ (Acryl aldehyde)
(2) $CH_2 = CH - CH_2 - OH$ (Acryl alcohol)
(3) $CH_2 = CH - CH_2 - COOH$ (Crotonic acid)
(4) All

14. Structural formula of isopropyl methanoate is:
- (1) $CH_3 - \overset{\overset{O}{\parallel}}{C} - O - \underset{\underset{CH_3}{|}}{CH} - CH_3$
(2) $H - \overset{\overset{O}{\parallel}}{C} - O - CH_2 - \underset{\underset{CH_3}{|}}{CH} - CH_3$
(3) $CH_3 - \overset{\overset{O}{\parallel}}{C} - O - CH_2 - \underset{\underset{CH_3}{|}}{CH_2}$
(4) $H - \overset{\overset{O}{\parallel}}{C} - O - \underset{\underset{CH_3}{|}}{CH} - CH_3$

15. Which of the following is not correctly matched:

- (1) Acetonitrile $CH_2 = CHCN$
(2) Allyl chloride $CH_2 = CH - CH_2Cl$
(3) s-Butyl group $CH_3 - \underset{\underset{|}{CH}}{CH} - C_2H_5$
(4) Ethylene chloride $CH_2Cl - CH_2Cl$

16. Common name of the given structure is

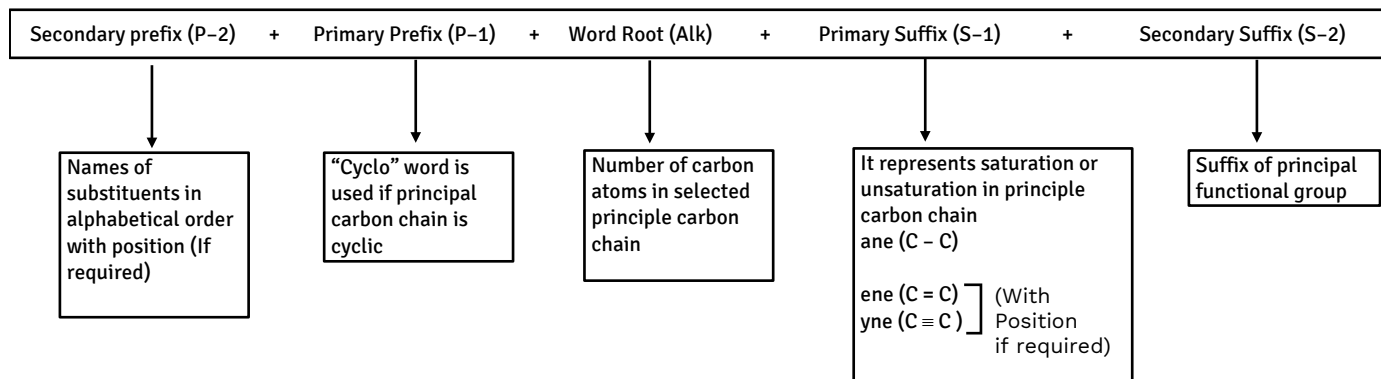


- (1) Ethylene Glycol
(2) Ethene dialcohol
(3) Glycerol
(4) Ethylene alcohol

IUPAC Name or Geneva Name System

(International union of pure and applied chemistry)

1. FORMAT :



No. of carbon in PCC

1C $\longrightarrow$
2C $\longrightarrow$
3C $\longrightarrow$
4C $\longrightarrow$
5C $\longrightarrow$
6C $\longrightarrow$
7C $\longrightarrow$
8C $\longrightarrow$
9C $\longrightarrow$
10C $\longrightarrow$

Word root

Meth
Eth
Prop
But
Pent
Hex
Hept
Oct
Non
Dec

No. of carbon in PCC

11C $\longrightarrow$
12C $\longrightarrow$
13C $\longrightarrow$
14C $\longrightarrow$
15C $\longrightarrow$
16C $\longrightarrow$
17C $\longrightarrow$
18C $\longrightarrow$
19C $\longrightarrow$
20C $\longrightarrow$

Word root

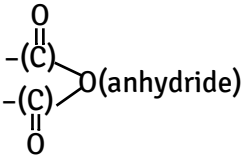
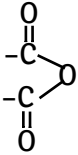
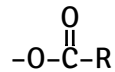
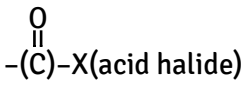
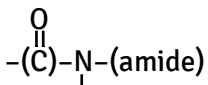
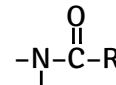
Undec
Dodec
Tridec
Tetradec
Pentadec
Hexadec
Heptadec
Octadec
Nonadec
Icos

2. **Selection of principal functional group (PFG)** : If compound has only one functional group then it is considered as principal functional group.

Note: Following functional group are always considered as substituent.

Functional group	Prefix name
- X	Halo
- OR	Alkoxy
	Epoxy
- NO ₂	Nitro
- NO	Nitroso

- If compound has more than one functional groups then according to IUPAC priority table, Higher priority functional group is considered as principal functional group (PFG) & rest of the functional groups are considered as substituent. (suffix is used for PFG and prefix is used for substituent)

S. No.	Functional group	Prefix	Suffix
1.	- (C) OOH (carboxylic acid)	×	oic acid
	- COOH	carboxy	carboxylic acid
2.	- SO ₃ H (sulphonic acid)	sulpho	sulphonic acid
3.	 (anhydride)	×	oic anhydride
		×	Carboxylic anhydride
4	- (C) OOR (ester)	×	alkyl oate
	- COOR	alkoxy carbonyl /  OR Alkanoyloxy 	alkyl..... carboxylate
5.	 (acid halide)	×	oyl halide
		Halo carbonyl/ halo formyl	carbonyl halide
6.	 (amide)	×	amide
		carbamoyl /  OR alkanoyl amino 	Carboxamide
7	- (C) N (cyanide)	×	Nitrile
	- CN	cyano	Carbonitrile

8.	$-\text{N}\equiv\text{C}$ (isocyanide)	isocyano/carbyl amino	Isonitrile
9.	$-(\text{C})\text{HO}$ (aldehyde)	oxo	Al
	$-\text{CHO}$	formyl	Carbaldehyde
10.	$-(\text{C})-$ (Ketone) O	keto/oxo	One
11.	$-\text{OH}$ (alcohol)	hydroxy	Ol
12.	$-\text{SH}$ (thio alcohol)	mercapto	Thiol
13.	$-\text{N}-$ (amine) $ $	amino	amine

Note: (C) atom written in brackets means it has been included in the parent chain

Ex.



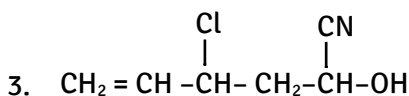
No. of F.G. = 2

P.F.G. = Not Present (Cl is a substituent)



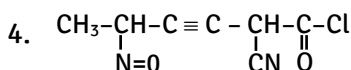
No. of F.G. = 2

P.F.G. = $-\text{OH}$



No. of F.G. = 4

P.F.G. = $-\text{CN}$ [$-\text{Cl}$, $-\text{OH}$ are substituent]



No. of F.G. = 4

P.F.G. = $-\text{COCl}$ [$-\text{N}=\text{O}$, $-\text{CN}$ are substituents]

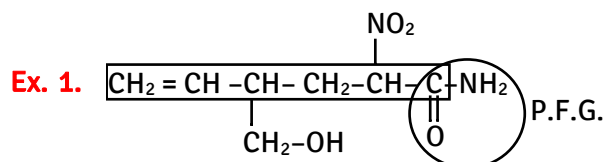
Note: $\text{C}=\text{C}$ & $\text{C}\equiv\text{C}$ are also considered as functional group

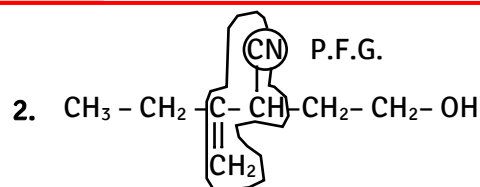
3. Selection of principal carbon chain (PCC)

- Select the longest continuous chain of carbon atoms which have maximum number of PFG, multiple bonds and substituents

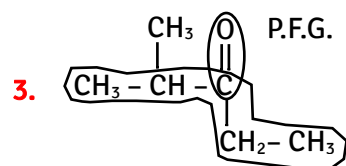
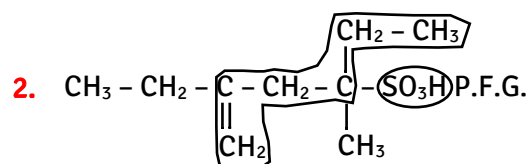
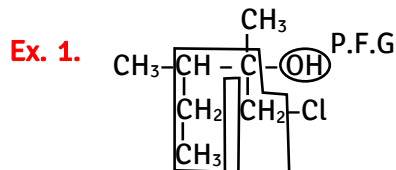
Priority order : PFG > Maximum no. of multiple bond > Maximum no. of carbon atoms > Maximum no. of substituents (Chain length)

Case - I : If P.F.G. has carbon then its carbon is included in principal carbon chain (P.C.C)





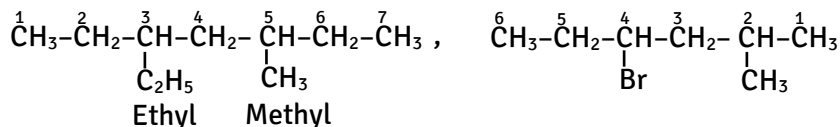
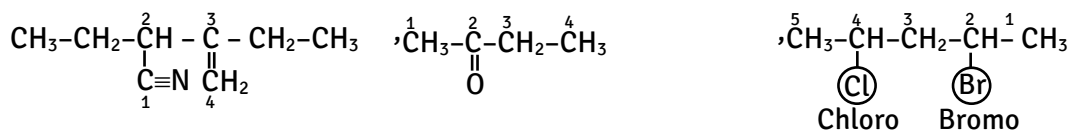
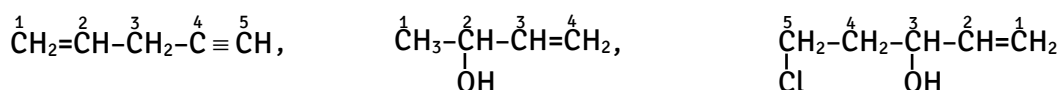
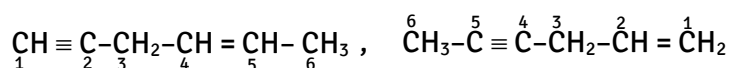
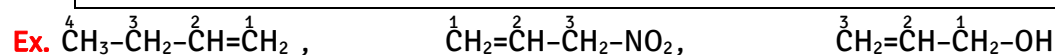
Case – II : If P.F.G. does not have carbon then including the carbon to which P.F.G. is directly attached, C atoms are converted on its both side to get free terminals.



4. Numbering of principal carbon chain (PCC)

- Selected principle carbon chain is numbered from the side where P.F.G, multiple bonds and substituents is getting lowest possible number

Priority order : P.F.G. > Multiple bond (= > ≡) > Lowest set of locants of Substituents > Alphabetical order



2, 3, 4 right numbering

3, 4, 5 wrong numbering

5. **Use of numerical prefixes :** These are used to show multiplicity of more than one identical substituents

- For simple substituents we use di, tri, tetra **Ex.** $\overset{1}{\text{CH}_3}-\overset{2}{\underset{\text{CH}_3}{\text{CH}}}-\overset{3}{\underset{\text{CH}_3}{\text{CH}}}-\overset{4}{\text{CH}_3}$ 2,3-dimethyl butane
- For complex substituents we use bis, tris, **Ex.** $\overset{1}{\text{CH}_2}=\overset{2}{\text{CH}}-\overset{3}{\underset{\text{CH}_2\text{Cl}}{\underset{\text{CH}_2\text{Cl}}{\text{C}}}}-\overset{4}{\text{CH}_2}-\overset{5}{\text{CH}_3}$ 3,3-bis (chloro methyl) pent-1-ene

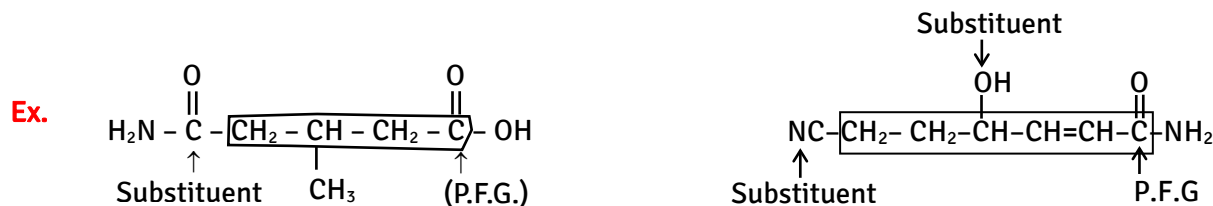
6. **Alphabetical order of substituents :**

Prefix cyclo, iso & Neo are considered in alphabetical order but other prefixes like n, sec, tert, di, tri ... etc are not considered in alphabetical order **Ex.** dimethyl, isopropyl, isobutyl, sec. butyl

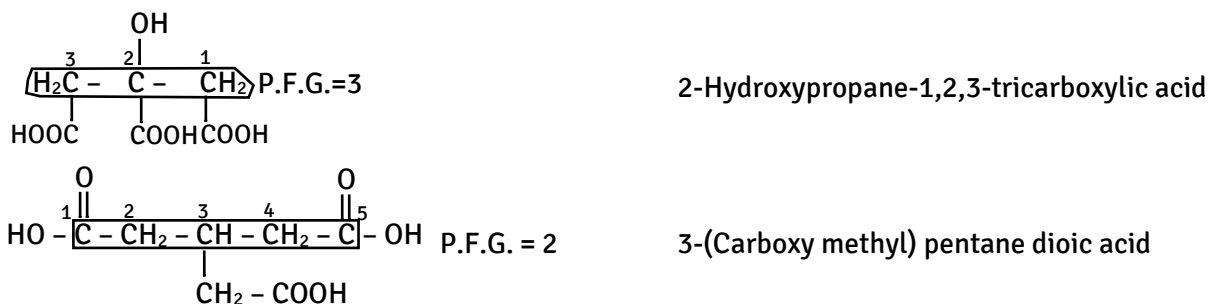
Note: If two or more than two double bonds/triple bonds are present in parent chain then "a" is added to word root (alka).

- Ex. (i)** $\overset{1}{\text{CH}_2}=\overset{2}{\text{CH}}-\overset{3}{\text{CH}}=\overset{4}{\text{CH}_2}$ Buta -1, 3-diene **(iii)** $\overset{1}{\text{CH}_2}=\overset{2}{\text{C}}-\overset{3}{\text{CH}_2}-\overset{4}{\text{C}}\equiv\overset{5}{\text{CH}}$ Pent-1-en-4-yne
- (ii)** $\overset{1}{\text{CH}}\equiv\overset{2}{\text{C}}-\overset{3}{\text{CH}_2}-\overset{4}{\text{C}}\equiv\overset{5}{\text{CH}}$ Penta -1,4-diyne

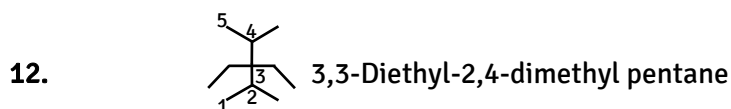
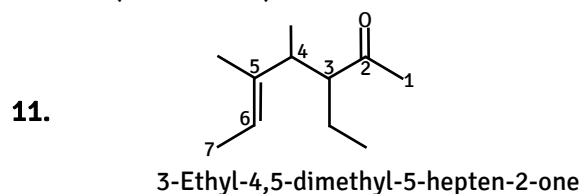
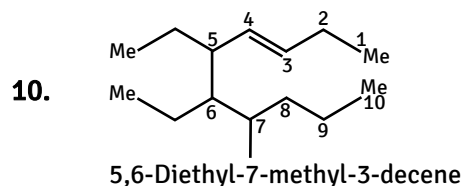
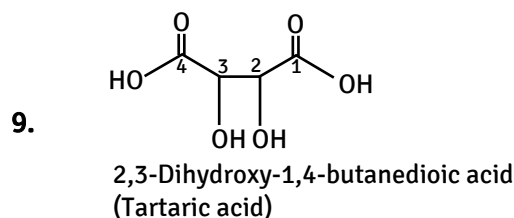
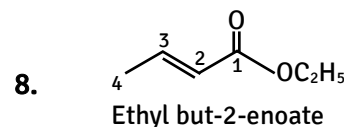
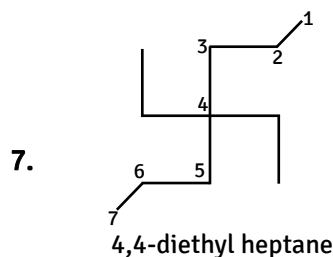
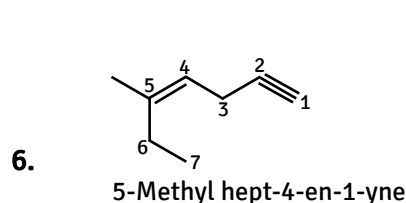
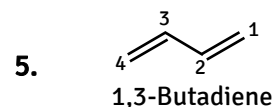
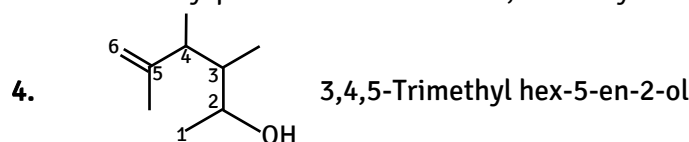
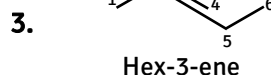
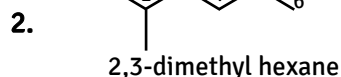
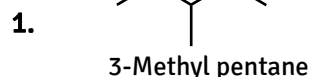
2. If carbon containing F.G. is present as a substituent in a compound (except aldehyde & ketone) then its carbon is not included in p.c.c.



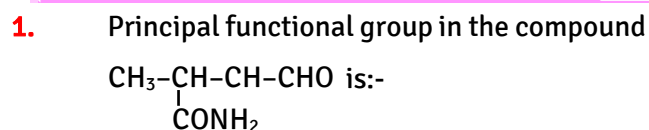
3. If aldehyde or ketone is present as substituent in a compound and if possible then include its carbon in p.c.c. but if not possible (as per p.c.c. selection rule), it is excluded.
4. Two prefixes [formyl and oxo] are suggested for $-\text{CH}=\text{O}$ and their use is conditional
- Formyl $\rightarrow$ When its C is not included in p.c.c.
 - Oxo $\rightarrow$ When its C is included in p.c.c.
5. If more than two carbon containing identical P.F.G are directly attached to principal carbon chain then their carbon is not included in principal carbon chain.
- When C of P.F.G. is out side the p.c.c. its additional suffix is used as secondary suffix.



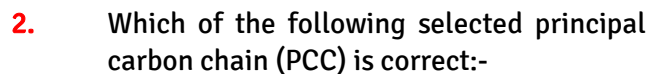
IUPAC name of open chain compounds

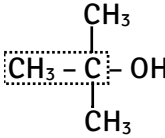
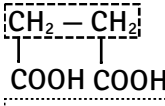
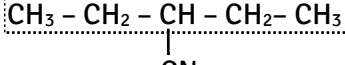
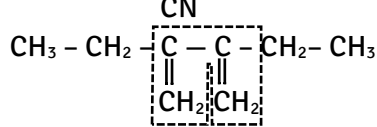


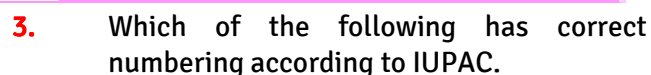
Exercise 1.3



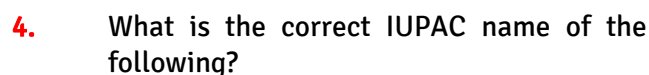
- (1) Aldehyde (2) Amine
(3) Amide (4) Ketone



- (1) 
(2) 
(3) 
(4) 

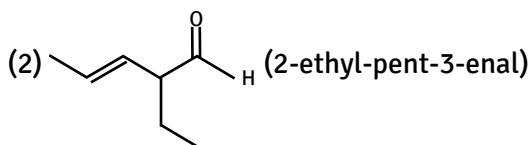
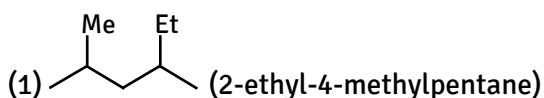


- (1) $\overset{1}{\text{CH}_2}=\overset{2}{\text{CH}}-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_2}-\overset{5}{\text{CN}}$
(2) $\overset{2}{\text{CH}}=\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}}=\overset{5}{\text{CH}}-\overset{6}{\text{C}}\equiv\overset{7}{\text{CH}}$
 ||
 CH_2
(3) $\overset{1}{\text{CH}}\equiv\overset{2}{\text{C}}-\overset{3}{\text{CH}}(\text{OH})-\overset{4}{\text{CH}}=\overset{5}{\text{CH}_2}$
(4) $\text{CH}_3-\text{CH}_2-\overset{2}{\text{CH}}(\text{CN})-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_3}$



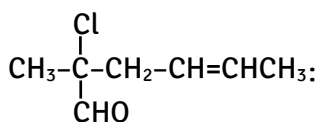
- $\text{CH}_3-\text{CH}=\text{CH}-\underset{\text{COOH}}{\text{CH}}-\text{CH}_2-\text{CH}_3$
(1) 4-Carboxy hex-2-ene
(2) Hex-4-ene-3-carboxylic acid
(3) 2-Ethyl pent-3-enoic acid
(4) 2-(prop-1-enyl) butanoic acid

5. Select the incorrect match of IUPAC name with following structure: _



- (3) CH_3COOMe (methyl ethanoate)
 (4) $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}-\text{CH}_3$ (pent-3-en-1-yne)

6. The IUPAC name of

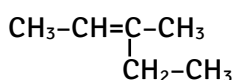


- (1) 5-chloro-5-Formyl hex-2-ene
 (2) 2-methyl-2-chloro hex-4-en-1-al
 (3) 2-chloro-2-methyl hex-4-en-1-carbaldehyde
 (4) 2-chloro-2-methyl hex-4-en-1-al

7. The IUPAC name of $\text{CH}_3-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}-\begin{matrix} \text{CH}_3 \\ \text{CH}_3 \end{matrix}$ is:

- (1) 2-Methylbutan-3-one
 (2) 3-Methylbutan-2-al
 (3) 2-Methylbutan-3-al
 (4) 3-Methylbutan-2-one

8. IUPAC name of the given compound is



- (1) 2-Ethyl-2-butene
 (2) 3-Ethyl-2-butene
 (3) 3-Methyl-3-pentene
 (4) 3-Methyl-2-pentene

9. The IUPAC name of the compound $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{OH}$ is

- (1) 1-Pentanol
 (2) Pentanol
 (3) 2-Methyl-4-butanol
 (4) 3-Methyl-1-butanol

10. IUPAC name of $\text{CH}_3-\text{CH}=\text{CH}-\text{C}\equiv\text{CH}$ is

- (1) Pent-2-en-4-yne
 (2) Pent-3-en-1-yne
 (3) Pent-3-yne-1-en
 (4) Pent-2-yne-1-en

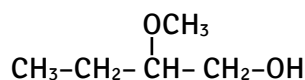
11. IUPAC name of the $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$ is

- (1) 1,1,2,3-Tetramethylethane
 (2) 1,2-Di-isopropylethane
 (3) 2,3-Dimethylbutane
 (4) 2,3,3-Trimethylbutane

12. The IUPAC name of $\text{CH}_3-\underset{\text{NH}_2}{\text{CH}}-\text{CH}_3$ is

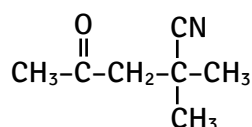
- (1) Dimethyl amine
 (2) 2-Amino propane
 (3) Isopropyl amine
 (4) 2-Propanamine

13. The IUPAC name of the compound



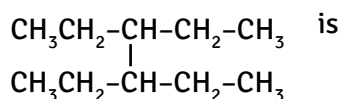
- (1) 2-Methoxy-1-butanol
 (2) 3-Methoxy-1-butanol
 (3) 3-Methoxy-4-butanol
 (4) 1-Hydroxy-2-methoxy-butane

14. The IUPAC name of the compound is



- (1) 4-Cyano-4-methyl-2-oxopentane
 (2) 2-Methyl-4-oxo pentanenitrile
 (3) 2,2-Dimethyl-4-oxo pentanenitrile
 (4) 4-Cyano-4-methylpentan-2-one

15. The IUPAC name of the following compound

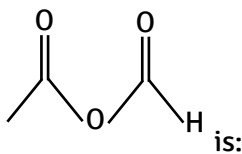


- (1) 3,4-Dimethyloctane
 (2) 3-sec. Pentylpentane
 (3) 3,4-Diethylhexane
 (4) 3,4-Dimethylhexane

16. The IUPAC name of $\text{CH}_3\text{CH}_2\text{NHCHO}$ is

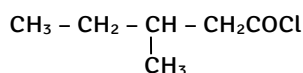
- (1) N-formylethanamine
- (2) Ethylaminomethanal
- (3) N-Ethylmethanamide
- (4) N-Ethylmethanol

17. The IUPAC name of



- (1) Acetic anhydride
- (2) Formyl anhydride
- (3) Butane-2,4-dione
- (4) Ethanoic methanoic anhydride

18. The correct IUPAC name of:

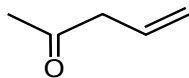


- (1) 3-Methylpentanoyl chloride
- (2) 3-Methylbutanoyl chloride
- (3) 1-Chloro-3-ethylbutanone
- (4) 1-Chloro-3-methyl pentanone

19. The IUPAC name of $\text{N}\equiv\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$ is

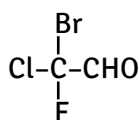
- (1) 3-Hydroxy propanecarbonitrile
- (2) 3-Hydroxy propanenitrile
- (3) 2-Hydroxy ethyl cyanide
- (4) 1-Hydroxy-2-cyanoethane

20. The IUPAC name for the given compound is:



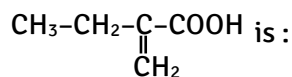
- (1) 2-Acetyl prop-1-ene
- (2) Pent-1-en-4-one
- (3) Pent-4-en-2-one
- (4) Formyl propene

21. The IUPAC name of the given compound is



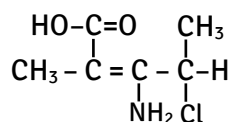
- (1) 2-Bromo-2-chloro-2-fluoroethanal
- (2) 1-Bromo-1-chloro-2-fluoroethanal
- (3) 2-Fluoro-2-chloro-2-bromoethanal
- (4) 1-Fluoro-1-chloro-1-bromoethanal

22. The correct IUPAC name of



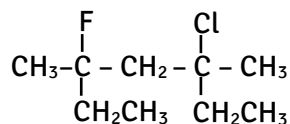
- (1) 2-Methyl butanoic acid
- (2) 2-Ethylprop-2-enoic acid
- (3) 2-Carboxy-1-butene
- (4) None of the above

23. The IUPAC name of compound



- (1) 2-amino-3-chloro-2-methyl-2-pentenoic acid
- (2) 3-amino-4-chloro-2-methyl-2-pentenoic acid
- (3) 4-amino-3-chloro-2-methyl-2-pentenoic acid
- (4) All of the above

24.



The IUPAC name of this compound is :

- (1) 2-chloro-2,4-diethyl-4-fluoropentane
- (2) 3-fluoro-5-chloro-3-methyl-5-ethylhexane
- (3) 3-chloro-5-fluoro-3,5-dimethylheptane
- (4) 3,5-dimethyl-5-fluoro-3-chloroheptane

25. The I.U.P.A.C. name of



- (1) 2-Ethyl butan-1-ol
- (2) 2-Methyl pentan-1-ol
- (3) 2-Ethyl pentan-1-ol
- (4) 3-Ethyl butan-1-ol

26. The correct IUPAC name of 2-ethyl-3-pentyne is :

- (1) 3-methyl hex-4-yne
- (2) 4-ethyl pent-2-yne
- (3) 4-methyl hex-2-yne
- (4) None of these

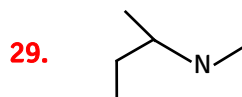
27. The IUPAC name of the given structure is :-

$$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{CH}-\text{CHO} \\ | \quad \quad | \\ \text{HOOC} \quad \text{COOH} \end{array}$$

- (1) 3-Amino-2-formyl butane-1, 4-dioic acid
- (2) 3-Amino-2, 3-dicarboxy propanal
- (3) 2-Amino-3-formyl butane-1, 4-dioic acid
- (4) 1-Amino-2-formyl succinic acid

28. I.U.P.A.C. name of $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}_2\text{Br}$ is :-

- (1) 1-bromopentane
- (2) 2-methyl-4-bromopentane
- (3) 1-bromo-3-methylbutane
- (4) 2-methyl-3-bromopropane



IUPAC name of above compound is :-

- (1) N, 1- di methyl propanamine
- (2) N-methyl-2-butanamine
- (3) Sec. butyl methyl amine
- (4) 1-methyl-1-methylamino propane

30. The correct name of 3,3-dimethyl propanamide is

- (1) 2-Methyl butanamide
- (2) 3-Methyl butanamide
- (3) Iso-propyl ethanamide
- (4) Iso propyl acetamide

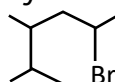
31. IUPAC name will be $\begin{array}{c} \text{CH}_2-\text{CH}-\text{CH}_2 \\ | \quad | \quad | \\ \text{CN} \quad \text{CN} \quad \text{CN} \end{array}$

- (1) 1,2,3-Tricyano propane
- (2) Propane 1,2,3- trinitrile
- (3) 1,2,3-cyano propane
- (4) Propane-1,2,3-tricarbonitrile

32. The IUPAC name of compound-

$$\begin{array}{c} \text{HOOC}-\text{CH}-\text{COOH} \\ | \\ \text{COOH} \end{array}$$

- (1) Tricarboxy methane
- (2) Propane trioic acid
- (3) Tributanoic acid
- (4) Methanetricarboxylic acid

33. The IUPAC name of  is -

- (1) 2-bromo-4-isopropylpentane
- (2) 2, 3-dimethyl-5-bromohexane
- (3) 2-bromo-4, 5-dimethylhexane
- (4) 5-bromo-2, 3-dimethylhexane

34. The IUPAC name of $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{COOH}$ is :

- (1) Cinnamic acid
- (2) 1-Phenyl-2-carboxy ethane
- (3) 3-Phenyl prop-2-enoic acid
- (4) Dihydroxy-3-phenyl propionic acid

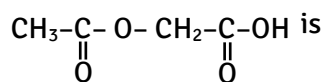
35. The IUPAC name of $\text{C}_6\text{H}_5-\underset{\text{C}_6\text{H}_5}{\text{CH}}-\text{CH}_2-\text{CCl}_3$

- is
- (1) 1,1,1-Trichloro-3,3-diphenyl propane
 - (2) 1,1-Diphenyl-3,3,3-trichloro propane
 - (3) Both (1) and (2)
 - (4) 3,3,3-trichloro-1,1-diphenyl propane

36. Which one of the following I.U.P.A.C. names is correct ?

- (1) 2-Methyl-3-ethyl pentane
- (2) 2-Ethyl-3-methyl pentane
- (3) 3-Ethyl-2-methyl pentane
- (4) 3-methyl-2-ethyl pentane

37. The IUPAC name of



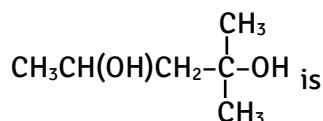
- (1) 1-acetoxy acetic acid
- (2) 2-acetoxy ethanoic acid
- (3) 2-ethanoyl oxyacetic acid
- (4) 2-ethanoyl oxyethanoic acid

38. $\text{CH}_3-\text{O}-\underset{\text{O}}{\text{C}}-\text{CH}_2-\text{COOH}$

The correct IUPAC systematic name of the above compound is :

- (1) 2-Acetoxy ethanoic acid
- (2) 2-Methoxy carbonyl ethanoic acid
- (3) 3-Methoxy formyl ethanoic acid
- (4) 2-Methoxy formyl acetic acid

39. The IUPAC name of

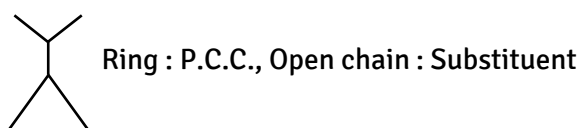
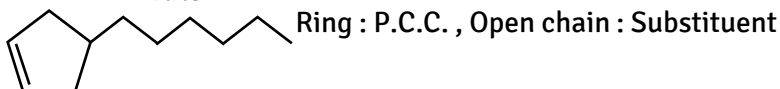
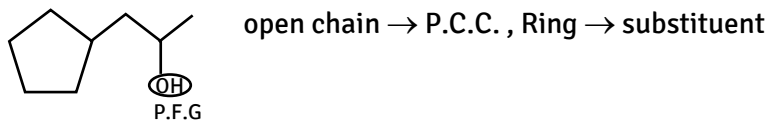


- (1) 1, 1-Dimethyl-1, 3-butanediol
- (2) 4-Methyl-2, 4-pentanediol
- (3) 1, 3, 3-Trimethyl-1, 3-propanediol
- (4) 2-Methyl-2, 4-pentanediol

IUPAC names of Alicyclic compounds

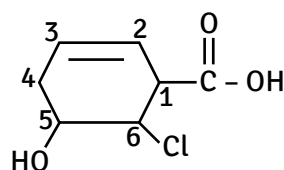
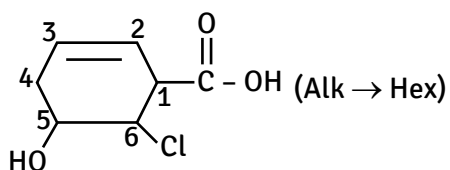
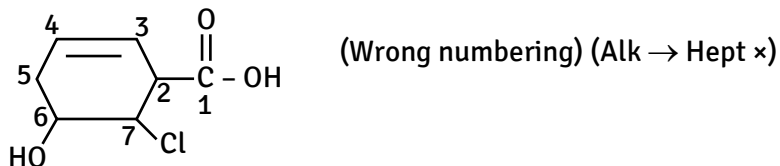
1. Format : P-2 + P-1(cylco) + alk + S-1 + S-2
2. If compound is composed of ring and open chain then principal carbon chain is selected according to given priority order.
Part containing (P.F.G.) > part containing (M.B.) > part containing more carbon in chain > part containing substituent.
If every thing is same in both part then priority is given to cyclic part.

Ex.

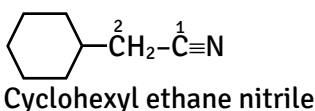


3. If carbon containing functional group is directly attached with cyclic ring then it is considered as part of parent chain but carbon of functional group is not included in parent chain.

Ex.

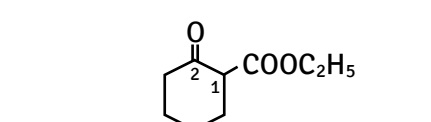
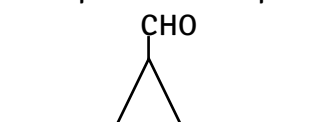
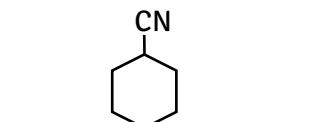


6-chloro-5-hydroxy cyclo hex-2-ene carboxylic acid
P-2 P-1 alk S-1 S-2

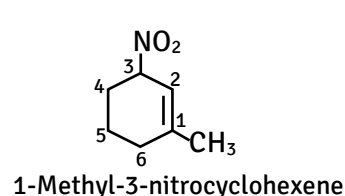
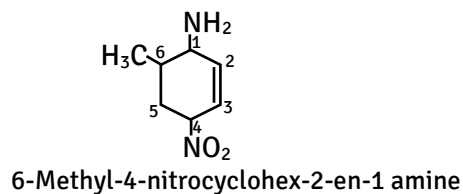
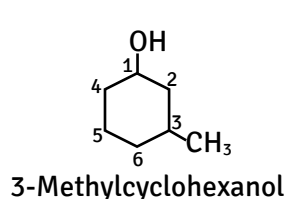


Note : Rest of the rules are same as open chain compound.

Ex.

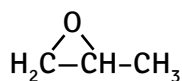


Some other examples



Exercise 1.4

1. The IUPAC name for the compound



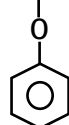
- (1) Propylene oxide
(2) 1,2-Oxopropane
(3) 1,2-Epoxypropane
(4) Methyloxirane

2. What is the structure of 3-Phenylpentane?

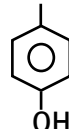
- (1) $\text{CH}_3\text{-CH-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$



- (2) $\text{CH}_3\text{-CH}_2\text{-CH-CH}_2\text{-CH}_3$



- (3) $\text{CH}_3\text{-CH}_2\text{-CH-CH}_2\text{-CH}_3$



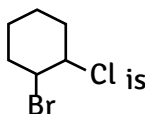
- (4) $\text{CH}_3\text{-CH}_2\text{-CH-CH}_2\text{-CH}_3$



3. Correct IUPAC name of is:

- (1) 1-Ethyl-3,3-dimethylcyclohexane
(2) 3-Ethyl-1,1-dimethylcyclohexane
(3) 1,1-Dimethyl-3-ethylcyclohexane
(4) 3,3-Dimethyl-1,1-ethylcyclohexane

4. The IUPAC name of the compound



- (1) 1-Chloro-2-bromocyclohexane
(2) 1,2-Bromochlorocyclohexane
(3) 4-Bromo-3-chlorocyclohexane
(4) 1-Bromo-2-chlorocyclohexane

5. The IUPAC name of is:

- (1) 3-Methylcyclo-1-buten-2-ol
(2) 4-Methylcyclo-2-buten-1-ol
(3) 4-Methylcyclo-1-buten-3-ol
(4) 2-Methylcyclo-3-buten-1-ol.

6. Which is correct IUPAC name:

- (1) $\text{CH}_3\text{-C(=O)-CH}_2\text{-Br}$ 1-Bromopropanone



- (2) Cyclohexane-1,3,5-trione

- (3) 2-Ketocyclopentanecarbaldehyde

- (4) All are correct

7. The IUPAC name of is

- (1) 3-Cyclopropyl-3-ethyl-2-propene
(2) 1-Cyclopropyl-1-ethylpropene
(3) 3-Cyclopropyl-2-pentene
(4) (1-Ethyl-1-propenyl) cyclopropane

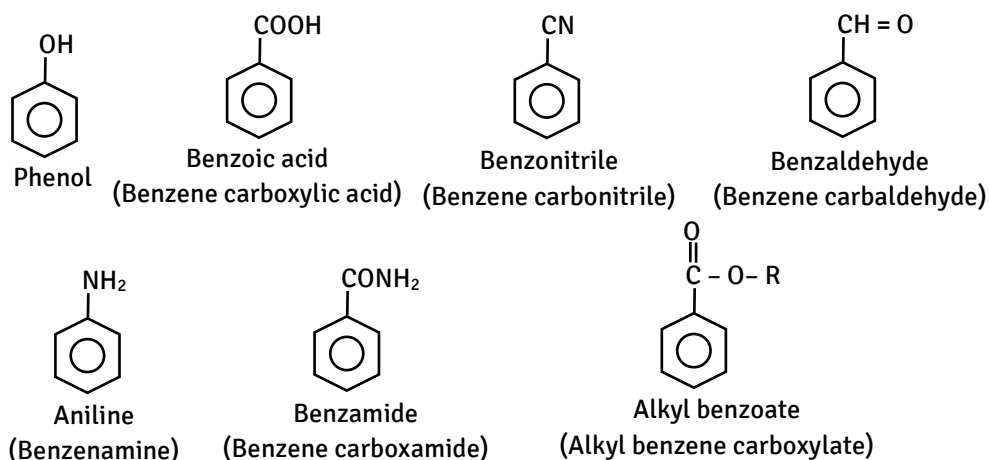
8. The IUPAC name of is

- (1) 1-Ethynyl-1-hydroxycyclohexane
(2) 1-(Hydroxycyclohexyl) ethyne
(3) 1-Ethynylcyclohexanol
(4) 1-Acetylenyl-1-hydroxycyclohexane

Nomenclature of Aromatic Compounds

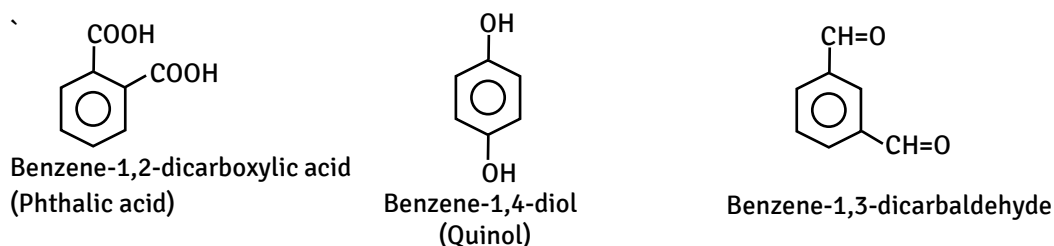
- If one principal functional group is present on benzene ring, its common name is accepted in IUPAC system.

Ex.



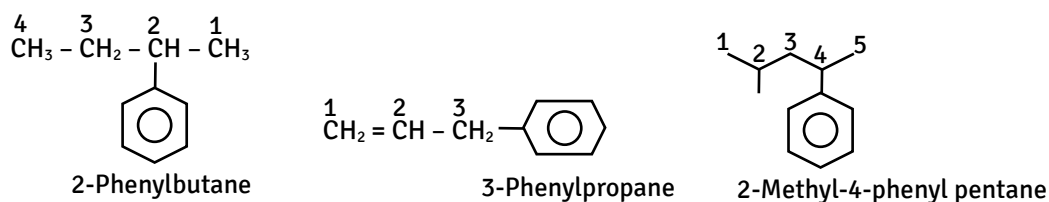
- If more than one principal functional groups are present on ring then they are named as derivative of benzene and suffix for P.F.G. is used as per IUPAC rules.

Ex.

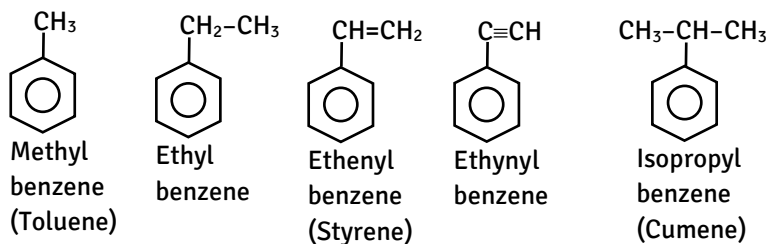


- In aromatic hydrocarbon if side chain is also present then except methyl benzene, ethyl benzene, ethenyl benzene, isopropyl benzene, ethynyl benzene, side chain is considered as main part and ring is considered as a substituent and is named as phenyl.

Ex.



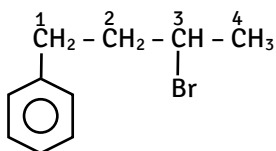
Exception



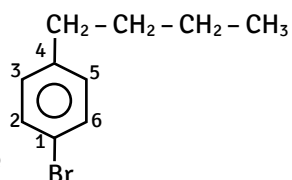
Note: Toluene is also valid in IUPAC

4. In Hydrocarbon derivative, the part having secondary functional group [Ring or side chain] is considered as main part and other is considered as substituent.

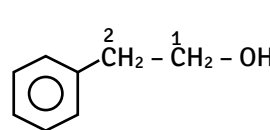
Ex.



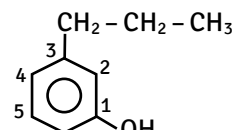
3-Bromo-1-phenyl butane



1-Bromo-4-butylbenzene



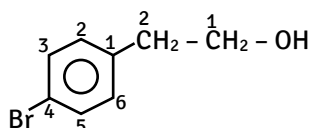
2-phenyl ethan-1-ol



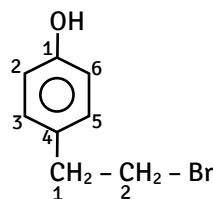
3-propylphenol

5. If functional group is present in both parts the part having principal functional group is considered as the main part.

Ex.

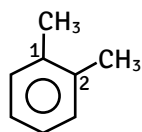


2-[4-bromophenyl]
ethanol

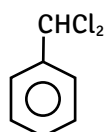


4-[2-Bromoethyl]
phenol

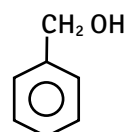
Ex.



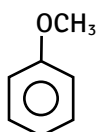
1,2-Dimethyl benzene
(o-Xylene)



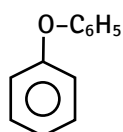
Phenyldichloromethane
(Benzal chloride)



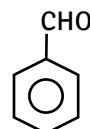
Phenyl methanol
(Benzyl alcohol)



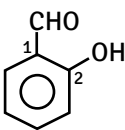
Methoxy benzene
(Anisole)



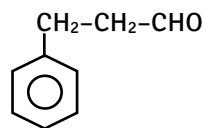
Phenoxy benzene
(Diphenyl ether)



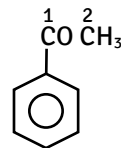
Benzaldehyde



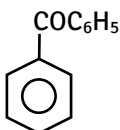
2-hydroxy benzaldehyde
(salicylaldehyde)



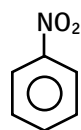
3-phenylpropanal



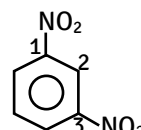
1-phenyl ethenone
(Acetophenone)



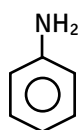
Diphenylmethanone
(Benzophenone)



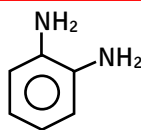
Nitrobenzene



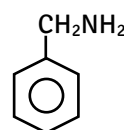
1,3- Dinitrobenzene
(m-dinitrobenzene)



Benzenamine
(Aniline)

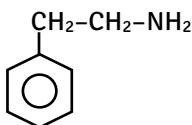


Benzene-1,2-diamine
(O-Phenylenediamine)

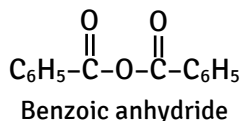


Phenyl methanamine
(Benzylamine)

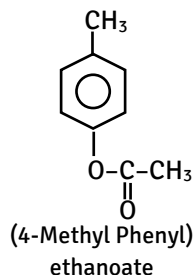
Ex.



2-Phenyl ethanamine



Benzoic anhydride



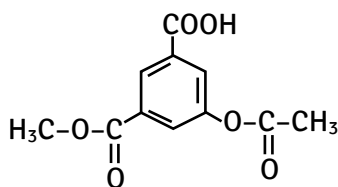
(4-Methyl Phenyl)
ethanoate

Exercise 1.5

1. Which of the following name will be incorrect?

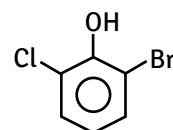
- (1) 3, 6-Dimethyl cyclohexene
 (2) 1, 6-Dimethyl cyclohexene
 (3) 6, 6-Dimethyl cyclohexene
 (4) 1, 5-Dimethyl cyclohexene

2. The correct IUPAC name of



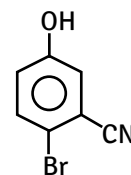
- (1) 3-methoxy carbonyl-5-ethanoyloxy benzene carboxylic acid
 (2) 5-methoxy carbonyl-3-ethanoyloxy benzene carboxylic acid
 (3) 5-ethanoyloxy -3- methoxy carbonyl-benzene carboxylic acid
 (4) 3-ethanoyloxy -5- methoxy carbonyl-benzene carboxylic acid

3. The correct IUPAC name of following compound is:



- (1) 2-Bromo -6-chloro phenol
 (2) 1-Bromo -2-hydroxy -3-chloro benzene
 (3) 2-chloro -6- bromo phenol
 (4) 1-chloro -2- hydroxy -3- bromo benzene

4. The correct IUPAC name of the following compound is

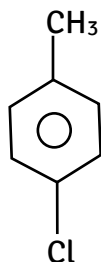


- (1) 4-Bromo-3-cyano phenol
 (2) 6-Bromo-3-hydroxy benzo nitrile
 (3) 2-cyano-4-hydroxy bromo benzene
 (4) 2-Bromo-5-hydroxy benzo nitrile

5. The IUPAC name of is

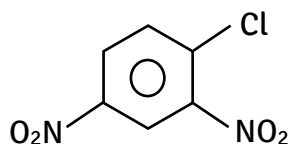
- (1) 4-hydroxy -2-pentanone
 (2) 4-oxo-2-Pentanal
 (3) Pentane-4-ol-2-one
 (4) Pentane-2-one-4-ol

6. Correct IUPAC name of following compound is:

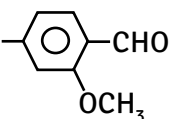


- (1) 1-Methyl-4-chlorobenzene
- (2) 1-Chloro-4-methylbenzene
- (3) 4-Chlorotoluene
- (4) Both (2) & (3)

7. Correct IUPAC name of following compound is:

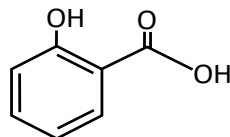


- (1) 4-Chloro-1,3-dinitrobenzene
- (2) 1-Chloro-2,4-dinitrobenzene
- (3) 2-Chloro-1,5-dinitrobenzene
- (4) 2,4-Dinitro-1-chlorobenzene

8. The IUPAC name of 

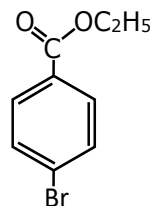
- (1) 2-Methoxy-4-nitrobenzaldehyde
- (2) 4-Nitro-2-methoxybenzaldehyde
- (3) 3-Methoxy-4-formylnitrobenzene
- (4) 2-Formyl-4-nitroanisole

9. The IUPAC name of



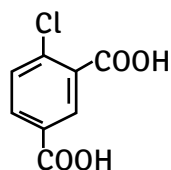
- (1) 2-Carboxyphenol
- (2) 2-Hydroxybenzoic acid
- (3) 1-Carboxy-2-hydroxybenzene
- (4) (2-Hydroxyphenyl) methanoic acid

10. Give IUPAC name of the following:



- (1) Ethyl-1-bromo 4-benzoate
- (2) Ethyl 4-bromobenzoate
- (3) Bromobenzoate
- (4) Bromobenzene ester

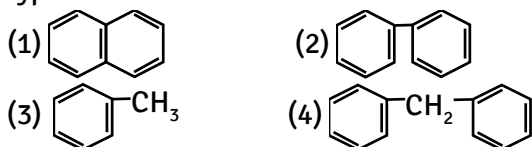
11. IUPAC name of given compound is :-



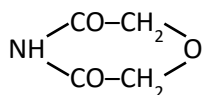
- (1) 4-chloro benzene-1,3-dicarboxylic acid
- (2) 5-carboxy-2-chloro benzoic acid
- (3) 4-chloro benzene-1,3-dioic acid
- (4) Both (1) and (3)

Exercise 2

1. Which of the following has more than one type of H-atom:

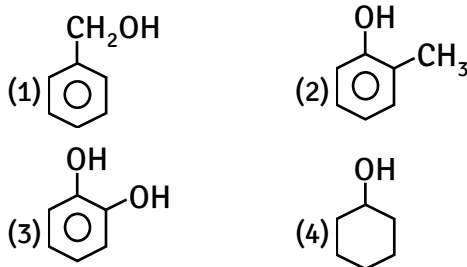


2. Number of hetero atoms in the ring is:



- (1) 1 (2) 2
(3) 4 (4) 3

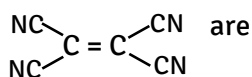
3. The homologue of phenol is:



4. Number of π electrons present in naphthalene is

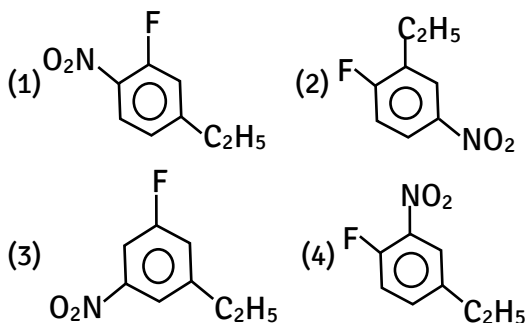
- (1) 4 (2) 5
(3) 10 (4) 14

5. Carbon atoms in the compound

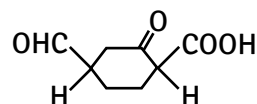


- (1) sp hybridized
 (2) sp^2 hybridized
 (3) sp and sp^2 hybridized
 (4) sp , sp^2 and sp^3 hybridized

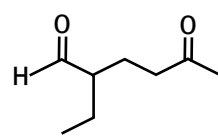
6. Structure formula of 4-Ethyl-1-fluoro-2-nitrobenzene is:



polyfunctional compound is:

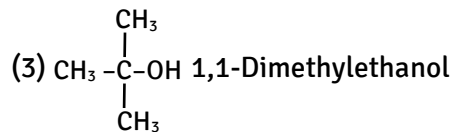
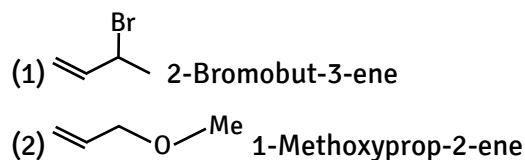


- (1) 2, 4-Dioxo cyclohexanoic acid
 (2) 2,4-Dioxocycloheptanoic acid
 (3) 4-Formyl-2-oxocyclohexane-1-carboxylic acid
 (4) 2,4-Dioxocyclohexane-1-carboxylic acid

8.  IUPAC name is:

- (1) 4-Ethyl-5-oxopentanoic acid
 (2) 4-Ethyl-4-formylbutanoic acid
 (3) 4-Formylhexanoic acid
 (4) 4-Oxohexanoic acid

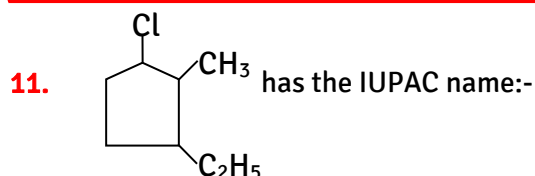
9. Incorrect match is:



- (4) All

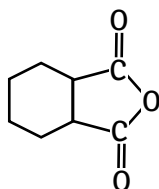
10. The IUPAC name of $\text{Cl}-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$ is:-

- (1) Ethoxyformyl chloride
 (2) Ethoxymethanoyl chloride
 (3) Ethylchloromethanoate
 (4) Ethoxycarbonyl chloride

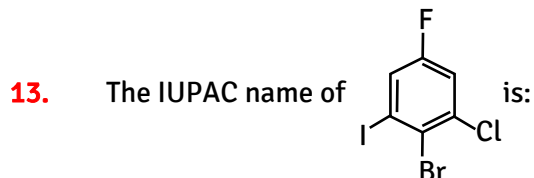


- (1) 3-Chloro-1-ethyl-2-methyl cyclopentane
- (2) 1-Chloro-3-ethyl-2-methyl cyclopentane
- (3) 4-Chloro-1-ethyl-5-methyl cyclopentane
- (4) 2-Chloro-5-ethyl-1-methyl cyclopentane

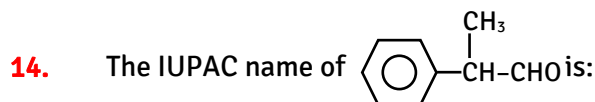
12. IUPAC name of the following compound:



- (1) Cyclohexane-1,2-dicarboxylic acid
- (2) Cyclohexane-1,2-dicarboxylic anhydride
- (3) Cyclohexane ethanoic anhydride
- (4) Cyclohexanebutanoic anhydride

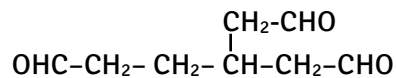


- (1) 1-Bromo-2-chloro-3-fluoro-6-iodo benzene
- (2) 2-Bromo-1-chloro-5-fluoro-3-iodo benzene
- (3) 4-Bromo-2-chloro-5-iodo-1-fluoro benzene
- (4) 2-Bromo-3-chloro-1-iodo-5-fluoro benzene



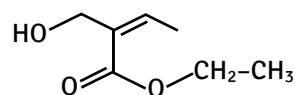
- (1) 2-Phenylpropan-3-al
- (2) Formylethylbenzene
- (3) 2-Phenylpropanal
- (4) Ethylformylbenzene

15. The IUPAC name of



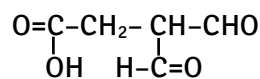
- (1) 4, 4-di(formylmethyl) butanal
- (2) 2-(oxoethyl) butane-1, 4-dicarbaldehyde
- (3) 3-Acetal hexane-1-6-dial
- (4) 3-(2-oxoethyl) hexane-1, 6-dial

16. The IUPAC name of :

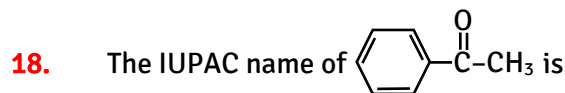


- (1) Ethoxy-2-methyl hydroxy-2butenoate
- (2) Ethyl-2-Hydroxy methyl-2-butenoate
- (3) Ethyl-2-hydroxy methyl-2-butene carboxylate
- (4) Ethoxy-2-Methyl hydroxy-2-butene Carboxylate

17. The correct IUPAC name of the following compound is :

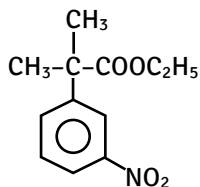


- (1) 3,3-diformyl propanoic acid
- (2) 3-formyl-4-oxo-butanoic acid
- (3) 3,3-dioxo propanoic acid
- (4) 3,3-dicarbaldehyde propanoic acid



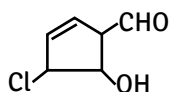
- (1) Phenyl ethanone
- (2) Methyl phenyl ketone
- (3) Acetophenone
- (4) Phenyl methyl ketone

19. IUPAC name of the given compound is



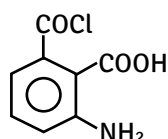
- (1) Ethyl-2-methyl-2-(m-nitro phenyl) propanoate
- (2) Ethyl-2-methyl-2-(o-nitro phenyl) propanoate
- (3) Ethyl-2-methyl-2-(3-nitro phenyl) propanoate
- (4) Ethyl-2-methyl-2-(3-nitro phenyl) propanoic acid

20. IUPAC name of given compound



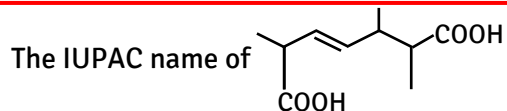
- (1) 2-hydroxy-3-chloro-cyclopent-4-enal
- (2) 3-Chloro-2-hydroxy-cyclopent-4-enal
- (3) 4-Chloro-5-hydroxy-cyclopent-2-enecarbaldehyde
- (4) 4-Chloro-5-hydroxy cyclohex-4-enal

21. IUPAC name of given compound is



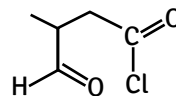
- (1) 6-amino-2-Chlorocarbonyl benzene Carboxylic acid
- (2) 6-amino-2-Chloroformyl benzoic acid
- (3) 1-amino-2-carboxy-3-Chloroformly benzene
- (4) 2-amino-6-Chlorocarbonyl-benzene Carboxylic acid

22.



- (1) 6-carboxy-2,3-dimethyl-4-heptenoic acid
- (2) 2, 3, 6- trimethyl-4-heptene-1,7-dioic acid
- (3) 2,5,6-trimethyl-3-heptene-1,7-dioic acid
- (4) 6-carboxy-2-5-dimethyl-3-heptenoic acid

23.

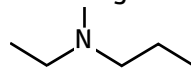


IUPAC name this compound is :-

- (1) 1-chloro-3-methyl butane-1, 4-dione
- (2) 3-formyl-butanoyl chloride
- (3) 3-methyl-4-oxo-butanoyl chloride
- (4) None

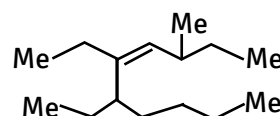
24.

Write the correct IUPAC name of the following bond line formula?



- (1) N-Methyl N-ethyl propan-1-amine
- (2) N-Ethyl N-methyl propan-1-amine
- (3) N-Methyl N-propyl ethan-1-amine
- (4) N-Ethyl N-propyl N-methyl amine

25.

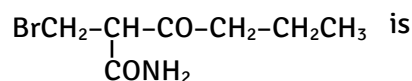


this compound is

- (1) 2, 4, 5-Triethyl-3-nonene
- (2) 5, 6-Diethyl-3-methyl-4-decene
- (3) 2, 4, 5-Triethyl-3-octene
- (4) 3-Heptyl-5-methyl-3-heptane

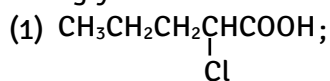
26.

The IUPAC name of

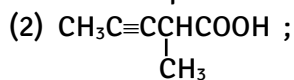


- (1) 2-(Bromomethyl)-3-oxohexanamide
- (2) 1-Bromo-2-amino-3-oxohexane
- (3) 1-Bromo-2-amino-n-propyl ketone
- (4) 3-Bromo-2-propyl-propenamide

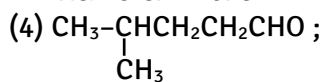
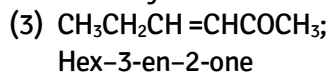
27. Which of the following compound is wrongly named?



2-chloro pentanoic acid

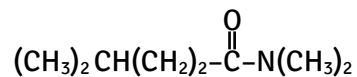


2-Methyl hex-3-enoic acid



4-Methyl pentanal

28. The IUPAC name of given compound is



(1) N, N, 4-Trimethylpentanamide

(2) Dimethylamino-4-methylpentanone

(3) N, N-Dimethylamino-4-methyl
pentan amide

(4) 2-Methyl-5-oxodimethylpentanamide

Exercise 3

1. Which of the following statements is incorrect?
- (1) Aromatic compounds may have hetero atom in the ring
 - (2) Tropolone is a non-benzenoid compound
 - (3) Pyridine is a heterocyclic aromatic compound
 - (4) Aniline is a non-benzenoid compound

2. Consider the following statements :
- I. The presence of functional group disables systematisation of organic compound into different classes.
 - II. In polyfunctional compounds, one of the functional group is chosen as the principal functional group.
 - III. The order of priority for functional group is $-\text{COCl} > -\text{COOR} > -\text{SO}_3\text{H} > -\text{COOH}$
 - IV. $\text{CH}_2(\text{OH})\text{CH}_2(\text{OH})$ is named as Ethane-1,2-diol.

Read the given statements and choose the correct option.

- | | |
|-------------------|--------------------|
| (1) Both I and II | (2) Both II and IV |
| (3) I, II and III | (4) All of these |

3. **Assertion :** $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CHO}$ is

2-chloro-4-methyl- pentanal.

Reason : Number is given according to the alphabetical order of substituents.

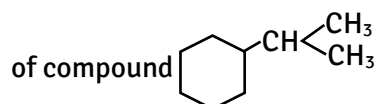
- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

4. **Assertion :** The IUPAC name for $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}_2$ is pent-4-en-1-yne

Reason : A tripe bond is always given a lower locant than a double bond.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

5. **Assertion :** The correct IUPAC name



[1-Methylethyl]cyclohexane.

Reason : It is derivative of cyclohexane because the no. of C-atoms in the ring is more than the side chain.

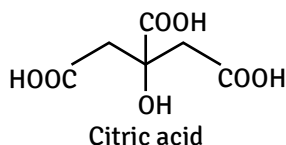
- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

6. **Assertion :** $\text{NC}-\text{CH}_2\text{CH}_2\text{COOH}$ is called 3-cyano propanoic acid.

Reason : While naming polyfunctional compounds, $-\text{COOH}$ group gets preference over cyano for selecting principal functional group.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

7. **Assertion :** The IUPAC name of citric acid is 2-hydroxy-propane-1,2,3-tricarboxylic acid.



Reason : When an unbranched C Chain is directly linked to more than two C-containing principal functional groups, then it is named as a derivative of the parent alkane which does not include the C atoms of the functional groups.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

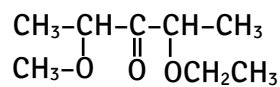
8. **Assertion :** IUPAC name of urea is aminomethanamide.



principal functional group is “amide” and “amino behave as substituent.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

9. **Assertion :** The IUPAC name of



2-Ethoxy-4- methoxy pentan -3- one.

Reason : large substituents are assigned lower locants.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

10. Which is correctly matched

- (1) Phenanthrene
- (2) Anthracene
- (3) Naphthalene
- (4) Benzene

11. **Assertion:** $\text{CH}_2=\text{CH}-\text{CH}_2-\text{NO}_2$ is named as 1-Nitroprop-2-ene in IUPAC.

Reason: $-\text{NO}_2$ group being secondary functional group gets priority over multiple bonds in numbering.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

12. Assertion: Benzene is an aromatic compound.

Reason: Benzene is homocyclic compound.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

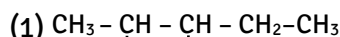
13. Assertion: Homologues can never be isomers.

Reason: Homologues always have different molecular formula while isomers always have same molecular formula.

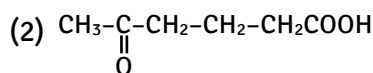
- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

Exercise 4 (Previous Year's Questions)

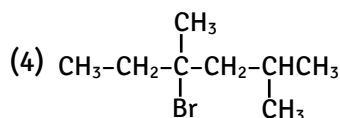
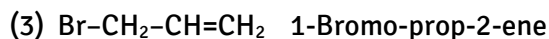
1. Which nomenclature is not according to IUPAC system [AIPMT-2012]



2-Methyl-3 phenylpentane

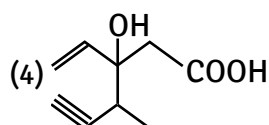
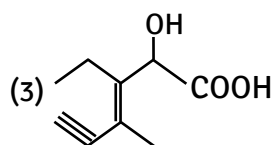
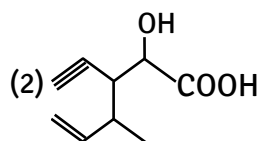
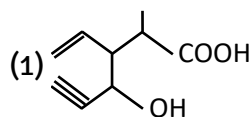


5-oxohexanoic acid



4-Bromo,2,4- dimethylhexane

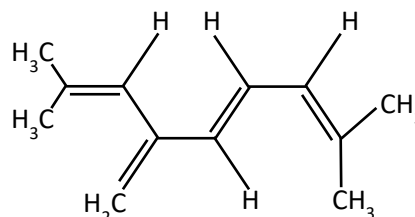
2. Structure of the compound whose IUPAC name is 3-Ethyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid is: [NEET UG-2013]



3. Which of the following organic compound has same hybridization as its combustion product CO_2 ? [AIPMT-2014]

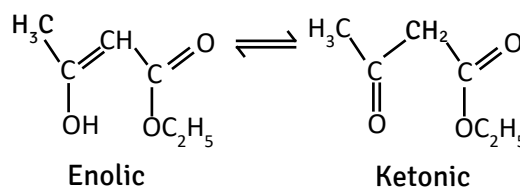
- (1) Ethane
(2) Ethyne
(3) Ethene
(4) Ethanol

4. The total number of π -bond electrons in the following structure is: [AIPMT-2015]



- (1) 8
(2) 12
(3) 16
(4) 4

5. The enolic form of ethyl acetoacetate as below has: [AIPMT-2015]

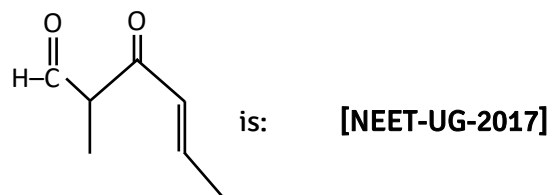


- (1) 16 sigma bonds and 1 pi-bond
(2) 9 sigma bonds and 2 pi-bonds
(3) 9 sigma bonds and 1 pi-bond
(4) 18 sigma bonds and 2 pi-bonds

6. The pair of electrons in the given carbanion, $\text{CH}_3\text{C}\equiv\text{C}^-$, is present in which of the following orbital? [NEET-I-2016]

- (1) 2p
(2) sp^3
(3) sp^2
(4) sp

7. The IUPAC name of the compound



- (1) 5-Formylhex-2-en-3-one
(2) 5-Methyl-4-oxohex-2-en-5-al
(3) 3-Keto-2-methylhex-5-enal
(4) 3-Keto-2-methylhex-4-enal

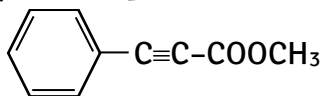
8. Which of the following molecules represents the order of hybridisation sp^2 , sp^2 , sp, sp from left to right atoms? [NEET-UG-2018]

- (1) $\text{HC}\equiv\text{C}-\text{C}\equiv\text{CH}$
(2) $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$
(3) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
(4) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$

9. The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is: [NEET-UG-2019]

- (1) 10σ bonds and 3π bonds
- (2) 8σ bonds and 5π bonds
- (3) 11σ bonds and 2π bonds
- (4) 13σ bonds and no π bonds

10. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound? [NEET-UG-2020 (COVID-19)]

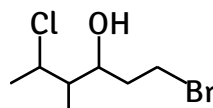


- (1) 7, 5
- (2) 8, 6
- (3) 7, 6
- (4) 8, 5

11. The correct structure of 2,6-Dimethyl dec-4-ene is: [NEET-UG-2021]

- (1)
- (2)
- (3)
- (4)

12. The correct IUPAC name of the following compound is: [NEET-UG-2022]

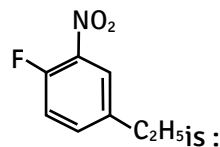


- (1) 1-bromo-5-chloro-4-methylhexan-3-ol
- (2) 6-bromo-2-chloro-4-methylhexan-4-ol
- (3) 1-bromo-4-methyl-5-chlorohexan-3-ol
- (4) 6-bromo-4-methyl-2-chlorohexan-4-ol

13. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is: [NEET-2024]

- (1) 2,3-dimethylbutane
- (2) 2,2-dimethylbutane
- (3) n-hexane
- (4) 2-methylpentane

14. The correct IUPAC name of the compound [RE-NEET-2024]



- (1) 4-Ethyl-1-fluoro-2-nitrobenzene
- (2) 4-Ethyl-1-fluoro-6-nitrobenzene
- (3) 3-Ethyl-6-fluoro-1-nitrobenzene
- (4) 1-Ethyl-4-fluoro-3-nitrobenzene

ANSWER KEY

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	3	2	3	3	1	2	3	4	2	3	2	2	3	1	4	1	4	3

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
Ans.	4	3	4	2	4	2	2	4	3	3	4	2	1	4	1	1				

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	4	3	1	4	4	4	4	2	3	4	1	3	3	3	4	1	2	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	
Ans.	1	2	2	3	1	3	3	3	2	2	4	4	4	3	1	3	4	2	4	

Exercise 1.4

Que.	1	2	3	4	5	6	7	8												
Ans.	3	4	2	4	2	4	3	3												

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11									
Ans.	3	4	1	4	1	4	2	1	2	2	1									

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	2	3	3	4	3	3	4	3	2	2	2	3	4	2	2	1	3	3
Que.	21	22	23	24	25	26	27	28												
Ans.	4	3	3	2	2	1	2	1												

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13							
Ans.	4	2	4	4	1	1	1	1	3	4	4	2	1							

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14						
Ans.	3	3	2	1	4	4	4	2	1	3	1	1	1	1						

Chapter 2

ISOMERISM STRUCTURAL, GEOMETRICAL AND CONFORMATIONAL

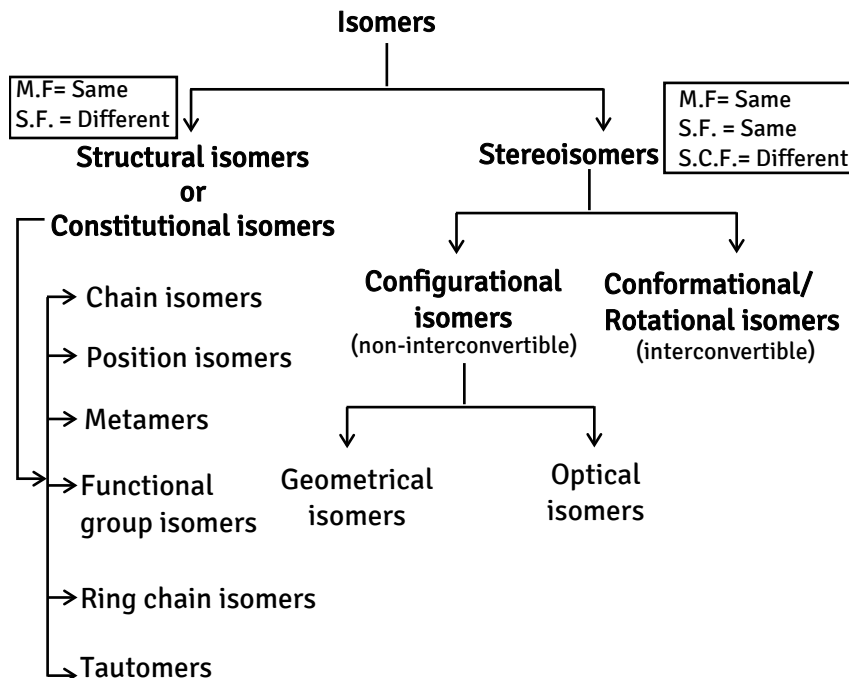
Chapter Summary

- Introduction
- Structural Isomerism
- Stereo Isomerism
- Geometrical Isomerism
- Configuration of geometrical Isomers
- Conformational Isomerism
- Gauche effect

Introduction

Two or more than two organic compounds having the same molecular formula but different physical, chemical or biological properties are called isomers and this phenomenon is called isomerism. The name was given by Berzelius.

Classification of Isomers:

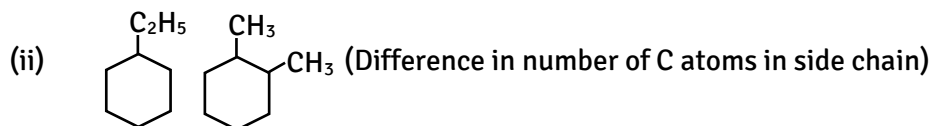
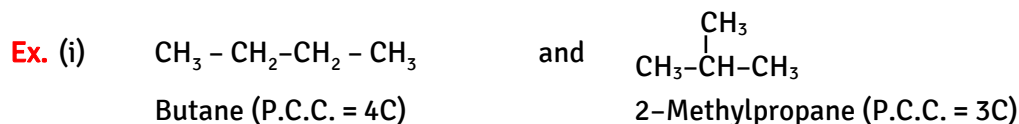


M.F. = molecular formula
S.F. = structural formula
SCF = stereochemical formula

Structural Isomerism

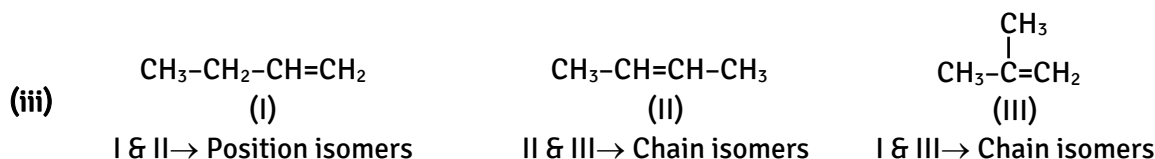
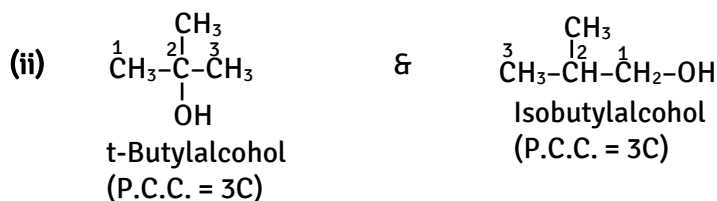
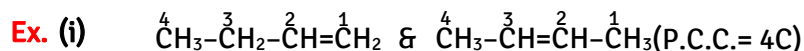
1. Chain Isomerism:

- Compounds belong to same homologous series (same functional group)
- Compounds have different number of carbon atoms in principal carbon chain or in side chain or both.



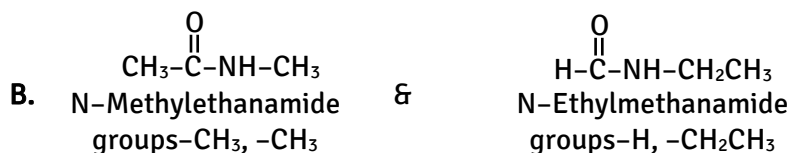
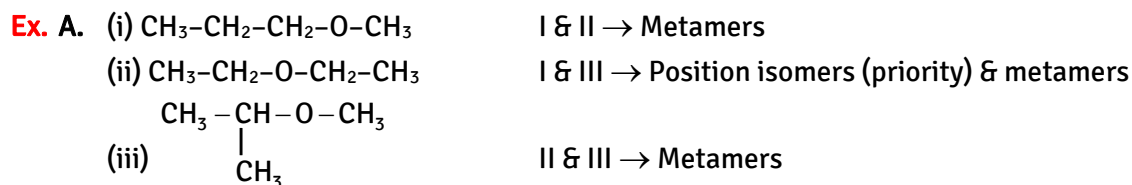
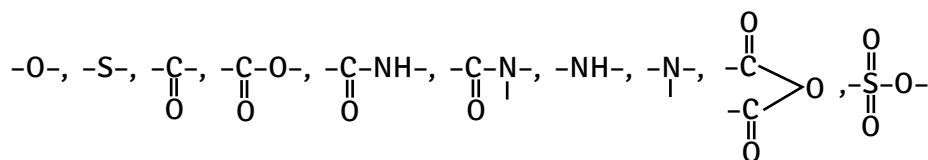
2. Position Isomerism:

- Compounds belong to same homologous series (same functional group)
- Compounds have difference in position of principal functional group, multiple bond or substituent in same carbon chain.



3. Metamerism:

- Compounds have same poly valent function group (P.V.F.G.)
 - Compounds have difference in units (groups) attached with P.V.F.G.
- Polyvalent Functional group [More than one valency] are:



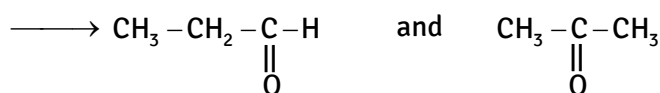
4. Functional Isomerism:

- Compounds having same molecular formula but different functional groups are called functional isomers.
- Compounds belong to different homologous series.
- Members of those different homologous series show functional isomerism, which have same general formula.

Ex. (i) Alcohol and ether
(G.F. $\rightarrow C_nH_{2n+2}O$)



(ii) Aldehydes and ketones
(G.F. $\rightarrow C_nH_{2n}O$)



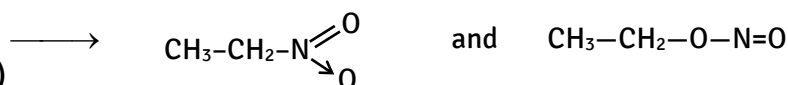
(iii) Acids and ester
(G.F. $\rightarrow C_nH_{2n}O_2$)



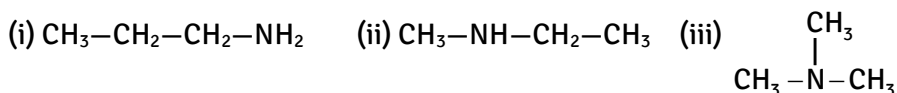
(iv) Cyanide and isocyanide
(G.F. $\rightarrow C_nH_{2n-1}N$)



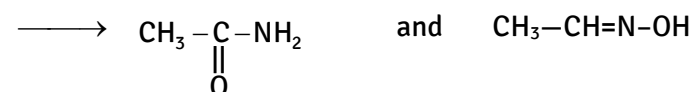
(v) Nitro and Nitrite
(G.F. $\rightarrow C_nH_{2n+1}O_2N$)



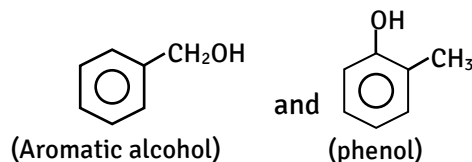
(vi) $1^\circ, 2^\circ, 3^\circ$ amines
(G.F. $\rightarrow C_nH_{2n+3}N$)



(vii) Amide and Oxime
(G.F. $\rightarrow C_nH_{2n+1}ON$)



(viii) Aromatic Alcohol and phenolic compounds:
($C_nH_{2n-6}O$)



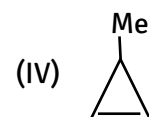
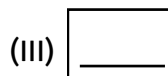
Note: Alkyl halides do not show Functional isomerism.

5. Ring Chain Isomerism (RCI):

- Compounds having same M.F. in which one has open chain and other has closed chain structure are called R.C.I.
- Compounds belong to different homologous series.

Ex. (I) $CH_3-C\equiv C-CH_3$

(II) $CH_2=CH-CH=CH_2$



I & II $\rightarrow$ F.G.I.

I & III → F.G.I. & R.C.I.

I & IV → F.G.I. & R.C.I.

II & III → F.G.I. & R.C.I.

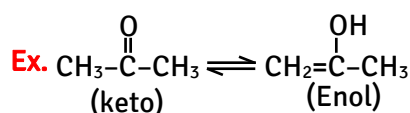
II & IV → F.G.I. & R.C.I.

III & IV → Chain isomers

Note: All R.C.I. are F.G.I. but converse is not true.

6. Tautomerism:

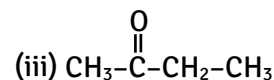
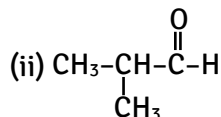
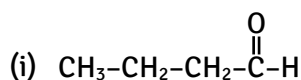
- Two compounds have different functional groups due to shifting of acidic H-atom and change in position of multiple bonds.
- The two isomers exist in dynamic equilibrium.
- The two forms in tautomeric equilibrium are called tautomers of each other.
- The interconvertibility of tautomers is a chemical reaction which can be categorised as acid base reaction.



Solved Example:

Ex. 1 Structural isomeric aldehydes & ketones with molecular formula $\text{C}_4\text{H}_8\text{O}$.

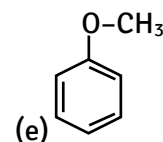
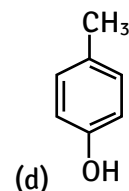
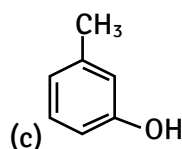
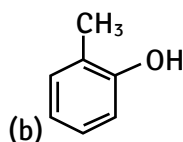
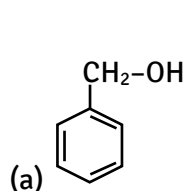
Ans. 3



$\text{C}_4\text{H}_8\text{O}$ (2 aldehyde and 1 ketone) total 3 isomers are possible.

Ex. 2 Benzenoid isomers of $\text{C}_7\text{H}_8\text{O}$.

Ans. 5



- a, b – Functional isomers
- c, d – Position isomers
- a, e – Functional isomers

- b, c – Position isomers
- a, d – Functional isomers
- b, e – Functional isomers

Note: Alcoholic and phenolic groups are Functional isomers.

Exercise 1.1

1. 1-Chlorobutane & 2-Chlorobutane are:

- (1) Position isomers
- (2) Chain isomers
- (3) Geometrical isomers
- (4) Ring-chain isomers

2. are called is :

- (1) Position isomers
- (2) Chain isomers
- (3) Functional isomers
- (4) Ring chain isomers

3. Possible number of disubstituted benzene isomers are :

- (1) 1 (2) 2
(3) 3 (4) 4

4. Which of the following are isomers:

- (1) Ethanol and ethoxyethane
(2) Methanol and methoxymethane
(3) Propanoic acid and ethylacetate
(4) Propionaldehyde and acetone

5. Functional group isomer of $\text{CH}_3\text{COOCH}_3$ is:

- (1) $\text{CH}_3\text{CH}_2\text{COOH}$ (2) $\text{HOCH}_2\text{-CH}_2\text{CHO}$
(3) HCOOC_2H_5 (4) Both (1) & (2)

6.  are:

- (1) Chain isomer (2) Functional isomer
(3) Position isomer (4) Metamers

7. CH_3CONH_2 & HCONHCH_3 are called:

- (1) Position isomer (2) Chain isomer
(3) Metamer (4) Functional isomer

8. The formula $\text{C}_4\text{H}_8\text{O}_2$ represents:

- (1) Only an carboxylic acid
(2) Only an ester
(3) Only an alcohol
(4) Both carboxylic acid and ester

9. Propene and cyclopropane are:

- (1) Chain isomers
(2) Position isomers
(3) Geometrical isomers
(4) Ring chain isomers

10. The minimum number of carbon atoms present in an organic compound to show chain isomerism are:

- (1) 2 (2) 3
(3) 5 (4) 4

11. The minimum number of carbon atoms present in an organic compound to be able to show position isomerism are:

- (1) 3 (2) 4
(3) 2 (4) 5

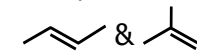
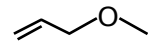
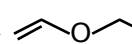
12. How many chain isomers are possible for C_6H_{14} :

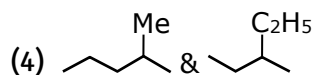
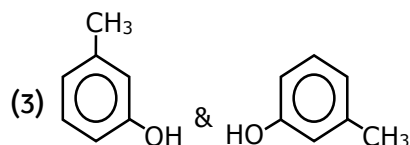
- (1) 5 (2) 4
(3) 3 (4) 2

13. How many benzenoid isomeric phenols are possible with the molecular Formula $\text{C}_7\text{H}_8\text{O}$:

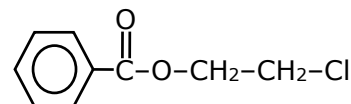
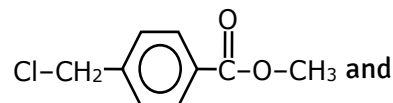
- (1) 3 (2) 4
(3) 2 (4) 5

14. A pair of position isomers is:

- (1)  & 
(2)  & 

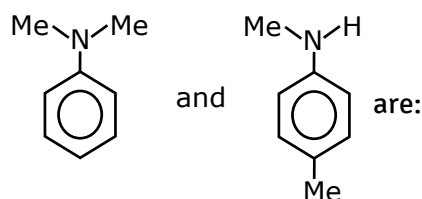


15. Given compounds are:

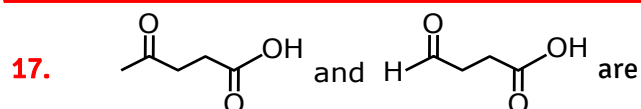


- (1) Chain isomers
(2) Metamers
(3) Position isomers
(4) Ring chain isomers

16.

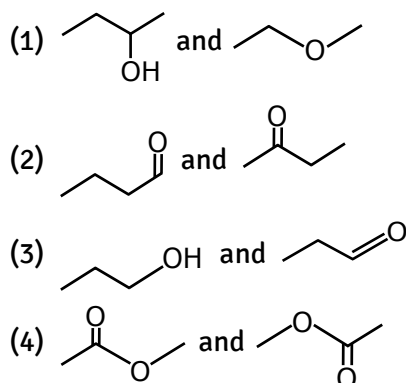


- (1) Functional isomers
(2) Metamers
(3) Position isomers
(4) Chain isomers



- (1) Homologues (2) Chain isomers
(3) Functional isomers (4) not a isomer

18. Which of the following pairs of compounds are functional group isomers?



19. $\text{CH}_3 - \text{CH}_2 - \text{CHO}$ & $\text{CH}_2 = \text{CH} - \text{CH}_2\text{OH}$ are:

- (1) Functional group isomers
(2) Chain isomers
(3) Position isomers
(4) Metamers

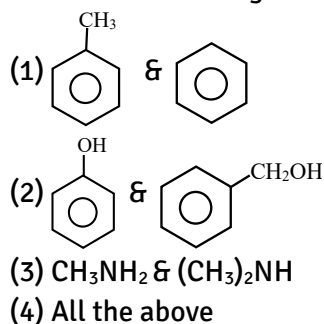
20. Alkynes are isomers of-

- (1) Cycloalkane
(2) Alkadiene
(3) Alkene
(4) All of the above

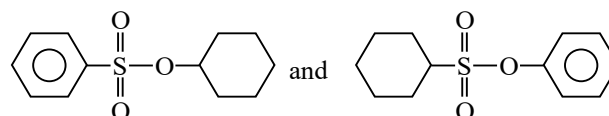
21. $\text{C}_7\text{H}_7\text{Cl}$ shows how many benzenoid aromatic isomers?

- (1) 4 (2) 3
(3) 5 (4) 6

22. Choose the homologues-



23. Given compound shows which type of isomerism



- (1) Chain isomerism
(2) Positional isomerism
(3) Metamerism
(4) Functional group isomerism

24. The compound which is not isomeric with diethyl ether is

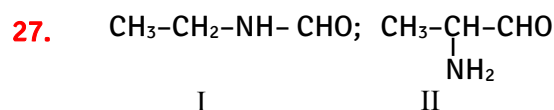
- (1) methyl n-propyl ether.
(2) 1-butanol.
(3) 2-methyl propan-2-ol.
(4) butanone.

25. Isomers have essentially identical.

- (1) Structural formula
(2) Chemical properties
(3) Molecular formula
(4) Physical properties

26. What is the relation between 3-Ethylpentane and 3-Methylhexane ?

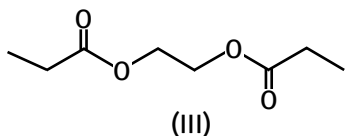
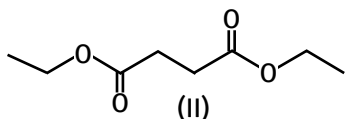
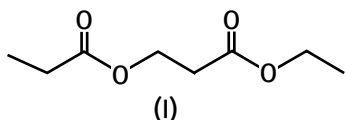
- (1) Chain isomers
(2) Position isomers
(3) Functional isomers
(4) Homologues



Which type of isomerism is observed between I and II ?

- (1) Chain isomers
(2) Position isomers
(3) Functional isomers
(4) Metamers

28. The correct relationship among the following pairs of given compounds is



	(I, II)	(II, III)
(1)	Functional Isomers	Metamers
(2)	Metamers	Functional Isomers
(3)	Metamers	Metamers
(4)	Functional Isomers	Functional Isomers

29. Molecular formula $C_4H_{10}O$ represent
 (1) Two primary alcohol
 (2) One secondary alcohol
 (3) One tertiary alcohol
 (4) All of these
30. How many positional isomers are possible for dimethyl cyclohexane.
 (1) 3
 (2) 4
 (3) 5
 (4) 6
31. How many aromatic isomers are possible for trichlorobenzene ($C_6H_3Cl_3$)?
 (1) 2
 (2) 3
 (3) 4
 (4) 5
32. The number of ether isomers represented by formula $C_4H_{10}O$ is (only structural)
 (1) 4
 (2) 3
 (3) 2
 (4) 1
33. Total number of 2° amine isomers of $C_4H_{11}N$ would be (only structural)
 (1) 4
 (2) 3
 (3) 5
 (4) 2
34. How many structural isomers of all the secondary alcohols with molecular formula $C_5H_{12}O$.
 (1) 2
 (2) 3
 (3) 4
 (4) 5
35. Relation between given compounds is:

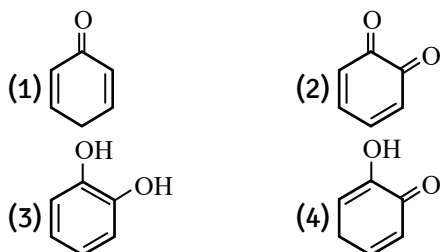
 (1) Position isomers
 (2) Chain isomers
 (3) Identical
 (4) Functional isomers
36. Which type of isomerism is observed between I and II?
 $CH_3-CH_2-C(=O)-OCH_3$
 (I)
 $CH_3-C(=O)-CH_2-O-CH_3$
 (II)
 (1) Functional isomerism
 (2) Metamerism
 (3) Position isomerism
 (4) Stereoisomerism
37. & show isomerism -
 (1) Chain
 (2) Position
 (3) Functional
 (4) None of these
38. The total number of cyclic compounds (neglecting stereoisomers) with the molecular formula C_5H_{10} is
 (1) 4
 (2) 5
 (3) 6
 (4) 7
39. Which of the following pairs of compounds are not isomers -
 (1) Propyne and cyclopropane
 (2) Propyne and propadiene
 (3) Propene and cyclopropane
 (4) 1-Propanol and methoxyethane
40. What is true for 1,2-pentadiene?
 (1) It is functional isomer of Pentyne
 (2) It is position isomer of Pentyne
 (3) It is chain isomer of 3-Methyl-1-butyne
 (4) It is metamer of cyclopentene

41. Functional isomerism is shown by –
 (1) o-Nitrophenol and p-nitrophenol
 (2) Dimethyl ether and ethanol
 (3) 2-Pentenoic acid and 3-pentenoic acid
 (4) Acetaldehyde and acetone

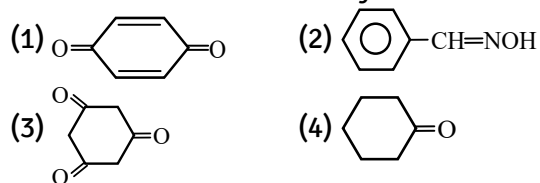
42. The minimum number of carbon atoms in ketone to show position isomerism –
 (1) 3 (2) 4
 (3) 5 (4) 6

43. The minimum number of carbon atoms in ketone to show metamerism:
 (1) 3 (2) 4
 (3) 5 (4) 6

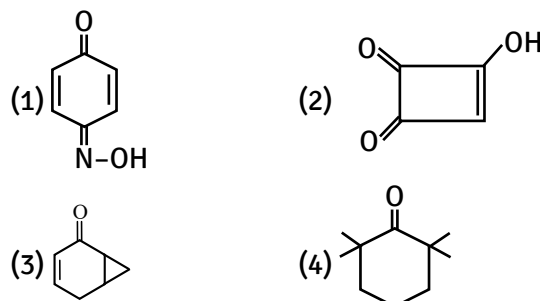
44. Which can't show Tautomerism?



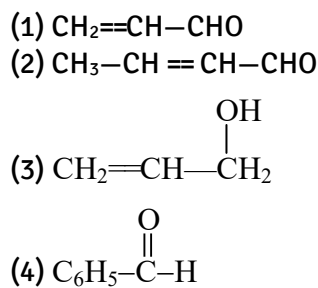
45. Tautomerism is not shown by -



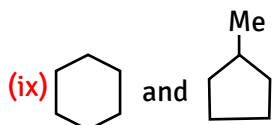
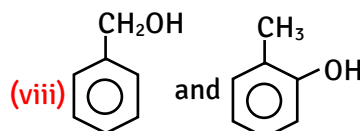
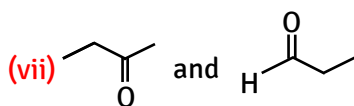
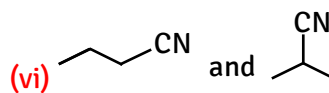
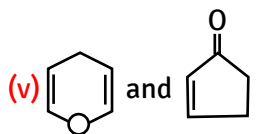
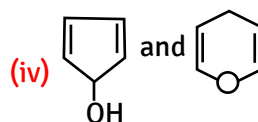
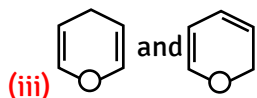
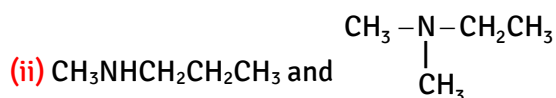
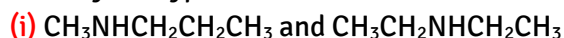
46. Which will not show tautomerism -

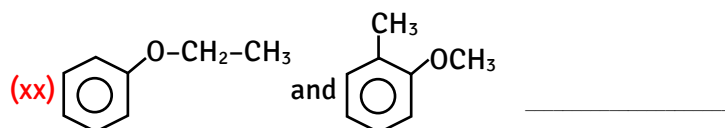
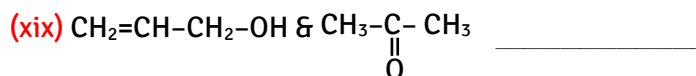
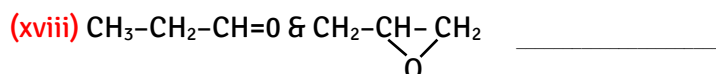
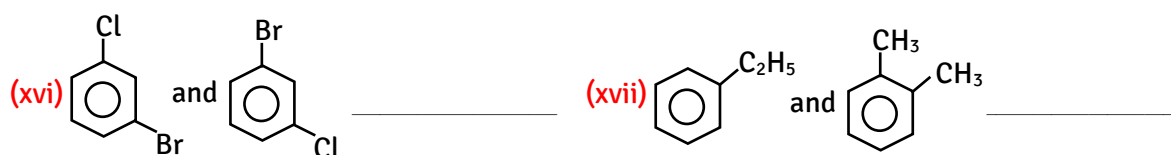
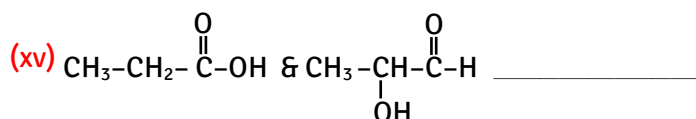
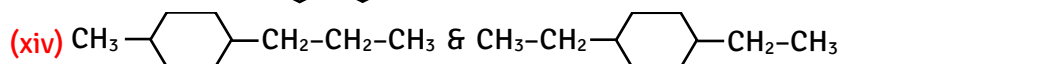
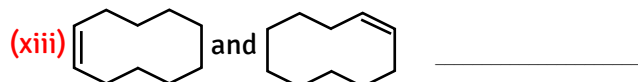
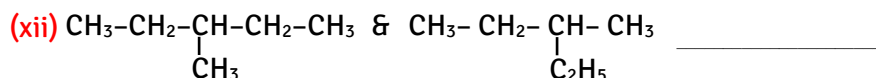
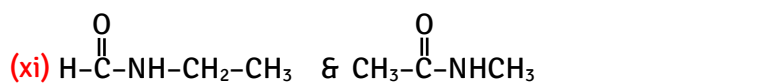
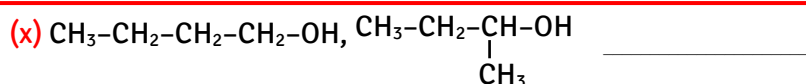


47. Tautomerism will be shown by



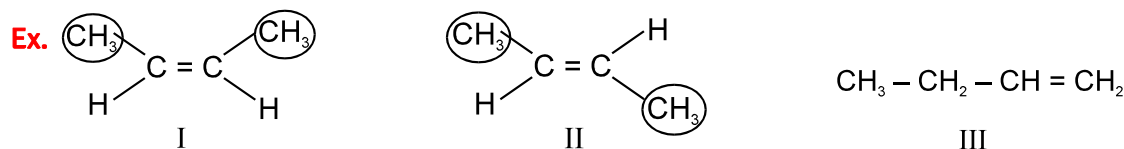
48. Identify the type of isomerism





Stereoisomerism

Stereoisomers: compounds having same structural formula & molecular formula but different stereochemical formula are called stereo isomer. These isomers have same connectivity of atoms and groups (same structural formula) but different orientation of groups about a stereo centre (system).



I, III → position Isomers

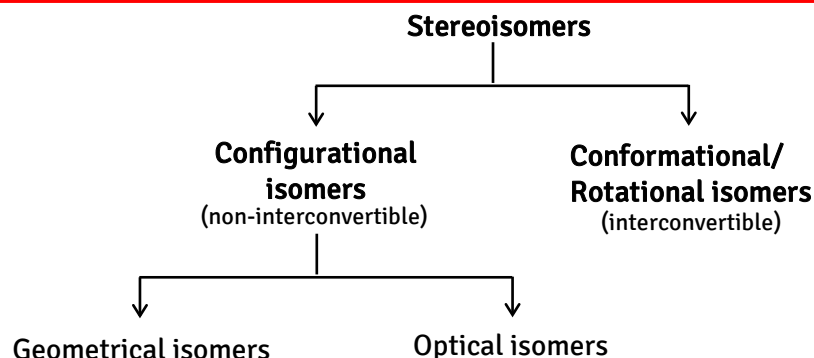
II, III → position Isomers

I, II → stereoisomers

same M.F. but diff. S.F.

same M.F. but diff. S.F.

same M.F. & S.F. but different S.C.F.



Configurational Isomers: It arises due to different orientations along a stereo centre and these isomers can be separated and these isomers do not convert into one-another at room temperature therefore, they are true isomers.

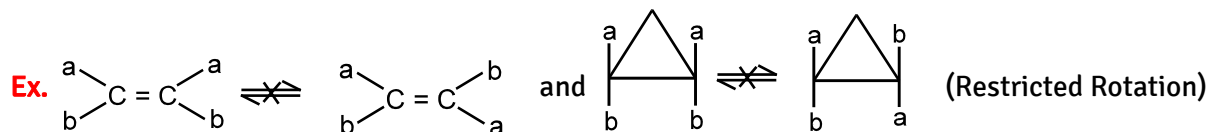
Ex. *cis-2-Butene*
& *trans-2-Butene*

Geometrical Isomerism

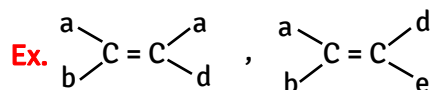
The compound having different 3-D arrangement which are not interconvertible by free rotation about any set of bonds are called geometrical isomers and the phenomenon is called geometrical isomerism.

(I) This isomerism arises due to restricted rotation about a system that is a set of two atoms where rotation of one atom with respect to other atom is not free.

- Restricted rotation is observed due to
 - double bond [$>C=C<$, $>C=N-$, $-N=N-$]
 - Ring structure
- Due to the restricted rotation about double bond or the ring structure the molecules exist in two or more orientations.

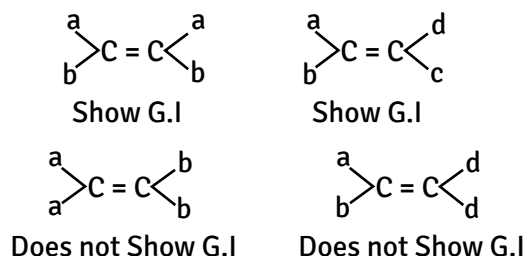


(II) Different groups should be attached at each atom of the system. (Set of two atoms where rotation is restricted)



Case (a): Geometrical Isomers in Alkenes

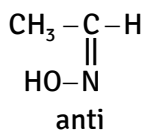
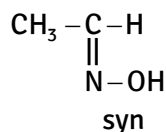
- Alkenes show G.I. due to restricted rotation about double bond
- Only those alkenes show G. I. in which "Each double bonded carbon individually have different atoms or groups"



Case (b): Geometrical Isomers in Oximes [$>C=N-OH$]

- Oximes show G. I. due to restricted rotation about double bond.
- Only those oximes show Geometrical isomerism in which double bonded carbon atom has two different units.

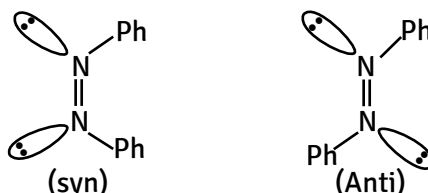
Ex. Acetaldoximes has two Geometrical isomers –



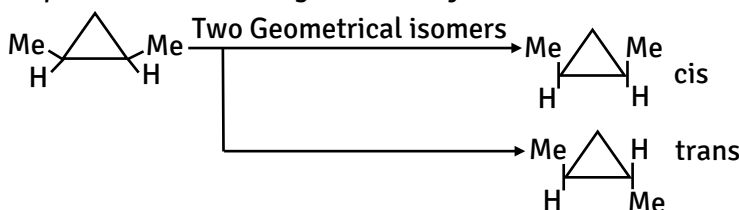
When H and OH are on the same side When H and OH are on the opposite side

Case (c): Geometrical Isomers in Azo Compounds ($-\ddot{N}=\ddot{N}-$)

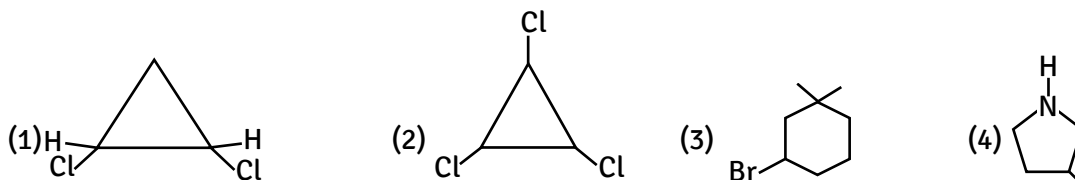
Ex. $\text{Ph}-\text{N}=\text{N}-\text{Ph}$ (Azo benzene)



Case (d): Geometrical Isomers in Cycloalkanes Cycloalkanes show Geometrical isomers due to restricted rotation about single bond. Only those cyclo alkanes show Geometrical isomers in which at least two different sp^3 carbons of the ring individually have two different units.



Ex. Which of the following show Geometrical isomerism–

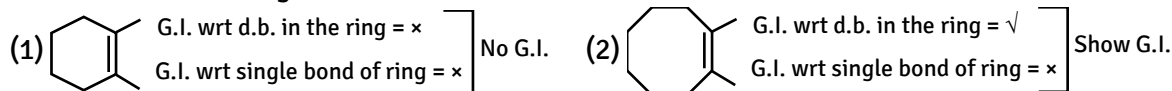


Ans. (1), (2) and (4)

Case (e): Geometrical Isomers in Cycloalkene

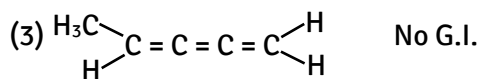
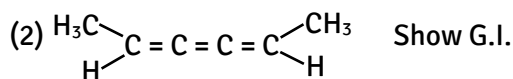
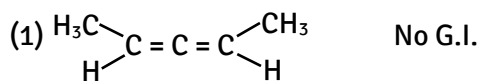
Condition - Minimum 8 membered ring is required to show G.I. w.r.t. double bond in the ring

Ex. Which of the following show G.I.

**Case (f): Geometrical Isomers in Allenes**

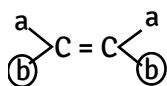
- In Allenes, continuous π bonds are perpendicular to each other.
- If no. of π bonds are even then terminal valencies will be in $\perp$ plane and compound will never show G.I.
- If number of π bonds are odd then terminal valencies will be in same plane and compound can show G.I. if both terminals have different units.

Ex. Which of the following show G.I.



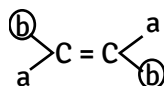
Configuration of geometrical isomers

(A) **cis-trans system** If same groups are present on same side of the system then *cis* and if same groups are present on opposite of the system then it is called *trans*.



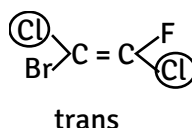
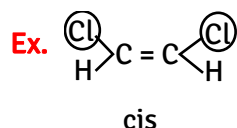
[Same groups, same side]

cis



[Same groups opposite side]

trans

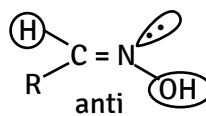
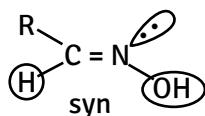


(B) **syn-anti system**

Condition Minimum one terminal should have lone pair.

Ex. (1) Oxime

(i) **Aldoxime** Oxime of aldehyde other than formaldehyde will show G.I. and exist in two forms Syn & Anti.

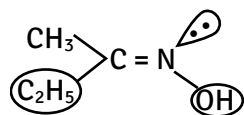


Note: H & OH same side → *syn*

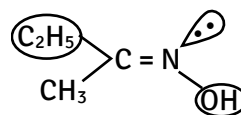
H & OH opposite side → *anti*

(ii) **Ketoxime**

- Ketoximes of symmetrical ketones do not show G.I.
- Ketoximes of unsymmetrical ketones show G.I.

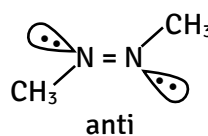
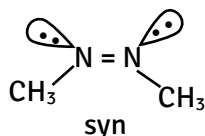


Syn-ethyl methyl ketoxime



Anti-ethyl methyl ketoxime

(2) **Azo Compounds**



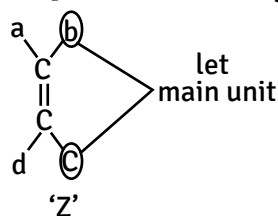
Note: lp-lp same side → *syn*

lp-lp opposite side → *anti*

(C) **E – Z System** : Main units are selected on both side of the system

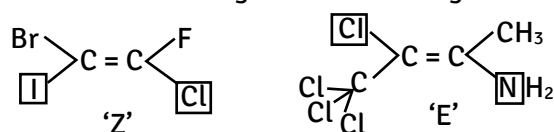
When main units are on opposite side → E [Entgegen i.e. opposite]

When main units are on same side → Z [Zusammen i.e. together]

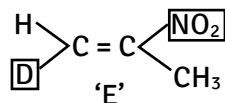


CIP sequence Rules (Given by Cahn, Ingold & Prelog) for selecting main unit:

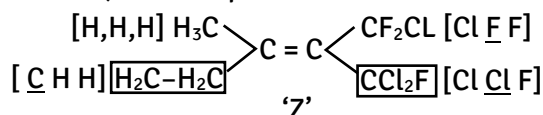
Rule I: Unit having 1st atom with higher atomic number gets priority.



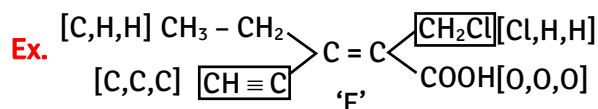
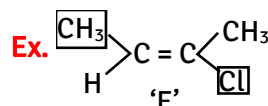
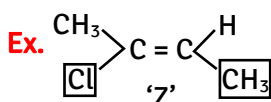
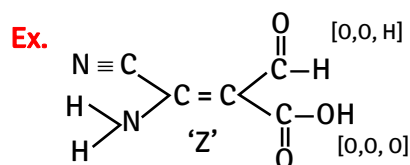
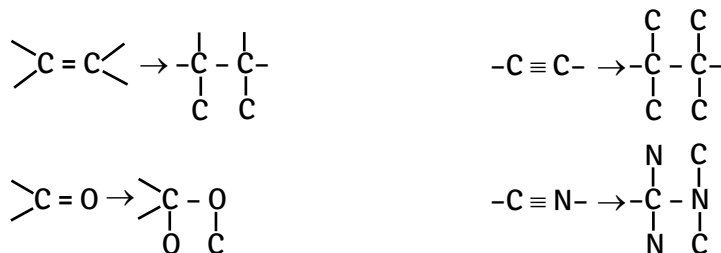
Rule II: If 1st atoms of two units are isotopes then heavier isotope gets priority.



Rule III: If 1st atom of two units are same then sets of next atoms (arranged in decreasing atomic number) are compared and at the first difference, atom with higher atomic number gets priority.



Rule IV: If multiple bond is present in unit then consider them as:-



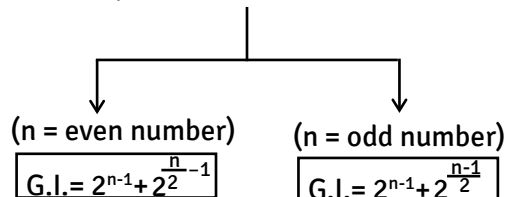
Number of Geometrical Isomers (G.I.) :

- (a) If compound is unsymmetrical (in respect of IUPAC numbering) or compound with dissimilar ends then,
number of G.I. = 2^n [n = number of double bonds or system where G.I. is possible]

Ex. $\text{CH}_3\text{-CH=CH-CH=CH-CH=CH-CH}_2\text{CH}_3$

As $n = 3 \rightarrow \text{number of G.I} = 2^3 = 8$

(b) If compound is symmetrical or compound with similar ends then



Ex. $\text{CH}_3\text{-CH=CH-CH=CH-CH=CH-CH}_3$

$n = 3$ (odd number)

$$\begin{aligned} \text{No of G.I.} &= 2^{n-1} + 2^{\frac{n-1}{2}} \\ &= 2^{3-1} + 2^{\frac{3-1}{2}} \\ &= 2^2 + 2^1 \\ &= 4 + 2 \\ &= 6 \text{ Ans.} \end{aligned}$$

Physical properties of Geometrical isomers

Physical properties	 I II	Remarks
Dipole moment	$I > II$	Cis-isomer has resultant of dipoles while in trans isomer dipole moments cancel out
Boiling point	$I > II$	Molecules having higher dipole moment have higher boiling point due to larger intermolecular force of attraction
Solubility (in H_2O)	$I > II$	More polar molecules are more soluble in H_2O (Polar solvent)
Melting point	$II > I$	More symmetric isomers have higher melting points due to better packing in crystalline lattice & trans isomers are more symmetric than cis.
Stability	$II > I$	The molecule having more vander waal strain are less stable. In cis isomer the bulky groups are closer so, they have larger vander waals strain.

Table

Physical properties	 I	 II
Dipole moment	I > II	
Boiling point	I > II	
Solubility (in H ₂ O)	I > II	
Melting point	I > II	
Stability	I > II	

Exercise 1.2

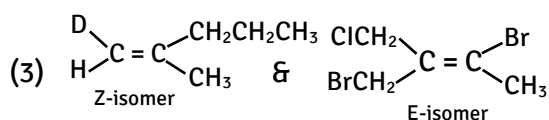
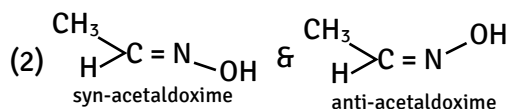
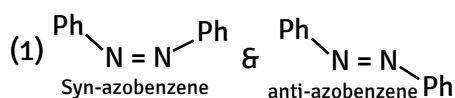
1. Fumaric acid shows geometrical isomerism with:

- (1) Malonic acid (2) Maleic acid
(3) Malic acid (4) Tartaric acid

2. Which of the following will show geometrical isomerism:

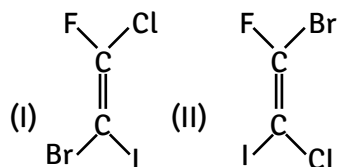
- (1) 1-Butene
(2) 1,2-Dibromoethene
(3) But-2-yne
(4) Isobutylene

3. Which of the following pair of compounds shows correct configuration:



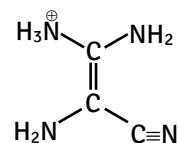
(4) All of these

4. Which is incorrect statement:



- (1) I has Z configuration
(2) II has E configuration
(3) I & II are geometrical isomers of each other
(4) None

5. Configuration of the given molecule is:



- (1) E (2) Z
(3) Trans (4) 2 & 3 both

6. Incorrect match is:

M.F.	No. of isomers
(1) C ₄ H ₈	3 alkenes
(2) C ₂ H ₇ N	2
(3) C ₂ H ₂ Br ₂	3
(4) C ₅ H ₁₂	3

7. Which of the following compound can show geometrical isomerism:

- (1) (CH₃)₂C = CH - C₂H₅
(2) H₂C = CBr₂
(3) C₆H₅ - CH = CH - CH₃
(4) CH₃ - CH = CCl₂

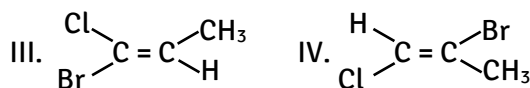
8. Which of the following is not true regarding hex-2-ene:

- (1) Boiling point of cis is higher
(2) Trans isomer has zero dipole moment
(3) Trans is more stable than cis
(4) Can show position isomerism

9. Which of the following compounds will show geometrical isomerism:

- (1) 2-Butene
(2) 1-Chloro-1-butene
(3) 1-Phenylpropene
(4) All of above

10. Which is a pair of geometrical isomers:



(1) I and II

(2) I and III

(3) II and IV

(4) III and IV

11. But-2-ene exhibits cis-trans isomerism due to:

- (1) Rotation around C_3-C_4 sigma bond
- (2) Restricted rotation around $\text{C}=\text{C}$ bond
- (3) Rotation around C_1-C_2 bond
- (4) Rotation around C_2-C_3 double bond

12. The number of geometrical isomers of $\text{CH}_3\text{CH}=\text{CHCH}=\text{CH}-\text{CH}=\text{CHCl}$ are:

(1) 2

(2) 4

(3) 6

(4) 8

13. Which of the following functional group containing compounds can exhibit geometrical isomerism:

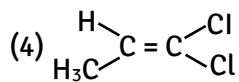
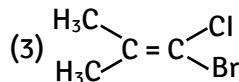
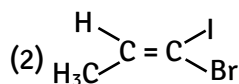
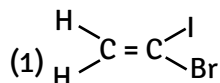
(1) $>\text{C}=\text{C}<$

(2) $>\text{C}=\text{N}-$

(3) $-\text{N}=\text{N}-$

(4) All of these

14. Geometrical isomerism is shown by



15. Among the following compounds, the one which can exhibit geometrical isomerism is:

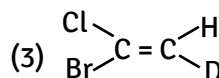
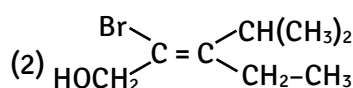
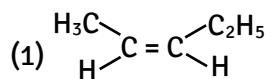
(1) 1,3-Butadiene

(2) 1,2-Butadiene

(3) 1,3-Pentadiene

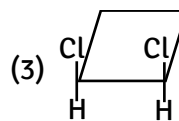
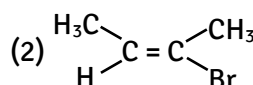
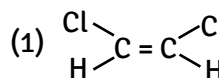
(4) 1,4-Pentadiene

16. Which of the following has Z-configuration:



(4) All the above

17. Which of the following is/are cis isomer?



(4) All of these

18. Which of the following cannot show geometrical isomerism?

(1) $\text{CH}_3\text{CH}=\text{CHC}_2\text{H}_5$

(2) $(\text{CH}_3)_2\text{C}=\text{CHC}_2\text{H}_5$

(3) $\text{CH}_3\text{CH}=\text{CHCH}_3$

(4) All the three

19. Which of the following will not show cis-trans isomerism?

(1) Butanedioic acid

(2) 1,2-dimethyl cyclopropane

(3) 1,2-dichloroethene

(4) Butenedioic acid

20. $\begin{array}{c} \text{CH}_3-\text{C}-\text{Cl} \\ \parallel \\ \text{Cl}-\text{C}-\text{Br} \end{array}$ is

(1) Trans

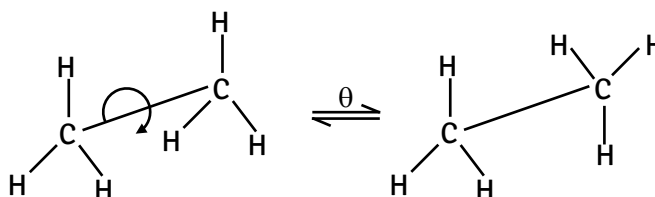
(2) Z

(3) Both Z & trans

(4) cis

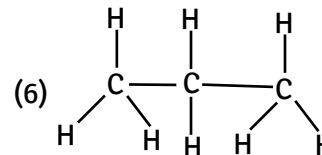
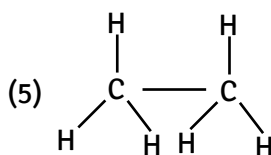
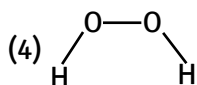
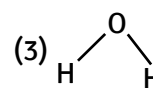
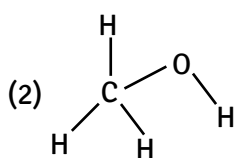
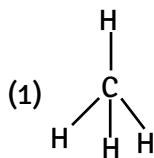
Conformational Isomerism

- The isomerism arises due to free rotation of atoms with respect to second atom in a molecule so it is also called rotational isomerism.
- Due to free rotation about single bond, a molecule has several 3-D temporary arrangements which are called conformers or rotamers or conformations.
- Infinite number of conformations are possible for the compound.
- It is to remember that C-C single bond is not completely free it is hindered by small barrier of 1-20 kJ/mol. It is due to
 - Repulsive interaction between adjacent bonds called torsional strain.
 - Repulsive interaction between adjacent atoms called steric strain.



Here $\theta(60^\circ)$ is dihedral angle, angle between two planes.

Ex. Which of the following show conformational isomerism.



Ans. 2, 4, 5, 6

In general conformations are represented by sawhorse and Newman projection.

1. Sawhorse projection:

The central C-C bond as a somewhat longer straight line. Upper end of the line is slightly tilted towards right or left hand side. The front carbon is shown at the lower end of the line, whereas the rear carbon is shown at the upper end. Each carbon has three lines attached to it corresponding to three hydrogen atoms. The lines are inclined at an angle of 120° to each other.

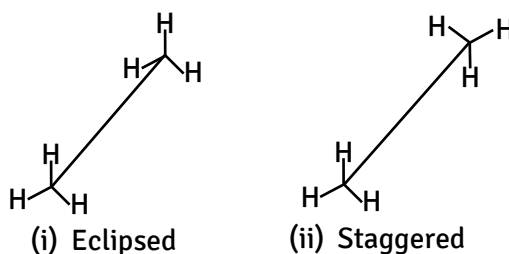


Fig. Sawhorse projection of ethane

2. Newman Projection:

In this projection, the molecule is viewed at the C-C bond head on. The carbon atom nearer to the eye is represented by a point. Three hydrogen atoms attached to the front carbon atom are shown by three lines drawn at an angle of 120° to each other. The rear carbon atom (the carbon away from the eye) is represented by a circle and the three hydrogen atoms are shown attached to it by the shorter lines drawn at an angle of 120° to each other.

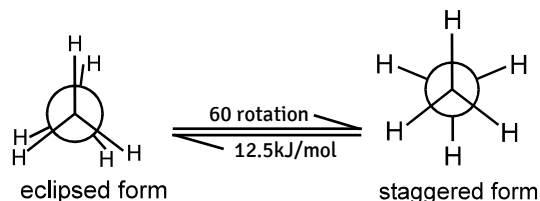


Fig. Newman's projection of ethane

Conformational Analysis of Ethane

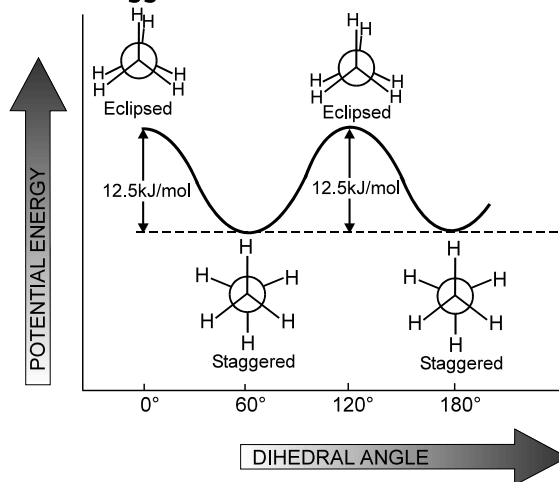
Ethane molecule contains a carbon-carbon σ bond and each carbon is further attached to 3 H-atoms. It exists in two extreme conformations are:

- Eclipsed Conformation [H-atoms attached with two carbons are as closed together as possible]
- Staggered Conformation [H-atoms attached with two carbons are as far apart as possible]



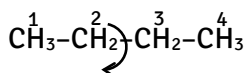
The potential energy barrier between the two extreme conformations of ethane is about 12.5 kJ/mol. The potential energy of ethane molecule is minimum for the staggered conformation, increase with rotation and reaches at maximum for the eclipsed conformation. Most of ethane molecules naturally exist in the most stable staggered conformation. Any other intermediate conformations or conformation other than eclipsed & staggered are called skew conformations

Energy Profile of Eclipsed and Staggered Forms of Ethane

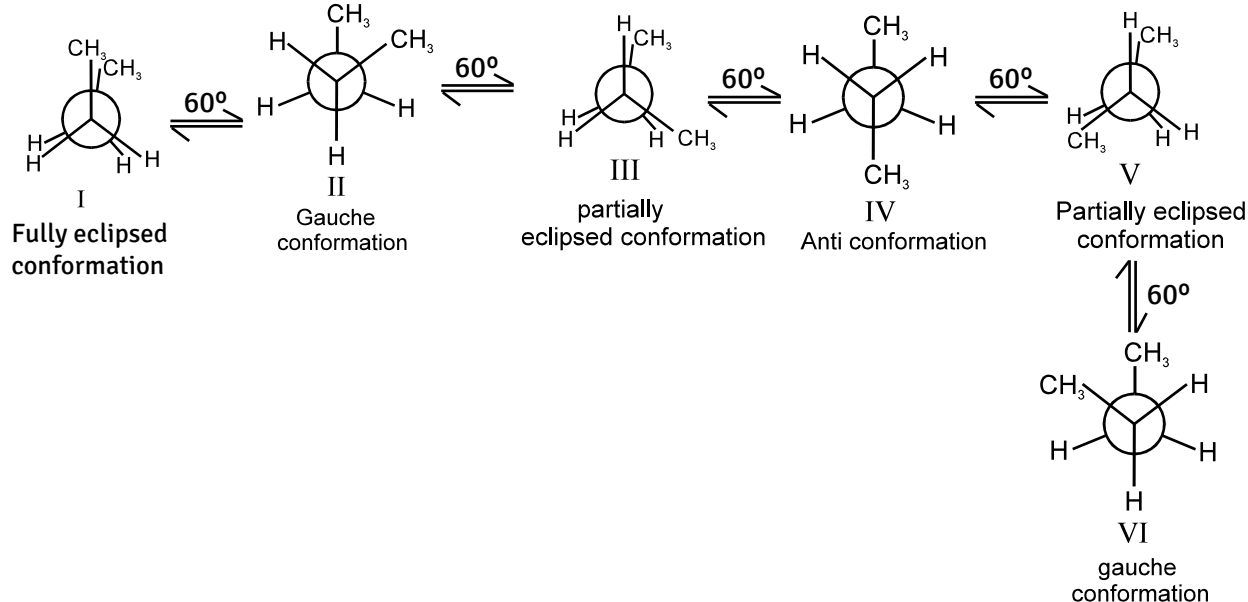


Stability Order Staggered > Skew > Eclipsed

Conformational Analysis of Butane



If we consider rotation about the $C_2 - C_3$ bond of butane, we find that there are four important conformations shown as:



The stability order $\rightarrow$ Anti > Gauche > Partially eclipsed > Fully eclipsed

(iv) (ii = vi) (iii = v) (i)

Order of potential energy $\rightarrow$ Anti < Gauche < Partially eclipsed < Fully eclipsed

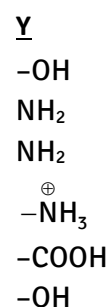
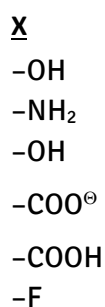
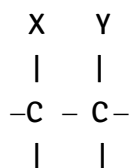
(iv) (ii = vi) (iii = v) (i)

Note: In all conformations, the bond angles, the bond lengths remain same.

Gauche Effect

Gauche conformation, in some cases, becomes even more stable than anti when intra-molecular hydrogen-bonding takes place.

- Cases where gauche is more stable than anti :

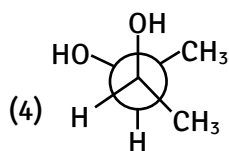
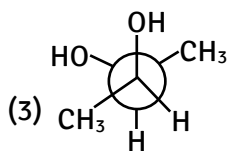
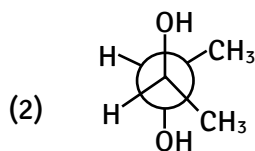
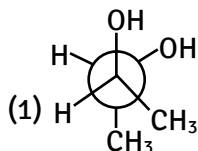


Exercise 1.3

1. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$. There is free rotation about ($\text{C}_2 \sigma \text{C}_3$) bond. The most stable form is obtained after rotation of _____, starting from fully eclipsed conformation.

(1) 60° (2) 120°
(3) 240° (4) 180°

2. Which one of the following is the most stable conformation of 2,3-butanediol:



3. How many conformations does ethane have:

(1) 1 (2) 2
(3) 3 (4) Infinite

4. The eclipsed and staggered conformation of ethane is due to:

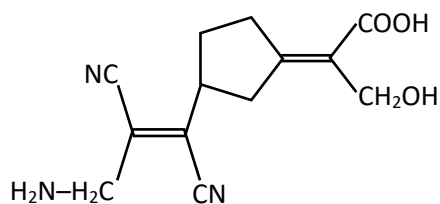
(1) Free rotation about C–C single bond
(2) Restricted rotation about C–C single bond
(3) Absence of rotation about C–C bond
(4) None of the above

5. Dihedral angle required to get maximum stable conformer from minimum stable conformer in $\text{C}_2 - \text{C}_3$ bond of n-butane is:

(1) 360° (2) 180°
(3) 120° (4) 240°

Exercise 2

1. Assign double bond configurations to the following:

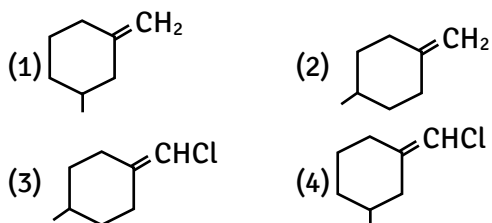


- (1) E, Z (2) Z, E
(3) E, E (4) Z, Z

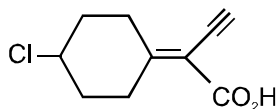
2. $\text{HO}-\text{CH}_2-\text{CH}_2-\text{F}$
Which conformer of above compound is most stable across C - C?

- (1) Anti staggered (2) Partially eclipsed
(3) Gauche (4) Fully eclipsed

3. The geometrical isomerism is shown by:



4. What are the correct designations for the structure below?

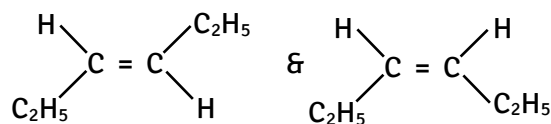


- (1) E, E
(2) Z, E
(3) E, Z
(4) Geometrical isomerism not possible.

5. Which of the following will not show geometrical isomerism?

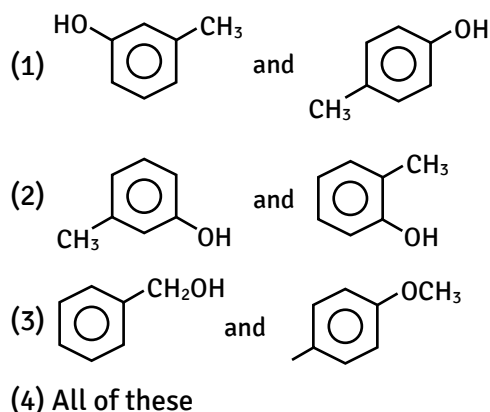
- (1) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2-\text{CH}_3$
(2) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$
(3) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
(4) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$

6. For geometrical isomers of 3-hexene:



- (1) M.P. is high and dipole moment high for trans
(2) M.P. is low and dipole moment low for trans
(3) M.P. is high and dipole moment low for trans
(4) M.P. is low and dipole moment high for trans

7. Which of the following pairs of compounds are not positional isomers?



8. $\text{CH}_3-\text{CH}=\text{C}(\text{Br})-\text{C}(\text{Br})=\text{CH}-\text{CH}_3$

How many geometrical isomers are possible for the above compound:

- (1) 2 (2) 3
(3) 4 (4) 6

9. $\text{CH}_3-\text{CH}=\text{C}(\text{Br})-\text{C}(\text{Cl})=\text{CH}-\text{CH}_3$

How many geometrical isomers are possible for this compound?

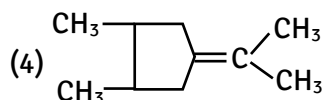
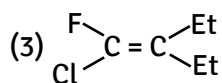
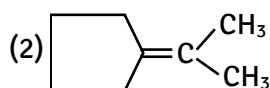
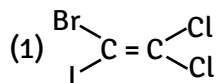
- (1) 2 (2) 3
(3) 4 (4) 6

10. $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$

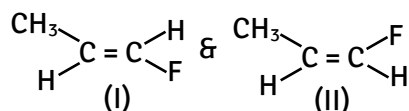
How many geometrical isomers are possible for this compound?

- (1) 2 (2) 3
(3) 4 (4) 8

11. Which of the following compound can show geometrical isomerism?



12. Which of the following is correct set of physical properties of the geometrical isomers?

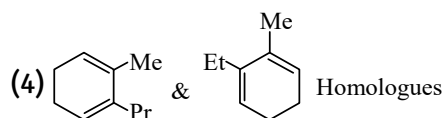
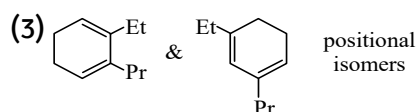
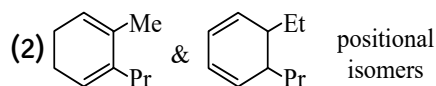
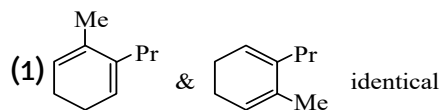


	Dipole moment	Boiling point	Melting point	Stability
(1)	I > II	I < II	II > I	I > II
(2)	II > I	II > I	II > I	II > I
(3)	I > II	I > II	I > II	I > II
(4)	II > I	II > I	I > II	I > II

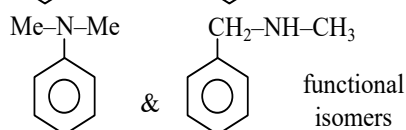
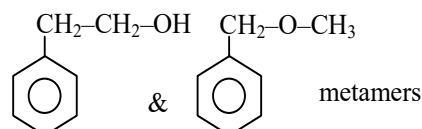
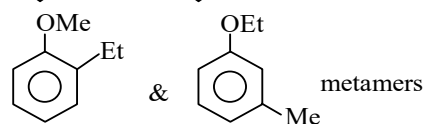
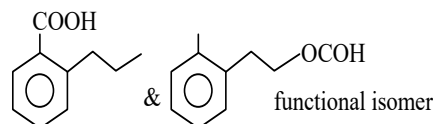
13. An alkane forms isomers if it possesses minimum no. of carbon atoms-

- (1) 1 (2) 2
(3) 3 (4) 4

14. Which of the following has incorrect relation?



15. Which of the following order is correct-



- (1) TTF (2) FTTF
(3) TTFT (4) TFFT

16. Which of the following cannot be written in an isomeric form?

- (1) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}_3$
(2) CH_3-CHO
(3) $\text{CH}_2=\text{CH}-\text{Cl}$
(4) $\text{Cl}-\text{CH}_2\text{CH}_2-\text{Cl}$

17. Only two isomeric monochloro derivatives are possible for (excluding stereo)

- (1) n-butane
(2) 2,2-dimethylpentane
(3) benzene
(4) neopentane

18. An alkene exists as geometrical isomers. The minimum number of carbon atoms in the molecule is -

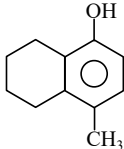
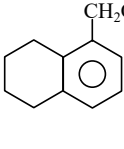
- (1) 1 (2) 2
(3) 3 (4) 4

19. Which of the following is not true for maleic acid and fumaric acid?

- (1) Configurational isomers
(2) Stereo isomers
(3) Z and E isomers
(4) Optical isomers

20. Functional isomers of $\text{CH}_3\text{C}(=\text{O})\text{O}-\text{H}$ is -

- (1) Amide (2) Ester
(3) Anhydride (4) All

21. Compound  and  are-

- (1) Functional isomers
(2) metamers
(3) Tautomers
(4) Ring chain isomers

22. Which will not show metamerism?

- (1) $\text{CH}_3-\text{O}-\text{C}_6\text{H}_4-\text{Et}$
(2) $\text{CH}_3-\text{O}-\text{CH}_3$
(3) $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{O}-\text{CH}_3$
(4) $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CH}_2-\text{CH}_3$

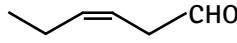
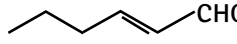
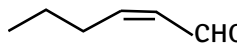
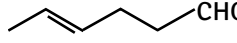
23. The functional isomers of 1-butyne are

- (I) $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3$ (II) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2$
(III) $\text{CH}_2=\text{C}=\text{CH}-\text{CH}_3$
(IV) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
(1) (I), (III) and (IV) only
(2) (II) and (III) only
(3) (II) and (IV) only
(4) (III) and (IV) only

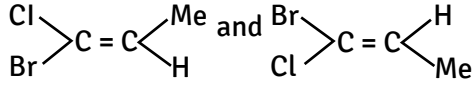
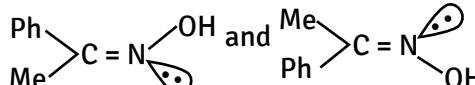
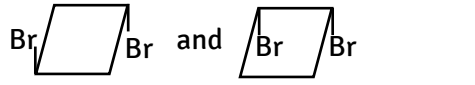
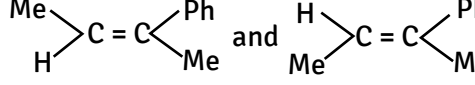
24. Which of the following group of compounds are not functional isomers?

- (1) Nitro alkane and alkyl nitrite.
(2) Aldehydes and ketones.
(3) Alcohols and aldehydes.
(4) Alcohols and ether (both saturated).

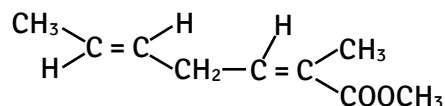
25. The correct structure of trans-hex-2-enal is

- (1) 
(2) 
(3) 
(4) 

26. Which is a pair of geometrical isomers -

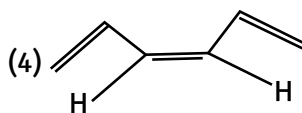
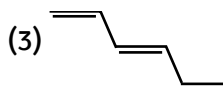
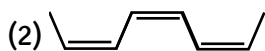
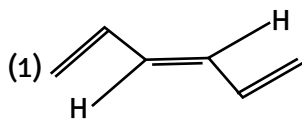
- (1) 
(2) 
(3) 
(4) 

27. The correct stereochemical name of -

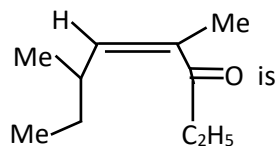


- (1) Methyl 2-methylhepta (2E, 5E) dienoate
(2) Methyl 2-methylhepta (2Z, 5Z) dienoate
(3) Methyl 2-methylhepta (2E, 5Z) dienoate
(4) Methyl 2-methylhepta (2Z, 5E) dienoate

28. Which of the following represents the structure having cis arrangement around each double bond?



29. The IUPAC name of compound



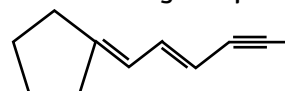
(1) (Z)-4, 6, 7- Trimethyl hept-4-en-3-one

(2) (E)-4, 6-Dimethyl oct-4-en-3-one

(3) (Z)-4, 6-Dimethyl oct-4-en-3-one

(4) (E)-4,6-Dimethyl oct-4-en-6-one

30. The number of cis-trans isomer possible for the following compound



(1) 2

(2) 4

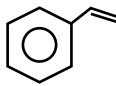
(3) 6

(4) 8

Exercise 3

1. **Assertion :** *Trans*-1-chloropropene has higher dipole moment than *cis*-1-chloropropene.
Reason : The resultant of the two vectors in *trans*-1-chloropropene is more than in *cis*-1-chloropropene.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

2. **Assertion :** All the carbon atoms in but-2-ene lie in one plane.
Reason : Carbon atoms in it are sp^2 and sp^3 -hybridized.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

3. **Assertion :**  is planar molecule but cannot show G.I.
Reason : In styrene every carbon is sp^2 hybridised.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

4. **Assertion :** Benzaldehyde forms two oximes on reacting with NH_2OH .
Reason : The two oximes arise due to geometrical isomerism around $C=N$ bond.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

5. **Assertion :** The boiling point of *cis* 1, 2-dichloro ethene is higher than that of corresponding *trans* isomer.
Reason : *Cis*-1,2-dichloro ethene has higher dipole moment as compared to that of the *trans*-isomer.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

6. **Assertion :** Ethanol cannot show position isomerism.
Reason : Ethanol cannot show isomerism.
 - (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
 - (3) If Assertion is True but the Reason is False
 - (4) If both Assertion & Reason are False

7. **Assertion** : $\text{CH}_3 - \text{CH}_2 - \text{Br}$ and $\text{CH}_3 - \text{CH}_2 - \text{I}$ are functional group isomers.

Reason : They have same molecular formula.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
- (3) If Assertion is True but the Reason is False
- (4) If both Assertion & Reason are False

8. **Assertion** : Boiling point of cis isomers are higher than trans-isomers.

Reason : dipole moments of cis isomers are higher than trans-isomers.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
- (3) If Assertion is True but the Reason is False
- (4) If both Assertion & Reason are False

9. **Assertion**: Acetone cannot show position isomerism.

Reason: Acetone cannot show isomerism.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

10. Match the column:

Column-I (Compound)		Column-II (Type of structural isomers)	
(a)	n-Butane, isobutane	(p)	Ring chain
(b)	Ethanol, methoxymethane	(q)	Position
(c)	But-1-ene, Cyclobutane	(r)	Chain isomerism
(d)	1-Propanol, 2-propanol,	(s)	Functional

(1) (a)-(p), (b)-(r), (c)-(s), (d)-(q)

(2) (a)-(q), (b)-(p), (c)-(s), (d)-(r)

(3) (a)-(r), (b)-(s), (c)-(p), (d)-(q)

(4) (a)-(p), (b)-(r), (c)-(q), (d)-(s)

11. Match the column:

Column-I (Conformation of n-butane)		Column-II (Property)	
(a)	Fully eclipsed	(p)	Has two methyl-hydrogen interactions
(b)	Partially eclipsed	(q)	Dihedral angle between two methyl group is 60°
(c)	Gauche	(r)	Has least energy
(d)	Anti	(s)	Has one methyl-methyl interaction

(1) (a)-(s), (b)-(p), (c)-(q), (d)-(r)

(2) (a)-(p), (b)-(q), (c)-(r), (d)-(s)

(3) (a)-(q), (b)-(p), (c)-(s), (d)-(r)

(4) (a)-(r), (b)-(p), (c)-(q), (d)-(s)

12. Match the column:

Column-I (Compound)		Column-II (No. of Geometrical isomers)	
(a)	1,2-dichloroethene	(p)	6
(b)	Chlorobromo-Fluoroiodoethene	(q)	4
(c)	2,4-Hexadiene	(r)	3
(d)	2,4-Heptadiene	(s)	2

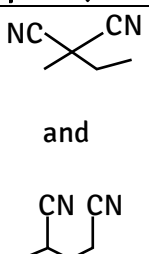
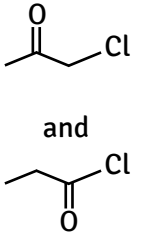
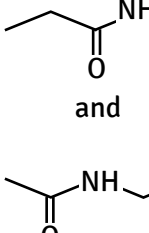
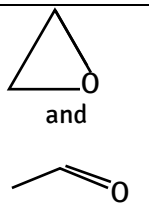
(1) (a)-(s), (b)-(q), (c)-(r), (d)-(p)

(2) (a)-(s), (b)-(p), (c)-(r), (d)-(q)

(3) (a)-(p), (b)-(q), (c)-(r), (d)-(s)

(4) (a)-(p), (b)-(r), (c)-(q), (d)-(s)

13. Match the column:

Column-I (Compound)		Column-II (Type of isomers)	
(a)		(p)	Metamerism
(b)		(q)	Chain isomerism
(c)		(r)	Ring chain isomerism
(d)		(s)	Functional isomerism

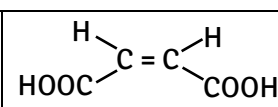
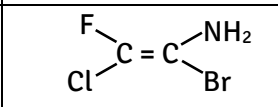
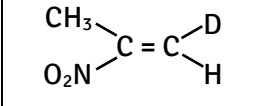
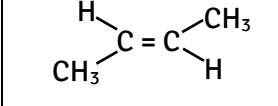
(1) (a)-(q), (b)-(s), (c)-(p), (d)-(r)

(2) (a)-(q), (b)-(s), (c)-(r), (d)-(p)

(3) (a)-(p), (b)-(q), (c)-(s), (d)-(r)

(4) (a)-(q), (b)-(p), (c)-(s), (d)-(r)

14. Match the column:

Column-I (Compound)		Column-II (Isomeric form)	
(a)		(p)	E
(b)		(q)	Z
(c)		(r)	Trans
(d)		(s)	Cis

(1) (a)-(r), (b)-(p), (c)-(q), (d)-(s)

(2) (a)-(s), (b)-(q), (c)-(p), (d)-(r)

(3) (a)-(p), (b)-(s), (c)-(r), (d)-(q)

(4) (a)-(q), (b)-(r), (c)-(s), (d)-(p)

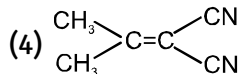
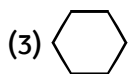
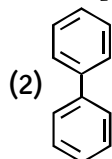
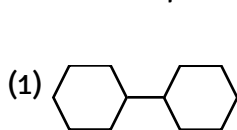
Exercise 4 (Previous Year's Questions)

1. The number of structural isomers possible from the molecular formula C_3H_9N is :

[AIPMT PRE-2015]

- (1) 2 (2) 3
(3) 4 (4) 5

2. In which of the following molecules, all atoms are coplanar? [NEET-2016]



3. The **correct** statement regarding the comparison of staggered and eclipsed conformation of ethane, is:- [NEET-I 2016]

- (1) The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain
(2) The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain
(3) The eclipsed conformation of ethane is more stable than staggered conformation even through the eclipsed conformation has torsional strain
(4) The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.

4. With respect to the conformers of ethane, which of the following statements is **true**?

[NEET-2017]

- (1) Bond angle remains same but bond length changes
(2) Bond angle changes but bond length remains same
(3) Both bond angle and bond length change
(4) Both bond angles and bond length remains same

5. Dihedral angle of least stable conformer of ethane is : [NEET(UG) 2021]

- (1) 120° (2) 180°
(3) 60° (4) 0°

6. The compound which shows metamerism is:

[NEET(UG) 2021]

- (1) C_5H_{12} (2) C_3H_8O
(3) C_3H_6O (4) $C_4H_{10}O$

7. Which one of the following compounds can exist as cis-trans isomers? [NEET-2025]

- (1) 1, 1-Dimethylcyclopropane
(2) 1, 2-Dimethylcyclohexane
(3) Pent-1-ene
(4) 2-Methylhex-2-ene

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	3	4	4	2	4	4	4	4	3	1	1	4	2	1	4	2	1	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	1	3	4	3	1	3	3	4	2	2	2	2	2	3	1	3	2	1	1
Que.	41	42	43	44	45	46	47	48												
Ans.	2	3	3	2	1	4	2	i	Metamer	vi	Chain	xi	Metamer	xvi	identical					
								ii	FGI	vii	Not isomer	xii	identical	xvii	Chain					
								iii	Position	viii	FGI	xiii	identical	xviii	RCl					
								iv	FGI	ix	Chain	xiv	Chain	xix	FGI					
								v	FGI	x	Position	xv	FGI	xx	Metamer					

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	3	4	1	3	2	4	3	2	4	4	2	3	4	4	2	1	3

Exercise 1.3

Que.	1	2	3	4	5															
Ans.	4	3	4	1	2															

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	4	1	3	3	2	3	1	4	3	4	2	3	3	1	4	4	2
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	1	2	4	3	2	4	4	2	3	1										

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14						
Ans.	1	2	2	1	1	3	4	1	3	3	1	2	1	2						

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7													
Ans.	3	2	4	4	4	4	2													

Chapter 3

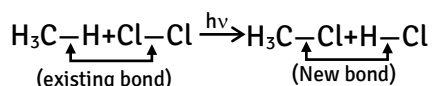
GENERAL ORGANIC CHEMISTRY

Chapter Summary

- Chemical Reaction
- Reaction intermediates
- Inductive effect
- Application of I-effect
- Resonance
- Application of Resonance effect
- Hyperconjugation effect
- Application of H-effect
- Electromeric effect
- Aromatic, Anti Aromatic & Non Aromatic compounds.

Chemical Reaction

Chemical reaction involve breaking of existing bonds and formations of new bonds.



A sequential account of each step, describing details of electron movement, energetics during bond cleavage and bond formation, and the rates of transformation of reactants into products (kinetics) is referred to as reaction mechanism.

Organic Molecule + Attacking Reagent $\rightarrow$ [Intermediate] $\rightarrow$ Product (s) + By Products (s)

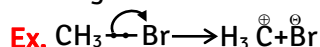
Substrate & Reagent : The reactant which supplies carbon to the new bond is called substrate and the other reactant is called reagent.

Concepts to Understand Reaction Mechanism:

- | | |
|------------------------------|-----------------------|
| (1) Fission of covalent bond | (2) Reagent |
| (3) Intermediate | (4) Electronic effect |

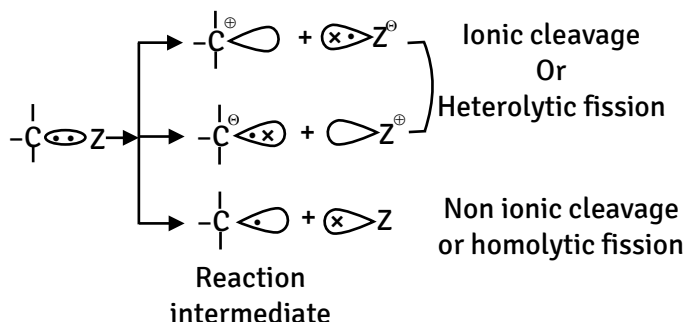
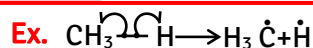
Type of covalent bond fission :

- (a) **Heterolytic Cleavage/Fission:** The bond fission in which unequal distribution of electrons takes place is known as heterolytic cleavage. due to unequal distribution of electrons, ions are formed, that's why it is also known as ionic bond cleavage.



- (b) **Homolytic Cleavage/Fission:** The bond fission in which equal distribution of electrons takes place is known as homolytic cleavage.

Due to equal distribution of electrons, neutral species is formed, which is known as free radical and cleavage is known as non ionic bond cleavage/homolytical fission.

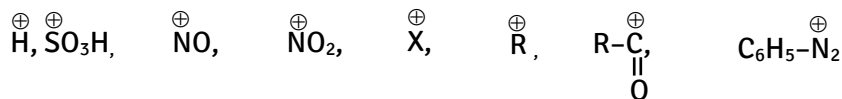


Types of Attacking Reagents: There are two types of reagents:

(a) Electrophilic Reagent or Electrophiles:

The reagents which accept e^- to form covalent bond are called electrophiles. Electrophiles may be positively charged or neutral.

(i) Positively Charged Electrophiles:



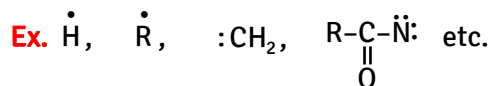
Ex. etc.

(ii) Neutral Electrophiles: central atom is e^- deficient

(a) All Lewis acids as:

Ex. BF_3 , AlCl_3 , ZnCl_2 , BeCl_2 , FeCl_3 , SnCl_2 , SnCl_4 etc.

(b) Free radicals, carbenes and nitrenes act as electrophiles.



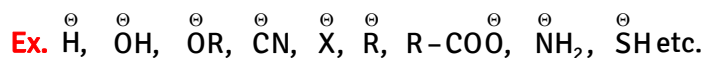
Note: Vacant orbital is required for a species to act as electrophile.

(b) Nucleophilic Reagent or Nucleophiles:

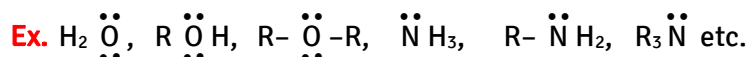
The reagent which donate electron to form covalent bond are called nucleophile.

Nucleophiles may be negatively charged ions or neutral & possess a lone pair or π electrons.

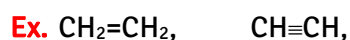
(i) Negatively charged nucleophiles.



(ii) All Lewis base which contains lone pairs



(iii) π - electron donor:



Ambident Nucleophile: species which have two different electron rich centre or species in which two or more different atoms bear a lone pair.

Ex. $^{\ominus}\text{O}-\text{N}=\text{O}$, $\ddot{\text{N}}\text{H}_2-\ddot{\text{O}}\text{H}$, $\overset{\ominus}{\text{C}}\equiv\ddot{\text{N}}$ etc.

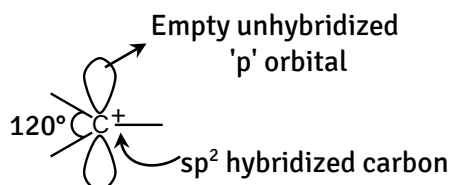
Ambiphile: Can act as both electrophile and nucleophile.

Ex. $-\text{C}\equiv\text{N}$, $-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ etc.

Reaction Intermediates:

Chemical species formed in the intermediate steps of a reaction are called reaction intermediates

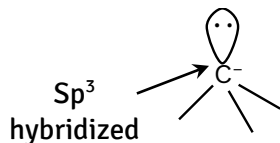
1. Carbocation:



(Trigonal planar structure)

- Positively charged carbon with three bond pairs is called carbocation.
- Due to electron deficiency it acts as an electrophile and always attack on electron rich site.
- It has incomplete octet.

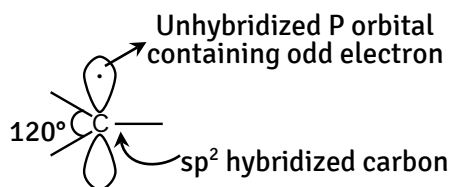
2. Carbanions:



Trigonal pyramidal structure

- Negatively charged carbon with three bond pairs and one lone pair is called carbanion.
- It has eight electron so has complete octet.
- Due to presence of non bonded electrons it acts as a nucleophile.

3. Free Radical:



Trigonal Planar structure

- Electrically neutral species in which carbon has three bond pairs and one unpaired $e^{\ominus}$ is known as carbon free radical.
- It has seven electron or odd electron in outermost shell i.e. $7e^-$ species.
- It has incomplete octet, hence it is electron deficient species.

Electron displacement effect in covalent bonds:

There are four effects which affect the properties of molecules.

- | | |
|--|---|
| (1) Inductive effect [I-effect] | (2) Resonance |
| (3) Hyperconjugation effect [H-effect] | (4) Electromeric effect or polarizability effect [E-effect] |

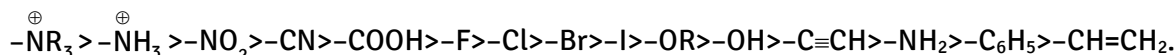
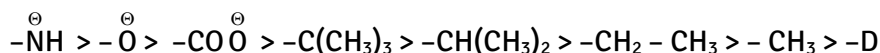
Inductive Effect:

- Polarisation of σ -bond caused by polarisation of adjacent σ -bond is called inductive effect.
- It causes permanent polarisation in molecule, hence it is a permanent effect.
- Partial displacement of σ e⁻ density takes place.
- The I- effect always operate through σ bond, it does not involve π bond electron.
- Its magnitude decreases with distance and it is negligible after 3rd carbon atom hence, it is distance dependant effect.

$$\text{I effect} \propto \frac{1}{\text{distance}}$$

Types of Inductive Effect:**(a) -I Effect:**

The groups having electron withdrawing power more than hydrogen.

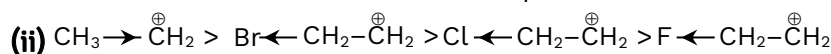
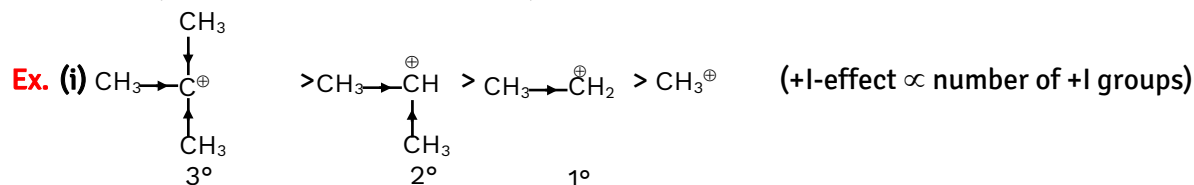
**(b) +I Effect:** The groups having electron releasing power more than hydrogen.

The hydrogen atom is reference for +I & -I series. The inductive effect of hydrogen is assumed to be zero.

Application of I-Effect:

- (1) Stability of Carbocation ($\text{C}^\oplus$) or Carbonium Ion.** Electron releasing group increases stability by decreasing positive charge on carbon.

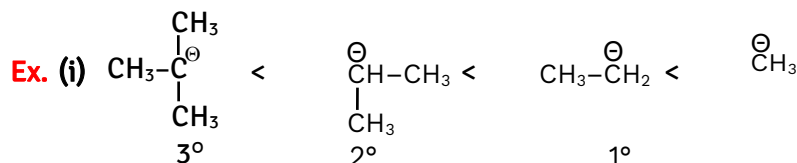
$$\text{Stability of } \text{C}^\oplus \propto \frac{1}{\text{Charge}} \propto \frac{+I}{-I}$$



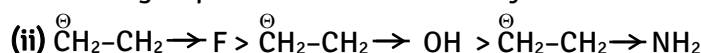
-I effect $\uparrow$, Stability of $\text{C}^\oplus \downarrow$

- (2) Stability of Carbanion ($\text{C}^\ominus$):** Electron withdrawing group increases stability by decreasing negative charge on carbon.

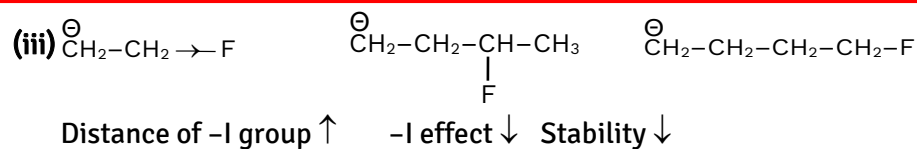
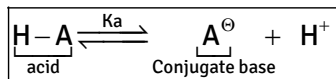
$$\left[\text{Stability of } \text{C}^\ominus \propto \frac{1}{\text{Charge}} \propto \frac{-I}{+I} \right]$$



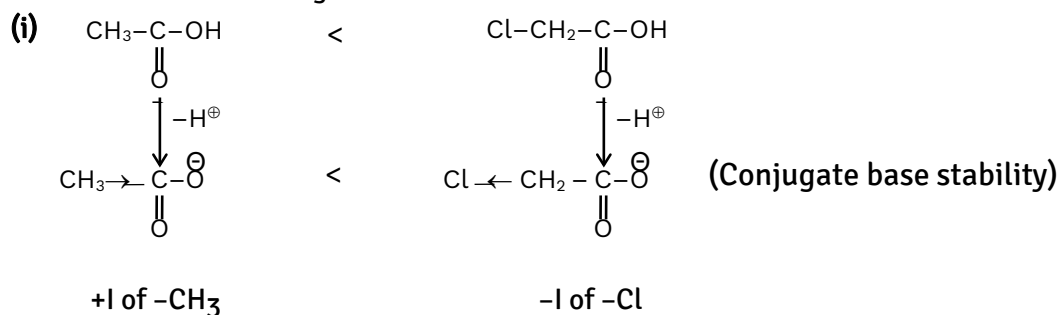
Number of +I group $\downarrow$ +I effect $\downarrow$ Stability of $\text{C}^\ominus \uparrow$



-I effect $\downarrow$ Stability of $\text{C}^\ominus \downarrow$

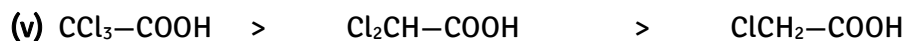
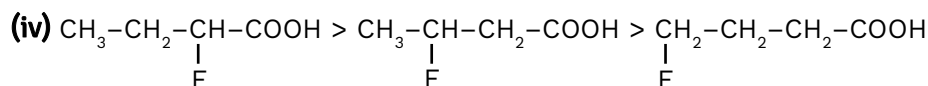
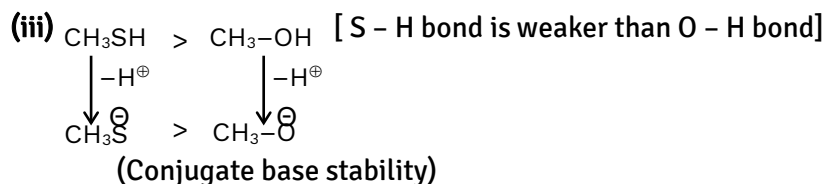
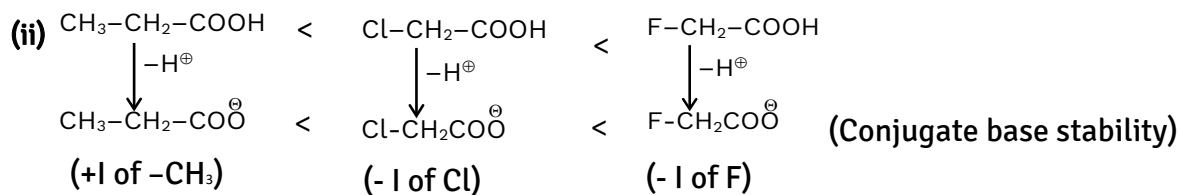
**(3) Acidic Strength:**

$$\left[\text{Acidic strength} \propto K_a \propto \frac{1}{pK_a} \propto \text{Stability of conjugate base (Anion)} \right]$$

Ex. Order of acidic strength:

$$\left[\text{Acidic strength} \propto \text{Stability of conjugate base (anion)} \propto \frac{-I}{+I} \right]$$

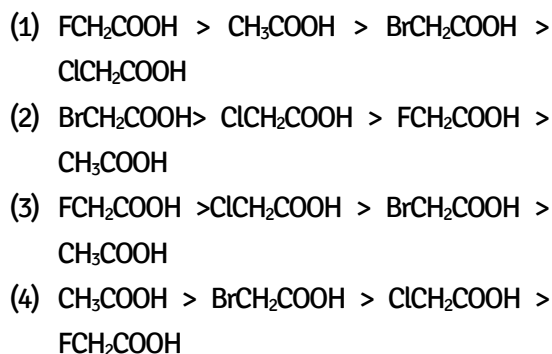
$$\boxed{\text{acidic strength} \propto \frac{-I}{+I}}$$



Exercise 1.1

- Which is most acidic compound:
 (1) $\text{C}_2\text{H}_5\text{COOH}$
 (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (3) CH_3COOH
 (4) HCOOH
- Maximum -I effect is exerted by the group
 (1) -F (2) $-\text{OCH}_3$
 (3) -Cl (4) $-\text{NO}_2$
- Which of the following is most stable:
 (1) $\text{CH}_3^\ominus$ (2) $\text{CH}_2=\text{CH}^\ominus$
 (3) $\text{CH}\equiv\text{C}^\ominus$ (4) $\text{CH}_3-\text{C}\equiv\text{C}^\ominus$
- Which of the following substituted carboxylic acids has the highest K_a value:
 (1) $\text{CH}_3-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-\text{COOH}$
 (2) $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\text{CH}_2-\text{COOH}$
 (3) $\underset{\text{Cl}}{\text{CH}_2}-\text{CH}_2-\text{CH}_2-\text{COOH}$
 (4) $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_2-\text{COOH}$
- Which of the following is electrophile
 (1) SO_3 (2) Na^+ (3) $\text{F}^\ominus$
 (4) NH_3
- Which of the following is nucleophile
 (1) H_2O (2) $\text{CH}_2=\text{CH}_2$
 (3) Both 1 & 2 (4) None of these
- The inductive effect:
 (1) Implies the atom's ability to cause bond polarization
 (2) Increases with increase of distance
 (3) Implies the transfer of lone pair of electrons from more electronegative atom to the lesser electronegative atom in a molecule
 (4) Implies the transfer of lone pair of electrons from lesser electronegative atom to the more electronegative atom in a molecule
- The correct order of increasing acidic strength of the compounds:
 (a) $\text{CH}_3\text{CO}_2\text{H}$ (b) $\text{MeOCH}_2\text{CO}_2\text{H}$
 (c) $\text{CF}_3\text{CO}_2\text{H}$ (d) $\text{Me}_2\text{CHCO}_2\text{H}$
 (1) $d < a < c < b$ (2) $d < a < b < c$
 (3) $a < d < c < b$ (4) $b < d < a < c$
- Which of the following groups has the highest +I effect:
 (1) CH_3- (2) CH_3CH_2-
 (3) $(\text{CH}_3)_2\text{CH}-$ (4) $(\text{CH}_3)_3\text{C}-$
- Pair of groups exerting (-I) effect is:
 (1) $-\text{NO}_2$ and $-\text{CH}_3$ (2) $-\text{NO}_2$ and $-\text{Cl}$
 (3) $-\text{Cl}$ and $-\text{CH}_3$ (4) $-\text{CH}_3$ and $-\text{C}_2\text{H}_5$
- Correct order of acidic strength is:
 (1) $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{C}_2\text{H}_5\text{COOH}$
 (2) $\text{C}_2\text{H}_5\text{COOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
 (3) $\text{HCOOH} > \text{C}_2\text{H}_5\text{COOH} > \text{CH}_3\text{COOH}$
 (4) $\text{CH}_3\text{COOH} > \text{HCOOH} > \text{C}_2\text{H}_5\text{COOH}$
- The number of electrons present in the valence shell of carbon of $\text{CH}_3\text{CH}_2^\oplus$ ion bearing +ve charge:
 (1) 8 (2) 7
 (3) 6 (4) 4
- The hybridisation of C of methylcarbanion is:
 (1) sp^3 (2) sp^2
 (3) sp (4) All of these
- Carbanion of $\text{CH}_3\text{CH}_2^\ominus$ has shape:
 (1) Planar
 (2) Pyramidal
 (3) Square bipyramidal
 (4) All of the above

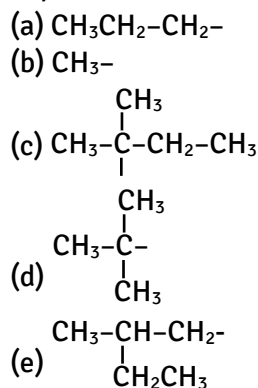
15. Which of the following presents the correct order of the acidic strength in the given compounds:



16. The correct order of decreasing acidic strength of trichloroacetic acid (A), trifluoroacetic acid (B), acetic acid (C) and formic acid (D) is:



17. Express in decreasing order of (+I) -



Correct answer is -

- (1) $(\text{c}) > (\text{d}) > (\text{e}) > (\text{a}) > (\text{b})$
 (2) $(\text{d}) > (\text{a}) > (\text{b}) > (\text{c}) > (\text{e})$
 (3) $(\text{a}) > (\text{b}) > (\text{c}) > (\text{d}) > (\text{e})$
 (4) $(\text{a}) > (\text{b}) > (\text{c}) > (\text{e}) > (\text{d})$

18. Which of the following has maximum pK_a :

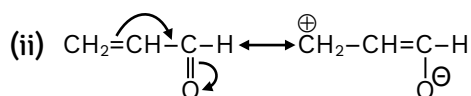
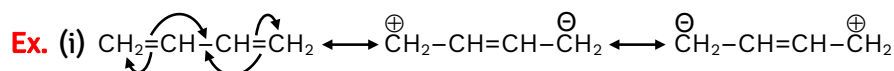


Resonance:

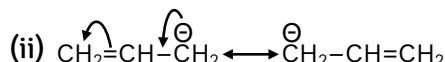
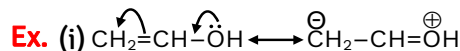
It involves complete delocalization of πe^- / $e^\ominus$ of lp /unpaired $e^\ominus$ is called as resonance or complete transfer of πe^- from one shell to another shell is called as Resonance.

Conditions For Resonance:

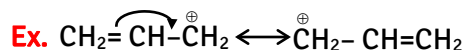
1. When π bonds are in conjugated system then e^- of one π bond are transferred towards another π bond.



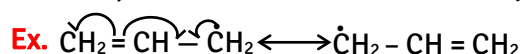
2. If lone pair and π -bond are in conjugated system then e^- of lone pair are transferred towards π bond.



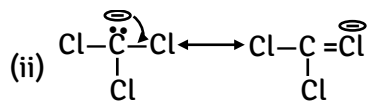
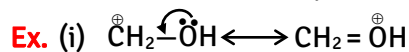
3. If positive charge (vacant orbital) and π bond are in conjugated system then e^- of π bond are transferred towards positive charge (vacant orbital).



4. If unpaired $e^\ominus$ and π bond are in conjugated system.



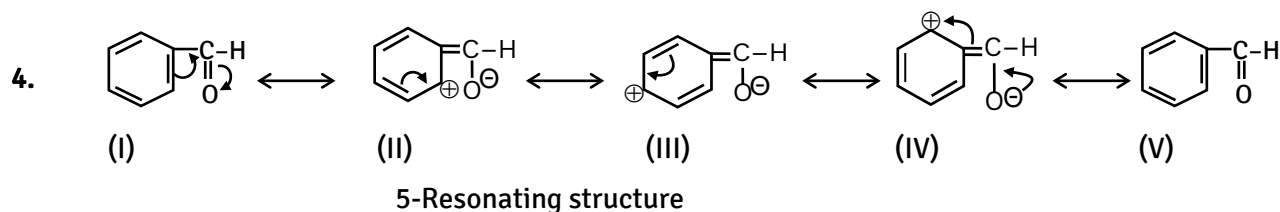
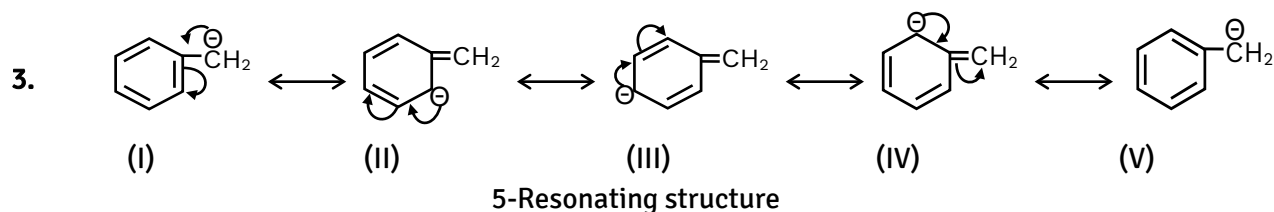
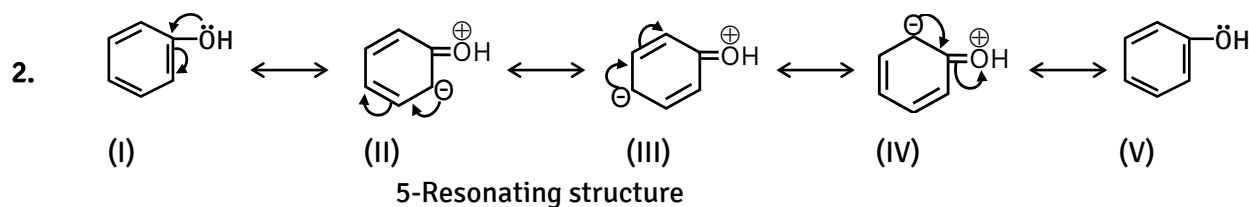
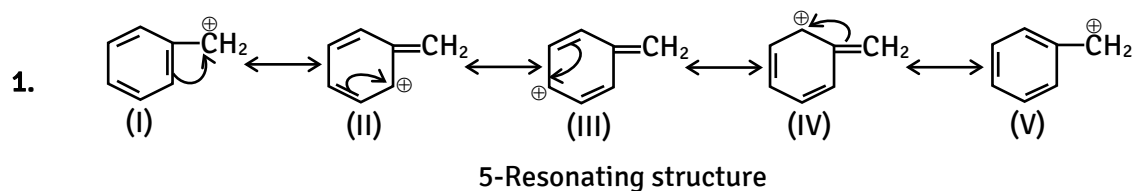
5. If lone pair and positive charge (vacant orbital) are in conjugated system then e^- of lone pair are transferred towards positive charge (vacant orbital).



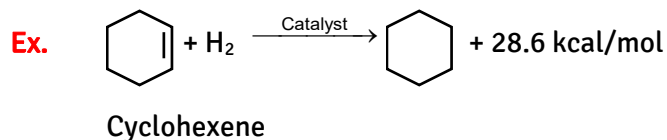
The Rules to draw resonating structure:

- The positions of the nuclei of the atoms must remain the same in all resonating structures.
- All canonical forms must have the same number of unpaired electron.
- All resonating structure must have same net charge/Sum of charge.
- All resonating structure must be according to lewis dot formula.

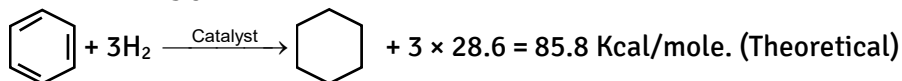
Draw Resonating Structures:



Resonance Energy: The difference in energy between the resonance hybrid and the most stable canonical structure is referred as the resonance energy. It is consumed in stabilising the molecule. The resonance energy of a resonance hybrid is the difference between the theoretical and experimental heats of hydrogenation of the compound.



Accordingly,



But practical value of heat of hydrogenation is 49.8 Kcal/mol

Therefore, benzene has $85.8 - 49.8$ Kcal/mol less energy than expected for a typical compound with three double bonds. Hence resonance energy of benzene molecule is $85.8 - 49.8 = 36.0$ Kcal/mole.

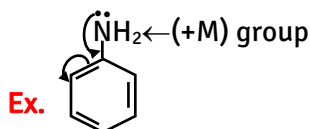
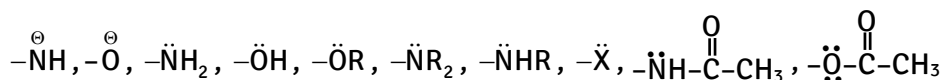
Types of Mesomeric Effect:

(a) +M effect

(b) -M effect

- (a) **Positive Mesomeric Effect (+M Effect):** When the group releases electron to the conjugated system it shows +M effect.

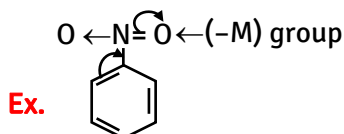
+M Group: Lone pair containing group like



order of +M Group: $-\text{NH}^- > -\text{O}^- > -\text{NH}_2 > -\text{NHR} > -\text{NR}_2 > -\text{OH} > -\text{OR} > -\text{NH}-\text{C}(=\text{O})-\text{CH}_3 > -\ddot{\text{O}}-\text{C}(=\text{O})-\text{CH}_3 > -\ddot{\text{F}} > -\ddot{\text{Cl}} > -\ddot{\text{Br}} > -\ddot{\text{I}}$

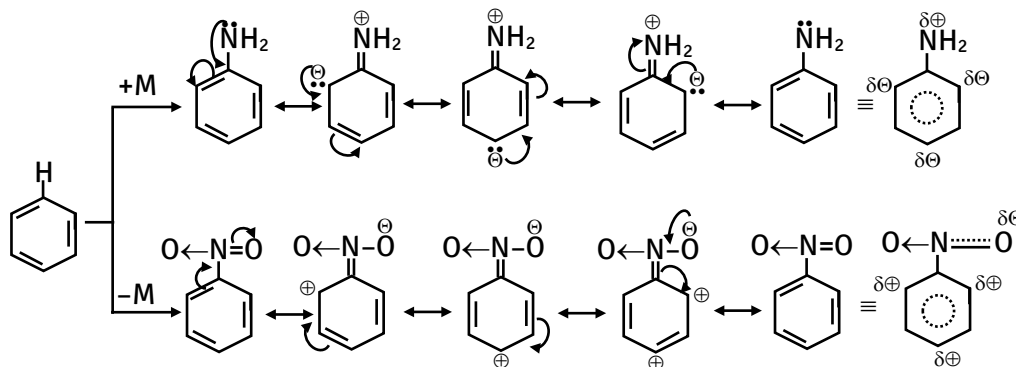
- (b) **Negative Mesomeric Effect (-M Effect):** When the group withdraws electron from the conjugated system it shows -M effect.

-M group: $-\text{CHO}, -\text{COOH}, -\text{COOR}, -\text{COR}, -\text{NO}_2, -\text{CN}, -\text{CONH}_2, -\text{SO}_3\text{H}$



order of -M Group: $-\text{NO}_2 > -\text{CN} > -\text{SO}_3\text{H} > -\text{CHO} > -\text{COR} > -\text{COOR} > -\text{COOH} > -\text{CONH}_2$

M-effect of a group on benzene ring :



- M-effect of a group affects ortho and para position while meta position remains unaffected.
- +M group increases πe^- density at ortho and para position and -M group decrease πe^- density at ortho and para position in both the cases meta position remains neutral.

- M-effect of a group affects ortho and para position equally while I-effect depends on distance so it is maximum at ortho and minimum at para.
- Comparison of M and I-effect of a group.
In general M-effect of a group dominates over its I-effect but in the following groups their I-effect dominates over their M-effect.



Comparison of stability of resonating structure:

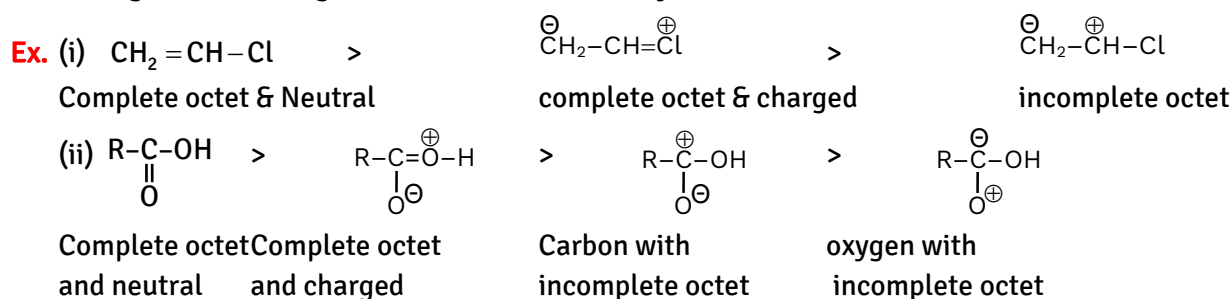
- Resonating structure having complete octate is more stable
- Neutral resonating structure is more stable than charged resonating structure

Or

Resonating structure having more covalent bond is more stable

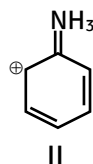
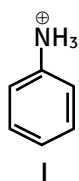
- Resonating structure having negative charge on more EN atom and positive charge on less EN atom is more stable
- Resonating structure having opposite charge closer and same charge away is more stable

Ex. Arrange the following in correct order of stability.



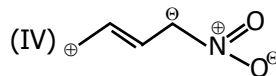
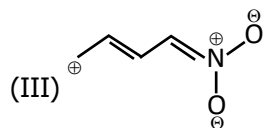
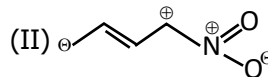
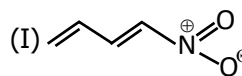
Exercise 1.2

- 1.** Examine the following two structures for the anilinium ion and choose the correct statement from the ones given below.



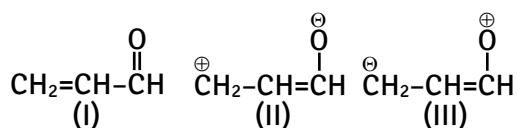
- II is not an acceptable canonical structure because carbonium ions are less stable than ammonium ion.
- II is not an acceptable canonical structure because it is non-aromatic.
- II is not an acceptable canonical structure because the nitrogen has 10 valence electrons.
- II is an acceptable canonical structure.

- 2.** Find the stability order of given resonating structures:



- (1) I > II > III > IV (2) I > III > II > IV
(3) I > III > IV > II (4) IV > I > III > II

3. The order of stability of the following resonating structures is:

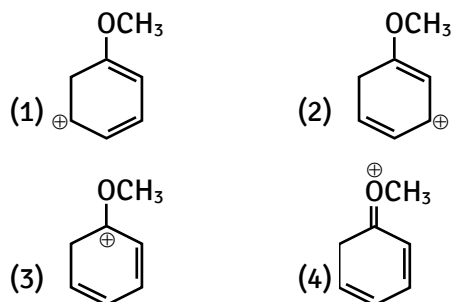


- (1) II > I > III (2) I > III > II
(3) I > II > III (4) III > II > I

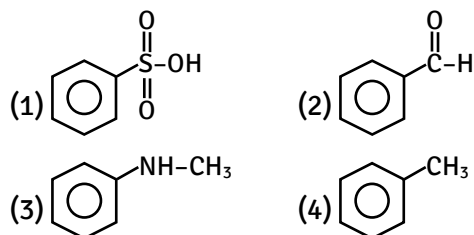
4. Mesomeric effect is the resonance of:

- (1) π - electrons and lone pair
- (2) σ - electrons only
- (3) π and σ -electrons both
- (4) π - electrons only

5. Which of the following resonating structures contributes the most to resonance hybrid?



6. Which of the following group has (+M) effect on benzene ring ?



7. $\text{CH}_2^{\ominus}\text{-C(=O)-CH}_3$ and $\text{CH}_2=\text{C}(\text{O}^{\ominus})\text{-CH}_3$ are:

- (1) Resonating structures
- (2) Tautomers
- (3) Geometrical isomers
- (4) Optical isomers

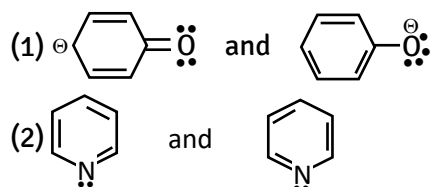
8. Which of the following is wrong about resonance:

- (1) Most stable resonating structures contribute the most to resonance hybrid.
- (2) All resonating structures have same number of unpaired electron.
- (3) All resonating structures have same number of bond pair electrons.
- (4) All resonating structures have same amount of net charge.

9. Which of the following contain + M & - I effect.

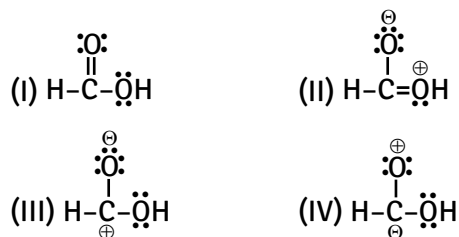
- (1) O = CH - (2) - NO₂
(3) - Cl (4) CH₃ -

10. Which of the following is not a pair of resonating structures?



- (3)  and 
- (4)  and 

11.



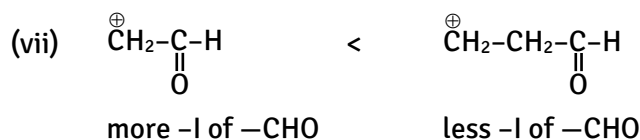
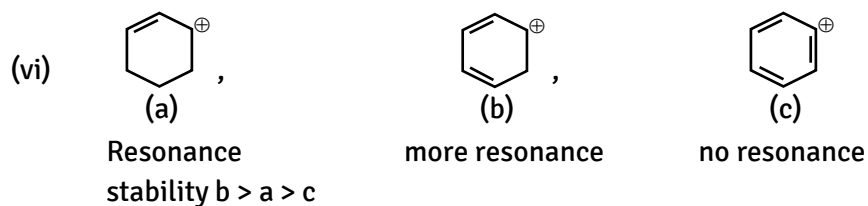
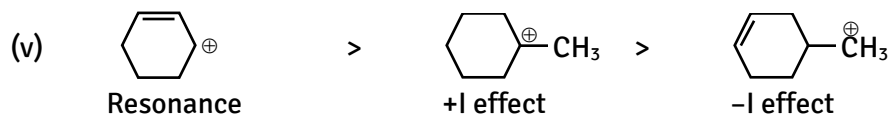
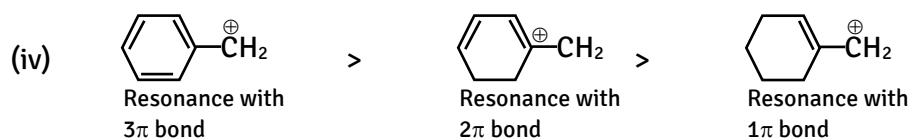
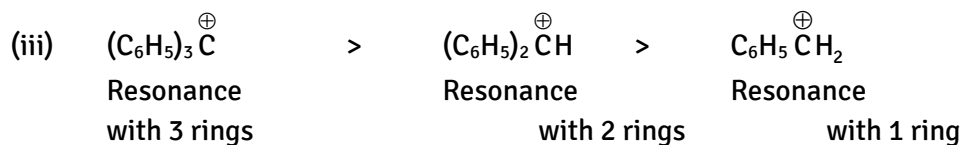
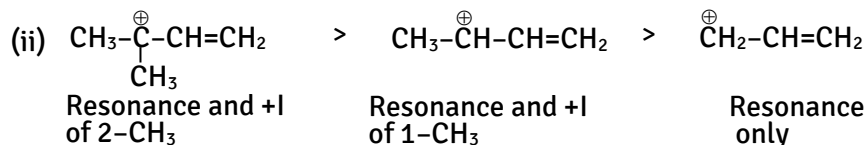
Increasing order of stability is -

- (1) I < III < II < IV
(2) IV < III < II < I
(3) III < IV < II < I
(4) II < IV < III < I

1. Stability of Carbocation:

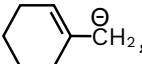
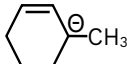
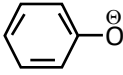
- Stability $\propto +M / +I \propto \frac{1}{-M / -I}$

(i) $\text{CH}_2=\text{CH}-\overset{\oplus}{\text{C}}\text{H}_2$ > $\text{CH}_3\rightarrow\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}_2$ > $\text{CH}_2=\text{CH}\leftarrow\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}_2$
 stable by +I of Alkyl group
 resonance -I of Vinyl group



- Stability $\propto -M / -I \propto \frac{1}{+M / +I}$

Ex. Give stability order of following compounds

- (i) $\text{CH}_2=\text{CH}-\overset{\ominus}{\text{CH}_2}$, $\text{CH}_2=\overset{\ominus}{\text{CH}}$, $\text{CH}_3-\overset{\ominus}{\text{CH}_2}$
 charge on sp^2C & resonance charge on sp^2C charge on sp^3C
 stability order: I > II > III
- (ii) $(\text{C}_6\text{H}_5)_3\overset{\ominus}{\text{C}} > (\text{C}_6\text{H}_5)_2\overset{\ominus}{\text{CH}} > \text{C}_6\text{H}_5\overset{\ominus}{\text{CH}_2} > \text{CH}_2=\text{CH}-\overset{\ominus}{\text{CH}_2} > \text{CH}_2=\overset{\ominus}{\text{CH}} > \overset{\ominus}{\text{C}}\text{H}_3 > \text{CH}_3-\overset{\ominus}{\text{CH}_2}$
 $> (\text{CH}_3)_2-\overset{\ominus}{\text{CH}} > (\text{CH}_3)_3\overset{\ominus}{\text{C}}$
- (iii) , , 
 (a) (b) (c)
 Resonance with 1 π resonance with 2 π no resonance
 stability order: b > a > c
- (iv) $\overset{\ominus}{\text{CH}_2}-\text{NO}_2$, $\overset{\ominus}{\text{CH}_2}-\text{CH}_2-\text{NO}_2$, $\text{CH}_3-\overset{\ominus}{\text{CH}}-\text{NO}_2$
 (a) (b) (c)
 Resonance No resonance Resonance but +I effect
 stability order: a > c > b
- (v) , , 
 (a) (b) (c)
 Resonance and +I effect Resonance Resonance and more +I effect
 stability order: b > a > c
- (vi) $\text{CH}_3-\overset{\text{O}}{\overset{\parallel}{\text{C}}}-\overset{\ominus}{\text{O}}$, 
 (a) (b)
 Resonance with oxygen Resonance with carbon
 (Equivalent R.S.) (Non equivalent R.S.)
 stability order: a > b

3. Stability of free radical :

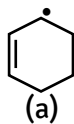
- Comparison similar to carbocation
- Stability $\propto$ Extent of resonance

$$\propto +M / +I \propto \frac{1}{-M / -I}$$

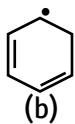
- Ex. (i)** , , , 
 (a) (b) (c) (d)
 Resonance and more +I effect Less +I effect More +I effect Resonance and less +I effect

Ans. a > d > c > b

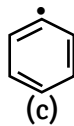
Ex. (ii)



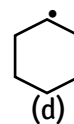
Resonance
with 1π bond



Resonance
with 2π bond



No resonance &
unpaired e^- on
unsaturated C
-I effect

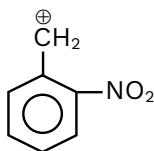


No resonance
+I -effect

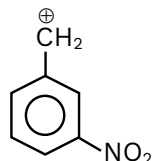
Ans. $b > a > d > c$

Question: Give stability order of following compounds

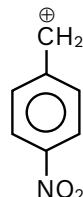
(i)



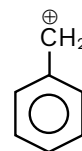
-M and
more -I
stability order is



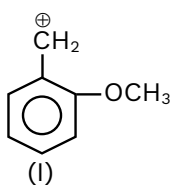
M = 0
only -I
 $IV > II > III > I$



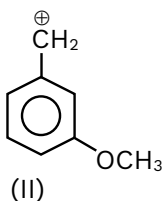
-M and
less -I



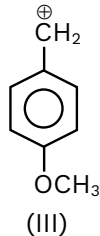
(ii)



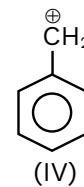
+M and
more -I
Stability order $III > I > IV > II$



M = 0
only -I

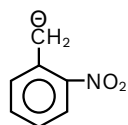


+M and
less -I

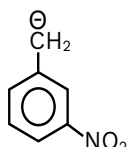


(Reference)

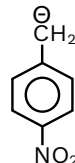
(iii)



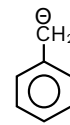
-M and
more -I
stability order $I > III > II > IV$



M = 0
only -I



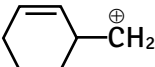
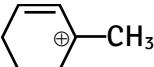
-M and
less -I



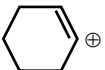
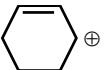
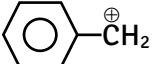
(Reference)

Exercise 1.3

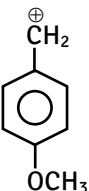
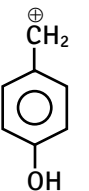
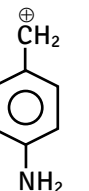
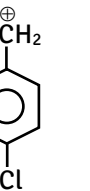
1. Most stable carbocation is:

- (1) $\text{CH}_2=\text{CH}^+$ (2) $\text{CH}_2=\text{CH}-\text{CH}_2^+$
 (3)  (4) 

2. Least stable carbonium ion in the following will be:

- (1)  (2) 
 (3)  (4) 

3. Arrange stability of the given carbocations in decreasing order:

-    
 (I) (II) (III) (IV)
 (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{III} > \text{II} > \text{I} > \text{IV}$
 (3) $\text{IV} > \text{I} > \text{II} > \text{III}$ (4) $\text{III} > \text{I} > \text{II} > \text{IV}$

4. Arrange the carbanions, $(\text{CH}_3)_3\text{C}^-$, CCl_3^- , $(\text{CH}_3)_2\text{CH}^-$, $\text{C}_6\text{H}_5\text{CH}_2^-$ in order of their decreasing stability:

- (1) $(\text{CH}_3)_2\text{CH}^- > \text{CCl}_3^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_3\text{C}^-$
 (2) $\text{CCl}_3^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_2\text{CH}^- > (\text{CH}_3)_3\text{C}^-$
 (3) $(\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > \text{CCl}_3^-$
 (4) $\text{C}_6\text{H}_5\text{CH}_2^- > \text{CCl}_3^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^-$

5. The increasing order of stability of the following free radicals is:

- (1) $(\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
 (2) $(\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
 (3) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H}$
 (4) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}}$

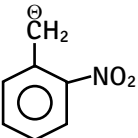
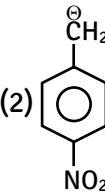
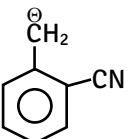
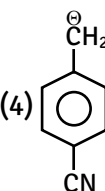
6. Which of the following is most stable:

- (1) CH_3^+
 (2) $\text{CH}_3-\text{CH}_2^+$
 (3) $\text{CH}_3-\text{CH}^+-\text{CH}_3$
 (4) $\text{CH}_2=\text{CH}-\text{CH}_2^+$

7. Which of the following is least stable carbocations:

- (1) $\text{CH}_2=\text{CH}^+$ (2) $\text{CH}_3-\text{CH}_2^+$
 (3) $(\text{Ph})_2\text{CH}^+$ (4) PhCH_2^+

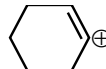
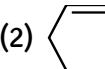
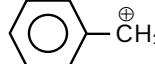
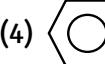
8. Which one of the carbanions is most stable:

- (1)  (2) 
 (3)  (4) 

9. Choose the most stable Carbocation:

- (1) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2^+$
 (2) $\text{CH}_2=\text{CH}-\text{CH}^+-\text{CH}_3$
 (3) $\text{CH}_2=\text{C}^+-\text{CH}_2-\text{CH}_3$
 (4) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2^+$

10. Most stable carbonium ion in the following will be:

- (1)  (2) 
 (3)  (4) 

11. Which free radical is the most stable:

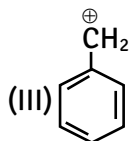
- (1) $\text{C}_6\text{H}_5-\dot{\text{C}}\text{H}_2$ (2) $\text{CH}_2=\text{CH}-\dot{\text{C}}\text{H}_2$
 (3) $\text{CH}_3-\dot{\text{C}}\text{H}-\text{CH}_3$ (4) $\text{CH}_3-\dot{\text{C}}(\text{CH}_3)-\text{CH}_3$

12. Most stable carbanion is :-

- (1) $\text{HC} \equiv \text{C}^-$ (2) $\text{H}_2\text{C}=\text{CH}^-$
 (3) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}_2^-$ (4) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}^-$

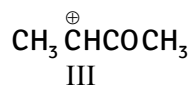
13. The order of stability of the following carbocations is-

- (I) $\text{CH}_2=\text{CH}-\text{CH}_2^+$
 (II) $\text{CH}_3-\text{CH}_2-\text{CH}_2^+$



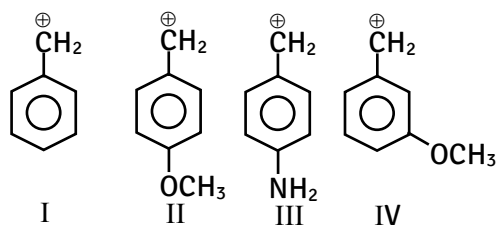
- (1) I > II > III (2) III > I > II
 (3) III > II > I (4) II > III > I

14. The order of decreasing stability of the Cations is-



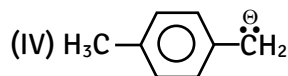
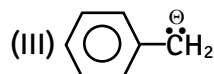
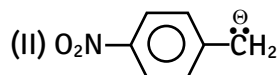
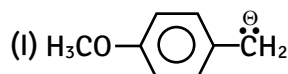
- (1) III > II > I (2) I > II > III
 (3) II > I > III (4) I > III > II

15. Correct order of stability is -



- (1) I > II > III > IV (2) III > II > I > IV
 (3) IV > I > II > III (4) II > III > I > IV

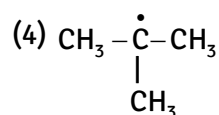
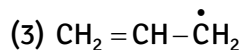
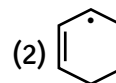
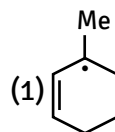
16. Consider the following carbanions :



Correct decreasing order of stability is-

- (1) II > III > IV > I (2) III > IV > I > II
 (3) IV > I > II > III (4) I > II > III > IV

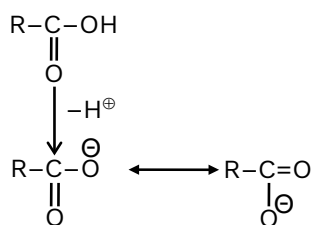
17. Which one is most stable free radical?



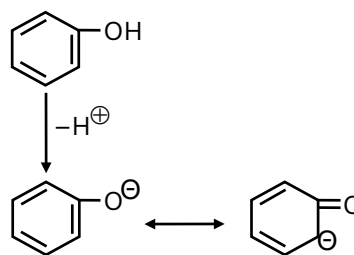
4. Acidic Strength comparison:

Acidic strength $\propto$ Stability of conjugate base

- Carboxylic acids are more acidic than phenols

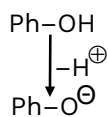


Two equivalent resonating structures make the anion stable so corresponding acid is more acidic.



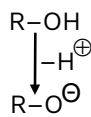
5, unequal R.S. less stable anion

- Phenol is more acidic than alcohols:



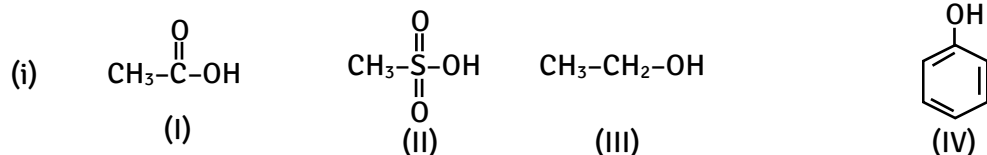
Conjugate base stable by resonance

So, it is more acidic.

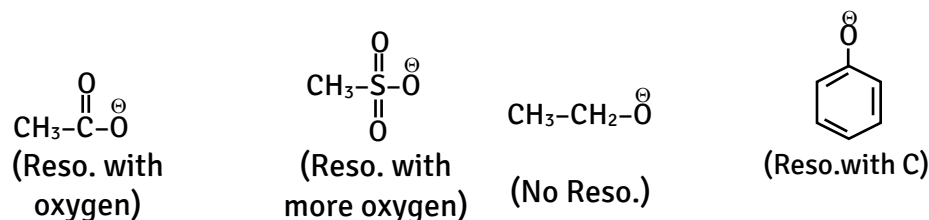


no resonance

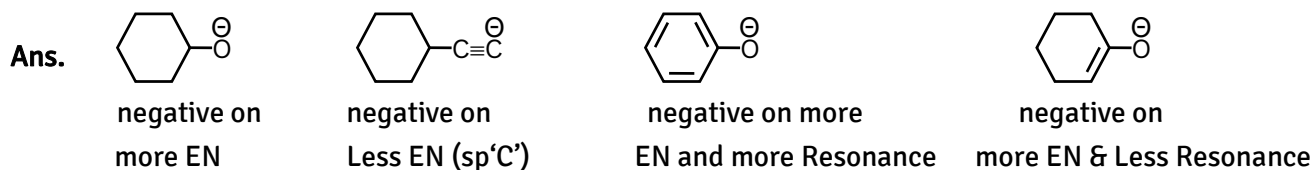
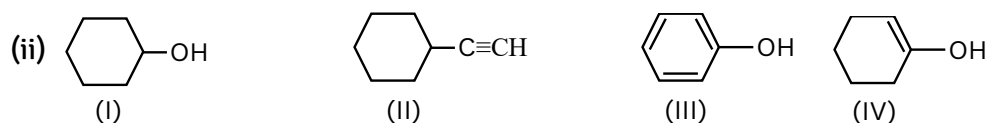
Question: Give acidic strength order for following compound



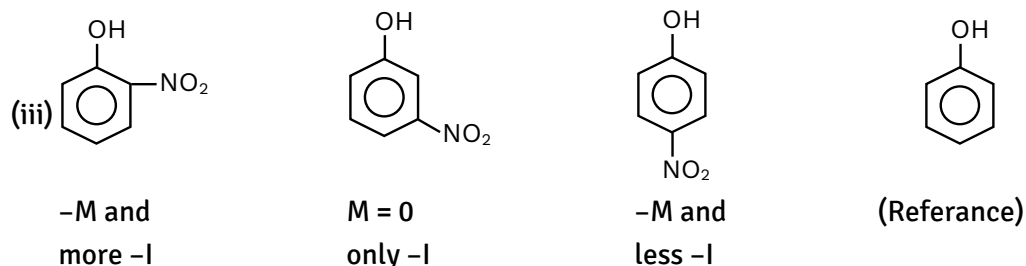
Ans. Acidic strength $\propto$ Stability of conjugate base



Acidic strength: II > I > IV > III



Acidic order – III > IV > I > II

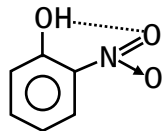


but intra molecular

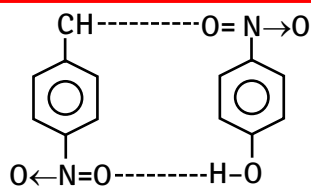
H-bonding

Acidic strength $\rightarrow$ III > I > II > IV

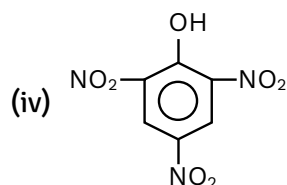
Note: Due to intramolecular H-bonding in ortho nitrophenol it is less acidic than para nitrophenol.



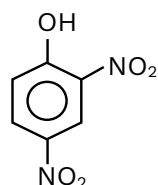
intramolecular H-bonding



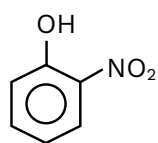
intermolecular H-bonding



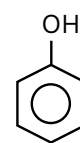
3 (-M)



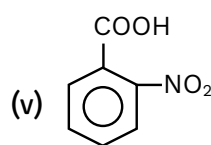
2 (-M)



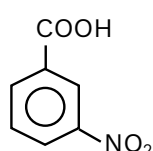
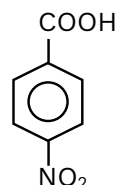
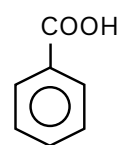
1 (-M)



(Reference)

Acidic strength $\rightarrow 1 > 2 > 3 > 4$ 

ortho effect

M = 0
only -I-M and
less -I

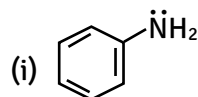
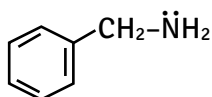
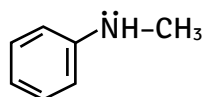
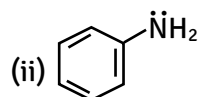
(Reference)

Acidic strength $\rightarrow 1 > 3 > 2 > 4$

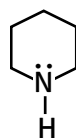
5. **Basic Strength Comparison:-** B.S. $\propto \text{H}^{\oplus}$ accepting tendency $\propto$ lone pair donating tendency.

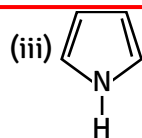
- Basic strength $\propto \frac{1}{\text{Resonance (l.p.)}}$
- Basic strength $\propto \frac{1}{\% \text{ S character of N}}$
- Basic strength $\propto +M, +I \propto \frac{1}{-M, -I}$

Question: Give basic strength order:

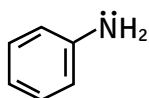
l.p. is in
resonancel.p. is not in
resonancel.p. is in resonance
and +I of CH_3 Basic strength $\rightarrow \text{II} > \text{III} > \text{I}$ 

l.p. in resonance

l.p. is not in
resonance & Nsp^2 l.p. is not in
resonance & Nsp^3 Basic strength $\rightarrow \text{III} > \text{II} > \text{I}$



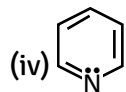
distribution of
l.p. among N & four C
(l.p. is part of aromaticity)
Basic strength $\rightarrow$ III > II > I



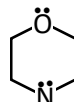
distribution of l.p.
among N & three C
(lp is not the part of aromaticity)



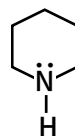
l.p. is not in
resonance



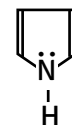
l.p. is not in
resonance & Nsp^2



l.p. is not in resonance
 Nsp^3 & -I effect of
oxygen



l.p. is not in resonance
& +I-effect



distribution of
l.p. among N & four C
(l.p. is part of aromaticity)

Basic strength $\rightarrow$ III > II > I > IV

Comparison of basic strength in aliphatic amines

(i) CH_3-NH_2 , $(CH_3)_2NH$, $(CH_3)_3N$, NH_3
B.S. in aqueous medium $2^\circ > 1^\circ > 3^\circ >$ ammonia

In gas phase : $3^\circ > 2^\circ > 1^\circ >$ ammonia

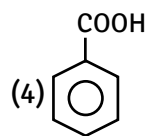
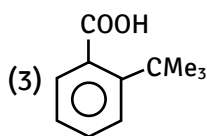
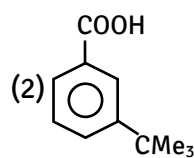
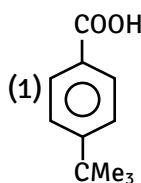
(ii) $C_2H_5-NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$, NH_3
B.S. in aqueous medium $2^\circ > 3^\circ > 1^\circ >$ ammonia

In gas phase : $3^\circ > 2^\circ > 1^\circ >$ ammonia

Note: if medium is not given then consider aqueous medium

Exercise 1.4

1. Acidic strength order is:-



(1) $4 > 1 > 2 > 3$
(3) $4 > 3 > 1 > 2$

(2) $3 > 1 > 2 > 4$
(4) $3 > 4 > 1 > 2$

2. The correct order of acidic strength of the following compounds is:

A. Phenol
C. m-Nitrophenol
(1) $D > C > A > B$
(3) $A > B > D > C$

B. p-Cresol
D. p-Nitrophenol
(2) $B > D > A > C$
(4) $C > B > A > D$

3. Amongst the following, the most basic compound is:

- (1) $C_6H_5NH_2$
- (2) $p-NO_2 - C_6H_4NH_2$
- (3) $m-NO_2 - C_6H_4NH_2$
- (4) $C_6H_5CH_2NH_2$

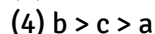
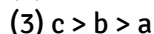
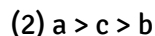
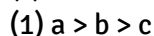
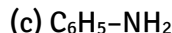
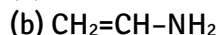
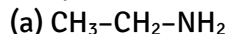
4. Amongst the following the most basic compound is:

- (1) aniline
- (2) benzylamine
- (3) p-nitroaniline
- (4) acetanilide

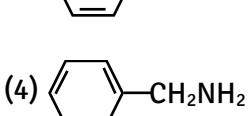
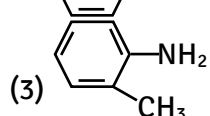
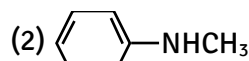
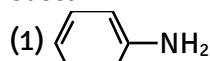
5. Correct acidic nature is :-

- (1) $CH_3OH < CH_3SH < CH_3NH_2$
- (2) $CH_3NH_2 < CH_3OH < CH_3SH$
- (3) $CH_3SH < CH_3OH < CH_3NH_2$
- (4) None of these

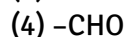
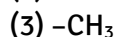
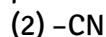
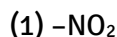
6. Arrange basic strength of the given compounds in decreasing order:



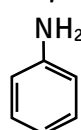
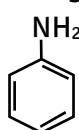
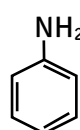
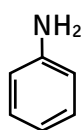
7. Which of the following is the strongest base:



8. Which of the following substituents will decrease the acidity of phenol:



9. Consider the following compounds:



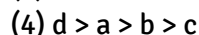
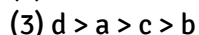
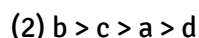
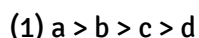
(a)

(b)

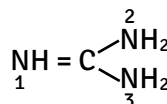
(c)

(d)

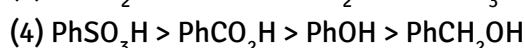
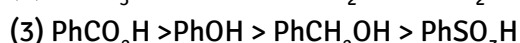
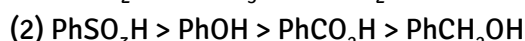
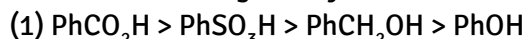
Arrange these compounds in decreasing order of their basicity:



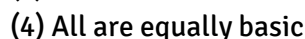
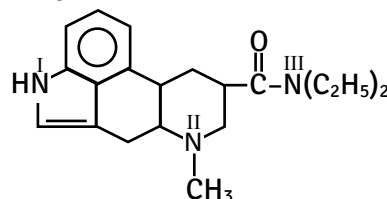
10. Which nitrogen is protonated readily in the guanidine?



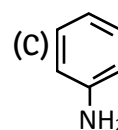
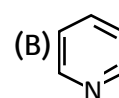
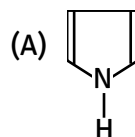
11. Which of the following shows the correct order of decreasing acidity:



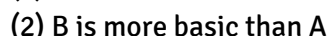
12. Which nitrogen in LSD (Lysergic acid and diethylamide) is most basic:



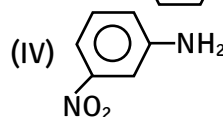
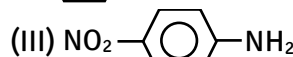
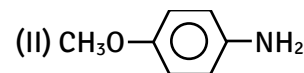
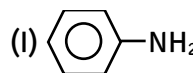
- 13.



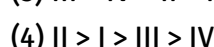
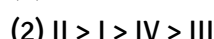
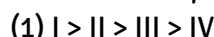
Choose the incorrect statement:



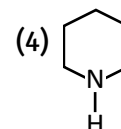
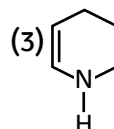
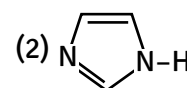
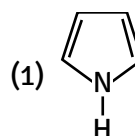
- 14.



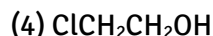
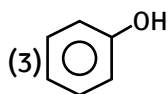
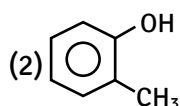
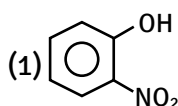
The correct order of decreasing basicity of the above compound is:



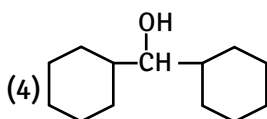
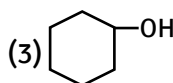
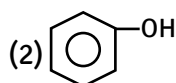
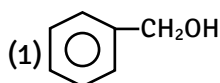
15. Which one of the following is least basic in character?



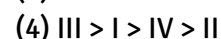
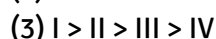
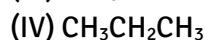
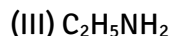
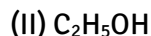
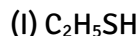
16. Which one of the following compounds is most acidic:



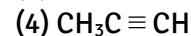
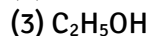
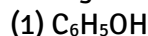
17. Which one of the following compounds has the most acidic nature



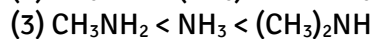
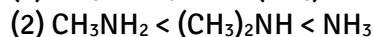
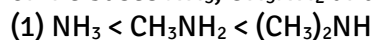
18. Arrange the following in correct order of acidic strength :



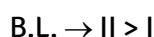
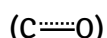
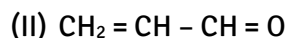
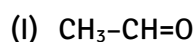
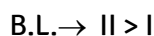
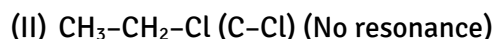
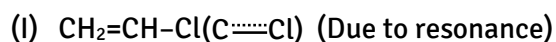
19. Among the following the $\text{p}K_a$ is minimum for :-



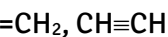
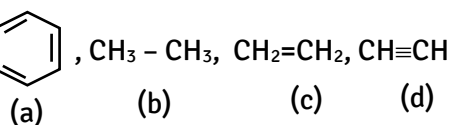
20. The correct order of increasing basic nature of the bases NH_3 , CH_3NH_2 and $(\text{CH}_3)_2\text{NH}$ is -



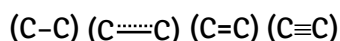
6. Bond length comparison:



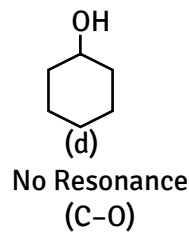
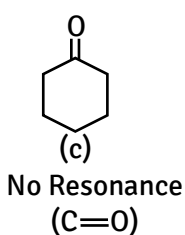
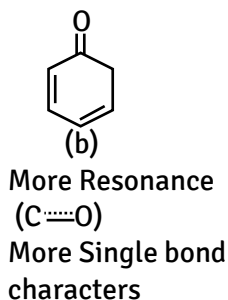
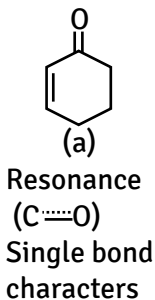
Ex. (i) Compare C-C bond length



Ans. $b > a > c > d$



Ex. (ii) Compare C–O bond length



Ans. $d > b > a > c$

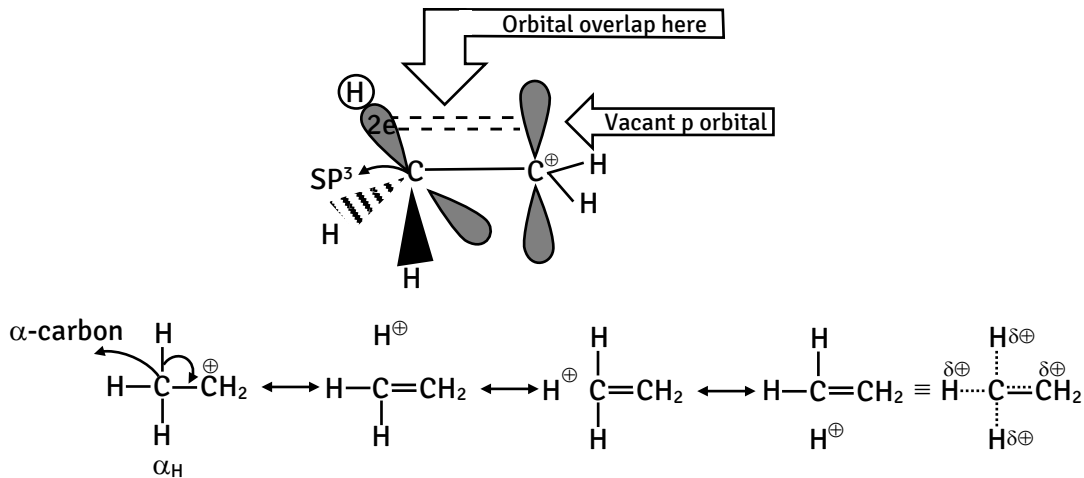
Hyperconjugation Effect (H-Effect)

H-Effect: Complete delocalisation of e^- of C-H σ bond towards carbon-carbon π bond or positive charge or free radical is called as H-effect.

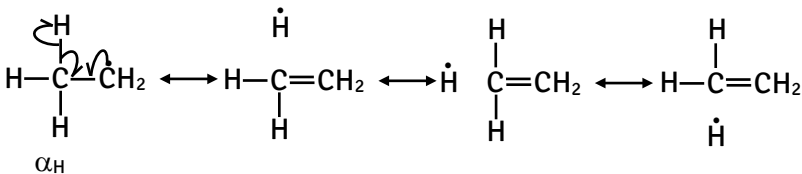
- It is also called as no bond resonance and Nathen & Baker effect.
- It is permanent effect
- It is distance independent effect

Conditions of H-Effect:

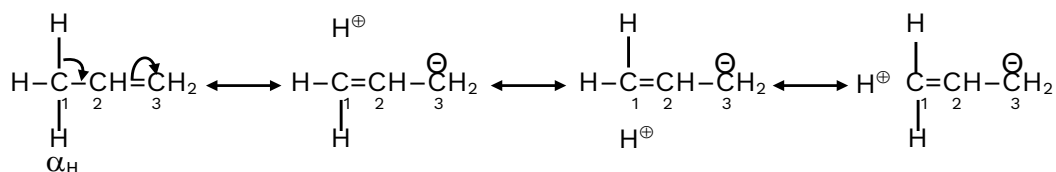
- 1.** When C-H σ bond of sp^3 'C' is in conjugation with +ve charge in carbocation.



2. When C-H σ bond of sp^3 'C' is in conjugation with unpaired e^- in free radical.

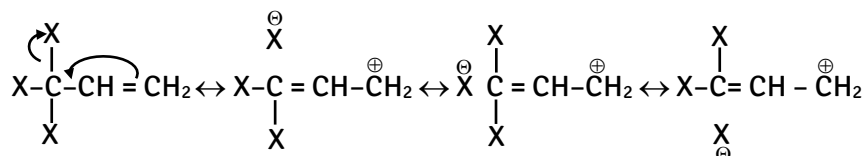


3. When C-H σ bond of sp^3 'C' is in conjugation with carbon-carbon double bond in alkene.



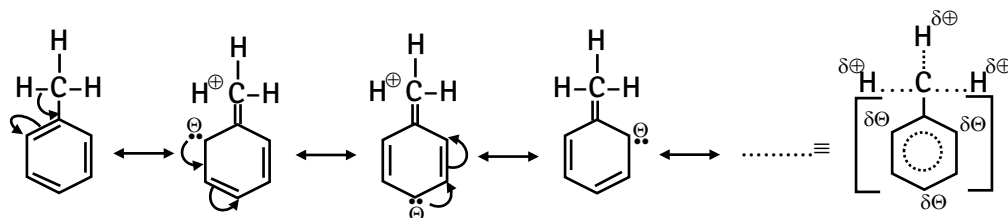
Note: In alkyl carbanion, H-effect is not observed.

4. When C-X σ bond of sp^3 'C' is in conjugation with carbon-carbon double bond then it is called **reverse hyper conjugation (-H effect)**

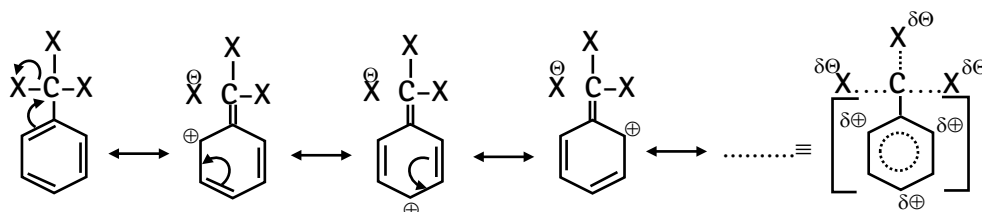


5. H-effect of a group on benzene ring:

+H-effect:



-H-effect:



- +H-effect of a group increases e^- density at o- & p-position.
- H-effect of a group decreases e^- density at o-&p-position.
- H-effect does not affect meta position.
- H-effect is equal at ortho and para position.

Order of +H-effect of a group on benzene ring.

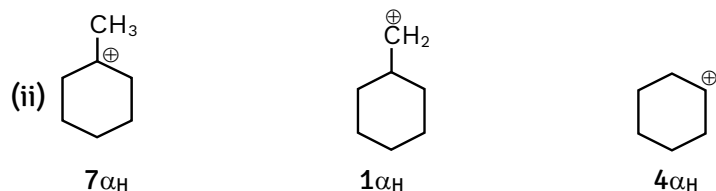
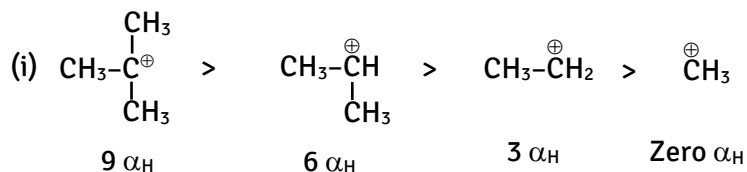
- $-\text{CH}_3 > -\text{CH}_2-\text{CH}_3 > -\text{CH}(\text{CH}_3)_2 > -\text{C}(\text{CH}_3)_3$
 $(3\alpha_{\text{H}}) \quad (2\alpha_{\text{H}}) \quad (1\alpha_{\text{H}}) \quad (\text{No } \alpha_{\text{H}})$
 (No H-effect)
- $-\text{CH}_3 > -\text{CD}_3$

Application of H-Effect

1. Stability of Carbocation:

- Stability of carbocation can be explained by M, I and H-effect.
- Stability $\propto$ H-effect $\propto$ No. of α_H

Ex. Give stability order for:-

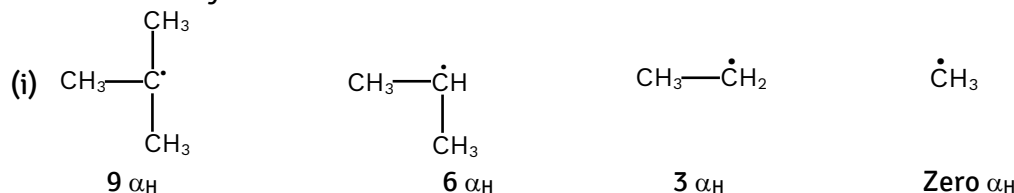


Stability order 1 > 3 > 2

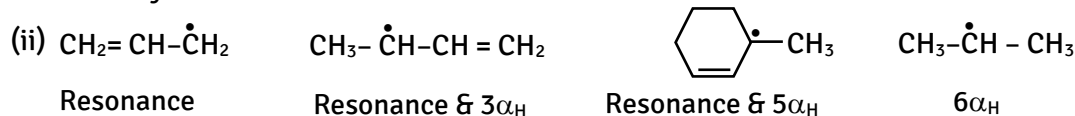
2. Stability of Carbon Free Radicals:

- Stability $\propto$ H-effect $\propto$ No. of α_H

Ex. Give stability order for:



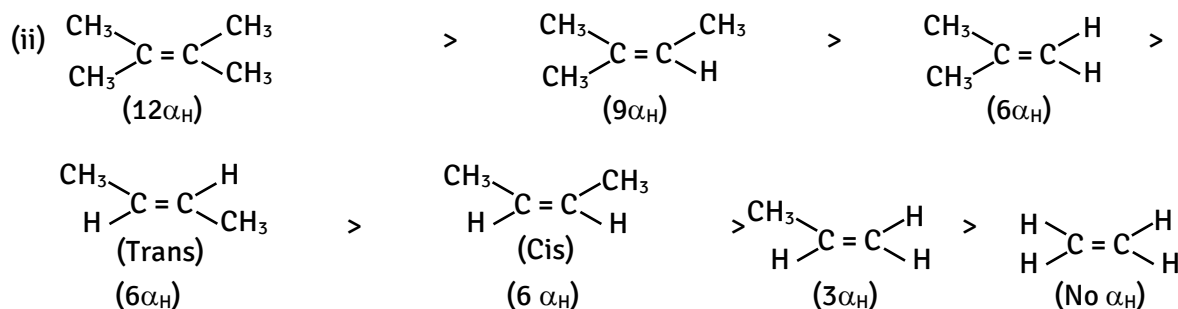
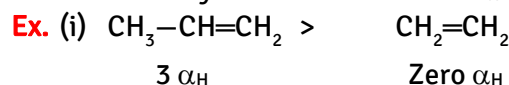
Stability order I > II > III > IV



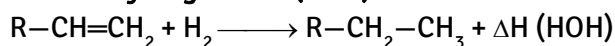
Stability order III > II > I > IV

3. Stability of Alkenes:

- Stability $\propto$ Resonance
- Stability $\propto$ H-effect $\propto$ No. of α_H



4. Heat of Hydrogenation(HOH):

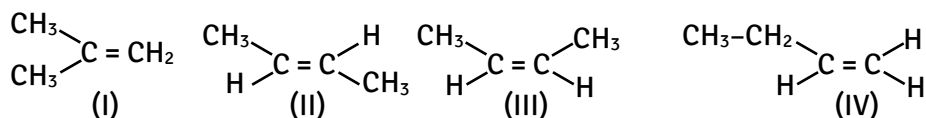


Heat evolved when one mole of any unsaturated hydrocarbon are hydrogenated is called heat of hydrogenation (HOH) If alkene is more reactive towards hydrogen then it will evolve more HOH.

$$HOH \propto \text{Number of } \pi \text{ bonds}$$

$$\propto \frac{1}{\text{stability of alkene}} \text{ (If } \pi \text{ bonds are equal)}$$

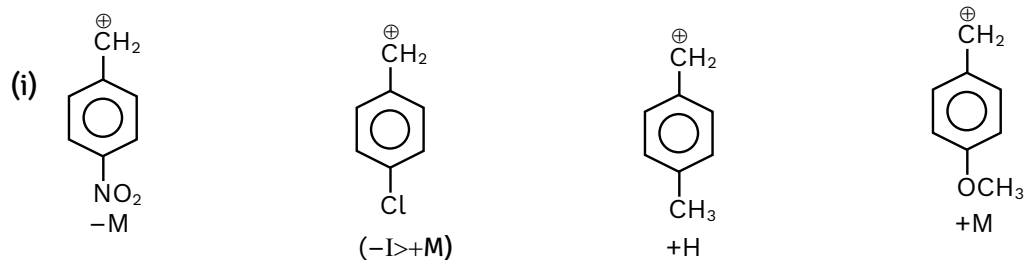
Question: Compare HOH



Stability: I > II > III > IV

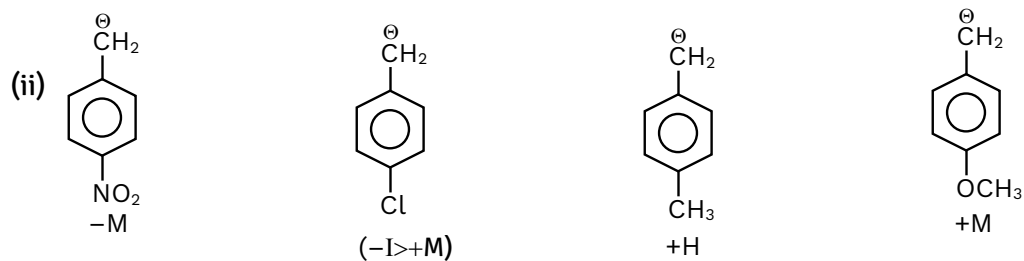
H.O.H. I < II < III < IV

Question: Give stability order of following compounds



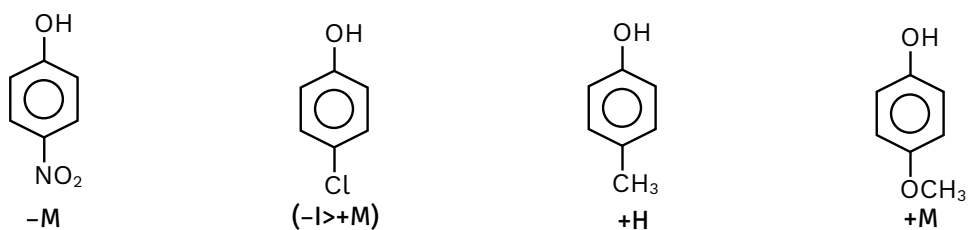
So stability order

IV > III > II > I



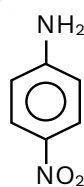
stability order I > II > III > IV

Question: Give acidic strength order for following compounds

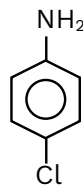


Acidic strength → I > II > III > IV

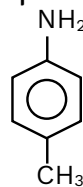
Question: Give basic strength order for following compounds



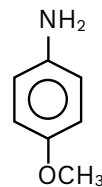
(-M)



(-I > +M)



(+H)



(+M)

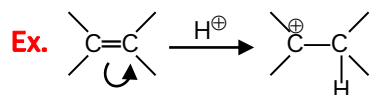
basic strength $\rightarrow$ IV > III > II > I

Electromeric Effect

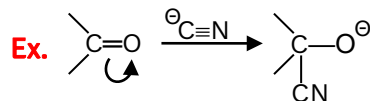
Electromeric Effect (E-effect): The complete transfer of the shared pair of π electron of a multiple bond to one of the shared atoms in the presence of an attacking reagent is called as E-effect.

It is a temporary effect and comes into play only when a reagent attacks on substrate having π electrons.

(a) **+E-effect:** When electrons of π bonds are transferred to atom of the multiple bond to which the reagent gets attached, the effect is called +E-effect.

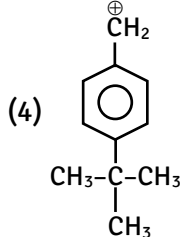
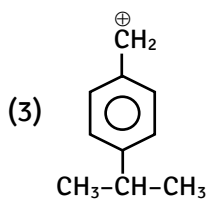
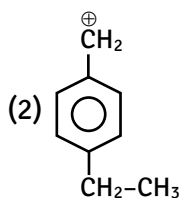
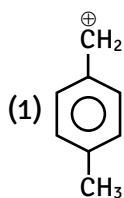


(b) **-E-Effect:** When electrons of π bond are transferred to atom of the multiple bond to which the attacking reagent is not attached, the effect is called as -E-effect.

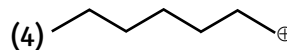
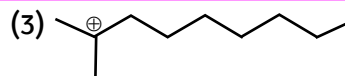
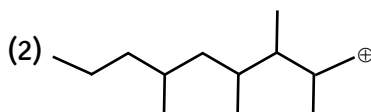
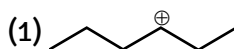


Exercise 1.5

1. Which of the following is most stable carbocation?



2. Select the most stable carbocation among the following:



3. Among the following alkenes

(I) 1-butene

(II) trans-2-butene

(III) cis-2-butene

(IV) Isobutylene

the order of decreasing stability is:

(1) II > I > III > IV

(2) III > IV > I > II

(3) IV > I > II > III

(4) IV > II > III > I

4. Correct order of stability is:

(1) $\text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}=\text{CH}_2 > (\text{CH}_3)_2\text{C}=\text{CH}_2$

(2) $\text{CH}_2=\text{CH}_2 < \text{CH}_3-\text{CH}=\text{CH}_2 < (\text{CH}_3)_2\text{C}=\text{CH}_2$

(3) $\text{CH}_2=\text{CH}_2 < (\text{CH}_3)_2\text{C}=\text{CH}_2 < \text{CH}_3-\text{CH}=\text{CH}_2$

(4) $\text{CH}_3-\text{CH}=\text{CH}_2 < \text{CH}_2=\text{CH}_2 < (\text{CH}_3)_2\text{C}=\text{CH}_2$

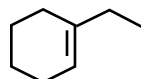
5. Arrange the Stability of following:



I



II



III

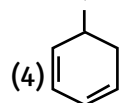
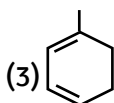
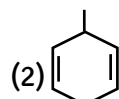
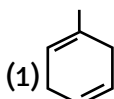
(1) $I < II < III$

(2) $II < I < III$

(3) $I < III < II$

(4) $II < III < I$

6. Which of the following alkene is most stable:



7. Give the decreasing order of hyperconjugative effect of R in $R-CH=CH_2$, where R is:

I. Me -

II. Et -

III. Me_2CH-

IV. Me_3C-

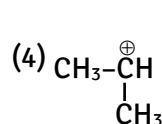
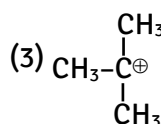
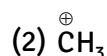
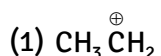
(1) $I > II > III > IV$

(2) $IV > III > II > I$

(3) $II > I > III > IV$

(4) $IV > III > I > II$

8. Which amongst the following is the most stable carbocation:



9. Electromeric effect:

(1) Comes into play at the demand of attacking reagent.

(2) involves displacement of electrons in a sigma bond.

(3) Comes into play in the molecule when at least one atom has unshared pair of electrons.

(4) is permanent effect.

10. Which statement is correct for electromeric effect:

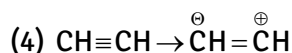
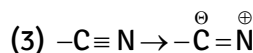
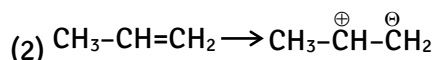
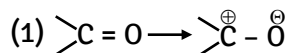
(1) It is a temporary effect

(2) It is the property of π bond

(3) It takes place in presence of reagent, i.e., electrophile or nucleophile

(4) All are correct

11. Which is wrong electromeric effect?



Aromatic, Anti Aromatic and Non Aromatic Compounds

Conditions for a compound to be aromatic:

(i) Cyclic

(ii) Planar ring

(iii) Cyclic resonance

(iv) According to Huckel's rule

$$\text{Number of } \pi e^- \text{ in cyclic resonance} = (4n + 2)\pi e^-$$

Where $n = 0, 1, 2, 4, \dots$

$$\pi e^- = 2, 6, 10, 14, \dots$$

Condition for a compound to be anti aromatic:

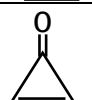
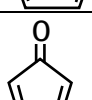
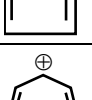
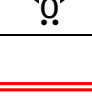
- (i) Cyclic
- (ii) Planar ring
- (iii) Cyclic resonance
- (iv) Number of πe^- in cyclic resonance = $4\pi e^-$

Where $n = 1, 2, 3, 4, \dots$

$\pi e^- = 4, 8, 12, 16, \dots$

Note: Compounds other than aromatic and antiaromatic are known as nonaromatic compounds

Stability : Aromatic > Non aromatic > Anti Aromatic

S.No.	Compound	Cyclic	Planar	Cyclic resonance	π Electron in cyclic Resonance	Aromatic (A)/ Anti-aromatic(AA)/ Non- Aromatic(NA)
1.		✓	✓	✓	$2\pi e^-$	A
2.		✓	✓	✓	$4\pi e^-$	AA
3.		✓	✓	✓	$4\pi e^-$	AA
4.		✓	✓	✓	$2\pi e^-$	A
5.		✓	✓	✓	$6\pi e^-$	A
6.		✓	✓	✓	$4\pi e^-$	AA
7.		✓	✓	✓	$6\pi e^-$	A
8.		✓	✗	✗	$4\pi e^-$	NA
9.		✓	✓	✓	$4\pi e^-$	AA
10.		✓	✓	✓	$6\pi e^-$	A
11.		✓	✓	✓	$6\pi e^-$	A

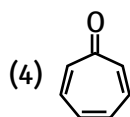
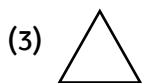
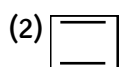
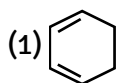
12.		✓	✓	✓	$6\pi e^-$	A
13.		✓	✓	✓	$6\pi e^-$	A
14.		✓	✓	✓	$6\pi e^-$	A
15.		✓	✓	✓	$6\pi e^-$	A
16.		✓	✓	✓	$6\pi e^-$	A
17.		✓	✓	✓	$6\pi e^-$	A
18.		✓	✓	✓	$10\pi e^-$	A
19.		✓	✓	✓	$14\pi e^-$	A
20.		✓	✗	✗	$6\pi e^-$	NA
21.		✓	✓	✓	$6\pi e^- + 6\pi e^-$	A
22.		✓	✗	✗	✗	NA

Exercise 1.6

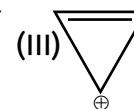
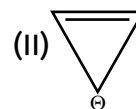
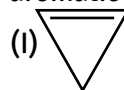
1. An aromatic molecule will not:

- (1) have $4n \pi$ electrons
- (2) have $(4n + 2) \pi$ electrons
- (3) be planar
- (4) be cyclic

2. Which of the following molecules, in pure form, is unstable at room temperature ?



3. Which of these cyclopropene systems is aromatic



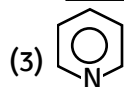
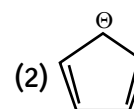
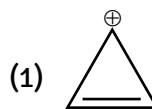
(1) I

(2) II

(3) III

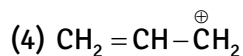
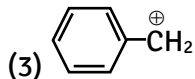
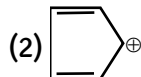
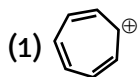
(4) all of these

4. Which is aromatic compound among the following?

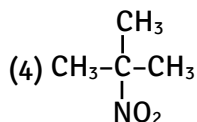
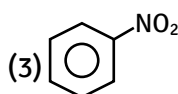
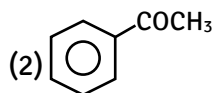
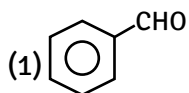


(4) All the above

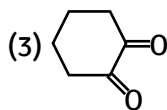
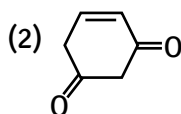
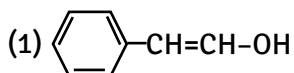
5. Which of the following is most stable carbocation?



6. Which of the following compounds can exhibit tautomerism



7. Tautomerism is exhibited by



(4) All the above

8. Tautomerism is due to :-

- (1) Delocalization of sigma electrons
- (2) Delocalization of pi electrons
- (3) Migration of active-H atom
- (4) None is correct

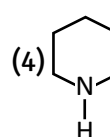
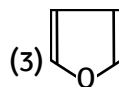
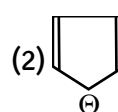
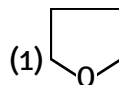
9. Which of the following does not show tautomerism ?

- (1) $\text{C}_6\text{H}_5\text{COCH}_3$
- (2) CH_3CHO
- (3) CH_3COCH_3
- (4) $\text{C}_6\text{H}_5\text{COC}(\text{CH}_3)_3$

10. The number of π electrons in benzene are:

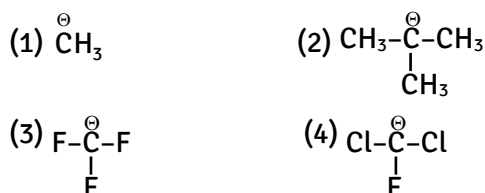
- (1) $8\pi e^-$
- (2) $3\pi e^-$
- (3) $6\pi e^-$
- (4) $2\pi e^-$

11. Which of the following is heterocyclic aromatic species ?



Exercise 2

1. Which of the following is most stable:



2. In the anion HCOO^- , the two carbon-oxygen bonds are found to be equal length. What is the reason for it:

- (1) the C=O bond is weaker than the C-O bond
- (2) the anion HCOO^- has two equivalent resonating structures
- (3) the electronic orbitals of carbon atom are hybridized
- (4) the anion is obtained by removal of proton from the acid molecule

3. Rank the following radicals in order of Decreasing stability:

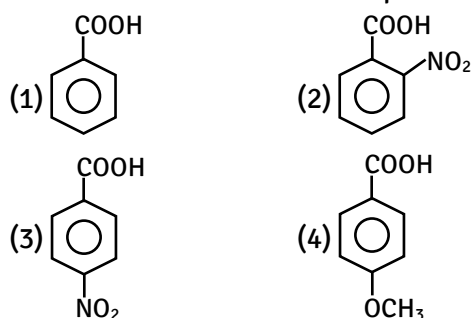


- (1) $\text{III} > \text{II} > \text{I} > \text{IV}$ (2) $\text{III} > \text{II} < \text{I} < \text{IV}$
- (3) $\text{II} > \text{III} > \text{II} > \text{IV}$ (4) $\text{III} < \text{II} < \text{I} < \text{IV}$

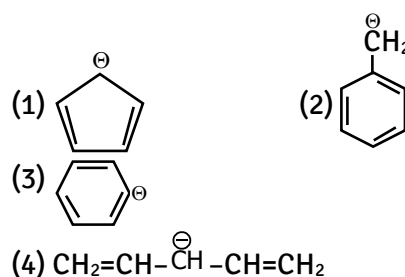
4. Which is the decreasing order of acidity in,

- (I) HCOOH (II) CH_3COOH
 (III) $\text{CH}_3\text{CH}_2\text{COOH}$ (IV) $\text{C}_6\text{H}_5\text{COOH}$
 (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{IV} > \text{III} > \text{II} > \text{I}$
 (3) $\text{IV} > \text{I} > \text{II} > \text{III}$ (4) $\text{I} > \text{IV} > \text{II} > \text{III}$

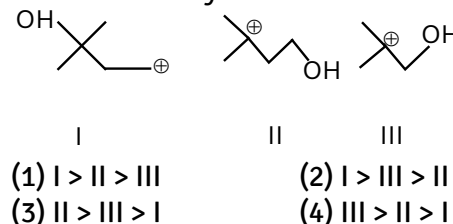
5. Which is maximum acidic compound:



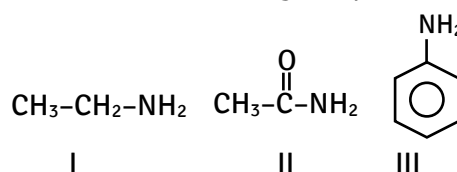
6. Which is most stable carbanion:



7. Arrange the Carbocations in decreasing order of stability:



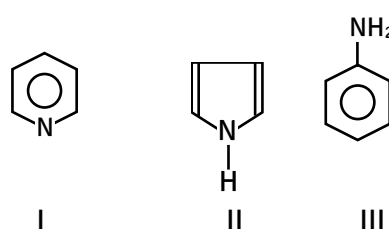
8. Consider the following compounds



Correct order of their basic strength is:

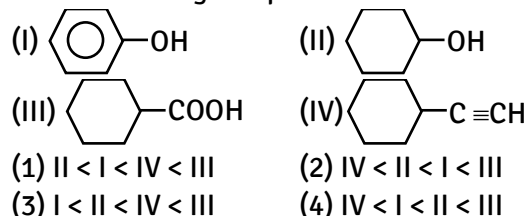
- (1) $\text{I} < \text{II} < \text{III}$ (2) $\text{II} > \text{I} > \text{III}$
- (3) $\text{III} > \text{II} < \text{I}$ (4) $\text{II} < \text{III} < \text{I}$

9. Arrange the following in decreasing of their basic nature:

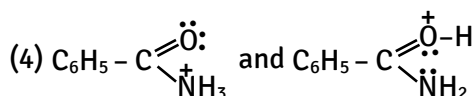
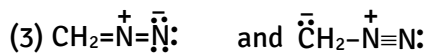
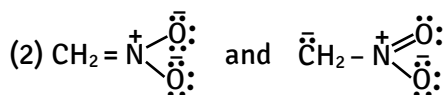
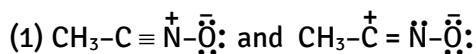


- (1) $\text{I} > \text{II} > \text{III}$ (2) $\text{I} > \text{III} > \text{II}$
- (3) $\text{III} < \text{II} > \text{I}$ (4) $\text{II} > \text{I} > \text{III}$

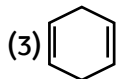
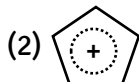
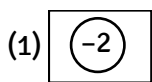
10. Give the correct order of increasing acidity of the following compounds:



11. Which of the following pairs does not represent canonical structures?



12. Which is aromatic species:



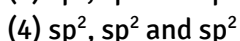
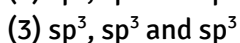
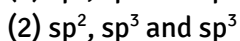
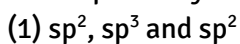
(4) All the above

13. The hybridization states of the nitrogen atoms in pyridine, piperidine and pyrrole are respectively:

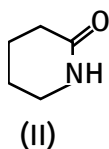
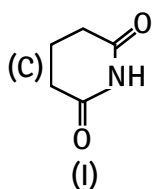
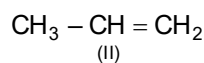
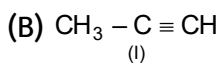
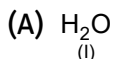


Pyridine Piperidine Pyrrole

are respectively:



14. Select an acid from each pair having stronger conjugate base:

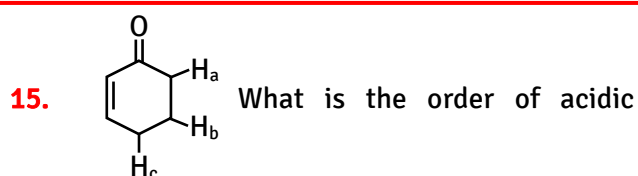


(1) I, I, II

(2) II, I, I

(3) I, II, I

(4) I, II, II



(1) $H_a > H_c > H_b$

(2) $H_b > H_c > H_a$

(3) $H_c > H_a > H_b$

(4) $H_a > H_b > H_c$

16. Consider the following bases:

(I) o-nitroaniline

(II) m-nitroaniline

(III) p-nitroaniline

Order of the base strength is:

(1) II > III > I

(2) II > I > III

(3) I > II > III

(4) I > III > II

17. Increasing order of basic strength of the following is:

I. CH_3NH_2

II. $(\text{CH}_3)_2\text{NH}$

III. $(\text{CH}_3)_3\text{N}$

IV. $\text{C}_6\text{H}_5\text{NH}_2$

V. $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

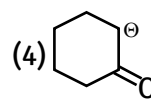
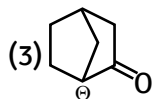
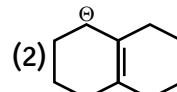
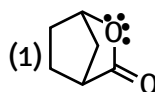
(1) IV < V < I < II < III

(2) IV < V < III < I < II

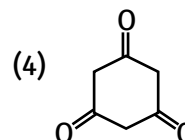
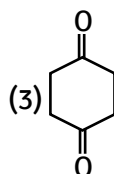
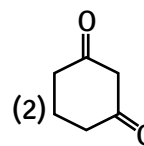
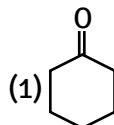
(3) IV < III < V < I < II

(4) IV < V < I < III < II

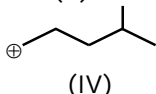
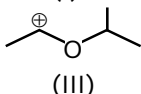
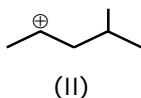
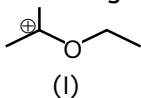
18. Which of the following compound is not resonance stabilized?



19. Which of the following ketonic compound is the least stable?

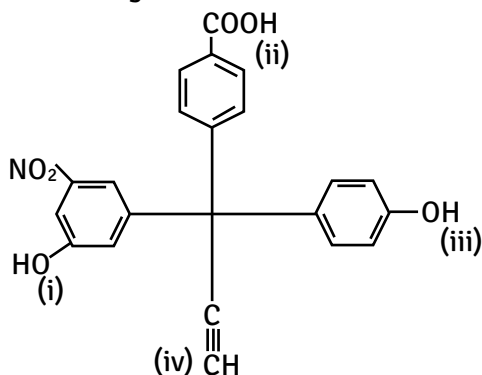


20. The correct order of stability for the following species is:



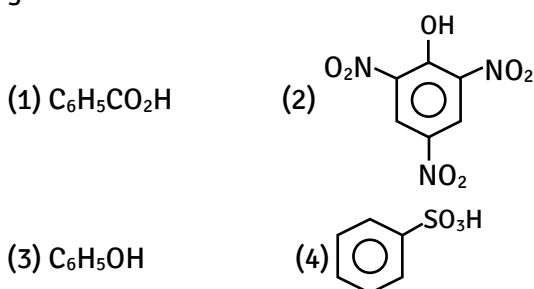
- (1) II > IV > I > III (2) I > II > III > IV
(3) II > I > IV > III (4) I > III > II > IV

21. Which of the following represent the decreasing order of K_a values?



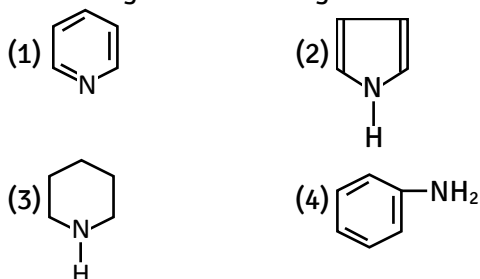
- (1) (ii) > (i) > (iii) > (iv)
(2) (ii) > (iii) > (i) > (iv)
(3) (i) > (ii) > (iii) > (iv)
(4) (ii) > (iv) > (i) > (iii)

22. Which of the following compounds will not give effervescence with sodium bicarbonate?



23. Among the following strongest acid is:
(1) CH_3COOH (2) C_6H_5COOH
(3) $m-CH_3OC_6H_4COOH$ (4) $p-CH_3OC_6H_4COOH$

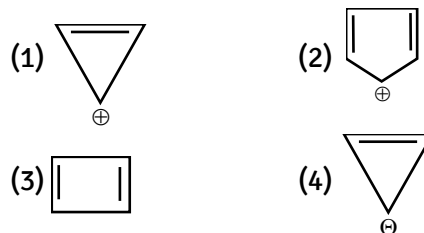
24. The strongest base among the following is:



25. Among the following the dissociation constant (K_a) is highest for:

- (1) C_6H_5OH (2) $C_6H_5CH_2OH$
(3) $CH_3C\equiv CH$ (4) $CH_3NH_3^+Cl^-$

26. Among the following the aromatic molecule is:



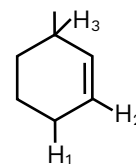
27. Pyridine is less basic than triethylamine because:

- (1) Pyridine has aromatic character
(2) Nitrogen in pyridine is sp^2 hybridized
(3) Pyridine is a cyclic system
(4) In pyridine, lone pair of nitrogen is delocalized

28. Among the following the weakest base is:

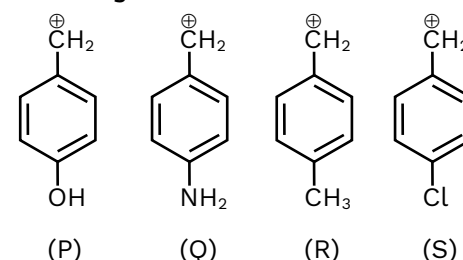
- (1) $C_6H_5CH_2NH_2$
(2) $C_6H_5CH_2NHCH_3$
(3) $O_2NCH_2NH_2$
(4) CH_3NHCHO

29. Which of the following orders is correct for the energy required for homolytic cleavage of indicated C-H bonds?



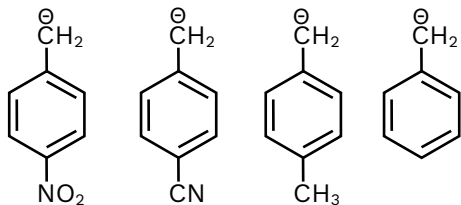
- (1) $H_1 > H_2 > H_3$ (2) $H_3 > H_1 > H_2$
(3) $H_2 > H_1 > H_3$ (4) $H_3 > H_2 > H_1$

30. The decreasing order of stability of following cations is



- (1) $P > Q > R > S$ (2) $Q > S > R > P$
(3) $Q > P > S > R$ (4) $Q > P > R > S$

31. The decreasing order of stability of following anions is



(P) (Q) (R) (S)

- (1) $P > Q > R > S$ (2) $S > R > Q > P$
 (3) $Q > P > R > S$ (4) $P > Q > S > R$

32. Polarization in acrolein as:

- (1) $\overset{+\delta}{\text{CH}_2} = \overset{-\delta}{\text{CH}} - \text{CHO}$
 (2) $\overset{-\delta}{\text{CH}_2} = \overset{+\delta}{\text{CH}} - \text{CHO}$
 (3) $\overset{-\delta}{\text{CH}_2} = \text{CH} - \overset{+\delta}{\text{CHO}}$
 (4) $\overset{+\delta}{\text{CH}_2} = \text{CH} - \overset{-\delta}{\text{CHO}}$

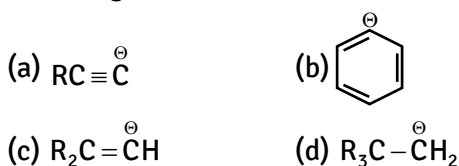
33. Which of the following order of acidic strength is correct:

- (1) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC}\equiv\text{CH}$
 (2) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC}\equiv\text{CH}$
 (3) $\text{RCOOH} > \text{HOH} > \text{HC}\equiv\text{CH} > \text{ROH}$
 (4) $\text{RCOOH} > \text{HC}\equiv\text{CH} > \text{HOH} > \text{ROH}$

34. Which of the following is more basic than aniline:

- (1) Diphenyl amine (2) Triphenyl amine
 (3) p-nitro aniline (4) Benzyl amine

35. The stability of carbanions in the following:



is in the order of:

- (1) $(d) > (b) > (c) > (a)$ (2) $(a) > (c) > (b) > (d)$
 (3) $(a) > (b) > (c) > (d)$ (4) $(b) > (c) > (d) > (a)$

36. Given are cyclohexanol (I), acetic acid (II) 2,4, 6-trinitrophenol (III) and phenol (IV). In these the order of decreasing acidic character will be:

- (1) $\text{III} > \text{II} > \text{IV} > \text{I}$ (2) $\text{II} > \text{III} > \text{I} > \text{IV}$
 (3) $\text{II} > \text{III} > \text{IV} > \text{I}$ (4) $\text{III} > \text{IV} > \text{II} > \text{I}$

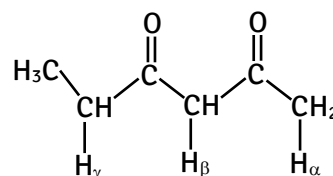
37. Which of the following will not be soluble in sodium hydrogen carbonate?

- (1) 2, 4, 6-trinitrophenol
 (2) Benzoic acid
 (3) o-nitrophenol
 (4) Benzenesulphonic acid

38. In benzene C-C bond length between all carbons are equal because of:

- (1) Tautomerism (2) sp^2 hybridization
 (3) Isomerism (4) Resonance

39. The order of acidic strength of the hydrogen atoms (H_α , H_β , H_γ) in the given molecule is:

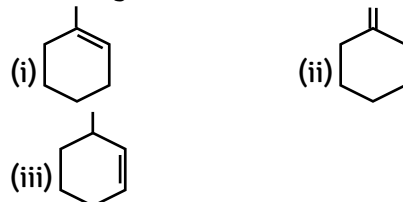


- (1) $\text{H}_\gamma > \text{H}_\alpha > \text{H}_\beta$ (2) $\text{H}_\gamma > \text{H}_\beta > \text{H}_\alpha$
 (3) $\text{H}_\beta > \text{H}_\gamma > \text{H}_\alpha$ (4) $\text{H}_\beta > \text{H}_\alpha > \text{H}_\gamma$

40. Among the following the compound having the most acidic alpha-hydrogen is:

- (1) CH_3CHO
 (2) CH_3COCH_3
 (3) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2\text{CHO}$
 (4) $\text{CH}_3-\text{CO}-\text{CH}_2-\text{CO}_2\text{CH}_3$

41. Compare heat of hydrogenation of the following:



- (1) $i > ii > iii$ (2) $iii > ii > i$
 (3) $ii > iii > i$ (4) $ii > i > iii$

42. Correct order of stability is:

- (1) 1-butene > trans-2-butene > cis-2-butene
 (2) trans-2-butene > 1-butene > cis-2-butene
 (3) trans-2-butene > cis-2-butene > 1-butene
 (4) cis-2-butene > trans-2-butene > 1-butene

43. $\text{CH}_3^\oplus$ is less stable than :-

- (1) $\text{CH}_3 - \text{CH}_2^\oplus$
- (2) $\text{CH}_3 - \text{CH}_2^\oplus - \text{CH}_3$
- (3) $\text{CH}_2^\oplus - \text{NO}_2$
- (4) $\text{CH}_3 - \text{CH}^\oplus - \text{C}_2\text{H}_5$

44. Arrange Increasing order of their K_a value.

I oxalic acid $\begin{pmatrix} \text{COOH} \\ | \\ \text{COOH} \end{pmatrix}$

II succinic acid,

$(\text{HOOC} - \text{CH}_2 - \text{CH}_2 - \text{COOH})$

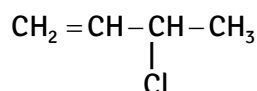
III malonic acid $(\text{HOOC} - \text{CH}_2 - \text{COOH})$

IV adipic acid $(\text{HOOC} - (\text{CH}_2)_4 - \text{COOH})$

- (1) $\text{III} < \text{II} < \text{I} < \text{IV}$
- (2) $\text{II} < \text{III} > \text{I} > \text{IV}$
- (3) $\text{I} > \text{III} > \text{II} > \text{IV}$
- (4) $\text{II} > \text{I} > \text{III} < \text{IV}$

45. Which of the following pairs represent resonating structures -

(1) $\text{ClCH}_2\text{CH}=\text{CHCH}_3$ and

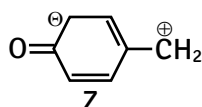
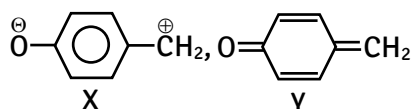


(2) $:\text{CH}_2^\oplus - \text{N}^\oplus \equiv \text{N}:$ and $\text{CH}_2 = \text{N}^\oplus = \text{N}^\oplus$

(3) $\text{CH}_3 - \text{C} \equiv \text{N}$ and $\text{CH}_3 - \text{N} \equiv \text{C}$

(4) All the above

46. Select the correct option for stability of following resonance structures :-



- (1) $x > y > z$
- (2) $z > y > x$
- (3) $y > x > z$
- (4) $y > z > x$

47. The correct order of stability of the following carbocations is -

(1) $\text{CH}_3\text{CH}_2\text{CH}_2^\oplus > \text{CH}_2=\text{CH}-\text{CH}_2^\oplus >$
 $\text{CH}_2=\text{CH}-\text{CH}^\oplus\text{CH}_3 > \text{CH}_2=\text{CH}-\text{C}^\oplus(\text{CH}_3)_2$

(2) $\text{CH}_2=\text{CH}-\text{CH}_2^\oplus > \text{CH}_3\text{CH}_2\text{CH}_2^\oplus >$
 $\text{CH}_2=\text{CH}-\text{C}^\oplus(\text{CH}_3)_2 > \text{CH}_2=\text{CH}-\text{CH}^\oplus\text{CH}_3$

(3) $\text{CH}_2=\text{CH}-\text{C}^\oplus(\text{CH}_3)_2 >$
 $\text{CH}_2=\text{CH}-\text{CH}^\oplus\text{CH}_3 > \text{CH}_2=\text{CH}-\text{CH}_2^\oplus >$

$\text{CH}_3\text{CH}_2\text{CH}_2^\oplus$
 (4) $\text{CH}_2=\text{CH}-\text{C}^\oplus\text{HCH}_3 > \text{CH}_2=\text{CH}-\text{CH}_2^\oplus >$
 $\text{CH}-\text{C}^\oplus(\text{CH}_3)_2 > \text{CH}_3\text{CH}_2-\text{CH}_2^\oplus >$
 $\text{CH}_2=\text{CH}-\text{CH}_2^\oplus$

48. (a) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$

(b) $\text{CH}_3-\text{CH}=\text{CH}_2$

(c) CH_3-CH_3

single bond length would be

- (1) a (146 pm) b (151 pm) c (153 pm)
- (2) a (151 pm) b (146 pm) c (153 pm)
- (3) a (146 pm) b (153 pm) c (151 pm)
- (4) a (153 pm) b (146 pm) c (151 pm)

49. Which of the following molecule has longest C = C bond length ?

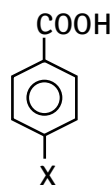
(1) $\text{CH}_2=\text{C}=\text{CH}_2$

(2) $\text{CH}_3-\text{CH}=\text{CH}_2$

(3) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$

(4) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$

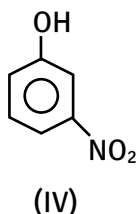
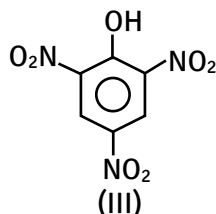
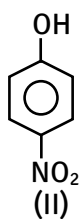
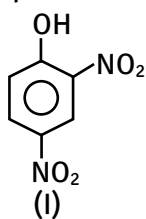
50.



$\text{p}K_a$ value of the compound decreases if X is:

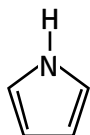
- (1) $-\text{NO}_2$
- (2) $-\text{NH}_2$
- (3) $-\text{OH}$
- (4) $-\text{OCH}_3$

51. The correct order of increasing dissociation constant [K_a] of the following compound is-



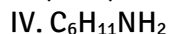
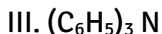
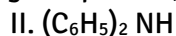
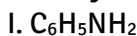
- (1) II < IV < I < III
(2) IV < III < I < II
(3) IV < II < I < III
(4) IV < I < II < III

52. Which type of hybrid orbitals do C and N use in the formation of C-N bond in pyrrole?



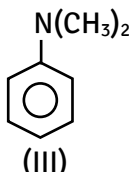
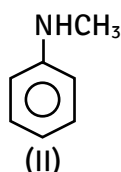
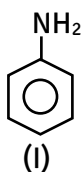
- (1) sp^2 and sp^2
(2) sp^2 and sp^3
(3) sp and sp^3
(4) sp and sp^2

53. Give the correct order of decreasing basicity of the following compounds,



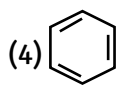
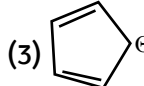
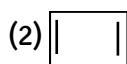
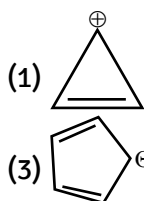
- (1) I > II > III > IV
(2) IV > III > II > I
(3) IV > I > II > III
(4) III > II > I > IV

54. Correct order of basic strength is-

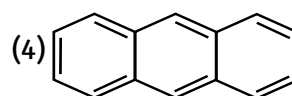
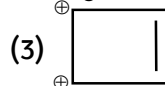
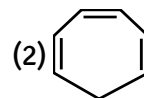
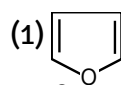


- (1) II < III < I
(2) I > II > III
(3) II > III > I
(4) I < II < III

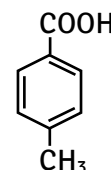
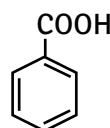
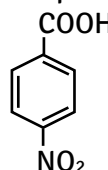
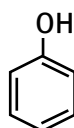
55. Which of the following species is anti aromatic?



56. Which one of the following compounds is non-aromatic?



57. The correct order of acidic character of the following compounds is:



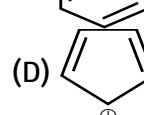
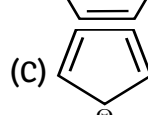
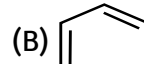
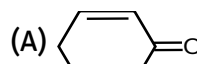
- (1) II > III > IV > I
(2) III > II > I > IV
(3) IV > III > II > I
(4) I > II > III > IV

58. A. Phenyl methanamine
B. N, N-Dimethylaniline
C. N-Methyl aniline
D. Benzenamine

Choose the correct order of basic nature of the above amines.

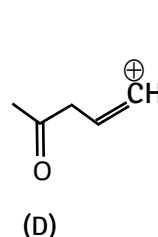
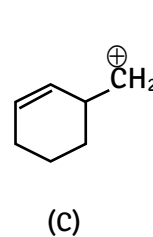
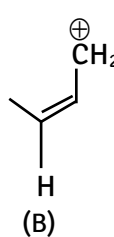
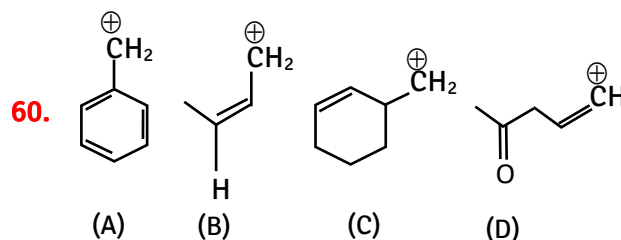
- (1) D > C > B > A
(2) D > B > C > A
(3) A > C > B > D
(4) A > B > C > D

59. Among the following, the aromatic compounds are:



Choose the correct answer from the following options:

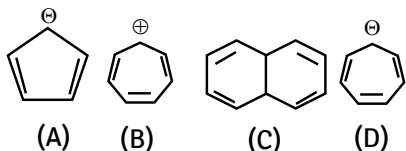
- (1) (A) and (B) only
(2) (A), (B) and (C) only
(3) (B), (C) and (D) only
(4) (B) and (C) only



Among the given species the Resonance stabilised carbocations are:

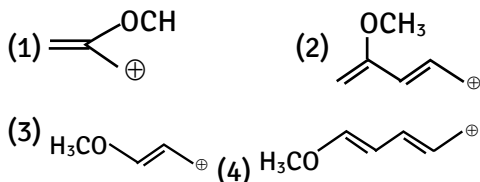
- (1) (C) and (D) only
(2) (A), (B) and (C) only
(3) (A), (B) and (D) only
(4) (A) and (B) only

61. Which of the following structures are aromatic in nature ?

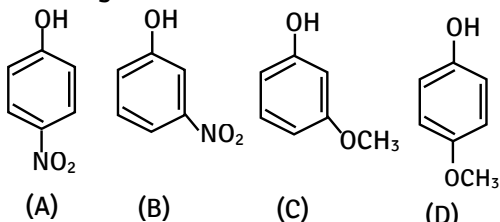


- (1) A, B, C and D (2) Only A and B
(3) Only A and C (4) Only B, C and D

62. Which of the following carbocations is most stable?

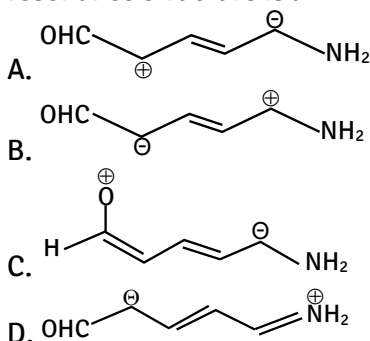


63. Arrange the following in decreasing acidic strength.



- (1) A > B > C > D (2) B > A > C > D
(3) D > C > A > B (4) D > C > B > A

64. Increasing order of stability of the resonance structure is :



- (1) C < D < B < A (2) C < A < B < D
(3) D < C < A < B (4) D < C < B < A

65. The increasing order of k_a for the following phenols is

- (a) 2, 4-Dinitrophenol
(b) 4 - Nitrophenol
(c) 2, 4, 5- Trimethylphenol
(d) Phenol
(e) 3-Chlorophenol

- (1) a < b < c < d < e (2) c < d < e < b < a
(3) d < b < e < c < a (4) e < c < d < b < a

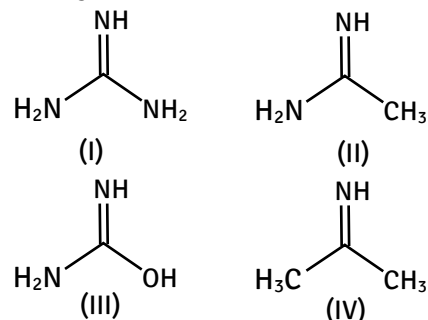
66. The functional group that shows negative resonance effect is :

- (1) -OH (2) -OR
(3) -COOH (4) -NH₂

67. The interaction between π bond and lone pair of electrons present on an adjacent atom is responsible for

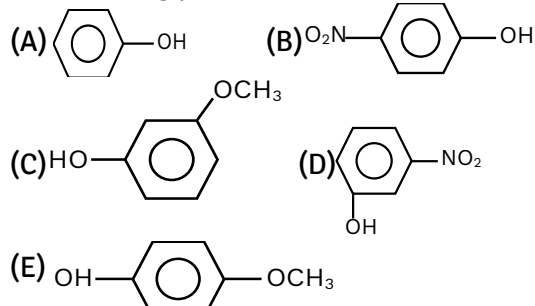
- (1) Resonance effect (2) Electromeric effect
(3) Inductive effect (4) Hyperconjugation

68. Arrange the following in increasing order of basicity



- (1) IV < II < III < I (2) IV < II < I < III
(3) IV < III < II < I (4) III < IV < II < I

69. For the given compounds, the correct order of increasing pK_a value



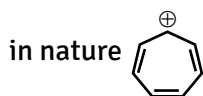
- (1) (E) < (D) < (C) < (B) < (A)
(2) (D) < (E) < (C) < (B) < (A)
(3) (E) < (D) < (B) < (A) < (C)
(4) (B) < (D) < (C) < (A) < (E)

70. Which among the following is incorrect statement?

- (1) Electromeric effect dominates over inductive effect
(2) The electromeric effect is, temporary effect
(3) The organic compound shows electromeric effect in the presence of the reagent only
(4) Hydrogen ion (H^+) shows negative electromeric effect

Exercise 3

1. **Statement I:** Tropylium cation is aromatic



Statement II: The only property that determines its aromatic behaviour is its planar structure.

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

2. **Statement I:** Me_3C^+ is more stable than Me_2CH^+ and Me_2CH^+ is more stable than the MeCH_2^+ .

Statement II: Greater the number of hyperconjugative structures, more is the stability of carbocation.

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

3. **Statement I:** Cyclopentadienyl anion is much more stable than allyl anion.
Statement II: Cyclopentadienyl anion is aromatic in nature.

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

4. **Statement I:** The pKa value of acetic acid is lower than that of phenol.

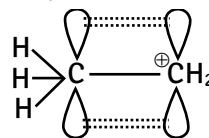
Statement II: Phenoxide ion is more resonance stabilized than acetate ion.

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

5. **Statement I:** $\text{CH}_3 - \text{CH}_2^+$ is more stable than CH_3^+ due to hyperconjugation.

Statement II: Hyperconjugation in

$\text{CH}_3 - \text{CH}_2^+$ involves following overlapping:



- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

6. Given below are two statements:

Statement I: Aniline is less basic than acetamide.

Statement II: In aniline, the lone pair of electrons on nitrogen atom is delocalised over benzene ring due to resonance and hence less available to a proton.

Choose the most appropriate option;

- (1) Both statement I and statement II are true.
- (2) Both statement I and statement II are false.
- (3) Statement I is true but statement II is false.
- (4) Statement I is false but statement II is true.

7. Given below are two statements:
Statement I: Hyperconjugation is a permanent effect.

Statement II: Hyper conjugation in ethyl cation $\left(\text{CH}_3-\overset{+}{\text{CH}_2}\right)$ involves the overlapping of $\text{C}_{\text{sp}^2}-\text{H}_{1s}$ bond with empty 2p orbital of other carbon.

Choose the correct option:

- (1) Both statement I and statement II are true
- (2) Statement I is incorrect but statement II is true
- (3) Statement I is correct but statement II is false
- (4) Both Statement I and statement II are false.

8. **Statement-I:** Resonance affects bond length.

Statement-II: Resonance stabilises molecule.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

9. **Statement I :** In an organic compound when inductive and electrometric effect operate in opposite direction, the electrometric effect Predominates.

Statement II : Polarisation of σ bond caused by the polarisation of adjacent σ bond is referred to as inductive effect.

- (1) Both statement I and statement II are true.
- (2) Both statement I and statement II are false.
- (3) Statement I is true but statement II is false.
- (4) Statement I is false but statement II is true.

10. Match the terms mentioned in Column I with the terms in Column II

Column I		Column II	
(i)	Carbocation	(a)	cyclohexane and 1- hexene
(ii)	Nucleophile	(b)	Conjugation of electrons of C-H σ bond with empty p-orbital present at adjacent positively charged carbon.
(iii)	Hyperconjugation	(c)	sp ² hybridised carbon with empty p-orbital
(iv)	Isomers	(d)	Ethyne
(v)	sp hybridization	(e)	Species that can receive a pair of electrons
(vi)	Electrophile	(f)	Species that can supply a pair of electrons

- (1) (i) $\rightarrow$ (c), (ii) $\rightarrow$ (f), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a), (v) $\rightarrow$ (d), (vi) $\rightarrow$ (e)
- (2) (i) $\rightarrow$ (f), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a), (v) $\rightarrow$ (d), (vi) $\rightarrow$ (e)
- (3) (i) $\rightarrow$ (c), (ii) $\rightarrow$ (f), (iii) $\rightarrow$ (a), (iv) $\rightarrow$ (b), (v) $\rightarrow$ (d), (vi) $\rightarrow$ (e)
- (4) (i) $\rightarrow$ (c), (ii) $\rightarrow$ (f), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (d), (v) $\rightarrow$ (a), (vi) $\rightarrow$ (e)

11. Match the intermediates given in Column I with their probable structure in Column II.

Column I		Column II	
(i)	Free radical	(a)	Trigonal planar
(ii)	Carbocation	(b)	Pyramidal
(iii)	Carbanion	(c)	Linear

- (1) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (a), (iii) $\rightarrow$ (b)
- (2) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (a)
- (3) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (c)
- (4) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (b)

12. Match the ions given in Column I with their nature given in Column II.

Column I		Column II	
i	$\text{CH}_3 - \ddot{\text{O}} - \overset{\oplus}{\text{C}}\text{H}_2$	(a)	Stable due to resonance
ii	$\text{F}_3\text{C} - \overset{\oplus}{\text{C}}$	(b)	Destabilised due to inductive effect
iii	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \overset{\oplus}{\text{C}} \\ \\ \text{CH}_3 \end{array}$	(c)	Stabilised by hyperconjugation
iv	$\text{CH}_3 - \overset{\oplus}{\text{C}}\text{H} - \text{CH}_3$	(d)	A secondary carbocation

- (1) (i) → (a), (b), (ii) → (b),
(iii) → (c), (iv) → (c), (d)
(2) (i) → (a), (b), (d), (ii) → (a),
(iii) → (b), (iv) → (c), (d)
(3) (i) → (a), (b), (d), (ii) → (b),
(iii) → (b), (iv) → (b), (d)
(4) (i) → (a), (c), (d), (ii) → (b),
(iii) → (b), (iv) → (c), (d)

13. Match List-I with List II:

List-I Mechanism steps		List-II Effect	
(A)		(I)	-E effect
(B)		(II)	-R effect
(C)		(III)	+E effect
(D)		(IV)	+R effect

Choose the correct answer from the options given below :

- (1) (A)- (IV), (B)- (III), (C)- (I), (D)- (II)
(2) (A)- (III), (B)- (I), (C)- (II), (D)- (IV)
(3) (A)- (II), (B)- (IV), (C)- (III), (D)- (I)
(4) (A)- (I), (B)- (II), (C)- (IV), (D)- (III)

14. Match the column

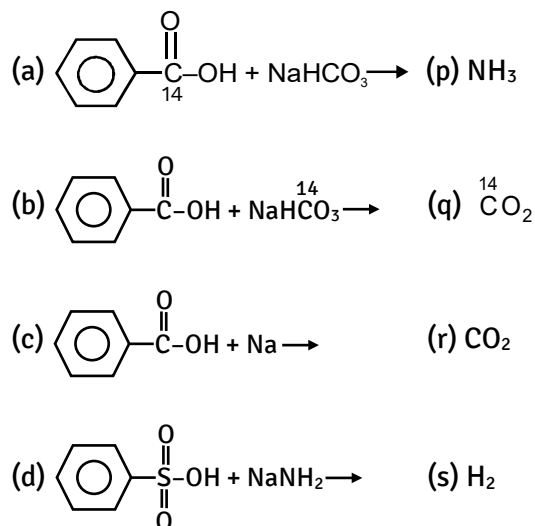
Column-I (Compound)		Column-II (P _{kb} value)	
(i)	Methan amine	(p)	4.75
(ii)	N-Methyl methanamine	(q)	3.38
(iii)	N, N-Dimethyl methanamine	(r)	3.27
(iv)	Ammonia	(s)	4.22

- (1) (i)-(p), (ii)-(q), (iii)-(r), (iv)-(s)
(2) (i)-(p), (ii)-(r), (iii)-(s), (iv)-(q)
(3) (i)-(s), (ii)-(r), (iii)-(q), (iv)-(p)
(4) (i)-(q), (ii)-(r), (iii)-(s), (iv)-(p)

15. Match the column-I and column-II and select the answer:

Column-I

Column-II



- (1) (a-r), (b-q), (c-s), (d-p)
(2) (a-q), (b-r), (c-s), (d-p)
(3) (a-s), (b-q), (c-r), (d-p)
(4) (a-p), (b-s), (c-q), (d-r)

- 16. Assertion A :** Enol form of acetone $[\text{CH}_3\text{COCH}_3]$ exists in $<0.1\%$ quantity. However, the enol form of the acetyl acetone $[\text{CH}_3\text{COCH}_2\text{COCH}_3]$ exists in approximately 15% quantity.

Reason R : Enol form of acetyl acetone is stabilized by intramolecular hydrogen bonding, which is not possible in enol form of acetone.

Choose the correct statement:

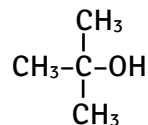
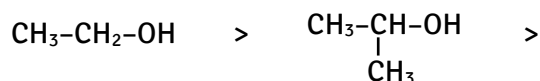
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

- 17. Assertion:** P-nitro benzoic acid is more acidic than p-hydroxy benzoic acid.

Reason: EDG increases acidity of benzoic acid while EWG decreases acidity of benzoic acid.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

- 18. Assertion :** The order of acidic strength for the following compound is

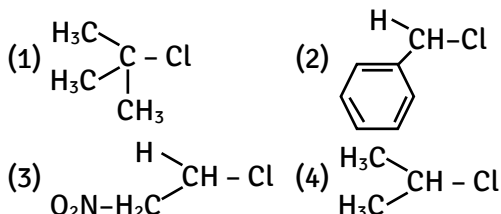


Reason : Electron releasing group ($-\text{CH}_3$) increases electron density on oxygen tending to decrease the polarity of O-H bond

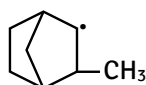
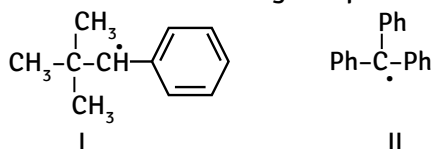
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

Exercise 4 (Previous Year's Questions)

1. In which of the following compounds, the C - Cl bond ionisation shall give most stable carbonium ion? [AIPMT - 2015]



2. Consider the following compounds



Hyperconjugation occurs in: [AIPMT - 2015]

- (1) II only (2) III only
(3) I and III (4) I only

3. The correct statement regarding the basicity of arylamines is:

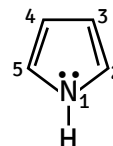
[NEET-I - 2016]

- (1) Arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring π electron system
 (2) Arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalized by interaction with the aromatic ring π electron system.
 (3) Arylamines are generally more basic than alkylamines because of aryl group.
 (4) Arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp -hybridized.

4. The pair of electron in the given carbanion, $\text{CH}_3\text{C}\equiv\text{C}^\ominus$, is present in which of the following orbitals? [NEET-I - 2016]

- (1) 2p (2) sp^3
(3) sp^2 (4) sp

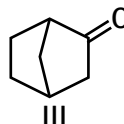
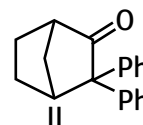
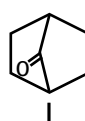
5. In pyrrole: [NEET-II - 2016]



The electron density is maximum on:

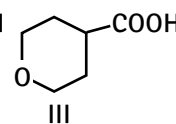
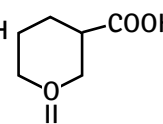
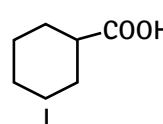
- (1) 2 and 4 (2) 2 and 5
(3) 2 and 3 (4) 3 and 4

6. Which among the given molecules can exhibit tautomerism? [NEET-II - 2016]



- (1) Both I and II (2) Both II and III
(3) III only (4) Both I and III

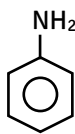
7. The correct order of strengths of the carboxylic acid: [NEET-II - 2016]



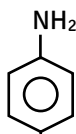
is:

- (1) III > II > I (2) II > I > III
(3) I > II > III (4) II > III > I

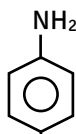
8. The correct increasing order of basic strength for the following compounds is: [NEET - 2017]



(I)



(II)



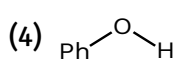
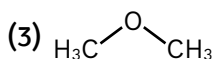
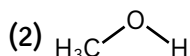
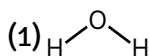
(III)

- (1) II < III < I
(2) III < I < II
(3) III < II < I
(4) II < I < III

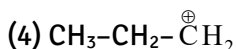
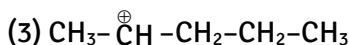
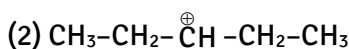
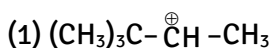
9. Which of the following is correct with respect to -I effect of the substituents? (R = alkyl) [NEET(UG) 2018]
- (1) $-\text{NH}_2 < -\text{OR} < -\text{F}$ (2) $-\text{NR}_2 < -\text{F} < -\text{OR}$
(3) $-\text{NR}_2 > -\text{OR} > -\text{F}$ (4) $-\text{NR}_2 < -\text{OR} > -\text{F}$

10. The correct order of the basic strength of methyl substituted amines in aqueous solution is [NEET(UG) 2019]
- (1) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$
(2) $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$
(3) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$
(4) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

11. The compound that is most difficult to protonate is [NEET(UG) 2019]



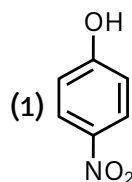
12. The most stable carbocation among the following is [NEET(UG) 2019(ODISHA)]



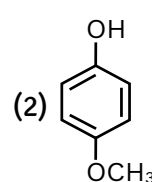
13. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following? [NEET(UG) 2020]

- (1) Hyperconjugation
(2) -I effect of $-\text{CH}_3$ groups
(3) +R effect of $-\text{CH}_3$ groups
(4) -R effect of $-\text{CH}_3$ groups

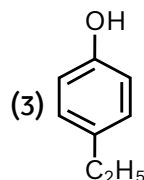
14. Which of the following substituted phenols is the strongest acid? [NEET(UG) 2020(COVID-19)]



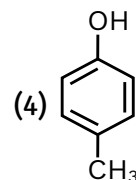
(1)



(2)

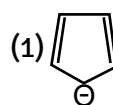


(3)

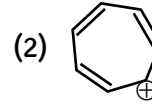


(4)

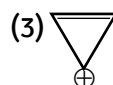
15. Which compound amongst the following is not an aromatic compound? [NEET - 2022]



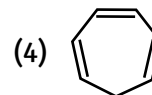
(1)



(2)



(3)



(4)

16. Given below are two statements: [NEET(UG) 2022]

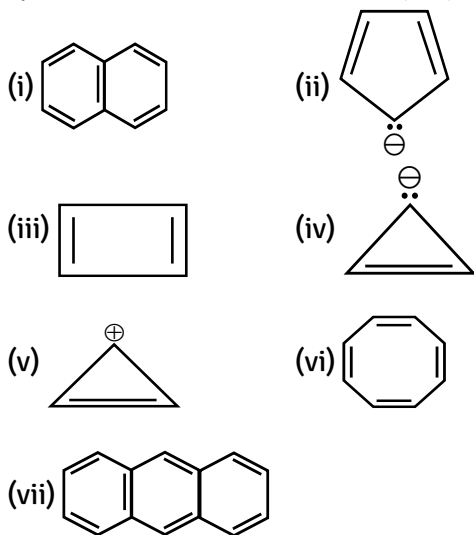
Statement I : The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II : o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

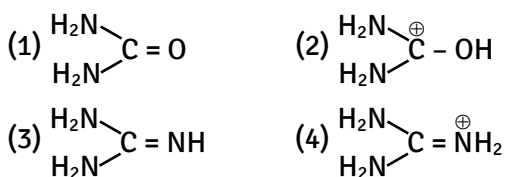
17. Consider the following compounds/species: [NEET(UG) 2023]



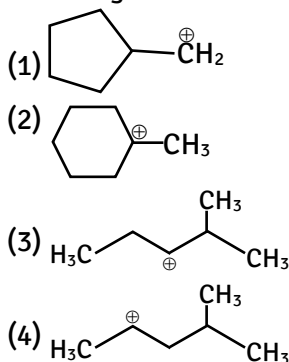
The number of compounds/species which obey Huckel's rule is _____.

- (1) 4 (2) 6
(3) 2 (4) 5

18. Which amongst the following Compounds /species is least basic? [NEET(UG) 2023]



19. The most stable carbocation among the following is: [NEET(UG)- 2024]



20. Methyl group attached to a positively charged carbon atom stabilizes the carbocation due to : [RE-NEET(UG) 2024]

- (1) -I inductive effect
(2) electromeric effect
(3) hyperconjugation
(4) mesomeric effect

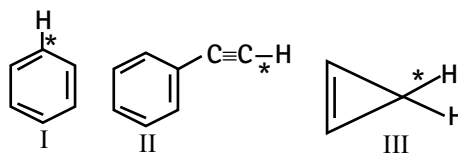
21. The correct order of decreasing acidity of the following aliphatic acids is :

[NEET(UG)-2025]

- (1) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
(2) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
(3) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
(4) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$

22. Among the given compounds I-III, the correct order of bond dissociation energy of C-H bond marked with * is:

[NEET(UG)-2025]



- (1) III > II > I (2) II > III > I
(3) II > I > III (4) I > II > III

23. The correct order of decreasing basic strength of the given amines is :

[NEET(UG)-2025]

- (1) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine
(2) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine
(3) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine
(4) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
Ans.	4	4	3	1	1	3	1	2	4	2	1	3	1	2	3	1	1	3	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	3	3	3	1	4	3	1	3	3	4	2	

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
Ans.	4	4	2	2	4	4	1	1	2	3	1	1	2	3	2	1	1	

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	4	2	2	1	4	3	3	1	4	2	1	2	1	1	2	3	2	1

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	1	3	4	2	3	3	1	3	1	4	3	

Exercise 1.6

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	1	2	3	4	1	2	4	3	4	3	3	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	1	4	2	1	3	4	2	2	4	1	1	4	3	1	2	3	4	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	3	3	3	4	1	2	4	3	4	4	4	2	4	3	1	3	4	4	3
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	3	3	2	3	3	1	4	1	3	1	3	4	2	2	1	4	4	4
Que.	61	62	63	64	65	66	67	68	69	70										
Ans.	2	4	1	2	2	3	1	1	4	4										

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
Ans.	3	1	1	3	3	4	3	4	1	1	1	1	1	4	1	1	3	1	

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	4	2	3	4	4	1	1	4	3	1	1	4	3	1	2	2	3
Que.	21	22	23																	
Ans.	1	1	1																	

Chapter 4 HYDROCARBONS

Chapter Summary

- Alkanes
- General Methods of Preparation of Alkanes
- Physical Properties of Alkanes
- Chemical Reactions of Alkanes
- Alkenes
- General Methods of Preparation of Alkenes
- Physical Properties of Alkenes
- Chemical Reactions of Alkenes
- Alkynes
- General Methods of Preparation of Alkynes
- Physical Properties of Alkynes
- Chemical Properties of Alkynes
- Aromatic Hydrocarbon
- General Methods of Preparation of Aromatic Hydrocarbon
- Physical Properties of Aromatic Hydrocarbon
- Chemical Properties of Aromatic Hydrocarbon

Alkanes

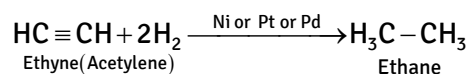
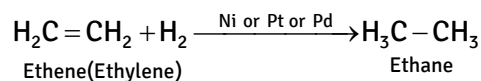
Introduction of Alkanes

- Alkanes are saturated hydrocarbons.
- Alkanes are inert under normal condition as they do not react with acids, bases and other reagents, so these are called paraffins i.e parum (little), affinis (affinity).
- General formula is C_nH_{2n+2} .
- Hybridisation state of carbon is sp^3 .
- Geometry is tetrahedral.
- Bond angle is $109^\circ 28'$.
- Petroleum and natural gas are the main sources of alkanes.

General Methods of Preparation of Alkanes

(1) From Unsaturated Hydrocarbons:

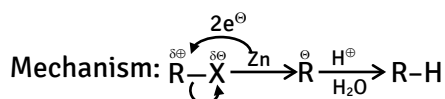
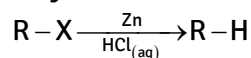
- Addition of dihydrogen gas on alkene, alkyne in presence of metal catalyst is called hydrogenation.
- Finely divided Pt, Pd or Ni is used as catalyst.
- Catalyst adsorbs dihydrogen gas and activates hydrogen - hydrogen bond,
- Pt or Pd catalyse the reaction at room temperature but relatively high temperature and pressure is required with Ni.
- Hydrogenation in presence of Ni is known as Sabatier-Senderen's reaction.



(2) From alkyl Halides

(i) By Reduction

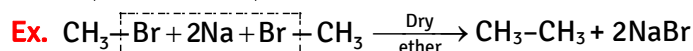
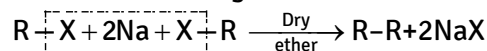
Alkyl Halides can be reduced into alkanes by using metal and dilute HCl.



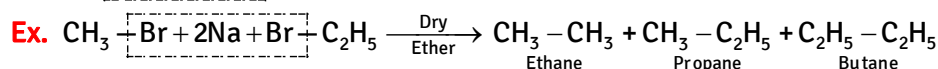
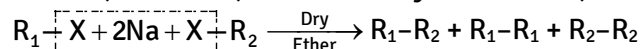
Note: alkyl fluorides are not used due to their less reactivity

(ii) By Wurtz Reaction:

Alkyl halide reacts with sodium metal in dry ether free from moisture solution to form symmetrical alkanes containing double the number of C atoms present in the alkyl halide.

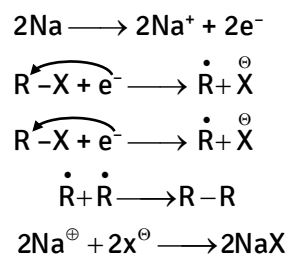


a. Limitations of the Reaction : The reaction is not an appropriate method to prepare unsymmetrical alkane (like R_1-R_2) since other hydrocarbons (R_1-R_1 & R_2-R_2) are also formed along with R_1-R_2 .

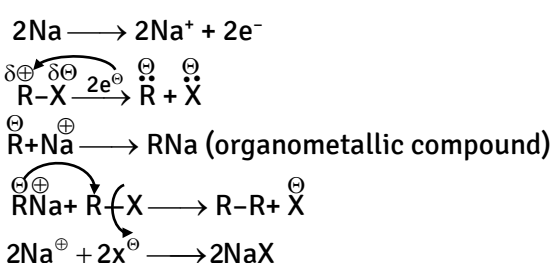


b. The reaction may complete through both free radical and ionic mechanism.

Free radical mechanism

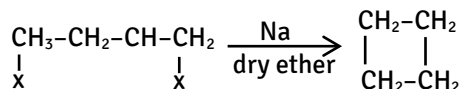
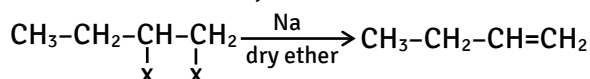


Ionic mechanism



c. Reactivity order of different alkyl halide for the reaction is $R-I > R-Br > R-Cl > R-F$

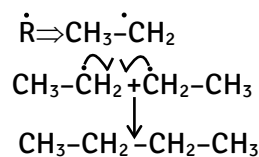
d. In case of di halides, intramolecular reaction is observed to give unsaturated or cyclic compound



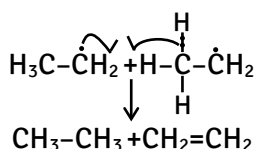
e. Side products of the reaction : When two alkyl free radicals approach each other two processes are observed between them

Ex.

Radical combination



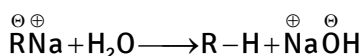
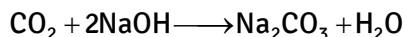
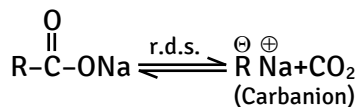
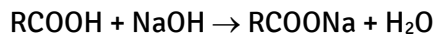
Radical Disproportionation



In case of 1° & 2° radical combination dominates but in 3° radical disproportionation dominates.

(3) From Carboxylic acid:**(a) Sodalime decarboxylation:**

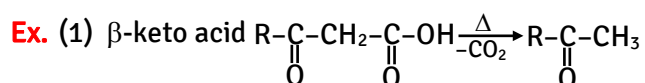
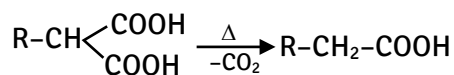
When a sodium salt of carboxylic acid is heated with soda lime $[\text{NaOH} + \text{CaO} (3 : 1)]$ an alkane with one C atom less than carboxylic acid is formed.



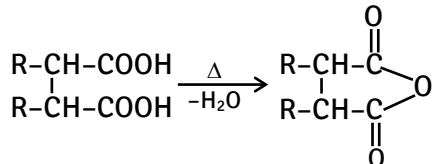
{Reactivity of $\text{R}-\text{COOH}$ for decarboxylation $\propto$ Stability of $\text{R}^\ominus$ }

CaO in soda lime is used to keep NaOH dry because it is hygroscopic in nature.

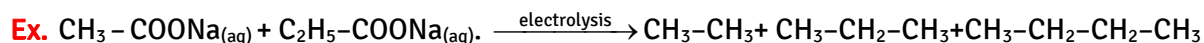
Note: Those acid which have multiple bond at beta position show high reactivity towards decarboxylation and the process is possible even in absence of Soda lime, only heat will be required to complete the process

**(2) Gem dicarboxylic acid**

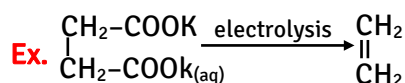
Note: Vicinal dicarboxylic acid on heating forms cyclic anhydride by dehydration

**(b) Kolbe's Electrolysis:**

- Electrolysis of an aqueous solution of sodium or potassium salt of a carboxylic acid gives higher alkane.
- Mixture of different acid salts gives mixture of hydrocarbons

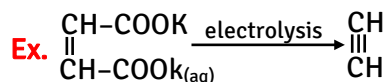


- Electrolysis of an aqueous solution of sodium or potassium salt of dibasic acid gives alkene

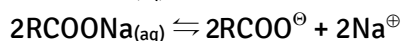
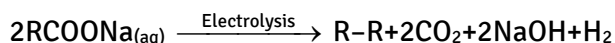


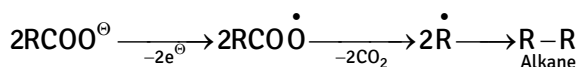
(Potassium succinate)

- Electrolysis of aqueous solution of sodium or potassium salt of dibasic unsaturated fatty acid gives alkyne.

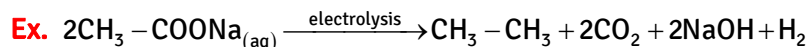
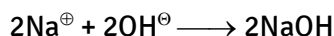


(Potassium maleate/potassium fumarate)

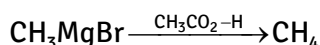
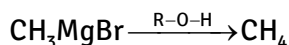
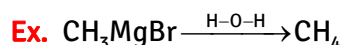
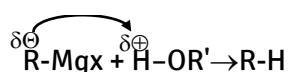
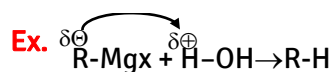
Mechanism:

At Anode (Oxidation):

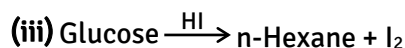
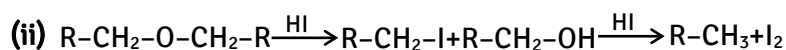
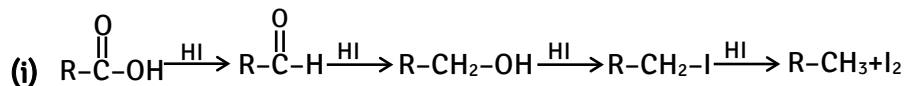
At Cathode (Reduction): Reduction of H_2O takes place since reduction potential of H_2O is greater than $\text{Na}^{\oplus}$.

**(4) From Grignard Reagent:**

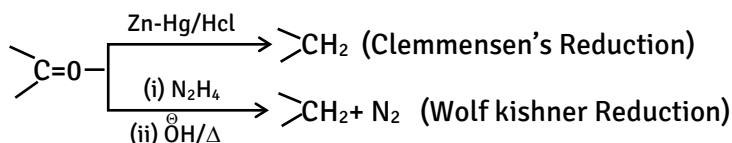
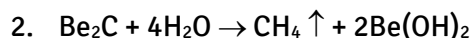
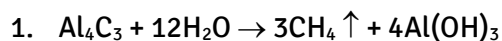
Grignard's reagents react with the compound having active hydrogen to form alkane by acid base reaction.

**(5) Reduction using (P + HI):**

It is a powerful reducing agent which converts aldehyde, ketone, alcohol, carboxylic acid, alkane iodide, ether and carbohydrates into alkanes with same number of carbon (except ether).



Note: P is used to remove I_2 from the end product.

(6) From Carbonyl compounds :**(7) Hydrolysis of Metal carbide:**

Physical Properties of Alkanes

Hydrocarbons are non-polar molecules. Therefore, the intermolecular forces of attraction are of van der Waals type, which mainly depends on the molecular weight and surface area of the contact between adjacent molecules. Greater the molecular weight and surface area, stronger are the van der Waals forces of attraction.

a. Boiling Point:

Due to weak van der Waals forces first four members, C_1 – C_4 are gases, C_5 – C_{17} are liquids, and C_{18} onwards are colourless, waxy solids.

$B.P \propto \text{Molecular weight} \propto \frac{1}{\text{Branching}}$ (Branching $\uparrow$, surface area $\downarrow$ van der Waals forces of attraction $\downarrow$)

b. Melting Point:

Melting point of alkanes increases with increase in carbon content. Melting point of alkane increases sharply when moving toward higher members specially from alkane with odd number of carbon to even number of carbon, but flat increment is observed when moving from lower alkane with even number of carbon to higher alkane with odd number of carbon.

This property is called alternation effect.

In n-alkane, due to sp^3 hybridisation, C atoms are arranged in a zig-zig chain. So, n-alkanes with even number of C atoms are more symmetrical than those containing odd number of C atoms and, hence, can pack closely in the crystal lattice. For example.

				
n-Propane	n-Butane	n-Pentane	n-Hexane	n-Heptane
M.P. 85.3K	134.6K	143.3K	178.5K	182.4K

Alternation effect is not observed in boiling points because in the liquid state the conformations of the molecules are fixed but keep on changing as a result of collisions.

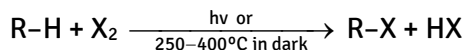
c. Solubility:

'Like dissolves like' is the general rule of solubility. Hydrocarbons (non polar molecules) are insoluble in polar solvents such as H_2O and alcohol, but are highly soluble in non-polar solvents such as petroleum, benzene, and CCl_4 .

Chemical Reactions of Alkanes or Chemical Properties

(A) **Free Radical Substitution Reactions:** Alkanes are non polar and have strongly bonded sigma electrons so reaction of alkanes are generally carried out in presence of light or at high temperature so, substitution reaction of alkanes follow free radical mechanism.

(a) **Halogenation :** substitution of H-atom by halogen atom



- The reactivity order for halogens towards the reaction:

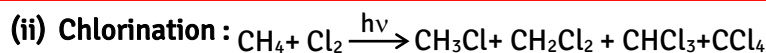


- Reactivity order of hydrogen atom present in alkane is :



(i) **Fluorination :**

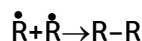
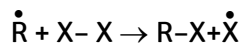
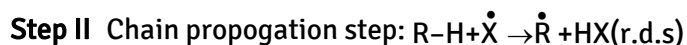
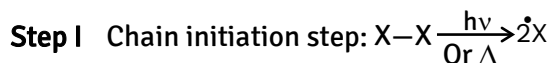
- Reaction is highly exothermic, so it is explosive reaction
- Reaction is possible even in dark.
- Direct halogenation gives carbon black, alkyl fluoride is formed if reaction is carried out in presence of N_2 .



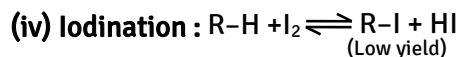
If $\text{CH}_4(\text{excess}) \rightarrow \text{CH}_3\text{Cl}$ (major)

$\text{Cl}_2(\text{excess}) \rightarrow \text{CCl}_4$ (major)

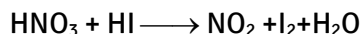
Mechanism:



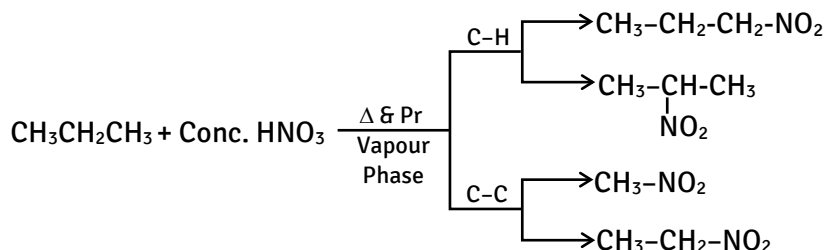
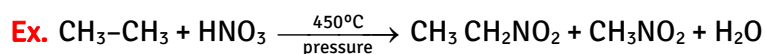
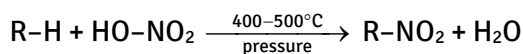
(iii) **Bromination** : Br_2 reacts with alkanes in a similar manner like Cl_2 .



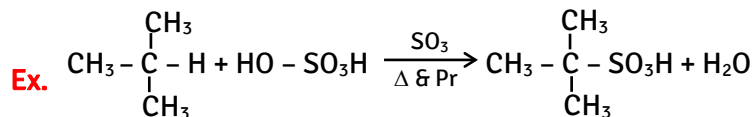
- The reaction is reversible since HI is a good reducing agent.
- The reaction is very slow since I_2 is least reactive in halogenation
- R-I can be formed with better yield if reaction is carried out in presence of HIO_3 or HNO_3 .



(b) **Nitration** : The reaction is carried out in vapour phase at high temperature and pressure, it involved breaking of C-C bond along with C-H, So mixture of all possible nitro alkanes is formed as product.

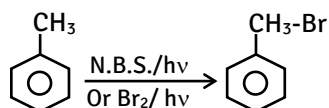
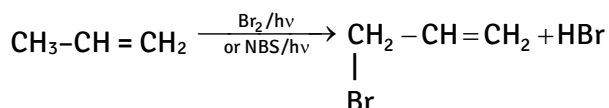


(c) **Sulphonation** : Replacement of H atom of alkane by $-\text{SO}_3\text{H}$ is known as sulphonation. Alkane react with fuming H_2SO_4 or oleum ($\text{H}_2\text{S}_2\text{O}_7$).



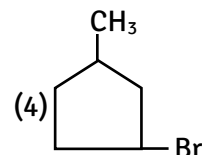
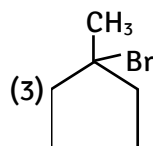
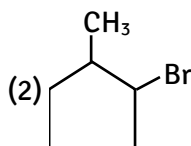
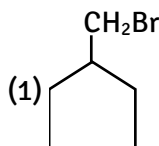
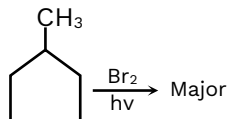
2-Methyl propane

Note: The reaction is observed in higher alkanes ($\geq 6\text{ C}$) or the alkanes having 3°H .

(d) Allylic or Benzylic Substitution by Br₂ (Low Concentration)/hv or NBS/hv

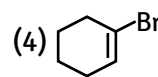
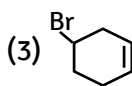
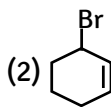
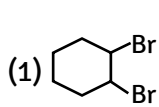
Note: The reaction is preferred at the site where free radical formed is most stable

Ex.1 In the following reaction, the major product is :-



Ans. (3)

Ex.2 $\xrightarrow[\text{hv}]{\text{Br}_2(\text{Low conc})}$ Major product



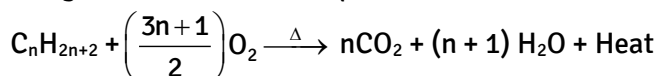
Ans. (2)

(B) Combustion

Alkanes on heating in excess of air or oxygen are completely oxidized (Combustion) to CO₂ and H₂O, with the evolution of large amount of heat.

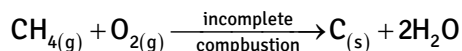


The general combustion equation for an alkane is-

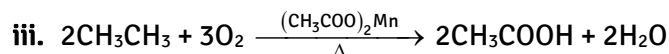
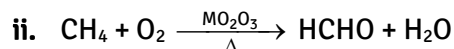
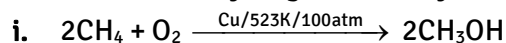


Note: Due to the evolution of large amount of heat during combustion, alkanes are used as fuels.

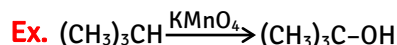
- During incomplete combustion of alkanes with insufficient amount of air or oxygen, carbon black is formed which is used in the manufacturing of ink, black pigments and as filters

**(C) Catalytic Oxidation
(Controlled oxidation)**

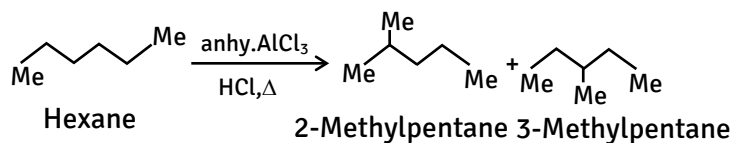
Alkanes on heating with a regulated supply of oxygen (air) at high pressure and in the presence of suitable catalysts give a variety of oxidation products



iv. With KMnO₄, alkanes containing tertiary H atom are oxidized to alcohol.

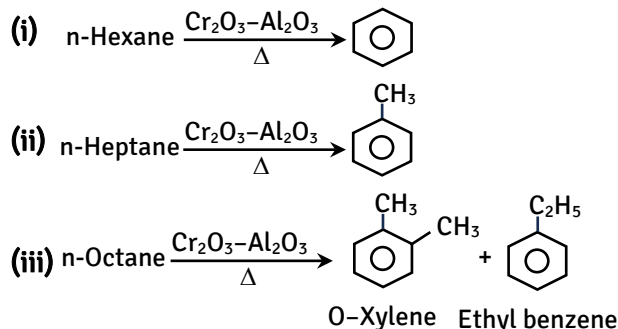


- (D) **Isomerisation:-** Unbranched alkanes are converted into branched alkane on heating in presence of anhydrous AlCl_3 and HCl gas



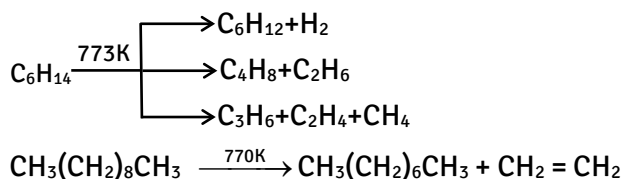
- (E) **Aromatisation**

Alkanes with six or more C atoms, when heated under pressure in the presence of a catalyst (e.g., Al_2O_3 + Cr_2O_3 or V_2O_5 or Mo_2O_3 at 773 K and 10-20 atm) first forms cyclo alkane and then aromatise to give benzene or its homologues by dehydrogenation.



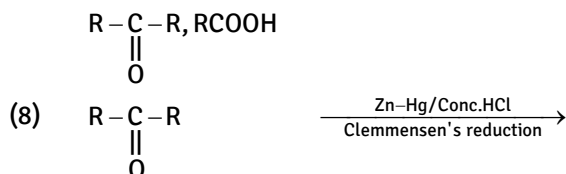
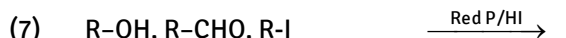
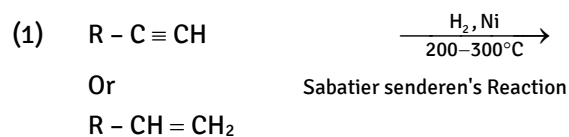
- (F) **Cracking (Pyrolysis or Thermal Decomposition)**

- Cracking may be defined as a process in which higher alkanes are converted into a mixture of lower alkanes and alkenes by the action of heat alone or heat
- Cracking involves breaking of (C-C) and (C-H) bonds resulting in the formation of a mixture of lower hydrocarbons.
- pyrolysis of alkanes involves free radical reaction. Preparation of oil gas or petrol gas from kerosene oil or petrol involves the principle of pyrolysis.



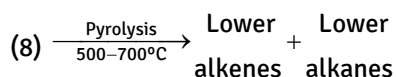
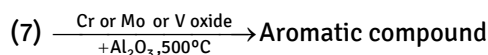
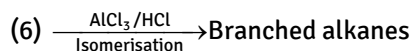
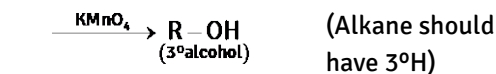
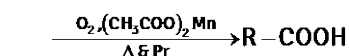
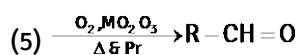
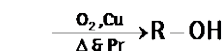
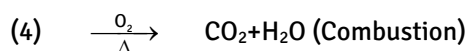
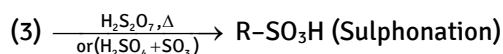
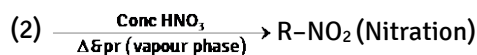
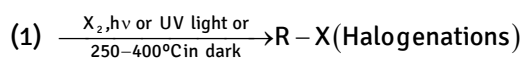
REACTION CHART FOR ALKANES

GNP



R-H
Or
R-R
or
 C_nH_{2n+2}

GR



Controlled Oxidation

Exercise 1.1

- An alkane of molecular weight 86 g/mol on monochlorination gives two products. The alkane is
 - 2-Methylbutane
 - n-butane
 - 2, 2-Dimethyl propane
 - 2, 3-Dimethyl butane
- Total number of products including stereoisomers formed during monochlorination of 2-methyl butane?

(1) 2	(2) 4
(3) 6	(4) 8
- Which of the following is a termination step in the free radical chlorination of methane?
 - $\text{Cl}_2 \rightarrow 2\dot{\text{Cl}}$
 - $\text{Cl}_2 + \dot{\text{Cl}} \rightarrow \dot{\text{Cl}} + \text{Cl}_2$
 - $\dot{\text{C}}\text{H}_3 + \dot{\text{Cl}} \rightarrow \text{CH}_3\text{Cl}$
 - $\text{CH}_4 + \dot{\text{Cl}} \rightarrow \text{HCl} + \dot{\text{C}}\text{H}_3$
- Which of the following is a chain propagation step in the free radical chlorination of methane?
 - $\text{CH}_4 + \dot{\text{Cl}} \rightarrow \dot{\text{C}}\text{H}_3 + \text{HCl}$
 - $\dot{\text{C}}\text{H}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \dot{\text{C}}\text{H}_3$
 - $\dot{\text{Cl}} + \dot{\text{C}}\text{H}_3 \rightarrow \text{CH}_3\text{Cl}$
 - $\text{Cl}_2 \rightarrow 2\dot{\text{Cl}}$
- Nitration of ethane gives
 - Nitro methane
 - Nitroethane
 - Ethyl nitrite
 - 1 & 2 both
- Arrange the correct sequence for mechanism of chlorination of methane

I $\text{CH}_4 + \text{Cl}^\bullet \xrightarrow{h\nu} \dot{\text{C}}\text{H}_3 + \text{H} - \text{Cl}$

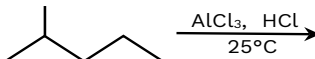
II $\text{Cl} - \text{Cl} \xrightarrow{h\nu} 2\dot{\text{Cl}}$

III $\dot{\text{C}}\text{H}_3 + \text{Cl}_2 \rightarrow \text{CH}_3 - \text{Cl} + \text{Cl}^\bullet$

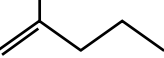
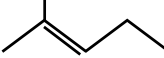
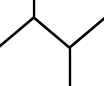
IV $\dot{\text{Cl}} + \dot{\text{Cl}} \rightarrow \text{Cl} - \text{Cl}$

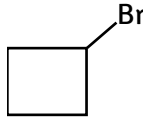
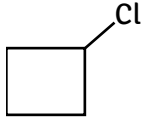
$\dot{\text{C}}\text{H}_3 + \dot{\text{C}}\text{H}_3 \rightarrow \text{CH}_3 - \text{CH}_3$

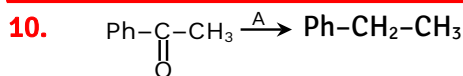
Choose the correct option.

(1) II, I, III and IV	(2) I, II, III and IV
(3) IV, III, II and I	(4) II, III, I and IV
- $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Cl} \xrightarrow[\text{dry ether}]{\text{Na}} \text{X} \xrightarrow[\Delta]{\text{V}_2\text{O}_5 - \text{Al}_2\text{O}_3} \text{Y}$
Y : Y is
 - 
 - 
 - $\text{CH}_3 - (\text{CH}_2)_4 - \text{CH}_3$
 - $(\text{CH}_3)_2\text{CH} - \text{CH}(\text{CH}_3)_2$
- 

Product is :

(1) 	(2) 
(3) 	(4) 
- 1-bromo-3-chlorocyclobutane is treated with two equivalents of Na in the presence of dry ether. Which of the following will be formed?

(1) 	(2) 
(3) 	(4) 



A could be :

- (1) NH_2NH_2 , glycol/ OH^-
- (2) Zn(Hg) /conc. HCl
- (3) Red P/HI
- (4) All

11. Which is correct product?

- (1) $\text{CH}_3-\text{CH}_3 \xrightarrow[\Delta, \text{pr}]{\text{O}_2, \text{Cu}} \text{CH}_3-\text{CH}_2-\text{OH}$
- (2) $\text{CH}_3-\text{CH}_3 \xrightarrow[\Delta, \text{pr}]{\text{O}_2, \text{MO}_2\text{O}_3} \text{CH}_3-\text{CHO}$
- (3) $(\text{CH}_3)_3\text{CH} \xrightarrow{\text{KMnO}_4} (\text{CH}_3)_3\text{C}-\text{OH}$
- (4) All

12. When aqueous solution of sodium ethanoate is electrolysed, the product(s) at anode is/are :

- (1) Ethane
- (2) CO_2
- (3) H_2
- (4) Both (1) & (2)

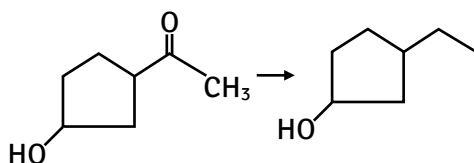
13. The no. of isomeric carboxylic acid that will be required to obtain neopentane by soda lime:

- (1) 3
- (2) 1
- (3) 4
- (4) 6

14. Which product is formed at the anode by the electrolysis of sodium butyrate :

- (1) Butane + CO_2
- (2) Pentane + CO_2
- (3) Hexane + CO_2
- (4) Hexane + H_2

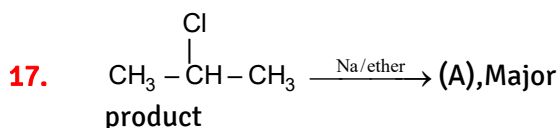
15. The appropriate reagent for the transformation is:



- (1) Zn(Hg) /conc. HCl
- (2) $\text{NH}_2\text{NH}_2/\text{OH}^-$ / Ethylene glycol
- (3) H_2/Ni
- (4) NaBH_4

16. Kolbe's electrolysis is convenient for the preparation of :

- (1) Methane
- (2) Symmetrical alkanes containing even number of carbon atoms
- (3) Alkanes containing even as well as odd number of carbon atoms
- (4) Alkanes containing odd number of carbon atoms



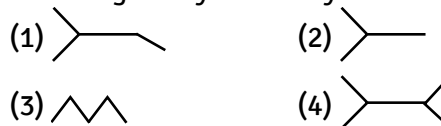
(A) is:

- (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$
- (2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
- (3) $\text{CH}_3\text{CH}_2-\text{CH}_2\text{CH}_3$
- (4) $\text{CH}_3-\text{CH}=\text{CH}_2$

18. The order of reactivity of alkyl halides in Wurtz reaction is :

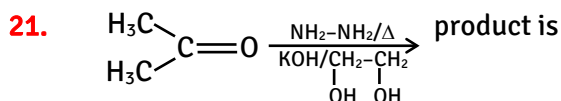
- (1) $\text{R-I} > \text{R-Br} > \text{R-Cl}$
- (2) $\text{R-I} < \text{R-Br} < \text{R-Cl}$
- (3) $\text{R-Br} \gg \text{R-I} < \text{R-Cl}$
- (4) $\text{R-I} \gg \text{R-Cl} > \text{R-Br}$

19. Which of the following alkane is synthesized from single alkyl halide by wurtz reaction :



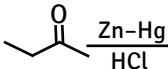
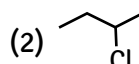
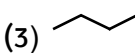
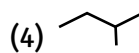
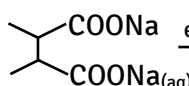
20. Which of the following reaction of alkanes involves free radical intermediates.

- (1) halogenation
- (2) Pyrolysis
- (3) Nitration
- (4) All of these



- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}_2$
- (2) $\text{CH}_3-\text{CH}_2-\text{CH}_3, \text{N}_2$
- (3) $\text{CH}_3-\text{CH}_2-\text{CH}_3, \text{NH}_3$
- (4) $\text{H}_3\text{C}-\underset{\text{OH}}{\text{CH}}\text{CH}_3, \text{N}_2$

22. The reaction, $\text{C}_2\text{H}_5\text{Br} \xrightarrow[(\text{ii}) \text{CuI}]{(\text{i}) \text{Li}} [\text{X}] \xrightarrow{\text{CH}_3\text{Br}} [\text{Y}]$ is called-
- (1) Wurtz synthesis
 - (2) Wolf-Kishner reduction
 - (3) Corey-House synthesis
 - (4) Kolbe's synthesis
23. What is the chief product obtained when n-butane is treated with bromine in the presence of light at 130°C ?
- (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Br}$
 - (2) $\text{CH}_3-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$
 - (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{Br}$
 - (4) $\text{CH}_3-\underset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2-\text{Br}$
24. If 1-chloropropane reacts with Na in presence of dry ether. Which alkane is obtained.
- (1) Hexane
 - (2) 2,3-Dimethyl butane
 - (3) Isopentane
 - (4) Neopentane
25. If isopropyl chloride and methyl chloride both react with Na in presence of dry ether which alkanes are obtained.
- (1) Ethane
 - (2) Isobutane
 - (3) 2,3-Dimethyl butane
 - (4) All of them

26. Which of the following compound can not be obtained from single alkyl halide by wurtz reaction.
- (1) ethane
 - (2) butane
 - (3) isopentane
 - (4) hexane
27.  Products is
- (1) 
 - (2) 
 - (3) 
 - (4) 
28.  Y is
- (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$
 - (2) $(\text{CH}_3)_2\text{CH}-\text{CH}(\text{CH}_3)_2$
 - (3) $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}_3$
 - (4) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
29. Give reactivity order for decarboxylation?
- (I) $\text{CH}_3-\text{CH}_2-\text{COOH}$
 - (II) $\text{CH}_2=\text{CH}-\text{COOH}$
 - (III) $\text{CH}\equiv\text{C}-\text{COOH}$
- (1) I > II > III
 - (2) III > II > I
 - (3) III > I > II
 - (4) None of these
30. Which of the following reaction can not be used to obtain propane in good yield.
- (1) Wurtz reaction
 - (2) Kolbe electrolysis
 - (3) Soda lime decarboxylation
 - (4) (1) & (2) both

Alkenes

Introduction of Alkenes

- Alkenes are unsaturated hydrocarbons.
- General formula : C_nH_{2n} .
- Alkenes are also called olefins (oil forming) since the first member, ethene forms an oily liquid on reaction with chlorine.
- The doubly bonded carbon atoms are sp^2 hybridized.
- Geometry is trigonal planar.
- Bond angle is 120° .

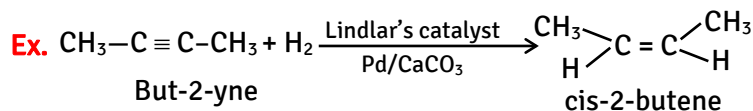
General Methods of Preparation of Alkenes

(1) By the Partial Hydrogenation of Alkynes:

Alkynes can be converted into alkenes by controlled or partial hydrogenation in two ways :

(i) by using Lindlar's Catalyst :

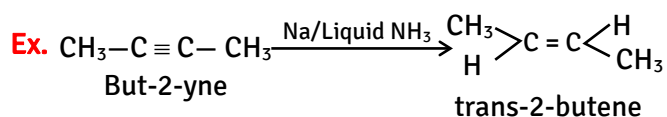
- Lindlar's catalyst is palladium deposited on barium sulphate or calcium carbonate. The catalytic is slightly poisoned with lead acetate, quinoline or sulphur.
- Alkynes are partially reduced with calculated amount of dihydrogen in presence of lindlar's catalyst into cis alkenes.



In place of Lindlar's catalyst Nickel-boride (Ni-B also called P-2 catalyst) can also be used.

(ii) By using Sodium in Liquid Ammonia :

This is known as Birch reduction and the major product is a trans alkene.



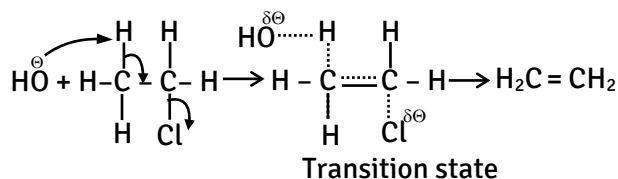
Note: Birch reduction fails in case of terminal alkynes due of acid base reaction

(2) From Alkyl Halide (By Dehydrohalogenation):

Alcoholic KOH or NaNH_2 is used as dehydrohalogenating agent



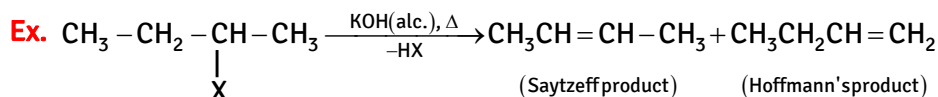
Mechanism : The reaction involves E_2 mechanism.



- Rate of reaction $\propto$ [substrate] [base]
- Order of reaction = 2
- Carbocation rearrangement is not observed during the reaction
- Since reaction is endothermic ($\Delta H = +Ve$), so high temperature is a favourable condition to get product with good yield
- It is an example of β -elimination
- The rate of dehydrohalogenation shows the order :

For alkyl group $\longrightarrow$ tertiary $>$ secondary $>$ primary

For halogen in alkyl halide $\longrightarrow$ Iodide $>$ Bromide $>$ Chloride $>$ fluoride

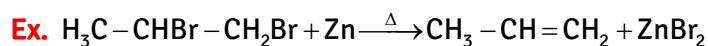
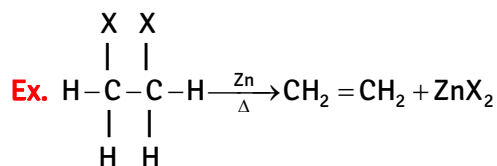


- Note:** (i) If $-\text{X} \Rightarrow -\text{Cl} \text{ or } -\text{Br}$, saytzeff's product is major (Based on stability of alkene)
 (ii) If $-\text{X} \Rightarrow -\text{F}$, Hoffman's product is major (Based on stability of carbanion)
 (iii) If attacking base is bulky (like t-butoxide ion) Hoffmann's product is major.

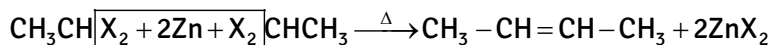
(3) From Alkyl Dihalide (By Dehalogenation of Vicinal or Gem Dihalide):

- Removal of X_2 from a substrate by Zn dust

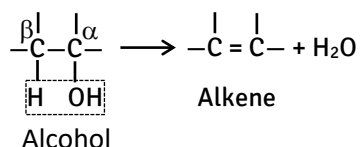
(i) From Vicinal Dihalide : alkene with same number of carbon is obtained.



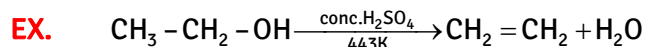
(ii) From gem Dihalide : Higher alkene obtained

**(4) From Alcohols: (By acidic dehydration)**

Alkenes can be prepared from monohydric alcohols or alkanols by the loss of H_2O in presence of an acid, the reaction is known as acidic dehydration of alcohols.



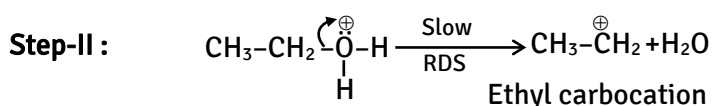
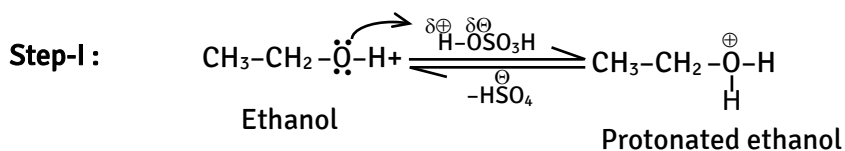
The dehydration can be carried out in the presence of heat with Conc. H_2SO_4 . Or Conc. H_3PO_4



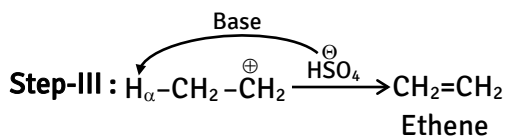
Ethanol

Ethene

Mechanism of Reaction : The reaction follows E_1 mechanism



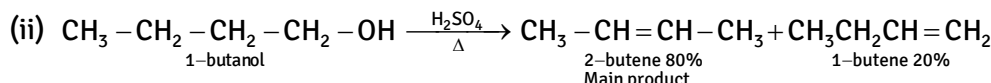
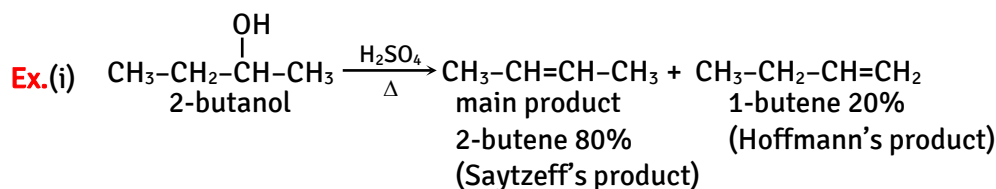
This is slow step and is regarded as **rate determining step** in E_1 reaction.



- The reaction involves carbocation intermediate.
- Rate of reaction $\propto$ [substrate]
- The reaction is an example of β -elimination.

Saytzeff Rule : When more than one alkenes are obtained by the elimination reaction then that alkene containing maximum number of alkyl group on double bonded C-atoms is called Saytzeff's product and formed as major product.

Note: The alkene having less number of alkyl groups on double bonded C-atoms is called Hofmann's product.

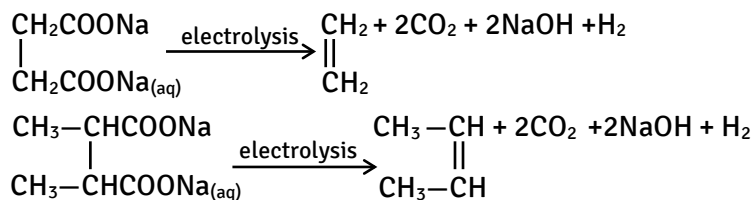


The order of acidic dehydration in different alcohols is: Tertiary > Secondary > Primary

Note: Al_2O_3 can also be used at high temperature for dehydration of alcohols.

(5) By Kolbe's Method :

Electrolysis of potassium or sodium salt of saturated dicarboxylic acid gives alkene.



Physical Properties of Alkenes

- From $C_2 - C_4$ they are colourless, odourless gases, from $C_5 - C_{17}$ they are colourless liquids, and higher alkenes are solids.
- Alkenes are insoluble in water but they are fairly soluble in organic solvents like benzene, chloroform, CCl_4 , petroleum ether etc. (Like dissolves like)
- B.P increase with increase in size and every $-CH_2$ group added increase B.P by $20-30K$.
- The melting points of cis isomers are lower than trans isomers because cis isomer is less symmetrical than trans. Thus trans packs more tightly in the crystal lattice and hence has a higher melting point.
- The boiling points of cis isomers are higher than trans isomers because cis-alkenes has greater polarity (Dipole moment) than trans one.
- The increase in branching in carbon chain decreases the boiling point among isomeric alkenes.

Chemical Reactions of Alkenes

Alkenes are more reactive than Alkanes this is because -

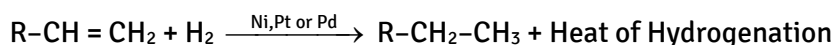
- (i) The π electrons of double bond are located much far from the carbon nuclei and are loosely bonded.
- (ii) π bond is weaker than σ bond

Following Reactions are observed in alkenes:

- (A) Addition reaction (B) Oxidation reaction (C) Free radical Substitution reaction
(D) Polymerization Reaction

(A) Addition reaction:

(a) Addition of H_2 :



- Reaction is exothermic, Heat released in reaction is called heat of hydrogenation.
- Heat of hydrogenation $\propto$ no. of π - bond

if no of π -bond are same then.

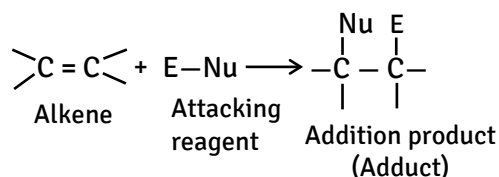
$$\text{Heat of hydrogenation} \propto \frac{1}{\text{stability of alkene}}$$

Note: Stability of alkene depends upon resonance, hyperconjugation and type of geometrical isomerism

- The process is used to obtain vegetable (saturated fats) ghee from hydrogenation of oil.

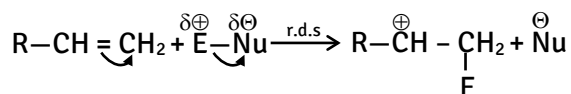
(b) Electrophilic Addition Reactions:

Characteristic reaction of alkenes are electrophilic addition reaction

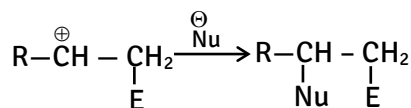


From mechanism point of view, the addition in alkenes is generally **electrophilic in nature** which means that attacking reagent which carries the initial attack is an electrophile (E^+). This is quite expected also as there is high electron density in the double bond. The mechanism proceeds in two steps.

Step-I: The reaction involves addition of E^+ in first step to form carbocation as intermediate.

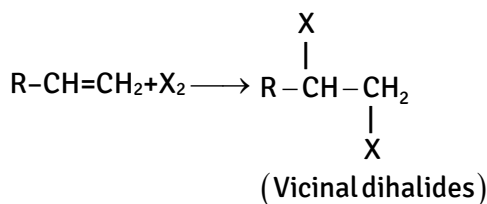


Step-II: The nucleophile (Nu^-) released in the slow step combines with the carbocation to give the addition product in the fast step.

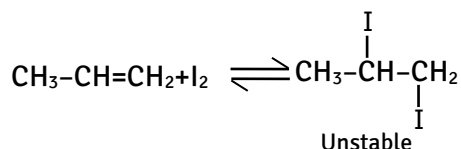


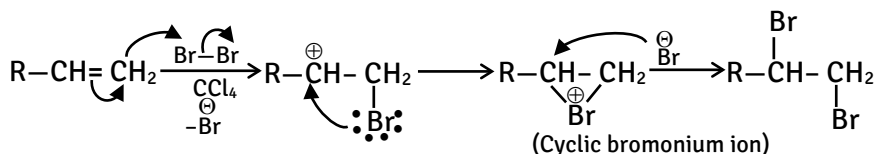
Reactivity of alkenes / alkynes for electrophilic addition $\propto$ stability of carbocation formed in r.d.s

(i) Addition of Halogen : It is an electrophilic addition reaction.

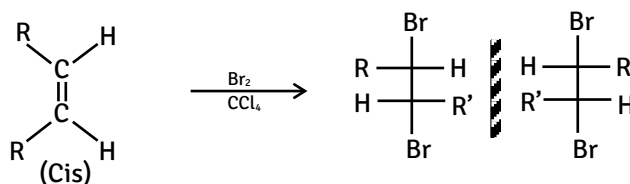
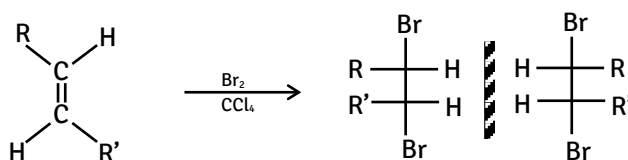
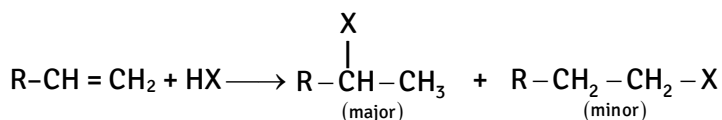


- The addition of Br_2 on alkenes is a useful test for unsaturation in molecule. The brown (or reddish orange) colour of the bromine in CCl_4 is discharged. Thus decolorization of Br_2 in CCl_4 by a compound suggest unsaturation in it.
- I_2 reacts with alkenes to form Vicinal di-iodides which are unstable and I_2 gets eliminated to give back alkene.



Mechanism:

- Electrophile involved in the reaction is Br_2 (due to vacant orbital)
- Cyclic bromonium ion is formed as intermediate
- Reaction involved anti addition of Br_2
- Carbocation rearrangement is not possible

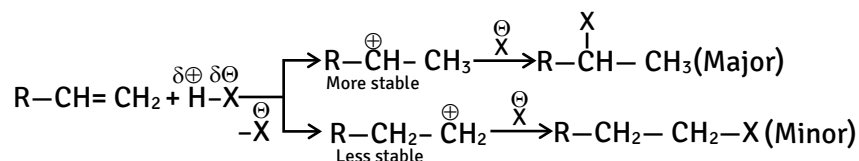
Stereochemistry of the reaction :If $\text{R}=\text{R}'$ (d+l) Racemic mixture $\text{R} \neq \text{R}'$ (d+l) Racemic mixtureIf $\text{R}=\text{R}'$ (Meso) $\text{R} \neq \text{R}'$ (d+l) Racemic mixture**(ii) Addition of Halogen acid : (hydrogen halides)**

- The order of reactivity of hydrogen halide is : $\text{HI} > \text{HBr} > \text{HCl}$
- Carbocation rearrangement is possible in the reaction.
- Addition on alkene proceeds via the formation of more stable carbonium ion.
- Addition of HX on unsymmetrical alkenes ($\text{R}-\text{CH}=\text{CH}_2$) takes place according to Markovnikov's rule.

- Addition of HX on unsymmetrical alkene : $\text{R}-\text{CH}=\text{CH}_2 + \text{H}-\text{X} \longrightarrow \begin{cases} \text{R}-\overset{\text{X}}{\underset{|}{\text{CH}}}-\text{CH}_3 \text{ (Major)} \\ \text{R}-\text{CH}_2-\text{CH}_2-\text{X} \text{ (Minor)} \end{cases}$

Markovnikov's Rule States :

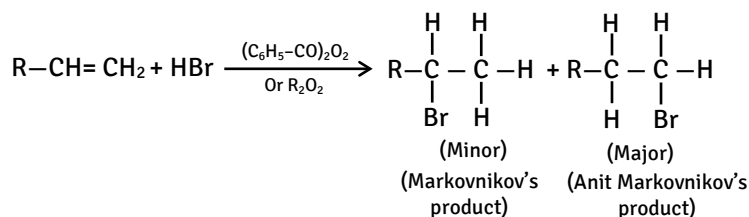
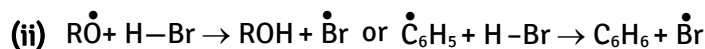
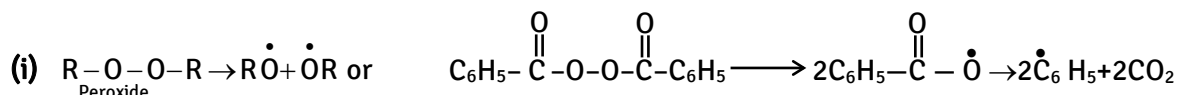
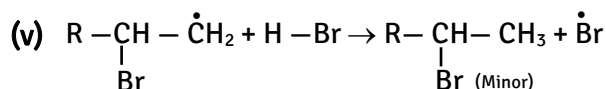
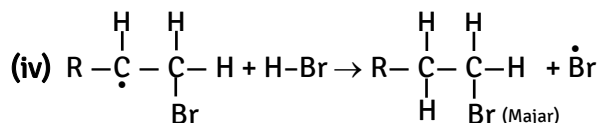
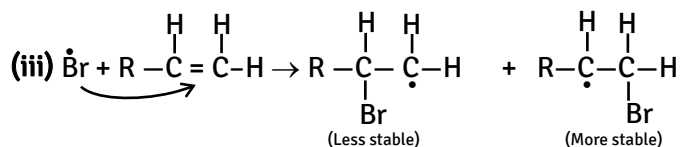
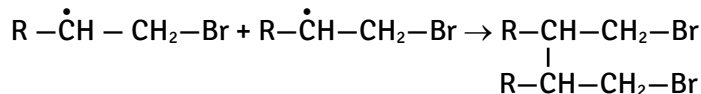
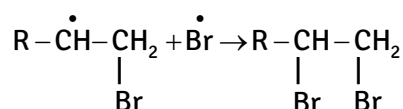
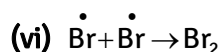
First Rule : In addition of HX on unsymmetrical alkene, the electrophile (H^+) mainly goes to the unsaturated carbon atom having more number of hydrogen atoms, or negative part of the reagent (addendum) mainly gets attached to the carbon with possesses less number of hydrogen atoms.

Mechanism :

- Markovnikov's rule is based on stability of carbocation.
- The electrophilic addition to unsymmetrical alkenes always occurs through the formation of a more stable carbocation intermediate.

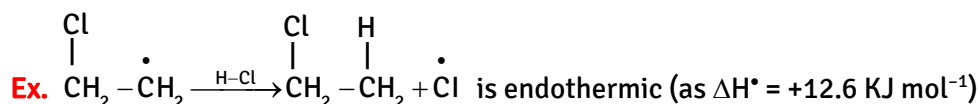
(iii) Anti Markovnikov's Rule / Kharasch Effect / Peroxide Effect –

Addition of HBr on unsymmetrical alkene in presence of peroxide follows free radical addition mechanism and major product is formed according to anti Markovnikov's rule.

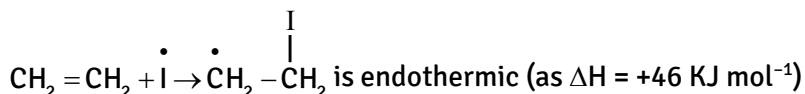
**Mechanism****(a) Chain initiation step:****(b) Chain propagation step:****(c) Chain termination step:**

Note: Anti Markovnikov addition in the presence of peroxide is not observed in HF, HCl and HI.

(i) In the case of H-F and H - Cl, one of the chain propagating steps is endothermic



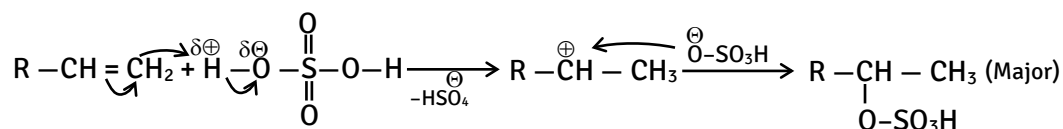
(ii) In the case of H – I, one of the chain propagating steps is endothermic



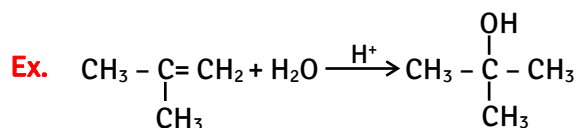
But in the case of HBr both of the chain propagating steps are exothermic, which results spontaneous reaction.

(iv) Addition of cold conc. H_2SO_4 :

Reaction follows electrophilic addition mechanism and forms major product according to Markovnikov's rule

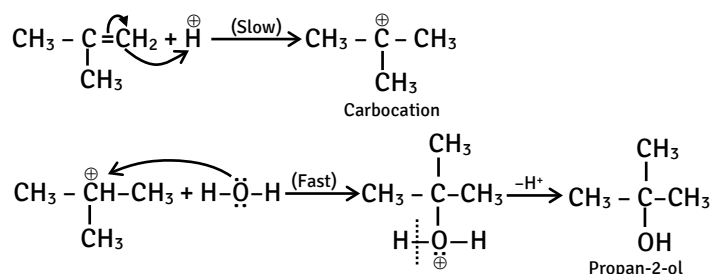


(v) **Addition of Water (Hydration of alkenes)** : Alkenes react with water in the presence of a few drops of concentrated sulphuric acid to form alcohol, this reaction is called acidic hydration.



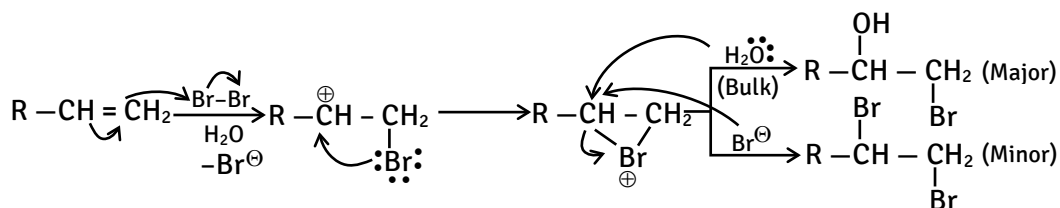
2-Methylpropene 2-Methylpropan-2-ol

Mechanism:



- Carbocation rearrangement is possible
- Addition of water molecule proceed via formation of more stable carbocation

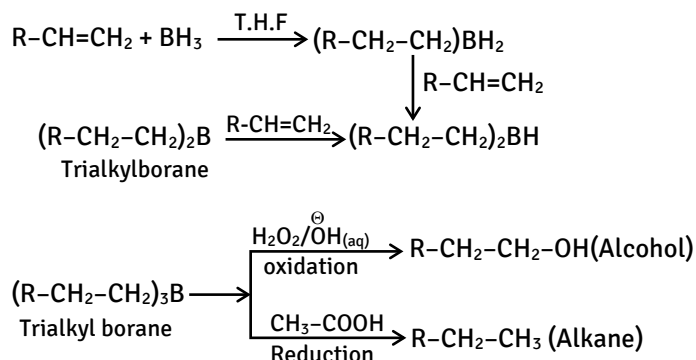
(vi) **Addition of Hypohalous Acid (or X_2/H_2O , or HOX) :**



- It is a electrophilic addition reaction.
- it follows Markovnikov's rule

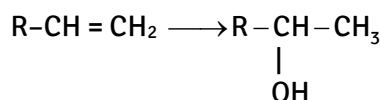
- Anti addition takes place
- Cyclic bromonium ion is formed as intermediate
- carbocation rearrangement is not possible

(vii) **Hydroboration Oxidation** : Borane readily reacts with alkenes to give trialkyl boranes. The reaction is called hydroboration.



Note: The overall process appears to be addition of water according to anti Markovnikov's rule without carbocation rearrangement and involves syn addition

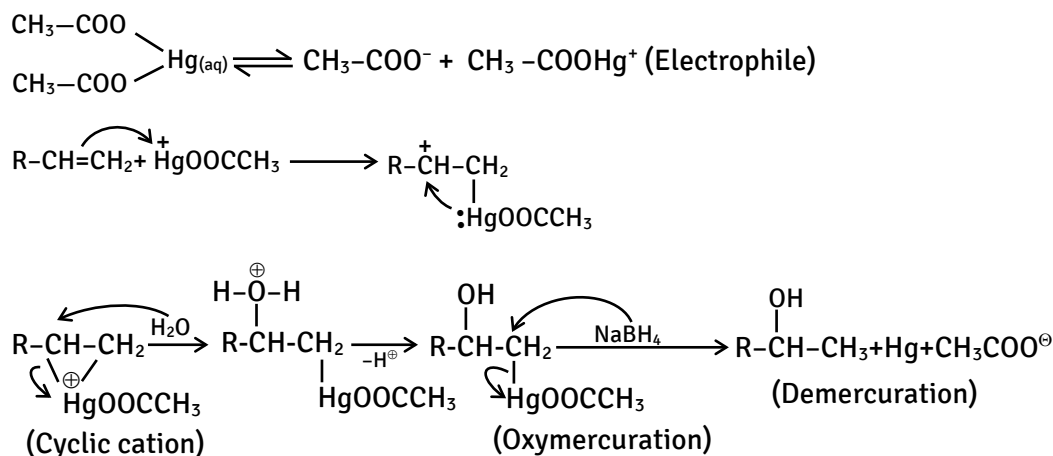
(viii) **Oxymercuration-Demercuration**: overall process appears to be addition of water molecule according to Markovnikov's rule without carbocation rearrangement.



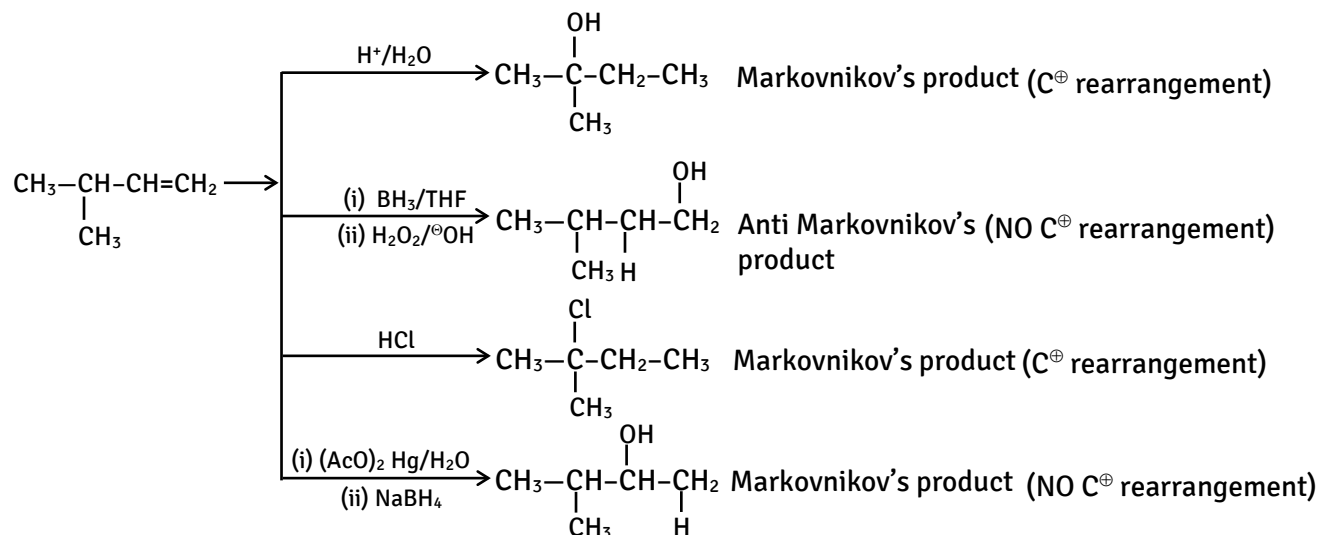
Reagents :

- (AcO)₂Hg/H₂O or (CH₃COO)₂Hg/H₂O (oxymercuration)
- NaBH₄ (demercuration)

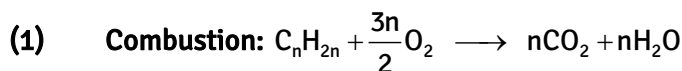
Mechanism :



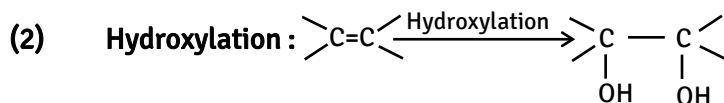
Ex.



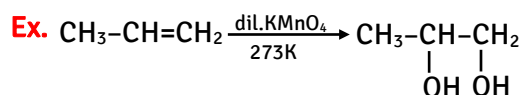
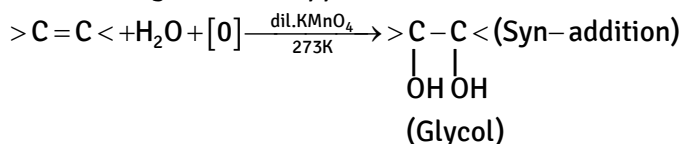
(B) Oxidation Reaction : Alkenes are easily oxidised by oxidising agents. Oxidising agents attack on double bond and product formed during oxidation depends on oxidising agents and experimental conditions.



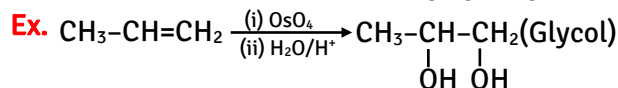
One mole of alkene requires $\frac{3n}{2}$ moles of O_2 for complete combustion.



(a) Oxidation by Baeyer's Reagent (A test for unsaturation) : Alkenes on reaction with cold, dilute, 1% alkaline KMnO_4 (i.e., Baeyer's reagent) produces vicinal glycol and decolourise the pink colour of KMnO_4 and gives brown ppt of MnO_2 . The reaction involves syn addition.

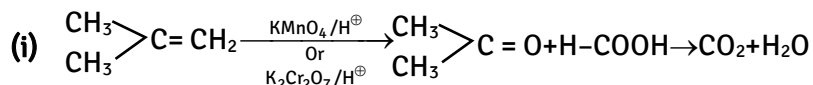


(b) reaction with OsO_4 followed by hydrolysis also form vicinal glycol and involves syn addition.



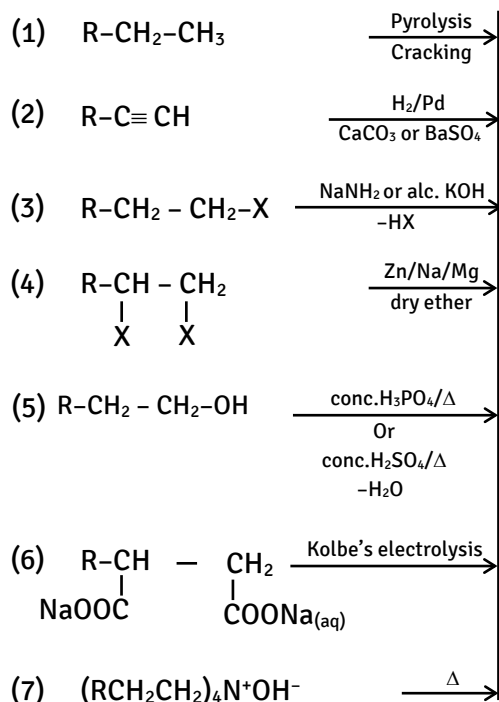
(3) Oxidation by Strong Oxidising Agent (Oxidative Cleavage) :- Acidic potassium permanganate or acidic potassium dichromate oxidises alkene to ketones and /or acids depending upon the nature of the alkene and the experimental condition.

Formic acid further gets oxidised to give $\text{CO}_2 + \text{H}_2\text{O}$.



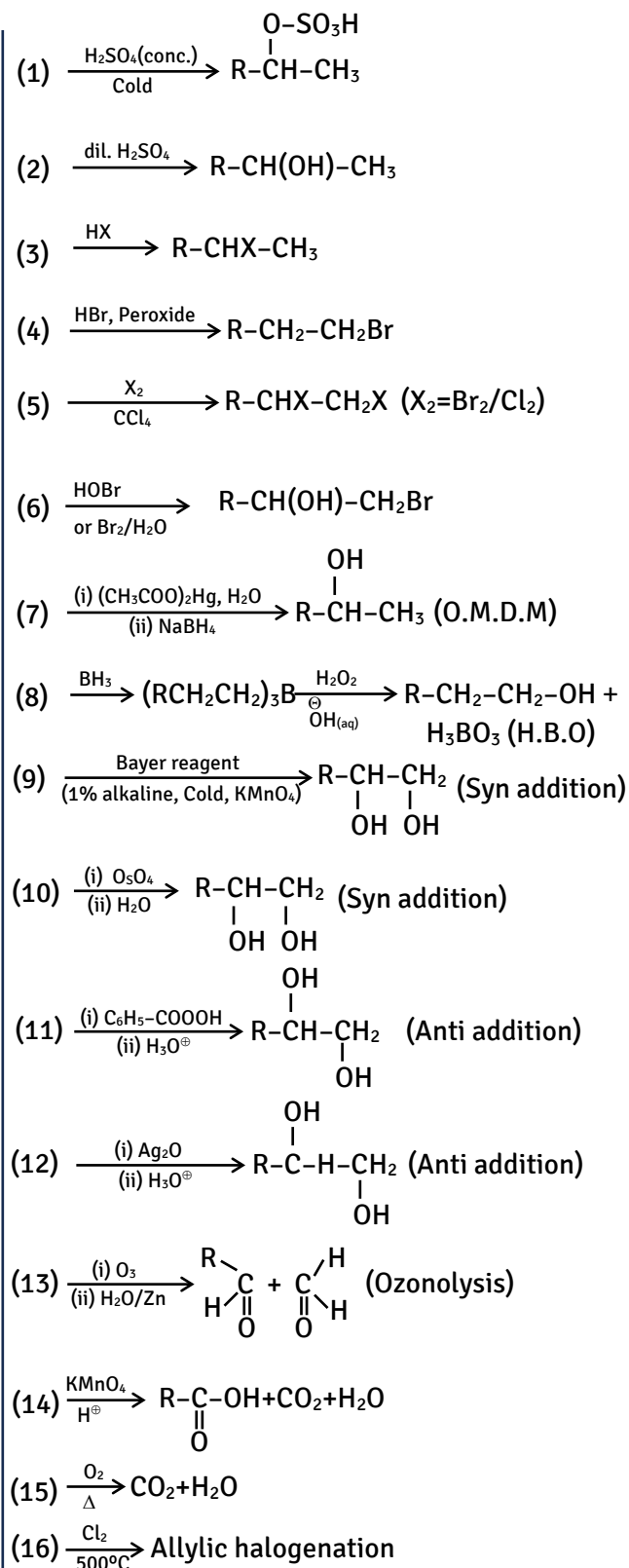
Reaction Chart for Alkenes

GMP



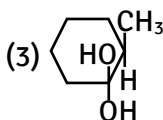
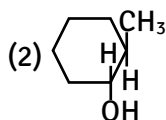
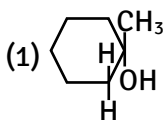
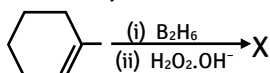
$R-CH=CH_2$
Or
 C_nH_{2n}

GR



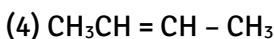
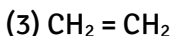
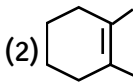
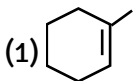
Exercise 1.2

1. The compound X is

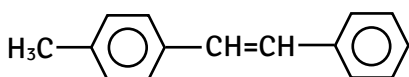


(4) All of these

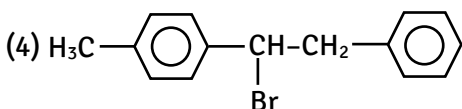
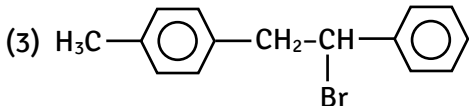
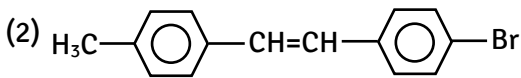
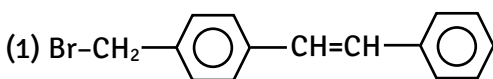
2. Hydroboration-oxidation and acidic hydration will not give the same products in case of



3. The major product(A) of the reaction given below is



A(major product)



4. How many of the following compounds will form acetic acid on reaction with acidic KMnO_4 ?

Prop-1-ene, 2-Methylbut-2-ene, 2-Methylpropene, But-2-ene, Cyclohexene.

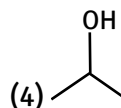
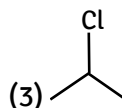
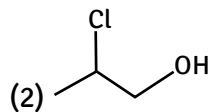
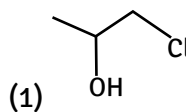
(1) 2

(2) 3

(3) 4

(4) 5

5. Propene on reaction with hypochlorous acid gives?



6. The olefin, which on ozonolysis give $\text{CH}_3\text{CH}_2\text{CHO}$ and CH_3CHO is

(1) But-1-ene

(2) But-2-ene

(3) Pent-1-ene

(4) Pent-2-ene

7. 2-Methylbutan-2-ol can be obtained by the acid catalyzed hydration of

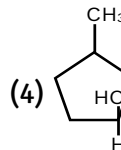
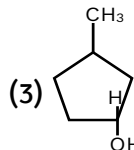
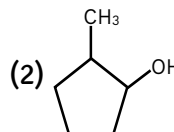
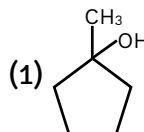
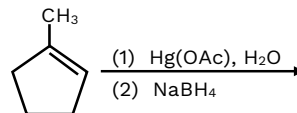
(1) $\text{CH}_3\text{CH}_2\text{CH} = \text{CH}_2$

(2) $\text{CH}_3\text{CH} = \text{CHCH}_2\text{CH}_3$

(3) $(\text{CH}_3)_2\text{C} = \text{CHCH}_3$

(4) Either of the three

8. The final product obtained in the reaction



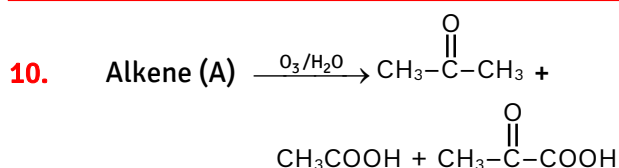
9. Which of the following will be the major product when 3-phenylpropene reacts with HBr ?

(1) $\text{C}_6\text{H}_5\text{CH}_2\text{CHBrCH}_3$

(2) $\text{C}_6\text{H}_5\text{-C}(\text{Br})=\text{CH}_2\text{-CH}_3$

(3) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

(4) $\text{C}_6\text{H}_5\text{CHBrCH}_2\text{CH}_3$



A can be :

- (1) $CH_3-C(=CHCH_3)-CH=CH_2$
 (2) $CH_3-C(=CH_2)-CH=CH-CH_3$
 (3) Both correct
 (4) None is correct

11. For the ionic reaction of hydrochloric acid with the following alkenes, predict the correct sequence of reactivity as measured by reaction rates :

- I. $ClCH=CH_2$ II. $(CH_3)_2C=CH_2$
 III. $OHC-CH=CH_2$ IV. $(NC)_2C=C(CN)_2$
 (1) $IV > I > III > II$ (2) $I > IV > II > III$
 (3) $III > II > IV > I$ (4) $II > I > III > IV$

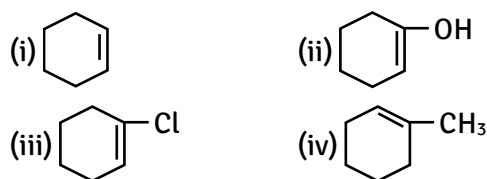
12. Which of the following will give different product with HBr in presence or absence of peroxide :

- (1) Cyclohexene
 (2) But-2-ene
 (3) 1, 2-dimethylcyclohexene
 (4) 1-butene

13. Reactivity of alkenes towards HX increases in the order :

- (1) Isobutene > Propene > Ethene
 (2) Isobutene > Ethene > Propene
 (3) Ethene > Propene > Isobutene
 (4) Propene > Ethene > Isobutene

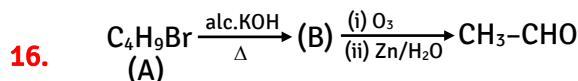
14. Give reactivity order towards EAR.



- (1) (i) > (ii) > (iii) > (iv)
 (2) (iv) > (iii) > (ii) > (i)
 (3) (ii) > (iv) > (i) > (iii)
 (4) (iii) > (ii) > (iv) > (i)

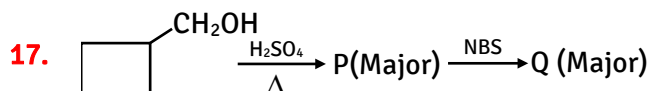
15. The compound formed when 2-butene is treated with hot acidic $KMnO_4$ is :

- (1) $CH_3-CH=O$
 (2) $CH_3-C(=O)-OH$
 (3) CH_2OH-CH_2OH
 (4) $CH_3CH_2-C(=O)-CH_3$

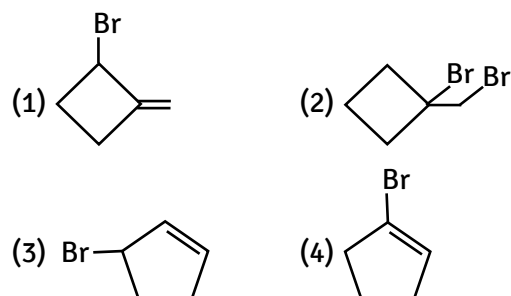


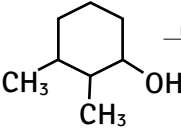
(A) and (B) in the above reaction sequence are:

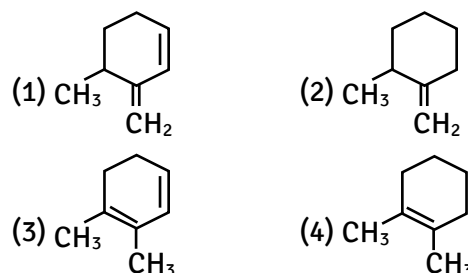
- (1) s-Butyl bromide, 1-Butene
 (2) t-Butyl bromide, 2-methyl propene
 (3) s-Butyl bromide, 2-Butene
 (4) n-Butyl bromide, 1-Butene

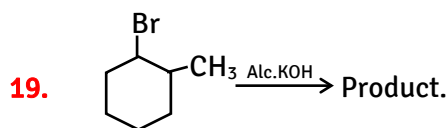


The structure of Q is :

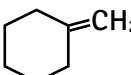
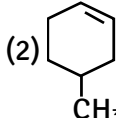
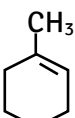
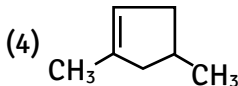


18.  $\xrightarrow[\Delta]{H^+} X(Major)$ 'A' is

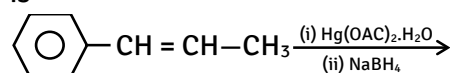


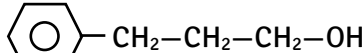
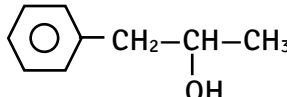
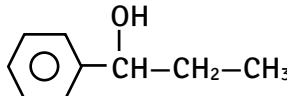
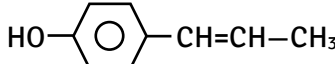


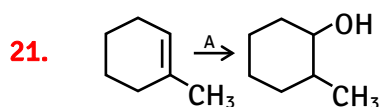
The product can be

- (1)  (2) 
 (3)  (4) 

20. The major product of the following reaction is



- (1) 
 (2) 
 (3) 
 (4) 

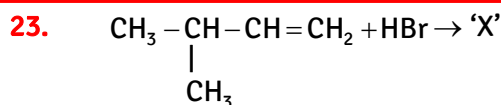


Reagent 'A' is

- (1) $\text{BH}_3 / \text{H}_2\text{O}_2 / \text{OH}^-$
 (2) $\text{H}_2\text{O} / \text{H}^+$
 (3) $\text{Hg}(\text{OCOCH}_3)_2 / \text{NaBH}_4$
 (4) $\text{Cl}_2 / \text{aq. NaOH}$

22. A hydrocarbon X adds on one mole of hydrogen to give another hydrocarbon and also decolourises bromine water. X reacts with KMnO_4 in presence of acid to give two moles of the same carboxylic acid. The structure of X is

- (1) $\text{CH}_3\text{CH}=\text{CH}.\text{CH}_2\text{CH}_2\text{CH}_3$
 (2) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_3$
 (3) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{CH}=\text{CHCH}_3$
 (4) $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_3$

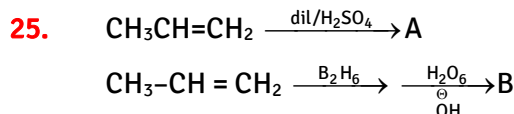


Major product X is-

- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_2\text{Br}$
 (2) $\text{CH}_3-\underset{\text{CH}_3}{\underset{\text{Br}}{\text{C}}}-\text{CH}_2-\text{CH}_3$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$
 (4) None is correct

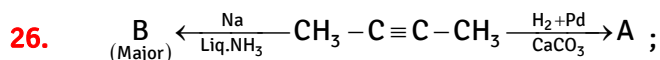
24. Two jars A and B are filled with hydrocarbon. Br_2 in CCl_4 is added to these jars. A does not decolourise the Br_2 solutions, but B decolourises. What are A and B respectively.

(1) Alkane and alkene
 (2) Alkene and alkane
 (3) Both 1 and 2
 (4) Alkene and Alkyne



Correct statement about the product is

- (1) A and B have different functional groups
 (2) A and B are position isomers.
 (3) A and B show metamerism
 (4) All of these



A and B are

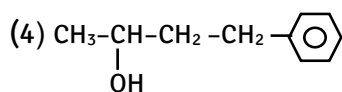
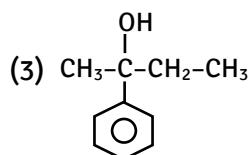
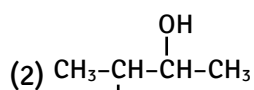
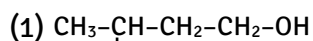
- (1) Homomers
 (2) Position isomers
 (3) Geometrical isomers
 (4) Homologues

27. When 3,3-dimethyl-2-butanol is heated with H_2SO_4 the major product obtained is

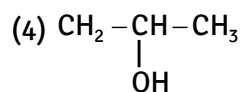
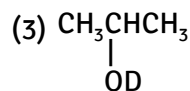
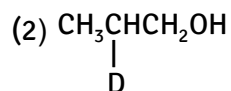
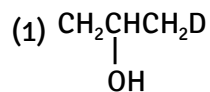
- (1) cis and trans isomers of 2,3-dimethyl-2-butene
 (2) 3,3-dimethyl-1-butene
 (3) 2,3-dimethyl-2-butene
 (4) 2,3-dimethyl-1-butene

28. But-2-ene can be obtained by electrolysis of an aqueous solution of
- (1) 2,3-Dimethyl maleic acid
 - (2) 2,2-Dimethylbutanedioic acid
 - (3) 2-Methylbutanedioic acid
 - (4) 2,3-Dimethylbutanedioic acid

29. $\text{CH}_3\text{-CH}(\text{C}_6\text{H}_5)\text{-CH=CH}_2 \xrightarrow[\text{Or dil. H}_2\text{SO}_4]{\text{H}_2\text{O}/\text{H}^+} \text{P (major)}$, P is:



30. $\text{CH}_3\text{CH=CH}_2 \xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{BD}_3/\text{THF}} \text{X}$,
Major product 'X' is



Alkynes

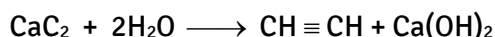
Introduction of Alkynes

- Like alkenes, alkynes are also saturated hydrocarbons.
- Alkynes are hydrocarbons that contain carbon-carbon triple bonds
- First member is ethyne popularly known as Acetylene.
- Their general formula is C_nH_{2n-2}
- Triply bonded carbon atoms are sp hybridized
- Bond angle in alkyne is 180° .
- Shape is linear
- Bond length order: $C-C$ (154 PM) $>$ $C=C$ (133 PM) $>$ $C\equiv C$ (120 PM)
- Bond strength order $C\equiv C$ (823 KJ mol^{-1}) $>$ $C=C$ (681 KJ mol^{-1}) $>$ $C-C$ (348 KJ mol^{-1})

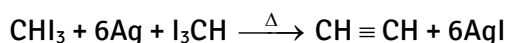
General Methods of Preparation of Alkynes

(1) Preparation of Ethyne or Acetylene :

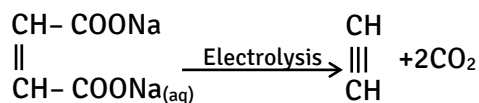
- (i) **From Calcium carbide [Industrial method]** : Acetylene is prepared on industrial scale by treating calcium carbide with water.



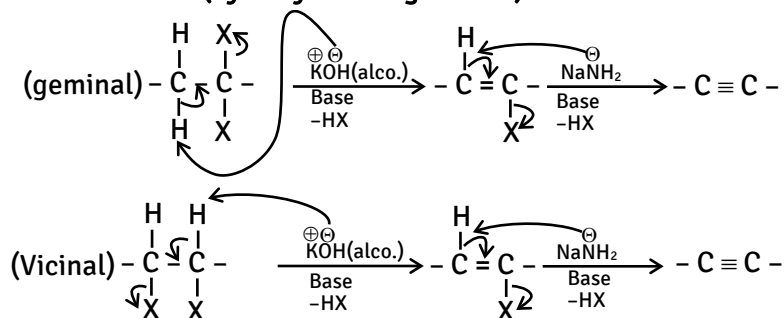
- (ii) **From Haloform [CHI_3 , $CHCl_3$]** : Pure acetylene is obtained when iodoform or chloroform is heated with Silver powder.



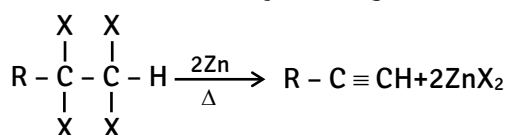
- (iii) **By Kolbe's Electrolysis** : By the electrolysis of aqueous Solution of sodium or potassium fumarate or maleate, acetylene is formed at anode.



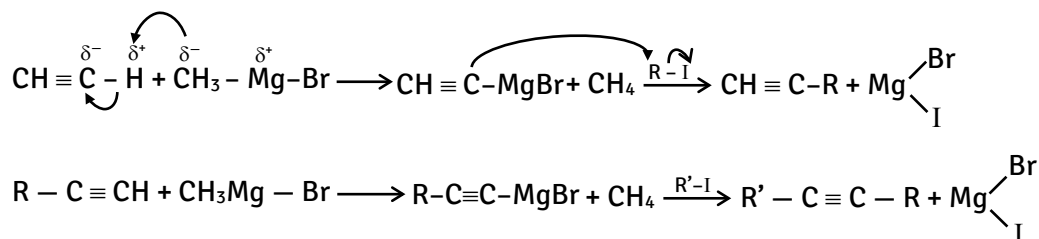
(2) From dihalides (by dehydrohalogenation):



(3) From tetrahalides (by dehalogenation)



- (4) **Preparation of Higher Alkynes by Grignard Reagent :** By this method lower alkyne is converted into higher alkyne



Physical Properties of Alkynes

- First three members [C_2 to C_4] are gases, from C_5 to C_{12} are liquid and higher ones are solid.
- Alkynes are immiscible with H_2O but soluble in organic solvents like CCl_4 , benzene, and ether.
- B.P., M.P. and densities of alkynes are comparatively more than alkenes and alkanes due to more polarisation, and increase with increase in molar mass.

$\text{B.P.} \propto \text{mol. wt.} \propto \frac{1}{\text{number of side chains}}$
--

- Terminal alkynes are acidic in nature. It is due to greater electronegativity of sp hybridised 'C'.

Chemical Properties of Alkynes

Alkynes show acidic nature, addition reaction and polymerisation reaction.

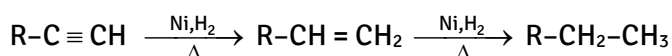
- (1) **Addition reaction :** Due to presence of loosely bonded π electrons, alkynes like alkenes, undergo electrophilic addition reaction.
Alkynes are less reactive than alkenes towards electrophilic addition reactions, it is due to less stable vinylic carbocation formed in r.d.s of electrophilic addition reaction of alkynes.
- (2) **Acidic character of alkynes :** The hydrogen atom attached to the triply bonded carbon can be removed by a strong base and hence acetylene and terminal alkynes show acidic behaviour

Explanation : The s-character in various types of C-H bonds is as –

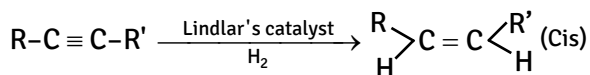
	Alkyne	Alkene	Alkane
	$\equiv \text{C}-\text{H}$	$=\text{C}-\text{H}$ H	$\begin{matrix} \text{H} \\ \\ -\text{C}-\text{H} \\ \\ \text{H} \end{matrix}$
S-Character	50%	33%	25%

Due to maximum % S character (50%), the sp hybridised orbitals of carbon atom in alkyne have highest electronegativity. Hence these attract the shared pair of electron of C-H bond of alkyne to a greatest extent, So H atom can be liberated more easily in alkynes as compared to alkenes and alkanes.

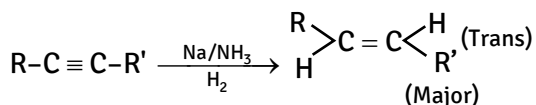
- (1) **Addition Reaction:**
- (a) **Addition of Hydrogen:**
- (i) Alkynes reacts with hydrogen in presence of a catalyst like of Pt, Pd or Ni to give corresponding alkanes with H_2



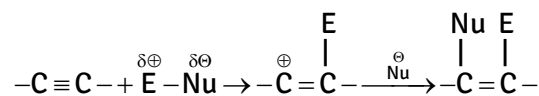
In presence of Lindlar's catalyst [$\text{Pd}/\text{CaCO}_3 + \text{quinoline}$] alkynes give cis-alkene



In presence of $\text{Na}/\text{liq. NH}_3$, alkynes give mainly trans-alkene. (Birch Reduction)



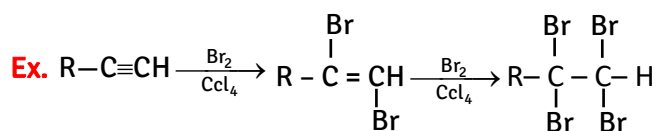
(b) Electrophilic Addition:



(Vinylic carbocation)

Addition product formed depends on stability of vinylic carbocation. Addition in unsymmetrical alkynes takes place according to Markovnikov's rule.

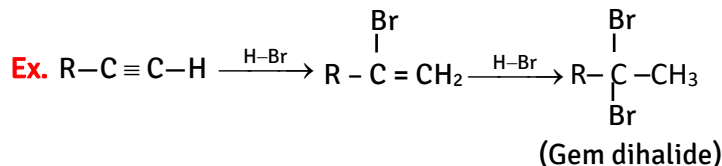
(i) **Addition of Halogens** : Alkynes react with Cl_2 or Br_2 in presence of CCl_4 to form di and tetra halo derivatives.



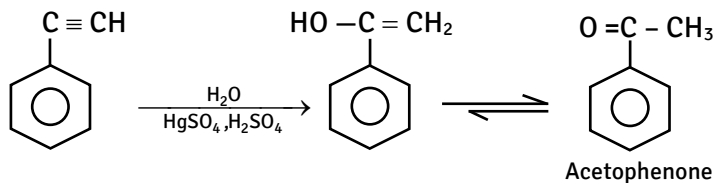
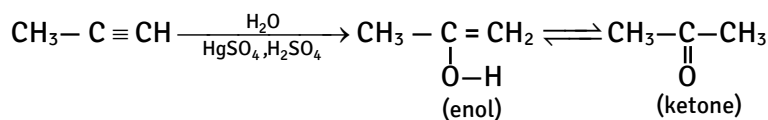
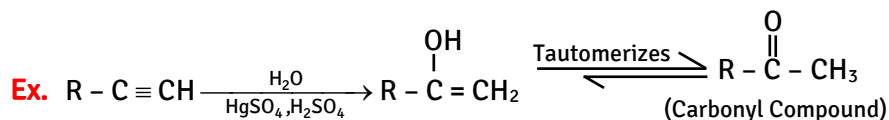
- Reddish orange colour of the solution of bromine in CCl_4 is decolorised. This is used as a test for unsaturation

(ii) **Addition of Hydrogen Halides (H-X)** : Addition takes place according to Markovnikov's Rule.

Reactivity order of H-X : $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$



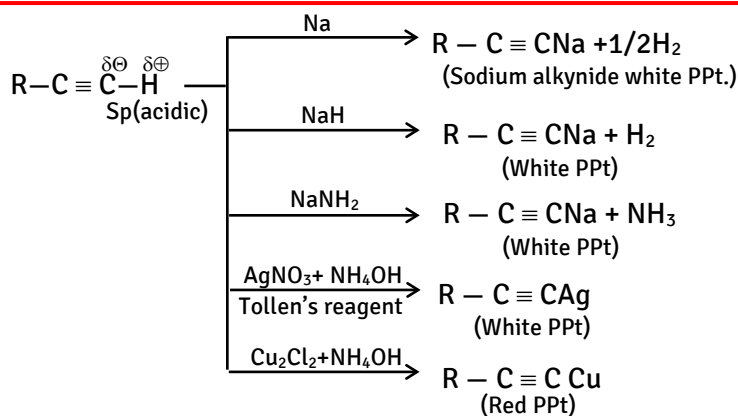
(iii) **Addition of water (Hydration)** : Addition of water to alkynes is carried out in the presence of dil. sulphuric acid and mercuric sulphate. This is known as Kucherov's reaction



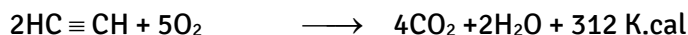
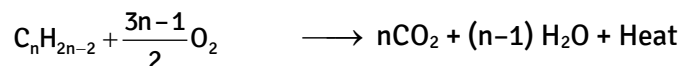
(2) Acidic Character of Terminal alkynes

In terminal alkyne or acetylene, the H which is attached with sp hybridised carbon is called acidic or active H.

So Acetylene and Terminal alkynes show acidic behavior when reacted with strong base.

**Used of the reaction:**

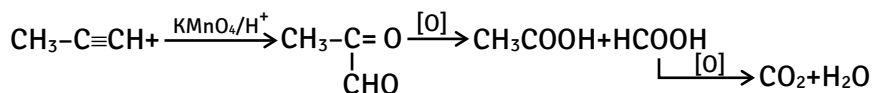
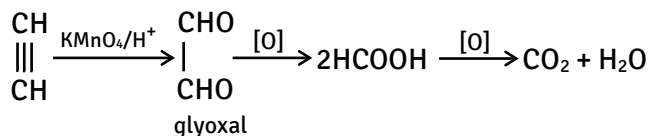
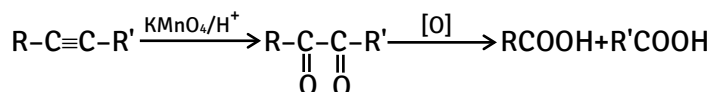
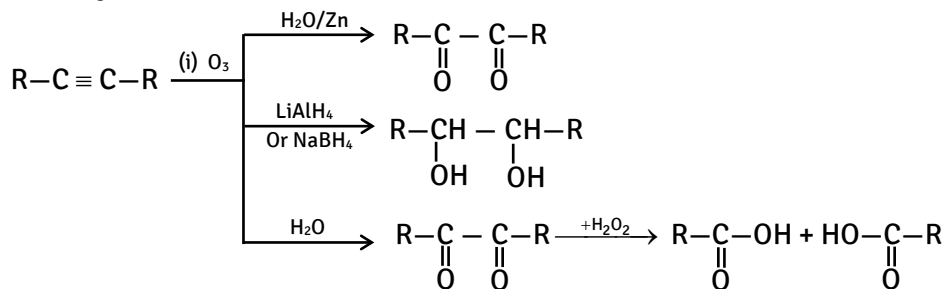
- The reaction can be used to differentiate terminal and Non terminal alkynes
- The reaction can be used to convert terminal alkynes into higher alkynes

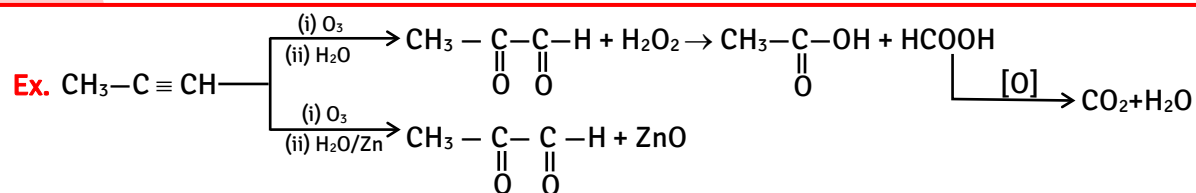
(3) Oxidation Reactions:**(a) Combustion :**

The combustion of acetylene is used for welding and cutting of metals in which oxy-acetylene flame having high temp (3000°C) is produced.

(b) Oxidation with Acidic KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$:

In presence of acidic KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$, alkynes are oxidised to monocarboxylic acids.

**(c) Ozonolysis :**

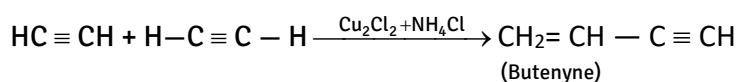


(4) Polymerisation Reactions :

Alkyne mainly shows addition polymerisation reactions.

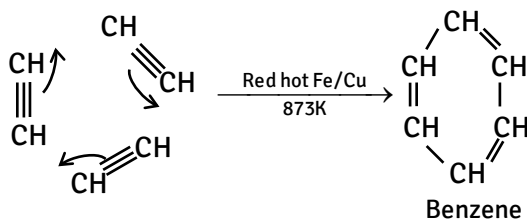
(a) Dimerisation :

Two mole acetylene reacts with Cu_2Cl_2 & NH_4Cl and forms butenyne

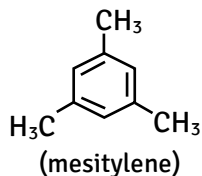


(b) Trimerisation :

If acetylene is passed through red hot iron or Cu tube, then polymerization will occur in which three molecules polymerise to form Benzene.



Note: Propyne gives mesitylene

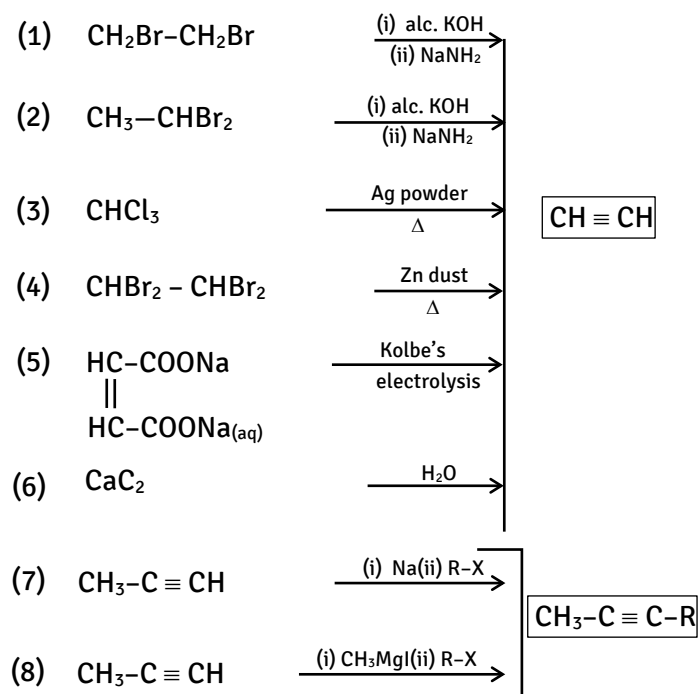


Test for Alkynes :

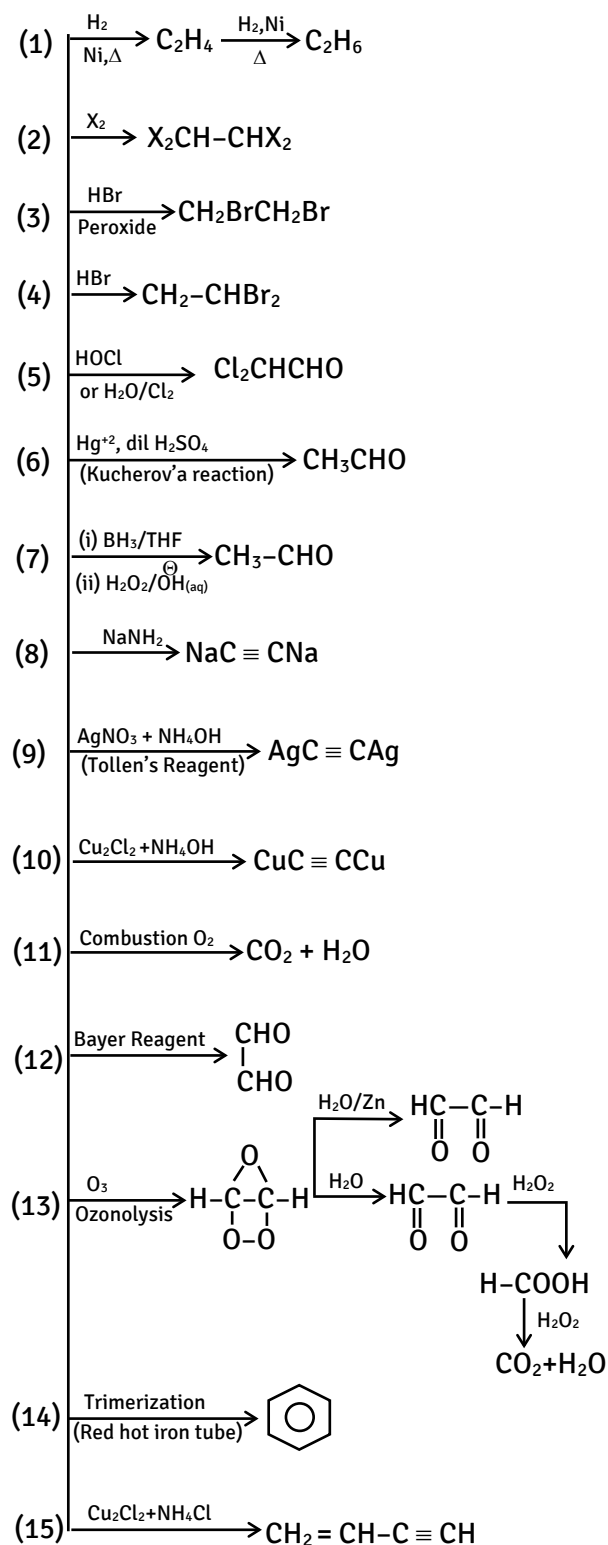
- Decolourization of Br_2 in CCl_4 Solution.
- Decolourisation of 1% alkaline KMnO_4 Solution.
- Terminal alkynes give white ppt. with ammoniacal AgNO_3 and red ppt with ammoniacal cuprous chloride Solution.

Reaction Chart for Alkynes

GMP

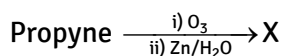


GR



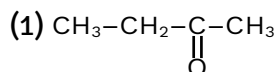
Exercise 1.3

1. The number of carbonyl carbon in the product X is



- (1) 2 (2) 4
(3) 6 (4) 8

2. The product obtained on acidic hydration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be :

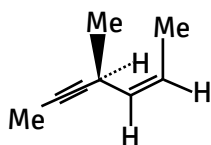


- (2) $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---CHO}$
(3) $\text{CH}_3\text{---CH}_2\text{---CHO} + \text{HCHO}$
(4) $\text{CH}_3\text{---CH}_2\text{---COOH} + \text{HCOOH}$

3. Propyne and propene can be distinguished by :

- (1) Conc. H_2SO_4 (2) Br_2 in CCl_4
(3) Dil. KMnO_4 (4) AgNO_3 in NH_4OH

4. Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives :



- (1) An optically active compound
(2) An optically inactive compound
(3) A racemic mixture
(4) A diastereomeric mixture

5. What is the chief product of reaction between 2-chloro but-2-ene and NaNH_2

- (1) 1,2-butadiene
(2) 1,3-butadiene
(3) 2-butyne
(4) 1-butyne

6. Acetylene can be prepared from

- (1) Potassium fumarate
(2) Calcium carbide
(3) Ethylene bromide
(4) All

7. Which of the following compounds will yield 2, 3-pentanedione on reductive ozonolysis?

- (1) 2, 3-Pentadiene (2) 2-Pentene
(3) Cyclopentene (4) 2-Pentyne

8. An alkyne which gives two moles of acetic acid on ozonolysis is :

- (1) 1-Butyne
(2) 2-Butyne
(3) Methyl acetylene
(4) 3-methyl-1-butyne

9. $\text{CH}_3\text{---CH}_2\text{---CH} \begin{matrix} \text{Cl} \\ \text{Cl} \end{matrix} \xrightarrow[\text{Excess}]{\text{NaNH}_2} \text{X} \xrightarrow{\text{CH}_3\text{Br}} \text{Y}$
 $\xrightarrow[\text{CaCO}_3]{\text{H}_2 + \text{Pd}} \text{Z}$; Z is :-

- (1) (2)
(3) (4)

10. $\text{CH}_3\text{---C}\equiv\text{CH} + 2\text{HBr} \longrightarrow \text{Product?}$

- (1) $\text{CH}_3\text{---}\underset{\text{Br}}{\text{CH}}\text{---}\underset{\text{Br}}{\text{CH}_2}$
(2) $\text{CH}_3\text{---CH---CH}_2 \begin{matrix} \text{Br} \\ \text{Br} \end{matrix}$
(3) $\text{CH}_2\text{---CH}_2\text{---CH}_2 \begin{matrix} \text{Br} \\ \text{Br} \end{matrix}$
(4) $\text{CH}_3\text{---}\underset{\text{Br}}{\underset{\text{Br}}{\text{C}}}\text{---CH}_3$

11. Acetylene can be obtained by the reaction?

- (1) $\text{HCOOK} \xrightarrow{\text{Electrolysis}}$
(2) $\text{CHI}_3 + \text{Ag} \longrightarrow$
(3) $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[443\text{K}]{\text{Conc. H}_2\text{SO}_4}$
(4) $\text{Be}_2\text{C} + \text{H}_2\text{O} \longrightarrow$

12. Acetylene may be prepared using Kolbe's electrolytic method employing :

- (1) Pot. acetate (2) Pot. succinate
(3) Pot. fumarate (4) None of these

- 13.** Which is/are true reactions ?
 (1) $\text{Al}_4\text{C}_3 + \text{H}_2\text{O} \longrightarrow \text{CH}_4$
 (2) $\text{CaC}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2$
 (3) $\text{Me}_3\text{C}-\text{H} + \text{KMnO}_4 \longrightarrow \text{Me}_3\text{C}-\text{OH}$
 (4) All
- 14.** For preparing methyl acetylene, we take :
 (1) $\text{CH}_3-\text{C}(\text{COOK})=\text{CHCOOK}$
 (2) $\text{CHCOOK}-\text{CHCOOK}$
 (3) $\text{CH}_3-\text{CH}_2-\text{C}(\text{COOK})=\text{CHCOOK}$
 (4) All of these
- 15.** Propyne is formed by heating 1, 2-dibromopropane with :
 (1) $\text{C}_2\text{H}_5\text{ONa}$
 (2) Alc. KOH and sodamide
 (3) Aqueous alkali
 (4) Sodalime
- 16.** An alkyne C_7H_{12} on reaction with Alk. KMnO_4 and subsequent acidification with HCl yields a mixture of $(\text{CH}_3)_2\text{CH}-\text{COOH} + \text{CH}_3-\text{CH}_2\text{COOH}$. The alkyne is
 (1) 3-Hexyne
 (2) 2-methyl-3-hexyne
 (3) 3-methyl-2-hexyne
 (4) 2-methyl-1-hexyne
- 17.** $\text{R}-\text{CH}_2-\text{CCl}_2-\text{R} \xrightarrow{\text{Reagent}} \text{R}-\text{C} \equiv \text{C}-\text{R}$. The reagent is
 (1) Na (2) $\text{KOH}_{(\text{aq})}$
 (3) NaNH_2 (4) Zn
- 18.** $\text{B} \xleftarrow[\text{H}_2\text{O}_2, \text{OH}]{\text{BH}_3, \text{THF}} \text{CH}_3-\text{C} \equiv \text{CH} \xrightarrow[\text{dil.}]{\text{HgSO}_4/\text{H}_2\text{SO}_4} \text{A}$,
 A and B are respectively
 (1) $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3-\text{C}(\text{O})-\text{CH}_3$
 (2) $\text{CH}_3-\text{C}(\text{O})-\text{CH}_3$, $\text{CH}_3\text{CH}_2\text{CHO}$
 (3) $\text{CH}_3\text{CH}_2\text{CHO}$ (both)
 (4) $\text{CH}_3-\text{C}(\text{O})-\text{CH}_3$ (both)
- 19.** $\text{B} \xleftarrow{\text{Lindlar}} \text{R}-\text{C} \equiv \text{C}-\text{R} \xrightarrow{\text{Na/liq. NH}_3} \text{A}$,
 A and B are
 (1) A is trans, B is cis
 (2) A and B both are cis
 (3) A and B both are trans
 (4) A is cis, B is trans
- 20.** Which is the most suitable reagent among the following to distinguish compound 'C' from the rest of the compounds?
 (A) $\text{CH}_3\text{C} \equiv \text{CCH}_3$ (B) $\text{CH}_3\text{CH}_2-\text{CH}_2\text{CH}_3$
 (C) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$ (D) $\text{CH}_3\text{CH}=\text{CH}_2$
 (1) Br_2 in CCl_4 (2) Br_2 in H_2O
 (3) Cold KMnO_4 (4) Tollen's reagent

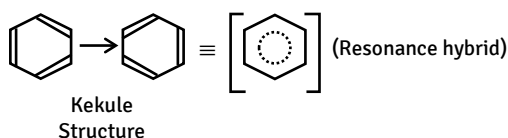
Aromatic Hydrocarbon

- Aromatic hydrocarbon are also called arenes.
- Characteristic reaction of arenes is Electrophilic substitution reaction (ESR).
- Arenes are cyclic unsaturated compounds but do not give test of unsaturation with Br_2/CCl_4 or cold KMnO_4 .
- Main source of Arenes is coal tar.
- They have higher percentage of carbon so burn with sooty flame.

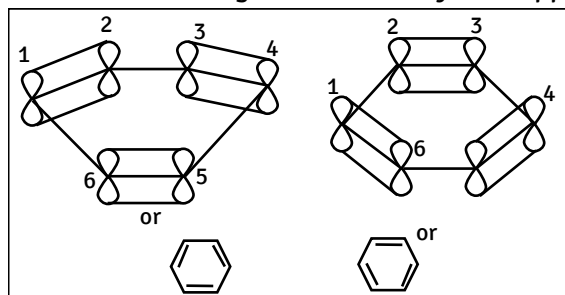
BENZENE

- [i] **Discoverer** : Michael Faraday (1825)
 [iii] Geometry – Hexagonal
 [v] C–C Bond length : $1.39 \text{ \AA}$
 [vii] Resonance in benzene

- [ii] C-hybridisation : sp^2
 [iv] Bond angle : 120°
 [vi] C–H Bond length : $1.09 \text{ \AA}$



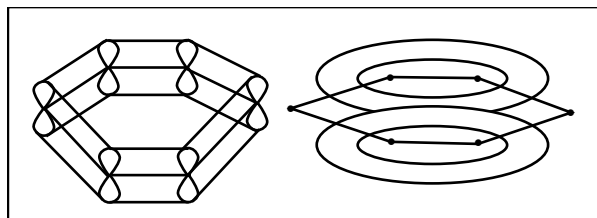
There are two equal possibilities of forming three π bond by overlapping of p orbitals



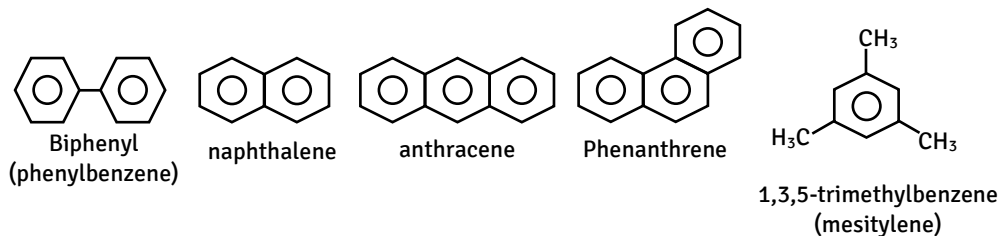
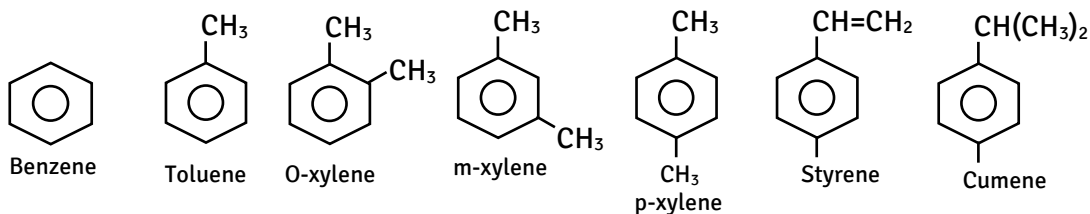
Overlapping Of p-orbital
 C_1-C_2, C_3-C_4 and C_5-C_6

Overlapping Of p-orbital
 C_2-C_3, C_4-C_5 and C_1-C_6

Delocalized π -electrons : There is equal probability for the p-orbital of each carbon atom to overlap with the p-orbital of adjacent carbon atoms. the six π -electron are thus delocalized and can move freely about the six carbon nuclei instead of any two kekule structure. the delocalized πe^- cloud is attracted more strongly by the nuclei of the carbon atoms than the electron cloud localized between two carbons. Therefore presence of delocalized πe^- makes the benzene more stable than the hypothetical cyclohexatriene

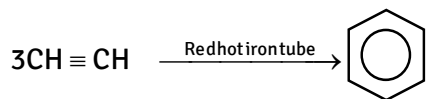


Aromatic hydrocarbons

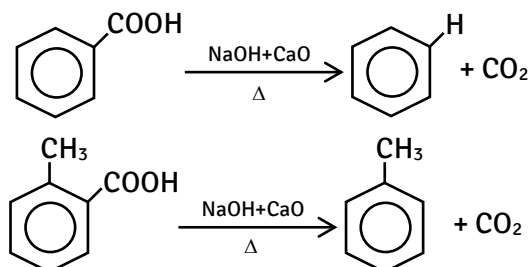


General Methods of Preparation of Aromatic Hydrocarbon

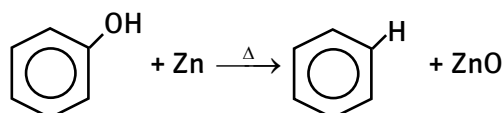
- (i) **From Alkyne (By cyclic polymerisation)** When acetylene is passed through red hot cu or Fe tube cyclic polymerisation takes place and benzene is formed.



- (ii) **From Carboxylic Acids (By decarboxylation) :**

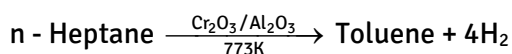
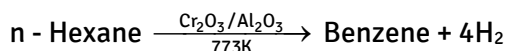


- (iii) **From Phenol (By reduction) :**

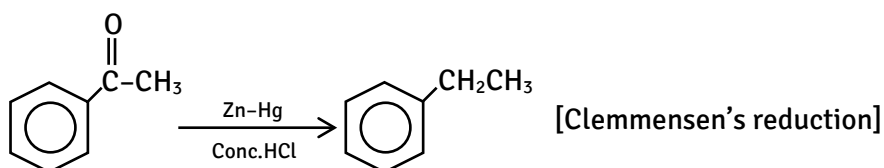
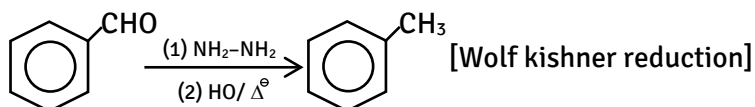


- (iv) **From Alkanes (By aromatization or reforming)**

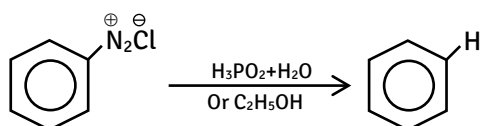
n-alkanes having six or more carbon atoms on heating to 773K at 10-20 atmospheric pressure in the presence of oxides of V, Mo or Cr supported over alumina get dehydrogenated and cyclised to benzene and its homologues. This is known as aromatization or reforming



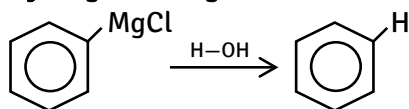
- (v) **From Carbonyl Compounds (By reduction) :**



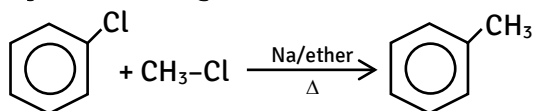
- (vi) **By Diazonium salts :**



(vii) By Grignard reagent :



(viii) By Wurtz Fitting Reaction :



Physical Properties of Aromatic Hydrocarbon

- Benzene is non polar molecule
- Benzene is colourless liquid [B.P. is 80°C]
- Benzene is insoluble in water and density less than water
- Benzene is used as a solvent and it is soluble in organic solvents.
- It is highly inflammable and burns with sooty flame.

Chemical Properties of Aromatic Hydrocarbon

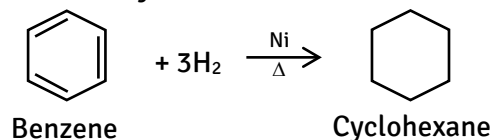
Benzene show following types of reaction -

- Addition reactions (under special condition)
- Electrophilic substitution reactions (Characteristic reaction)
- Oxidation reactions (under special condition)

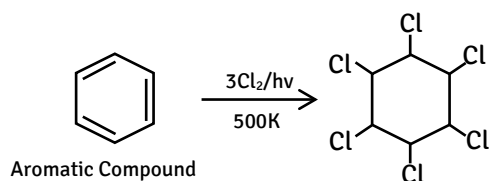
(A) Addition Reaction :

(i) Addition of H_2 :

Reaction completes under vigorous conditions i.e at high temperature and /or pressure in the presence of Ni catalyst



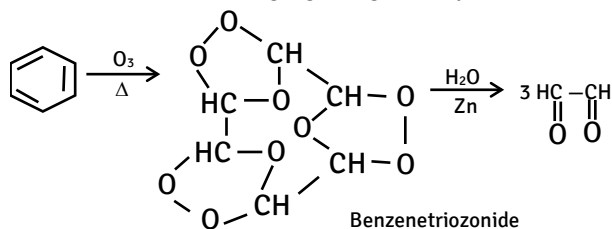
(ii) Addition of X_2



Alicyclic compound ($\text{C}_6\text{H}_6\text{Cl}_6$) or Benzene hexachloride (BHC)
or Gammexane or lindane or 666
*BHC is used as insecticide

(iii) Addition of O_3 :

Addition of O_3 followed by hydrolysis in presences of Zn (ozonolysis)

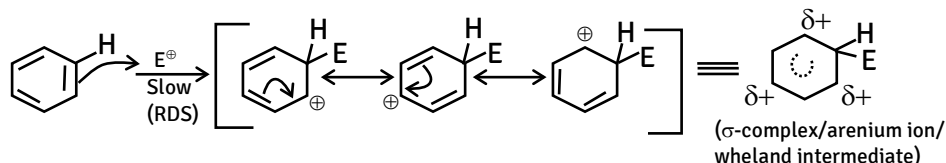


Note: Addition of 3 mole Cl_2 or 3 mole H_2 or 3 mole O_3 on benzene show presence of 3 double bonds in benzene.

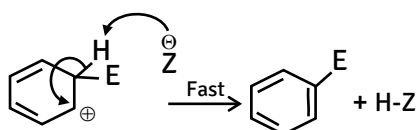
(B) Electrophilic Substitution Reaction [ESR] :

- Characteristic reaction of arenes is ESR.
- The reaction involves attack of $\text{E}^\oplus$ on benzene ring to replace H in the form of $\text{H}^\oplus$.
- ESR reactions occur via a two steps mechanism.

Step 1. Addition of the electrophile



Step 2. Loss of a proton to reform the aromatic ring



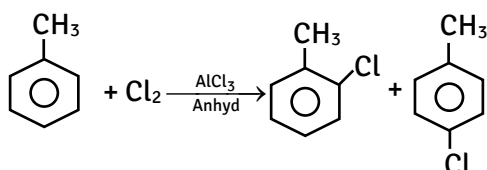
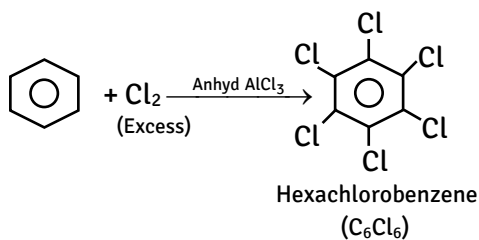
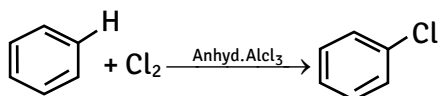
Note: In general 1st step of the reaction is r.d.s. but in case of Iodination and sulphonation 2nd step (removal of $\text{H}^\oplus$) is r.d.s.

Example of ESR:

- (i) Halogenation
- (ii) Nitration
- (iii) Sulphonation
- (iv) Friedel craft reaction (F.C.R)
 - (a) Alkylation
 - (b) Acylation

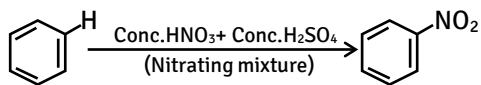
(1) Halogenation: Source of electrophile ($\text{E}^\oplus$) →

- (i) $\text{X}-\ddot{\text{X}} + \text{AlCl}_3 \rightleftharpoons \text{X}^\oplus + \text{AlCl}_4^\ominus$
(Lewis Acid)
- (ii) $\text{X}_2 + \text{Fe} \rightarrow \text{FeX}_3 \xrightarrow{\text{X}_2} \text{X}^\oplus + \text{FeX}_4^\ominus$
(Lewis Acid)

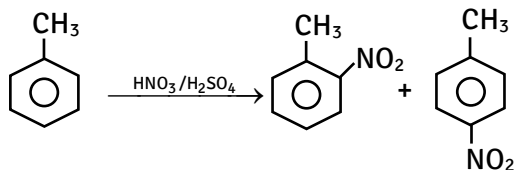
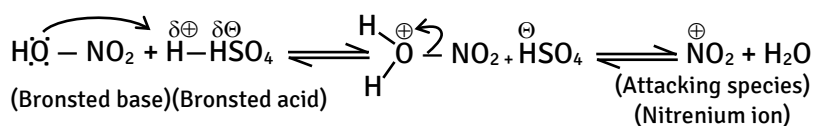


Note: CH_3 group in toluene is o/p directing and activating group.

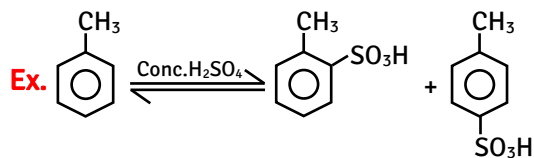
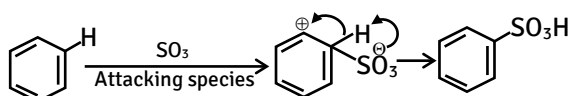
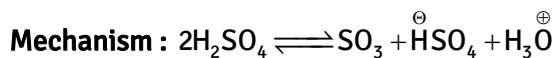
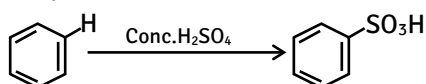
(2) Nitration :



Formation of E :



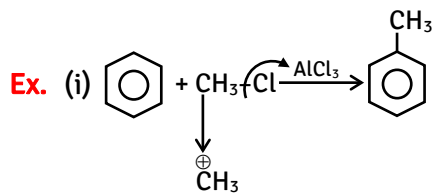
(3) Sulphonation :



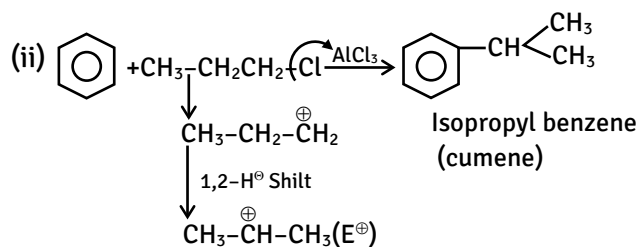
Note: Fuming sulphuric acid $\text{H}_2\text{SO}_4(\text{SO}_3)$ acts as a better source for sulphonation.

(4) Friedel Craft's Reaction [FCR] : Alkylation or acylation of arenes in presence of lewis acid [FeCl_3 , AlCl_3 or ZnCl_2 ] is called as FCR.

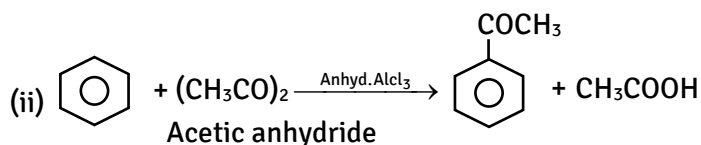
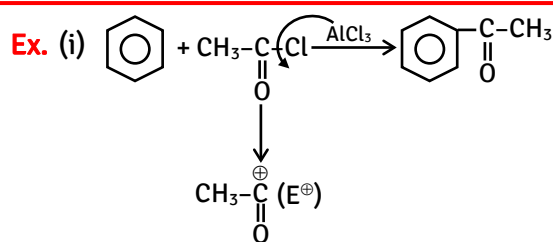
(a) Friedel craft alkylation



Intermediate carbocation is formed in Friedel craft alkylation so rearrangement is possible in presence of strong Lewis acid like AlCl_3 .



(b) Friedel craft acylation

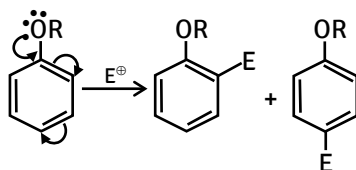
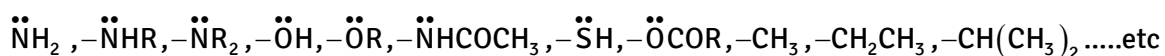


Direction influence of a functional group in monosubstituted benzene:

(1) **Ortho/Para Directing Group :** Group which directs incoming electrophile to ortho and para position is called ortho /directing group

- +M/+H effect of these group increases electron density at ortho and para positions. hence the substitution take place mainly at these position
- The groups which have net electron releasing effect increase electron density in benzene ring, activates the benzene ring easily for the attack by an electrophile. So these groups are called activating groups.

These group are:

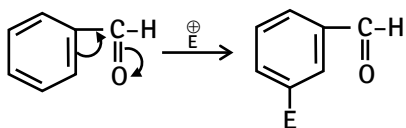


Note: Although Halogens are deactivating groups ($-I > +m$ effect) but they are ortho and para directing.

(2) **Meta Directing:** Due to $-M/-H$ of groups electron density at ortho and para position is decreased, so electrophile mainly attack at meta position. So, groups which direct electrophile to meta position are called meta directing groups.

- The groups which have net electron withdrawing effect decrease electron density in benzene ring and deactivate the benzene ring for the attack by electrophile are called deactivating groups.

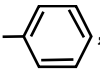
These groups are :



Note: (i) M and H effect does not depend on distance so attack at ortho and meta position equally while I-effect depends on distance .

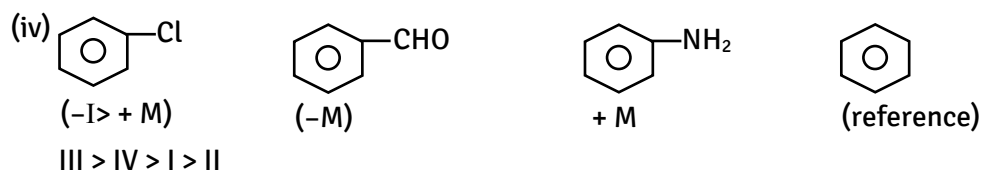
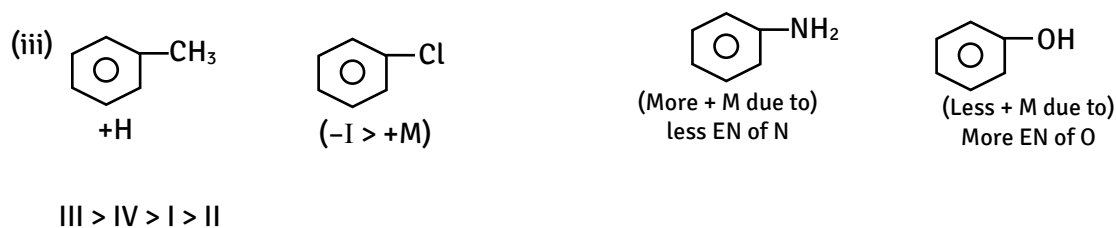
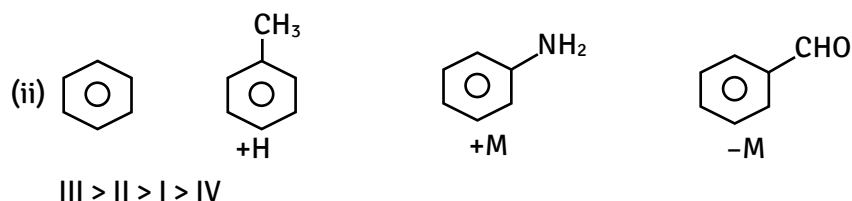
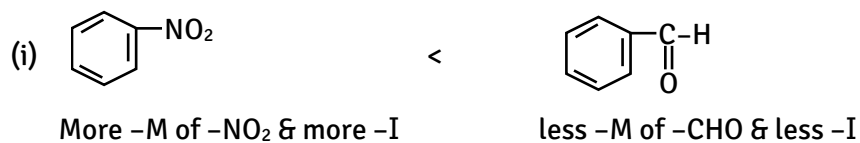
(ii) M-effect does not affect meta position.

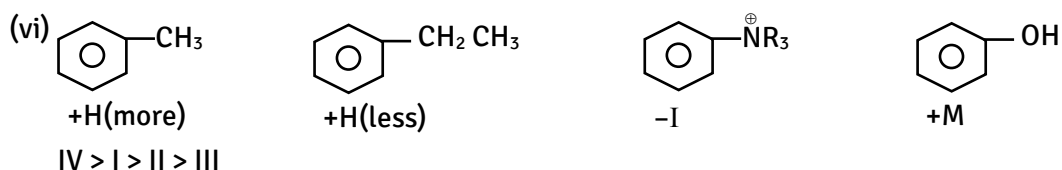
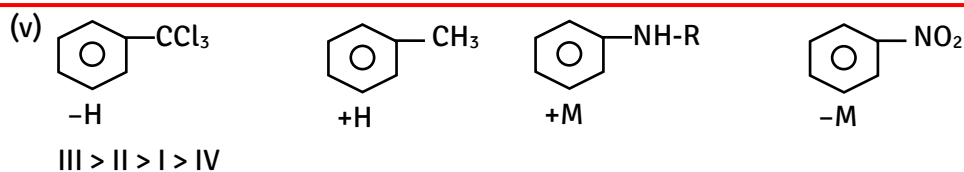
Relative Reactivity of Substituted Benzene

Ortho para directors	Mata directors
Strongly activating $-\text{O}^\ominus-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2, -\ddot{\text{O}}\text{H}, -\ddot{\text{O}}\text{R}$	Strongly deactivating $-\text{NH}_3^\oplus, -\text{NR}_3^\oplus, -\text{NO}_2, -\text{C}\equiv\text{N}, -\text{CF}_3, -\text{CCl}_3$
Moderately activating $-\ddot{\text{N}}\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3, -\ddot{\text{O}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$	Moderately deactivating $-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}, -\overset{\text{O}}{\parallel}{\text{C}}-, -\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}, -\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2, -\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}, -\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}$
Weakly activating $-\text{CH}_3, -\text{CH}_2\text{CH}_3,$  , $-\text{CH}=\text{CH}_2$	
Note: Halogens ($-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$) are weakly deactivating but they are ortho para directing	

Ex. Give reactivity order for electrophilic substitution reaction.

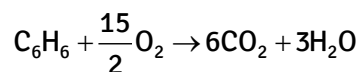
$$\left(\text{Reactivity} \propto \frac{1}{-I / -H / -M} \right)$$



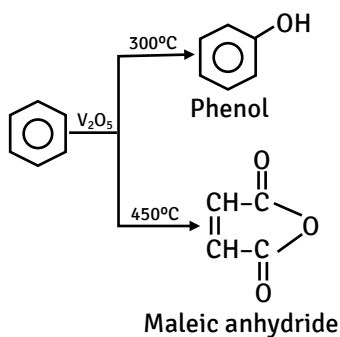


(c) **Oxidation reactions :**

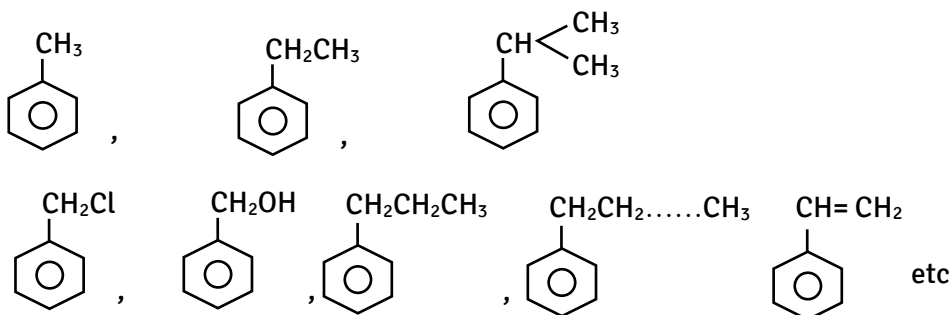
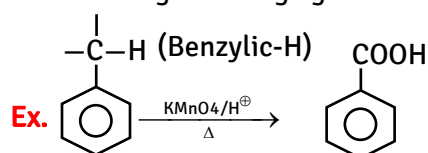
(i) **Combustion :**



(ii) **Catalytic Oxidation :**



(iii) **Side Chain Oxidation :** At least one benzylic -H containing alkyl benzene gives benzoic acid in presence of strong oxidising agent.



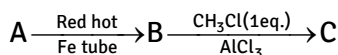
Note: t-butyl benzene does not give benzoic acid in presence of H^+/KMnO_4 , since t-butyl benzene does not contain benzylic-H

Exercise 1.4

1. O-xylene on ozonolysis will give :

- (1) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$ & $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{CHO} \end{array}$
 (2) $\begin{array}{c} \text{CH}_3-\text{C}=\text{O} \\ | \\ \text{CH}_3-\text{C}=\text{O} \end{array}$ & $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{CHO} \end{array}$
 (3) $\begin{array}{c} \text{CH}_3-\text{C}=\text{O} \\ | \\ \text{CH}_3-\text{C}=\text{O} \end{array}$ & $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$
 (4) $\begin{array}{c} \text{CH}_3-\text{C}=\text{O} \\ | \\ \text{CH}_3-\text{C}=\text{O} \end{array}$, $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{CHO} \end{array}$ & $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$

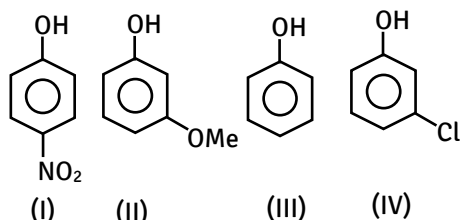
2. In the following sequence of reactions the maximum number of atoms present in product 'C' in sp^3 hybridised state is



(A is a lowest molecular weight alkyne)

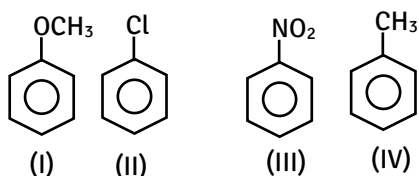
- (1) 1 (2) 3
 (3) 5 (4) 7

3. The correct increasing order of reactivity for following molecules towards electrophilic aromatic substitution.



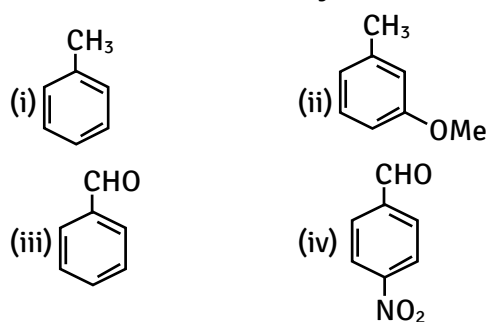
- (1) $\text{I} < \text{IV} < \text{II} < \text{III}$ (2) $\text{I} < \text{IV} < \text{III} < \text{II}$
 (3) $\text{I} < \text{III} < \text{II} < \text{IV}$ (4) $\text{I} < \text{III} < \text{IV} < \text{II}$

4. Increasing order of reactivity towards E.S.R. for following compounds is ?



- (1) $\text{III} < \text{II} < \text{IV} < \text{I}$
 (2) $\text{II} < \text{III} < \text{IV} < \text{I}$
 (3) $\text{III} < \text{II} < \text{I} < \text{IV}$
 (4) $\text{III} < \text{I} < \text{II} < \text{IV}$

5. Correct order of reactivity towards ESR is



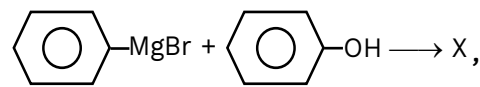
- (1) $\text{i} > \text{iii} > \text{ii} > \text{iv}$
 (2) $\text{ii} > \text{i} > \text{iii} > \text{iv}$
 (3) $\text{i} > \text{iv} > \text{iii} > \text{ii}$
 (4) $\text{iv} > \text{ii} > \text{iii} > \text{i}$

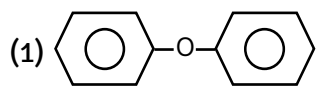
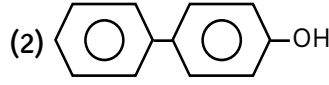
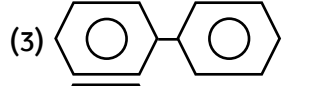
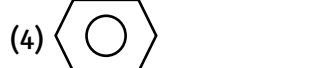
6. Consider the following statements :

- I. The $-\text{OH}$ group present in phenol is ortho and para-directing.
- II. Directive influence of a functional group in monosubstituted benzene depends on the nature of the substituent already present in the benzene ring.
- III. The $-\text{OH}$ group activates the benzene ring for the attack by an electrophile.
- IV. Groups such as $-\text{NH}_2$, $-\text{NHR}$, $-\text{NHCOCH}_3$, $-\text{OCH}_3$, $-\text{CH}_3$, $-\text{C}_2\text{H}_5$ etc, are examples of activating groups.

Select the correct option

- (1) I and II (2) II and III
 (3) II, III and IV (4) All of these

7.  $\longrightarrow \text{X}$,
 X is:

- (1) 
 (2) 
 (3) 
 (4) 

8. In the sulphonation, acetylation and formylation [$\text{CO} + \text{HCl}/\text{AlCl}_3$] of benzene the group of effective electrophiles would be

- (1) SO_3^+ , $\text{CH}_3\text{C}\equiv\text{O}^+$, $\text{HC}=\text{O}^+$
- (2) SO_3 , $\text{CH}_3-\text{C}\equiv\text{O}^+$, $\text{H}-\text{C}^+=\text{O}$
- (3) SO_3 , $\text{CH}_3-\text{C}\equiv\text{O}^+$, $\text{CO} + \text{HCl}$
- (4) HSO_3 , CH_3CO , $\text{HC}=\text{O}$

9. The electrophile involved in chlorination of benzene is

- (1) Chloride ion
- (2) Chloronium ion
- (3) Cl_2 molecule
- (4) AlCl_3

10. Which of the following is ortho and para directing.

- (1) $-\text{CCl}_3$
- (2) $-\text{NHCOCH}_3$
- (3) $-\text{SO}_3\text{H}$
- (4) $-\text{NO}_2$

11. In the nitration of benzene with concentrated HNO_3 and H_2SO_4 , the attack on ring is made by :

- (1) $\text{NO}_2^\ominus$
- (2) $\text{NO}_2^\oplus$
- (3) $\text{NO}_3^\ominus$
- (4) HNO_3

12. $\text{C}_6\text{H}_6 + \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Cl} \xrightarrow{\text{AlCl}_3} \text{'X'}$

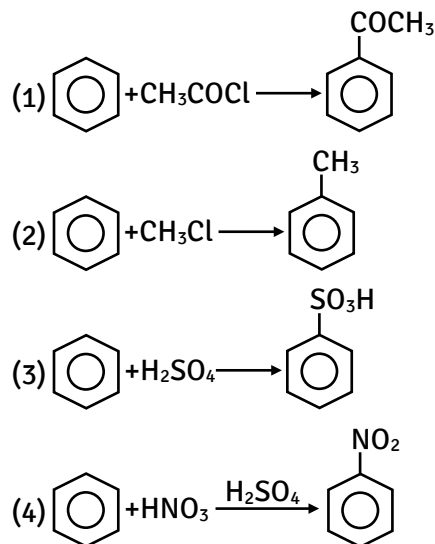
Major product X is

- (1)
- (2)
- (3)
- (4)

13. Chlorobenzene is O, P-directing in electrophilic substitution reaction. The directing influence is explained by

- (1) +M of Ph
- (2) -I of Cl
- (3) -I of Ph
- (4) +M of Cl

14. In which reaction, polysubstitution takes place :



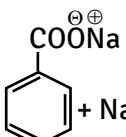
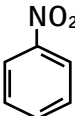
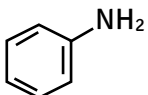
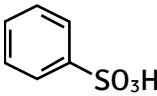
15. Identify correct statement/s :

- (A) $-\text{OCH}_3$ and $-\text{NHCOCH}_3$ are activating group
- (B) $-\text{CN}$ and $-\text{OH}$ are meta directing group
- (C) $-\text{CN}$ and $-\text{SO}_3\text{H}$ are meta directing group
- (D) Activating groups act as ortho directing groups and para
- (E) Halides are activating groups

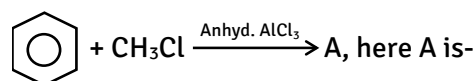
Choose the correct answer from the options given below:

- (1) (A), (C) and (D) only
- (2) (A), (B) and (E) only
- (3) (A) only
- (4) (A) and (C) only

Exercise 2

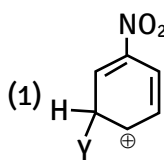
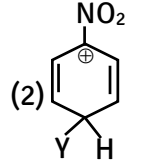
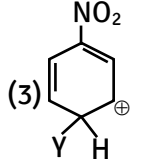
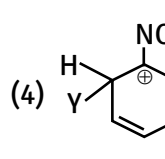
1.  + NaOH $\xrightarrow[\Delta]{\text{CaO}}$ A + Na₂CO₃, A is
 (1) C₆H₆ (2) C₆H₅ - CH₃
 (3) C₆H₅ - ONa (4) C₆H₅ - OH
2. On passing vapours of phenol over heated zinc, dust it gets reduced to
 (1) toluene (2) aniline
 (3) benzene (4) phenol
3. For an electrophilic substitution reaction, the presence of a halogen atom in the benzene ring
 (1) deactivates the ring by inductive effect
 (2) deactivates the ring by resonance
 (3) increases the charge density at ortho position relative to para-position by resonance
 (4) directs the incoming electrophile to meta -position by increasing the charge density relative to ortho and para -position
4. Consider the reaction given below.
 + Conc. HNO₃ + Conc. H₂SO₄ $\xrightarrow{(323-333)K}$ A + H₂O,
 Here, A refers to
 (1)  (2) 
 (3)  (4) All of these
5. In an electrophilic substitution reaction of nitrobenzene, the presence of nitro group
 (1) activates the ring by resonance effect
 (2) activates the ring by inductive effect
 (3) decreases the charge density at ortho and para -position of the ring relative to meta -position by resonance
 (4) increases the charge density at meta -position relative to the ortho and para -positions of the ring by resonance

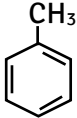
6. Consider the reaction given below.



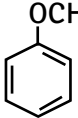
- (1) methylbenzene (2) Halo benzene
 (3) benzyl chloride (4) P-Chloro toluene
7. In the reaction,

$$X \xleftarrow[\text{AlCl}_3]{\text{Cl}_2} \text{Benzene Ring} \xrightarrow[\text{Fuming sulphuric acid}]{\text{H}_2\text{SO}_4/\text{SO}_3} Y$$

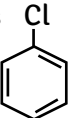
 X and Y respectively are
 (1) C₆H₅Cl and C₆H₅SO₃H
 (2) C₆H₅Cl and C₆H₅OH
 (3) C₆H₄Cl₂ and C₆H₅SO₂
 (4) C₆H₅Cl and C₆H₅CHO
8. Acetophenone (C₆H₅COCH₃) is prepared from reaction of benzene (C₆H₆) with A in the presence of anhyd. AlCl₃. Here, A refers to
 (1) CH₃COCH₃ (2) CH₃CHO
 (3) CH₃CH₂CHO (4) CH₃COCl
9. Which of the following carbocations is expected to be most stable?
- (1)  (2) 
 (3)  (4) 
10. Benzene hexachloride is obtained by addition of X moles of Cl₂ with C₆H₆ in the presence of UV/500 K. The value of X is
 (1) 2 (2) 4
 (3) 3 (4) 8
11. Which of the following deactivates the benzene ring towards electrophilic substitution and is o/p directing?
 (1) -Cl (2) -OCH₃
 (3) -CHO (4) All of these

12. Which of the following set of functional groups is meta-directing?
 (1) $-\text{NO}_2, -\text{NH}_2, -\text{COOH}, -\text{COOR}$
 (2) $-\text{NO}_2, -\text{CHO}, -\text{SO}_3\text{H}, -\text{COR}$
 (3) $-\text{CN}, -\text{CHO}, -\text{NHCOCH}_3, -\text{COOR}$
 (4) $-\text{CN}, -\text{NH}_2, -\text{NHR}, -\text{OCH}_3$
13. Correct order of electrophilic substitution reaction is
- 

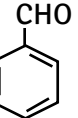
(A)

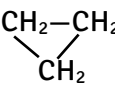


(B)



(C)

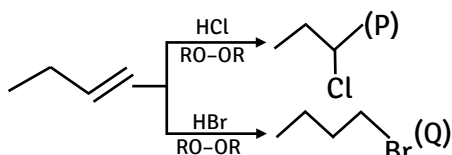


(D)
- (1) $A > B > C > D$ (2) $D > B > A > C$
 (3) $B > A > C > D$ (4) $B > A > D > C$
14. Which of the following statements are correct regarding Kolbe's electrolytic process?
 I. Alkanes containing even number of C-atoms are prepared.
 II. Sodium or potassium salts of carboxylic acids are taken as substrate.
 III. PH decreases with reaction progress
 IV. H_2 gas is liberated at anode
 (1) Both I and II (2) Both II and III
 (3) Both I and III (4) All of these
15. The reactivity of the halogens towards methane decreases in the order -
 (1) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
 (2) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 (3) $\text{F}_2 > \text{Cl}_2 > \text{I}_2 > \text{Br}_2$
 (4) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$
16. An aqueous solution containing sodium acetate and sodium propionate is electrolyzed. Which of the following alkanes is expected as product -
 (1) Ethane (2) Propane
 (3) Butane (4) All of these
17. A mixture of ethyl iodide and n-propyl iodide is subjected to Wurtz reaction. The hydrocarbon that will not be formed is-
 (1) n-Butane (2) Propyne
 (3) n-Pentane (4) n-Hexane
18. Amongst the following hydrocarbons, the one having lowest boiling point is-
 (1) n-Hexane (2) n-Pentane
 (3) Isopentane (4) Neopentane
19. Which of the following on reaction with RMgX does not give RH ?
 (1) Ethylamine (2) Dimethylamine
 (3) Trimethylamine (4) Phenol
20. How many different alkene including geometrical isomer on catalytic hydrogenation produces 2-methyl pentane.
 (1) Three (2) Two
 (3) Five (4) One
21. $\text{BrCH}_2 - \text{CH}_2 - \text{CH}_2\text{Br}$ reacts with Na in the presence of ether at 100°C to produce -
 (1) $\text{BrCH}_2 - \text{CH} = \text{CH}_2$
 (2) $\text{CH}_2 = \text{C} = \text{CH}_2$
 (3) 
 (4) All of these
22. What is the slow, rate-determining step, in the acid-catalyzed dehydration of 2-methyl-2-propanol?
 $(\text{CH}_3)_3\text{COH} \xrightarrow{\text{H}_2\text{SO}_4}$
 (1) Protonation of the alcohol to form an oxonium ion
 (2) Loss of water from the oxonium ion to form a carbocation.
 (3) Loss of a β -hydrogen from the carbocation to form an alkene.
 (4) The simultaneous loss of a β -hydrogen and water from the oxonium ion.
23. Propene when heated with chlorine at about 500°C forms majorly :
 (1) $\text{CH}_2\text{Cl}.\text{CH} = \text{CH}_2$
 (2) $\text{CH}_3\text{CHCl}.\text{CH}_2\text{Cl}$
 (3) $\text{CH}_2\text{Cl}.\text{CHClCH}_2\text{Cl}$
 (4) All the three

24. When two alkenes maybe formed by dehydrohalogenation of an alkyl halide, the alkene which is more substituted is the major or preferred product. This generalization is known as :

(1) Markownikoff's rule
 (2) Anti-Markownikoff's rule
 (3) Saytzeff rule
 (4) Hofmann's rule

25. Which is correct statement for P & Q



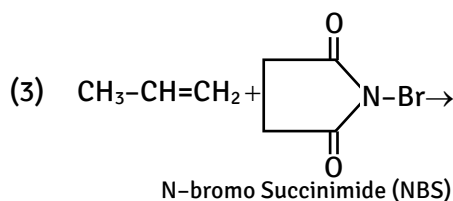
- (1) P is product of ionic reaction & Q is product of radical reaction.
 (2) P & Q both are product of ionic reaction.
 (3) P & Q both are product of free radical reaction
 (4) P is free radical reaction & Q is ionic reaction.

26. The compound formed when 2-butene is treated with hot alk KMnO_4 is -

(1) Acetaldehyde (2) Acetic acid
 (3) $\text{CH}_2\text{OH} \cdot \text{CH}_2\text{OH}$ (4) $\text{CH}_3\text{CH}_2\text{COCH}_3$

27. Which of the following do not give Allyl halide ?

(1) $\text{CH}_2 = \text{CH} - \text{CH}_3 + \text{SO}_2\text{Cl}_2 \xrightarrow[\text{traces of peroxide}]{475\text{K}}$
 (2) $\text{CH}_2 = \text{CH} - \text{CH}_3 + \text{Cl}_2 \xrightarrow{400^\circ\text{C} - 500^\circ\text{C}}$

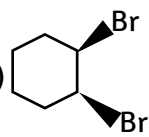
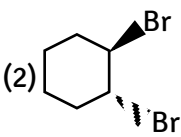


(4) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}}$

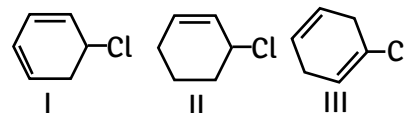
28. The simplest branched chain alkene cannot be prepared by heating which of the following with ethanolic potassium hydroxide -

(1) t-Butyl halide
 (2) s-Butyl halide
 (3) Isobutyl halide
 (4) 2-Halo-2-methylpropane

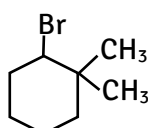
29.  + $\text{Br}_2 \rightarrow \text{A}$; A will have configuration

(1)  (2) 
 (3) both true (4) none is true

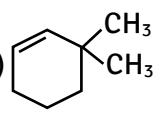
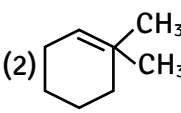
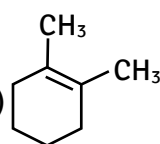
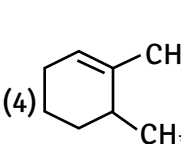
30. Which is most easily dehydrohalogenated -



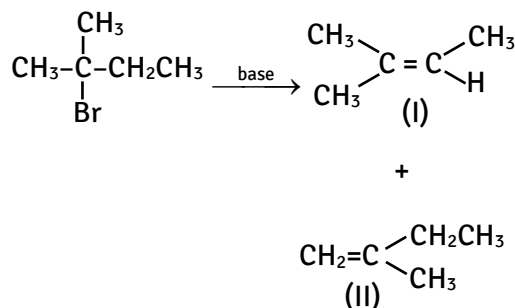
(1) I
 (2) II
 (3) III
 (4) all with same rate

31.  $\xrightarrow{\text{alc. KOH}}$ Product.

The product can be—

(1)  (2) 
 (3)  (4) 

32. Consider the following reaction

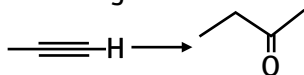


Which of the following base will give the best yield of the alkene II as the major product -

(1) CH_3O^-
 (2) $\text{C}_2\text{H}_5\text{O}^-$
 (3) $(\text{CH}_3)_3\text{CO}^-$
 (4) $(\text{C}_2\text{H}_5)_3\text{CO}^-$

33. HCl does not form Anti-Markownikoff's product with propene, because -
 (1) HCl is a polar covalent molecule
 (2) Dipole-Dipole attraction exists between HCl molecules
 (3) Formation of $\text{Cl}^\cdot$ is energetically unfavourable
 (4) Chlorine has the highest electron affinity
34. Choose the correct statement -
 (1) Addition of HCl to vinyl chloride yield vicinal dichloride
 (2) There is difference between peroxide effect and Kharash effect
 (3) Peroxide is a ready source of free radical in the Markownikoff's reaction
 (4) Propene with HCl in presence of peroxide gives isopropyl chloride
35. Diborane reacts with terminal alkenes to form trialkylboranes. These react with alkaline hydrogen peroxide to form -
 (1) Secondary alcohols
 (2) Tertiary alcohols.
 (3) Isobutyl alcohol
 (4) Primary alcohols
36. 2-Butyne and 1-Butyne show resemblance in all except -
 (1) Both decolourise alkaline KMnO_4
 (2) Both turn bromine water colourless
 (3) Both undergo addition reaction
 (4) Both form white precipitate with Tollen's reagent
37. Consider the reaction :
 $\text{CH} \equiv \text{CH}(\text{g}) + 2\text{HCl}(\text{g}) \rightarrow \text{product}.$
 The product of this reaction is :
 (1) $\text{CH}_3\text{CHCl}_2(\text{g})$ (2) $\text{CH}_2 = \text{CHCl}(\text{g})$
 (3) $\text{CHCl} = \text{CHCl}(\text{g})$ (4) $\text{CH}_2\text{ClCH}_2\text{Cl}(\text{g})$
38. The end product of the following sequence is :
 $\text{CaC}_2 \xrightarrow{\text{H}_2\text{O}} \text{A} \xrightarrow[\text{HgSO}_4]{\text{H}_2\text{SO}_4} \text{B}$
 (1) ethanol
 (2) ethyl hydrogen sulphate
 (3) acetaldehyde
 (4) ethylene glycol
39. The catalyst required for the reaction :
 $\text{HC} \equiv \text{CH} + \text{dil. H}_2\text{SO}_4 \xrightarrow{\text{Catalyst}} \text{CH}_3\text{CHO}$ is
 (1) HgSO_4 (2) Pd
 (3) Pt (4) AlCl_3
40. Which of the following statements is correct?
 (1) Alkynes are more reactive than alkenes towards halogen additions
 (2) Alkynes are less reactive than alkenes towards halogen additions.
 (3) Both alkynes and alkenes are equally reactive towards halogen additions
 (4) Primary vinylic cations ($\text{RC} = \text{CH}^+$) is more stable than secondary vinylic cation ($\text{RC}^+ = \text{CH}_2$)
41. A gas which reacts with cold aqueous alkaline KMnO_4 solution but does not give precipitate with ammoniacal Cu_2Cl_2 solution is:
 (1) ethylene (2) methane
 (3) ethane (4) acetylene
42. Addition of 2 mol of HCl to 1-butyne would yield :
 (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHCl}_2$
 (2) $\text{CH}_3\text{CH}_2\text{CCl}_2\text{CH}_3$
 (3) $\text{CH}_3\text{CH}_2\text{CHClCH}_2\text{Cl}$
 (4) $\text{CH}_3\text{CH}_2\text{CH} = \text{CHCl}$
43. Acetylene gives-
 (1) White ppt with AgNO_3 and red ppt with Cu_2Cl_2
 (2) White ppt with Cu_2Cl_2 and red ppt with AgNO_3
 (3) White ppt with both
 (4) Red ppt with both
44. Which of the following reductions of an alkyne is not correct :
 (1) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_3 \xrightarrow[\text{Pt}]{2\text{H}_2} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 (2) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_3 \xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2} \text{CH}_3\text{CH}_2\text{CH} = \text{CHCH}_3$
 (3) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_3 \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{CH}_3\text{CH}_2\text{CH} = \text{CHCH}_3$
 (4) All of the above are correct

45. Which reagents would be best to perform the following reaction?



- (1) 1) BH_3 , 2) NaOH , H_2O_2
 (2) HgSO_4 , H_2O , H_2SO_4
 (3) 1) NaNH_2 , 2) CH_3I , 3) HgSO_4 , H_2SO_4
 (4) 1) $(\text{CH}_3\text{COO})_2\text{Hg}$, H_2O 2) NaBH_4

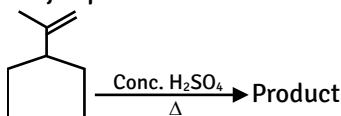
46. Indicate the expected structure of the organic product when ethyl magnesium bromide is treated with heavy water (D_2O) -

- (1) $\text{C}_2\text{H}_5 - \text{C}_2\text{H}_5$ (2) $\text{C}_2\text{H}_5\text{OD}$
 (3) C_2H_6 (4) $\text{C}_2\text{H}_5\text{D}$

47. $\text{C}_3\text{H}_8 + \text{Cl}_2 \xrightarrow{\text{Light}} \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$ is an example of which of the following types of reactions -

- (1) Substitution (2) Elimination
 (3) Addition (4) Rearrangement

48. Major product of the reaction is



- (1) (2)
 (3) (4)

49. $\xrightarrow[(2) \text{H}_2\text{O}/\text{Zn}]{(1) \text{O}_3} \text{X} \xrightarrow[\text{HCl}]{\text{Zn-Hg}} \text{Y}$; Y is

- (1) (2)
 (3) (4)

50. Consider the following reaction sequence.
 $\text{CH}_3\text{C}\equiv\text{CH} \xrightarrow{\text{CH}_3\text{MgBr}} \text{I} \xrightarrow{\text{CH}_3\text{CH}_2\text{Br}} \text{II}$
 The final product (II) formed is :

- (1) $\text{CH}_3\text{C}\equiv\text{CCH}_3$
 (2) $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$
 (3) $\text{CH}_3\text{C}\equiv\text{CMgBr}$
 (4) $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CHCH}_2\text{CH}_3$

51. Which one of these is incorrect for benzene?
 (1) It forms only one type of monosubstituted product

- (2) There are three C-C single bonds and three C-C double bonds
 (3) The heat of hydrogenation of benzene is less than the theoretical value
 (4) The bond-angle between c-c-c bond is 120°

52. $\text{A} \xrightarrow{\text{H}_2\text{O}} \text{B} \xrightarrow{\text{Cl}_2} \text{C} \xrightarrow[\text{Ether}]{\text{Na}} \text{C}_2\text{H}_6$

In the above reaction sequence A and B respectively are-

- (1) CaC_2 , C_2H_2 (2) Al_4C_3 , C_2H_6
 (3) Al_4C_3 , CH_4 (4) CaC_2 , CH_4

53. $\xrightarrow{\text{H}_2/\text{Ni}} \text{A}$
 A is

- (1) $\text{CH}_3 - (\text{CH}_2)_4 - \text{CH}_3$ (2)
 (3) (4)

54. $\text{MeCH}_2\text{C}\equiv\text{CH} \xrightarrow{\text{NH}_3/\text{NaNH}_2} \text{A} \xrightarrow{\text{EtBr}} \text{B}$
 , A and B respectively are-

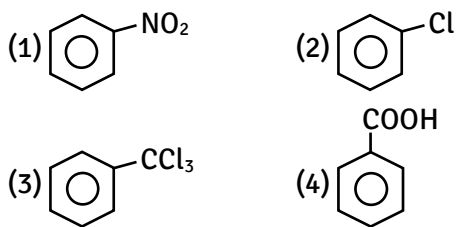
- (1) $\text{MeCH}_2\text{C}\equiv\text{C} - \text{Na}$, $\text{MeCH}_2\text{C}\equiv\text{C} - \text{Et}$
 (2) $\text{MeCH}_2\text{CH}=\text{CH}_2$, $\text{MeCH}_2 - \text{CH}(\text{Et}) - \text{CH}_3$
 (3) $\text{MeCH}_2\text{CH}=\text{CHNH}_2$, $\text{MeCH}_2\text{CH}=\text{CH} - \text{NHBr}$
 (4) $\text{MeCH}_2\text{C}\equiv\text{C} - \text{NH}_2$, $\text{MeC}\equiv\text{C} - \text{NH} - \text{Br}$

55. Hydration of in

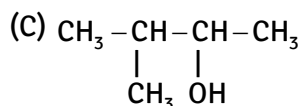
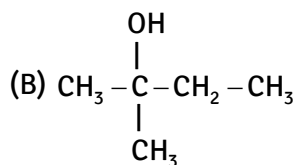
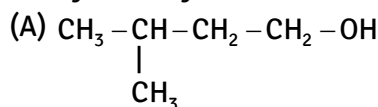
Presence of dil. $\text{H}_2\text{SO}_4/\text{HgSO}_4$ gives-

- (1) COCH_3
 (2) CH_2CHO
 (3) COCH_3
 (4) CH_2CHO

56. In which of the following compound the electrophile attack on O^- and P^- positions

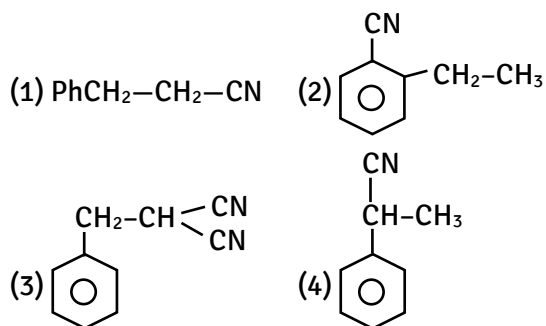


57. Arrange the following alkanols A, B and C in order of their reactivity towards acid catalysed dehydration:-

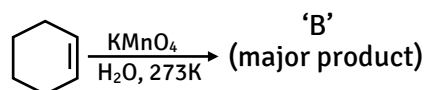
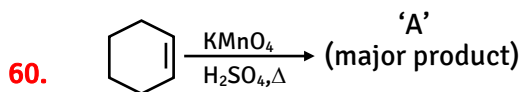
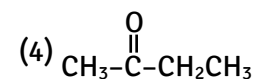
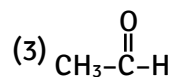
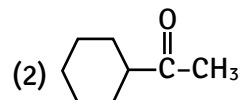
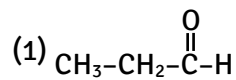


- (1) $A > B > C$ (2) $B > A > C$
 (3) $B > C > A$ (4) $C > B > A$

58. $PhC_2H_5 \xrightarrow{Br_2/h\nu} A \xrightarrow{KCN} B$; B is



59. Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of $HgSO_4$ and H_2SO_4 ?



For above chemical reactions, identify the correct statement from the following:

- (1) Compound 'A' is dicarboxylic acid and compound 'B' is diol
 (2) Compound 'A' is diol and compound 'B' is dicarboxylic acid
 (3) Both compound 'A' and compound 'B' are diols
 (4) Both compound 'A' and compound 'B' are dicarboxylic acids

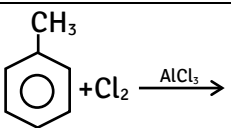
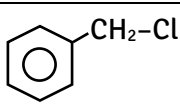
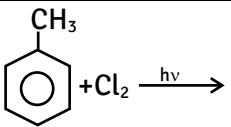
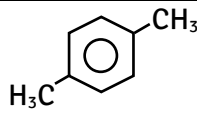
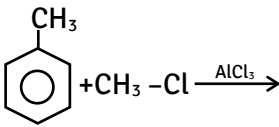
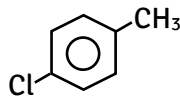
Exercise 3

1. Match the column :

	Column I (Reactants)		Column II (Product)
(A)	$\text{CaC}_2 + \text{H}_2\text{O}$	(p)	CH_3-CH_3
(B)	$\text{CH}_3-\text{CH}_2\text{MgBr} + \text{H}_2\text{O}$	(q)	H_2
(C)	$\text{HC} \equiv \text{CH} + \text{Na}$	(r)	CH_4
(D)	$\text{CH}_3\text{COOH} + \text{NaOH} + \text{CaO}$	(s)	C_2H_2

(1) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$ (2) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{s}); \text{D} \rightarrow (\text{r})$ (3) $\text{A} \rightarrow (\text{s}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{q}); \text{D} \rightarrow (\text{r})$ (4) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$

2. Match the column :

	Column I (Reaction)		Column II (Product)
(A)		(p)	
(B)		(q)	
(C)		(r)	

(1) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r})$ (2) $\text{A} \rightarrow (\text{r}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{q})$ (3) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{r}); \text{C} \rightarrow (\text{p})$ (4) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{r})$

3. Match the column-I with column-II

	Column I		Column II
(A)	Lindlar's catalyst	(p)	$\text{Na} / \text{liq. NH}_3$
(B)	Birch catalyst	(q)	$\text{H}_2 / \text{Pd} - \text{CaCO}_3$
(C)	Baeyer's Reagent	(r)	$\text{HBr} / (\text{PhCO})_2\text{O}_2$
(D)	Kharasch effect	(s)	$\text{Dil. KMnO}_4 / 273\text{K}$

(1) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$ (2) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{s}); \text{D} \rightarrow (\text{r})$ (3) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{s}); \text{D} \rightarrow (\text{r})$ (4) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$

4. Match the column-I with column-II

	Column I (Reaction)		Column II (Product)
(A)	$2\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{dry ether}]{2\text{Na}}$	(p)	
(B)	$\text{CH}_3\text{COO}^\ominus\text{Na}^+ \xrightarrow[\Delta]{\text{CaO} + \text{NaOH}}$	(q)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
(C)	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3 \xrightarrow[10-20 \text{ atm}]{\text{V}_2\text{O}_5, 773\text{K}}$	(r)	$2\text{CH}_3\text{OH}$
(D)	$2\text{CH}_4 + \text{O}_2 \xrightarrow[100 \text{ atm}]{\text{Cu}/523\text{K}}$	(s)	CH_4

(1) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$ (2) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{s}); \text{C} \rightarrow (\text{p}); \text{D} \rightarrow (\text{r})$ (3) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{s}); \text{D} \rightarrow (\text{r})$ (4) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$

5. Match the column-I with Column II

	Column I (Reaction)		Column II (Product)
(A)	$\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mo}_2\text{O}_3}$	(p)	$(\text{CH}_3)_3\text{C}-\text{OH}$
(B)	$2\text{CH}_3\text{CH}_3 + 3\text{O}_2 \xrightarrow{(\text{CH}_3\text{COO})_2\text{Mn}}$	(q)	$\text{HCHO} + \text{H}_2\text{O}$
(C)	$(\text{CH}_3)_3\text{CH} \xrightarrow[\text{Oxidation}]{\text{KMnO}_4}$	(r)	$\text{CO} + 3\text{H}_2$
(D)	$\text{CH}_4 + \text{H}_2\text{O} \xrightarrow[\Delta]{\text{Ni}}$	(s)	$2\text{CH}_3\text{COOH} + 2\text{H}_2\text{O}$

(1) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{p}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$ (2) $\text{A} \rightarrow (\text{r}); \text{B} \rightarrow (\text{s}); \text{C} \rightarrow (\text{p}); \text{D} \rightarrow (\text{r})$ (3) $\text{A} \rightarrow (\text{q}); \text{B} \rightarrow (\text{s}); \text{C} \rightarrow (\text{p}); \text{D} \rightarrow (\text{r})$ (4) $\text{A} \rightarrow (\text{p}); \text{B} \rightarrow (\text{q}); \text{C} \rightarrow (\text{r}); \text{D} \rightarrow (\text{s})$

6. Match the column-I with column-II

	Column I (Reaction)		Column II (Type of reaction)
(A)	$\text{HC} \equiv \text{CH} \xrightarrow[\text{Pd}]{\text{H}_2}$	(p)	Pyrolysis
(B)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow[\text{AlCl}_3+\text{HCl}]{\text{Anhydrous}}$	(q)	Catalytic hydrogenation
(C)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow{773\text{K}}$	(r)	Free radical substitution
(D)	$\text{CH}_3\text{CH}_3 \xrightarrow[\text{h}\nu]{\text{Cl}_2}$	(s)	Isomerization

- (1) A → (q); B → (s); C → (p); D → (r)
 (2) A → (p); B → (q); C → (s); D → (r)
 (3) A → (q); B → (p); C → (s); D → (r)
 (4) A → (p); B → (q); C → (r); D → (s)

7. Match the column-I with column-II

	Column I		Column II
(A)	$\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{H}_2/\text{Pd}}$	(p)	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$
(B)	$\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{Na/liq. NH}_3}$	(q)	$\text{CH}_3 - \text{C} \equiv \text{C}^\ominus \text{Na}^\oplus$
(C)	$\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow[\text{Quinoline}]{\text{H}_2/\text{Pd}-\text{C}}$	(r)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
(D)	$\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow{\text{Na/liq. NH}_3}$	(s)	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

- (1) A → (s); B → (r); C → (q); D → (p)
 (2) A → (p); B → (s); C → (r); D → (q)
 (3) A → (r); B → (s); C → (p); D → (q)
 (4) A → (p); B → (q); C → (r); D → (s)

8. Match the column-I with column-II

	Column I (Compound)		Column II (Boiling point)
(A)	2-Methyl propane	(p)	309.1 K
(B)	Pentane	(q)	282.5 K
(C)	2-Methyl butane	(r)	261 K
(D)	2,2-Dimethylpropane	(s)	300.9 K

- (1) A → (r); B → (p); C → (s); D → (q)
 (2) A → (q); B → (r); C → (s); D → (p)
 (3) A → (r); B → (s); C → (q); D → (p)
 (4) A → (s); B → (r); C → (q); D → (p)

9. Match the column-I with column-II

	Column I		Column II
(A)	$\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{H}_2\text{O}]{\text{H}^\oplus}$	(p)	Catalytic Hydrogenation
(B)	$\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{R}_2\text{O}_2]{\text{HBr}}$	(q)	Acidic Hydration
(C)	$\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{CCl}_4]{\text{Br}_2}$	(r)	Halogenation
(D)	$\text{CH}_2 = \text{CH}_2 \xrightarrow[\text{Pd}]{\text{H}_2} \text{CH}_3 - \text{CH}_3$	(s)	Hydrohalogenation

- (1) A → (s); B → (r); C → (q); D → (p)
 (2) A → (q); B → (s); C → (r); D → (p)
 (3) A → (r); B → (s); C → (p); D → (q)
 (4) A → (p); B → (q); C → (r); D → (s)

10. Match List-I with List-II and select the correct answer.

List-I (Reactions)

- (A) $\text{CH}_3 - \text{CH} = \text{CH}_2 \rightarrow \text{CH}_3 - \text{CHBr} - \text{CH}_3$
 (B) $\text{CH}_3 - \text{CH} = \text{CH}_2 \rightarrow \text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Br}$
 (C) $\text{CH}_3 - \text{CH} = \text{CH}_2 \rightarrow \text{BrCH}_2 - \text{CH} = \text{CH}_2$
 (D) $\text{CH}_3 - \text{CH} = \text{CH}_2 \rightarrow \text{CH}_3 - \text{CHBr} - \text{CH}_2\text{Br}$

List-II (Reagents)

- (P) HBr
 (Q) Br_2/CCl_4
 (R) HBr/Peroxide
 (S) NBS
 (1) A-P, B-R, C-S, D-Q
 (2) A-R, B-P, C-Q, D-S
 (3) A-P, B-S, C-R, D-Q
 (4) A-Q, B-P, C-S, D-R

- 11. Statement-I :** Alkenes decolourise KMnO_4 solution
Statement-II: Alkenes are unsaturated hydrocarbons.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and Statement-II are correct,
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.
- 12. Statement-I :** Ozonolysis of alkenes involves the addition of ozone molecule to alkene to form ozonide and then cleavage of the ozonide by $\text{Zn-H}_2\text{O}$ to smaller molecules.
Statement-II : Ozonolysis reaction is highly useful in detecting the position of double bond in alkenes or in other unsaturated compounds. In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II both are correct.
(2) Statement-I and II both are incorrect.
(3) Statement-I is wrong but Statement-II is correct.
(4) Statement-I is correct Statement-II is wrong.
- 13. Statement-I:** Cis-But-2-ene has higher dipole moment than trans-but-2-ene.
Statement-II: $\text{R}-\text{C}\equiv\text{C}-\text{R}'$ reacts with H_2 in presence of Na/liq. NH_3 to form mainly cis-alkene.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct
(3) Statement-I and Statement-II are correct.
(4) Both Statement-I and II are incorrect.
- 14. Statement-I :** Dihydrogen gas adds to alkenes and alkynes in the presence of finely divided catalyst like platinum, palladium or Nickel to form alkanes.
Statement-II : Metals adsorb dihydrogen gas on their surfaces. and activate the hydrogen hydrogen bond. In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statements-I and II are correct.
(2) Both Statement-I and II are incorrect
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.
- 15. Statement-I :** Sodamide reacts with Ethyne to form sodium acetylide with liberation of hydrogen gas.
Statement-II : Acetylene is more acidic than ethene . In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statements-I and II both are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Statement-I and II both are correct
- 16. Statement-I:** Ethyne is formed from 1,2-dibromoethane by using alcoholic KOH followed by NaNH_2 .
Statement-II: 1,2- dibromo ethane on reaction with Na in dry ether gives ethene. In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and Statement-II both are correct.
(2) Statement-I and II both are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

17. Statement-I : Alkyl halides on treatment with sodium metal in moist ether gives higher alkane.

Statement-II : Wurtz reaction is used for the preparation of higher alkanes containing even number of carbon atoms.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I and Statement-II both are correct.
- (2) Statement-I and II both are incorrect.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

18. Given below are two statements

Statement-I : Ozonolysis of o-xylene (1,2-dimethyl benzene) gives three products. The result supports kekule structure of benzene.

Statement-II : Benzene have $6\pi e^-$ in cycle and have cyclic delocalisation. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Statement-I and Statement-II are incorrect.
- (4) Both Statement-I and Statement-II are correct.

19. Assertion (A): 1-butene on reaction with HBr in the presence of a peroxide, produces 1-bromobutane as main product

Reason (R): It involves the formation of more stable free radical.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

20. Assertion (A): Propene reacts with HI in the presence of peroxide to give 1-iodopropane.

Reason (R): 1° free radical is less stable than 2° free radical.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

21. Assertion (A): Nitration of benzene requires the use of nitrating mixture.

Reason (R): The mixture of concentrated sulphuric acid and concentrated nitric acid produces the NO_2^+ .

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

22. Assertion: 2, 3-dimethyl but-2-ene decolorizes Br_2 water

Reason: 2, 3-dimethyl but-2-ene is an unsaturated compound

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

23. Assertion: Iodination of alkane is carried out in presence of iodic acid.

Reason: Iodic acid is an oxidizing agent.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

24. Assertion: benzene does not decolourise Br_2 -water.

Reason: Benzene is stabilized by aromaticity and normally does not under addition reaction.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

25. Assertion: Phenol is more reactive than benzene towards electrophilic substitution.

Reason: In case of phenol, the intermediate carbocation is more resonance stabilised.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

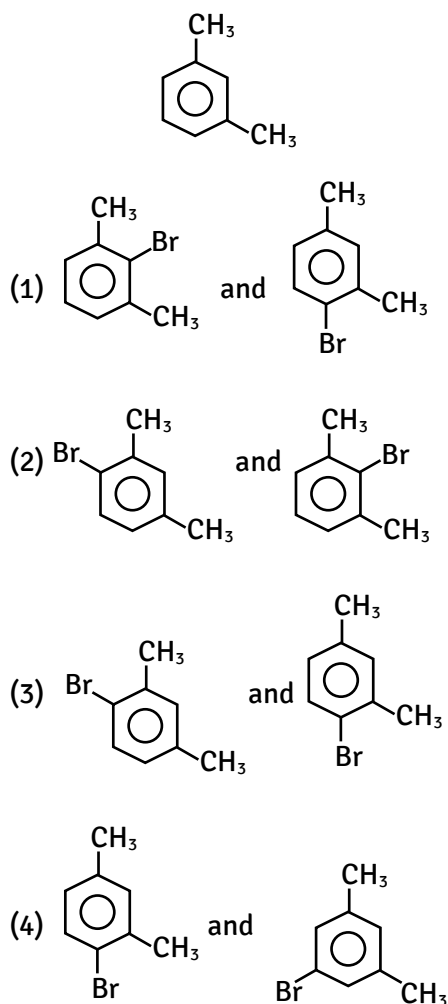
Exercise 4 (Previous Year's Questions)

1. Which of the following compounds will not undergo Friedel-Crafts' reaction:

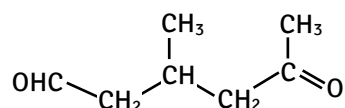
[NEET-UG - 2013]

- (1) Toluene
- (2) Cumene
- (3) Xylene
- (4) Nitrobenzene

2. What products are formed when the following compound is treated with Br_2 in the presence of FeBr_3 ? [AIPMT - 2014]

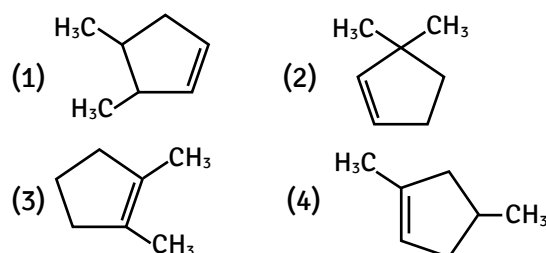


3. A single compound of the structure :-

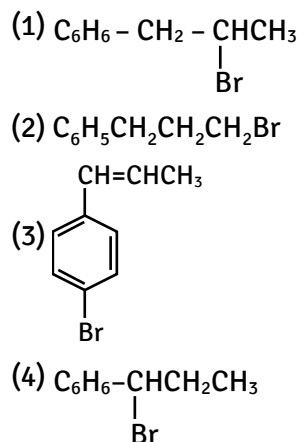


is obtainable from ozonolysis of which of the following cyclic compounds?

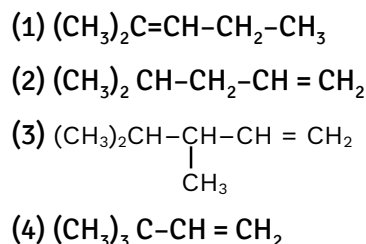
[AIPMT - 2015]



4. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3$ with HBr produces: [AIPMT - 2015]



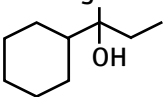
5. 2,3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid? [Re-AIPMT - 2015]



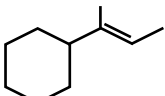
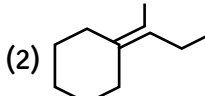
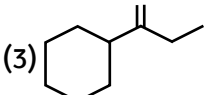
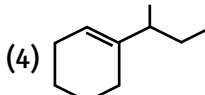
6. The oxidation of benzene by V_2O_5 in the presence of air produces :

[Re-AIPMT - 2015]

- (1) benzoic acid
- (2) benzaldehyde
- (3) benzoic anhydride
- (4) maleic anhydride

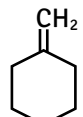
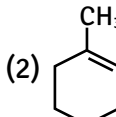
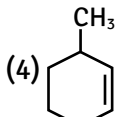
7. Which of the following is not the product of dehydration of  ?

[Re-AIPMT - 2015]

- (1) 
- (2) 
- (3) 
- (4) 

8. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1- methylcyclohexane. The possible alkene is:

[Re-AIPMT - 2015]

- (1) 
- (2) 
- (3) (1) and (2) Both
- (4) 

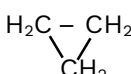
9. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3 . If a large amount of $KHSO_4$ is added to the mixture, the rate of nitration will be:

[NEET-I - 2016]

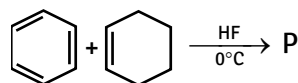
- (1) Faster
- (2) Slower
- (3) Unchanged
- (4) Doubled

10. Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only by elimination reaction?

[NEET-I-2016]

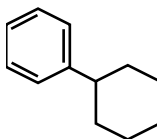
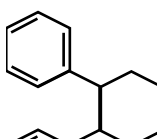
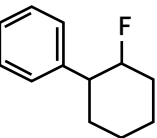
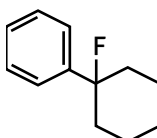
- (1) $H_2C = C = O$
- (2) $H_3C-CH_2-CH_2Br$
- (3) 
- (4) $H_3C-CH_2-CH_2OH$

11. In the given reaction



the product P is :

[NEET - I - 2016]

- (1) 
- (2) 
- (3) 
- (4) 

12. The compound that will react most readily with gaseous bromine has the formula

[NEET - I - 2016]

- (1) C_4H_{10}
- (2) C_2H_4
- (3) C_3H_6
- (4) C_2H_2

13. Which of the following can be used as the halide component for Friedel-Crafts reaction?

[NEET - I - 2016]

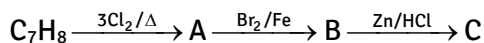
- (1) Chloroethene
- (2) Isopropyl chloride
- (3) Chlorobenzene
- (4) Bromobenzene

14. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is

[NEET-2018]

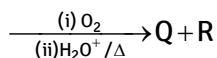
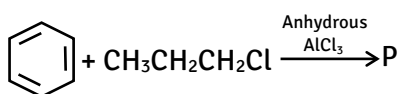
- (1) $CH \equiv CH$
- (2) $CH_2 = CH_2$
- (3) CH_3-CH_3
- (4) CH_4

15. The compound C_7H_8 undergoes the following reaction: [NEET-2018]



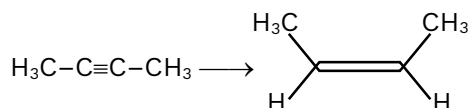
The product 'C' is

- (1) m-bromotoluene
(2) o-bromotoluene
(3) 3-bromo-2, 4, 6-trichlorotoluene
(4) p-bromotoluene
16. Identify the major products P, Q and R in this following sequence of reactions: [NEET-2018]



	P	Q	R
(1)			$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
(2)			
(3)			$\text{CH}_3\text{CH}_2\text{-OH}$
(4)			$\text{CH}_3\text{CO-CH}_3$

17. The most suitable reagent for the following conversion is: [NEET-2019]

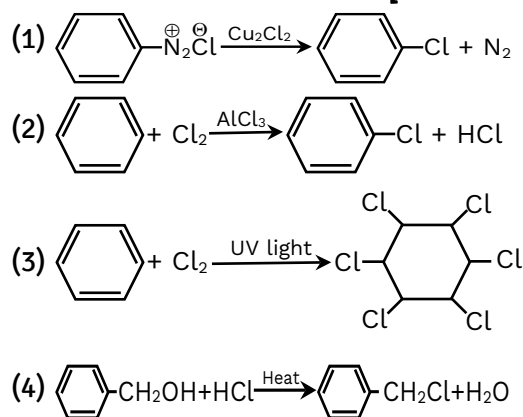


- (1) Na/liquid NH_3
(2) H_2 , Pd/C, quinoline
(3) Zn/HCl
(4) $\text{Hg}^{2+}/\text{H}^+$, H_2O

18. An alkene "A" on reaction with O_3 and $\text{Zn-H}_2\text{O}$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is: [NEET-2019]

- (1) $\text{Cl-CH}_2\text{-CH}_2\text{-CH(CH}_3)_2$
(2) $\text{H}_3\text{C-CH}_2\text{-CH(CH}_3\text{)-CH}_2\text{Cl}$
(3) $\text{H}_3\text{C-CH}_2\text{-C(CH}_3)_2\text{-Cl}$
(4) $\text{H}_3\text{C-CH(Cl)-CH(CH}_3)_2$

19. Among the following the reaction that proceeds through an electrophilic substitution is [NEET-2019]



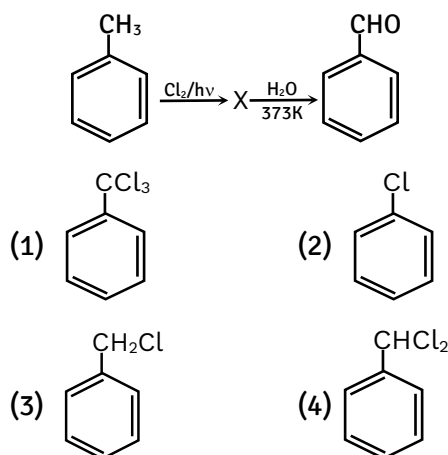
20. The alkane that gives only one mono-chloro product on chlorination with Cl_2 in presence of diffused sunlight is- [NEET-(Odisha)2019]

- (1) 2, 2-dimethylbutane
(2) neopentane
(3) n-pentane
(4) Isopentane

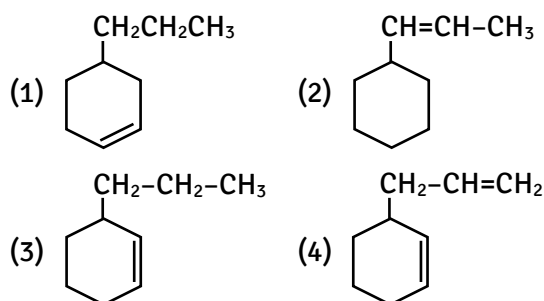
21. Which of the following alkane cannot be made in good yield by Wurtz reaction? [NEET-2020]

- (1) n-Butane
(2) n-Hexane
(3) 2,3-Dimethylbutane
(4) n-Heptane

22. Identify compound X in the following sequence of reactions: [NEET-2020]



23. An alkene on ozonolysis gives methanal as one of the product. Its structure is [NEET-2020]



24. $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow[\text{(ii) H}_2\text{O}_2/\text{OH}^-]{\text{(i) B}_2\text{H}_6} \text{Z}$
What is Z? [NEET-2020(Covid-19)]

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (2) $(\text{CH}_3)_2\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2$
 (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
 (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

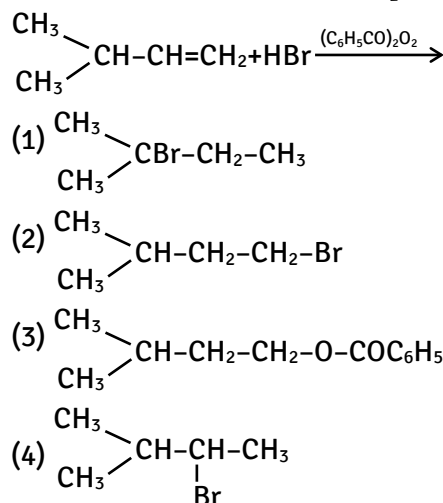
25. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on? [NEET-2021]

- (1) Saytzeff's Rule
 (2) Hund's Rule
 (3) Hofmann Rule
 (4) Huckel's Rule

26. $\text{CH}_3\text{CH}_2\text{COO}^-\text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, +?}}$
 $\text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$.
 Consider the above reaction and identify the missing reagent/chemical. [NEET-2021]

- (1) B_2H_6 (2) Red Phosphorus
 (3) CaO (4) DIBAL-H

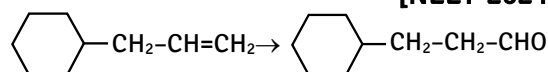
27. The major product of the following chemical reaction is: [NEET-2021]



28. Compound X on reaction with O_3 followed by $\text{Zn}/\text{H}_2\text{O}$ gives formaldehyde and 2-methyl propanal as products. The compound X is: [NEET-2022]

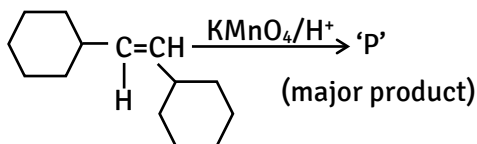
- (1) 2-Methylbut-1-ene
 (2) 2-Methylbut-2-ene
 (3) Pent-2-ene
 (4) 3-Methylbut-1-ene

29. Identify the correct reagents that would bring about the following transformation. [NEET-2024]



- (1) (i) BH_3 (ii) $\text{H}_2\text{O}_2/\text{OH}^-$ (iii) alk. KMnO_4 (iv) H_3O^+
 (2) (i) $\text{H}_2\text{O}/\text{H}^+$ (ii) PCC
 (3) (i) $\text{H}_2\text{O}/\text{H}^+$ (ii) CrO_3
 (4) (i) BH_3 (ii) $\text{H}_2\text{O}_2/\text{OH}^-$ (iii) PCC

30. For the given reaction : [NEET-2024]



'P' is :

- (1)
- (2)
- (3)
- (4)

31. Given below are two statements:

[NEET-2024]

Statement I : The boiling point of three isomeric pentanes follows the order n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below:

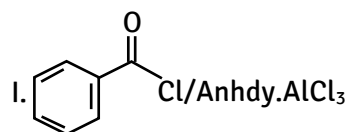
- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

32. Match List I With List II. [NEET-2024]

List-I (Reaction)

- A.
- B.
- C.
- D.

List-II reagents/Condition



II. CrO_3

III. $\text{KMnO}_4/\text{KOH}, \Delta$

IV. (i) O_3

(ii) $\text{Zn}-\text{H}_2\text{O}$

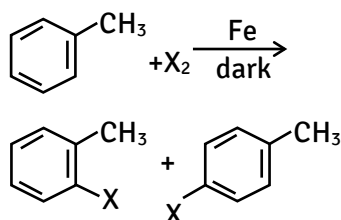
Choose the correct answer from the options given below.

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-II, D-IV

33. Baeyer's reagent is: [RE-NEET-2024]

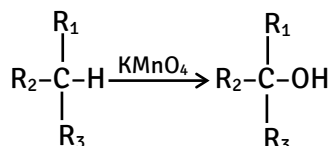
- (1) acidic potassium permanganate solution
- (2) acidic potassium dichromate solution
- (3) cold, dilute, aqueous solution of potassium permanganate
- (4) hot, concentrated solution of potassium permanganate

34. The following reaction method :
[RE-NEET-2024]



Is not suitable for the preparation of the corresponding haloarene products, due to high reactivity of halogen, when X is :

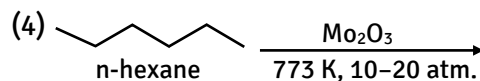
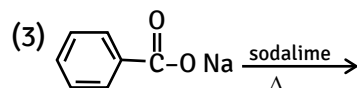
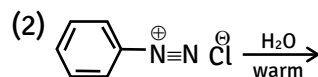
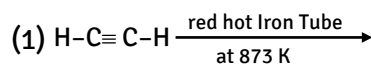
- (1) F (2) I
(3) Cl (4) Br
35. The alkane that can be oxidized to the corresponding alcohol by KMnO_4 as per the equation :
[RE-NEET-2024]



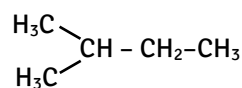
is, when :

- (1) $\text{R}_1 = \text{H}$; $\text{R}_2 = \text{H}$; $\text{R}_3 = \text{H}$
(2) $\text{R}_1 = \text{CH}_3$; $\text{R}_2 = \text{CH}_3$; $\text{R}_3 = \text{CH}_3$
(3) $\text{R}_1 = \text{CH}_3$; $\text{R}_2 = \text{H}$; $\text{R}_3 = \text{H}$
(4) $\text{R}_1 = \text{CH}_3$; $\text{R}_2 = \text{CH}_3$; $\text{R}_3 = \text{H}$

36. Which one of the following reactions does NOT give benzene as the product?
[NEET-2025]



37. How many products (including stereoisomers) are expected from monochlorination of the following compound?
[NEET-2025]



- (1) 5 (2) 6
(3) 2 (4) 3

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	1	4	1	2	3	4	4	4	4	2	3	2	2	2	1	4	4
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	2	3	2	1	4	3	3	1	2	4										

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	3	2	1	4	3	1	4	3	4	4	1	3	2	3	3	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	1	2	2	1	2	3	3	4	3	2										

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	4	2	3	4	4	2	4	4	2	3	4	1	2	2	3	2	1	4

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15					
Ans.	4	1	2	1	2	4	4	2	2	2	2	2	4	2	1					

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	1	1	3	1	1	4	1	3	1	2	3	1	1	4	2	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	2	1	3	1	2	4	2	2	1	1	4	3	4	4	4	1	3	1	2
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	1	3	3	4	1	1	2	2	2	3	2	1	3	2	3	4	1	1

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	2	3	1	3	1	2	1	1	1	1	1	3	1	4	4	1	4
Que.	21	22	23	24	25															
Ans.	1	1	1	1	1															

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	4	4	4	4	4	3	2	1	1	3	2	4	1	4	2	3	2	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37			
Ans.	4	4	4	1	1	3	2	4	4	4	3	1	3	1	2	2	2			

Chapter 5

PURIFICATION AND ANALYSIS OF ORGANIC COMPOUNDS

Chapter Summary

- Methods of Purification of Organic Compounds
- Qualitative analysis of organic compounds
- Quantitative analysis of organic compounds

Methods of Purification of Organic Compounds

The common techniques used for purification are as follows:

- (i) Sublimation
- (ii) Crystallisation
- (iii) Distillation
- (iv) Differential extraction and
- (v) Chromatography

Sublimation

Some solid substance changes from solid to vapour state without passing through liquid state

The purification technique based on the above principle is known as sublimation and is used to separate sublimable compounds from non-sublimable impurities.

Naphthalene and anthraquinone can be separated from non volatile impurities using sublimation method. Camphor and NaCl can also be separated by this method.

Crystallisation

This is one of the most commonly used techniques for the purification of solid organic compounds. It is based on the difference in the solubilities of the compound and the impurities in a suitable solvent.

The impure compound is dissolved in a solvent in which it is sparingly soluble at room temperature but appreciably soluble at higher temperature. The solution is concentrated to get a nearly saturated solution. On cooling the solution, pure compound crystallises out and is removed by filtration. The filtrate (mother liquor) contains impurities and small quantity of the compound. If the compound is highly soluble in one solvent and very little soluble in another solvent, crystallisation

can be satisfactorily carried out in a mixture of these solvents. Impurities, which impart colour to the solution are removed by adsorbing over activated charcoal. Repeated crystallisation becomes necessary for the purification of compounds containing impurities of comparable solubilities.

Distillation :

This important method is used to separate

- (i) Volatile liquids from nonvolatile impurities and
- (ii) The liquids having sufficient difference in their boiling points. Liquids having different boiling points vaporise at different temperatures. The vapours are cooled and the liquids so formed are collected separately. Chloroform (b.p. 334 K) and aniline (b.p. 457 K) are easily separated by the technique of distillation figure.

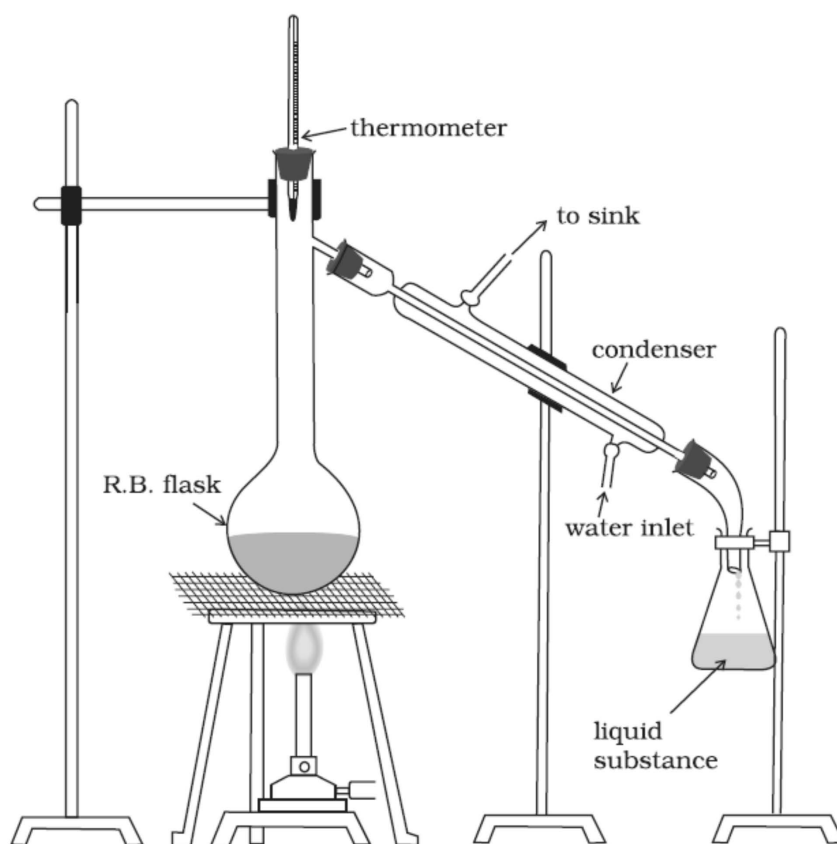


Fig. Simple distillation. The vapours of a substance formed are condensed and the liquid is collected in conical flask.

Fractional Distillation:

If the difference in boiling points of two liquids is not much, simple distillation cannot be used to separate them. The vapours of such liquids are formed within the same temperature range and are condensed simultaneously. The technique of fractional distillation is used in such cases. In this technique, vapours of a liquid mixture are passed through a fractionating column before condensation. The fractionating column is fitted over the mouth of the round bottom flask.

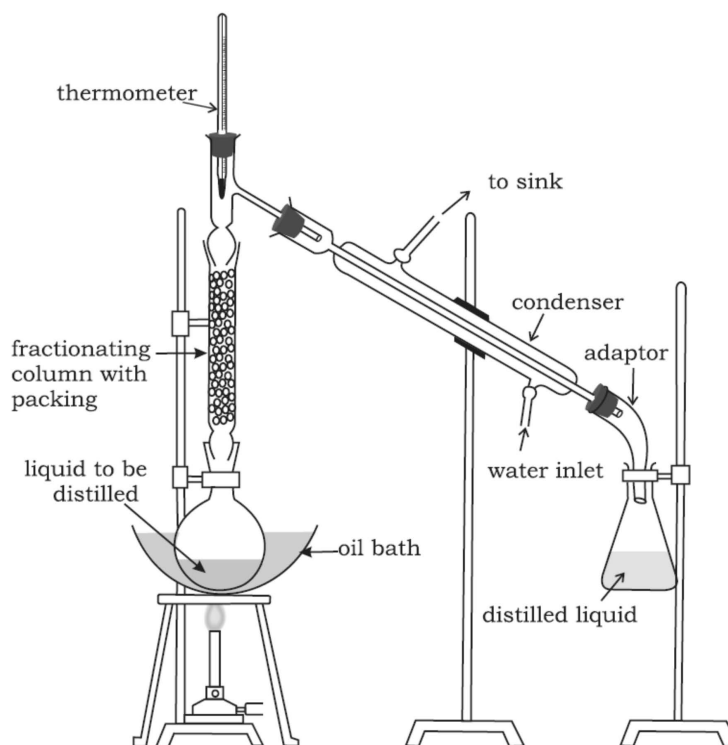


Fig. Fractional distillation. The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.

Vapours of the liquid with higher boiling point condense before the vapours of the liquid with lower boiling point. The vapours rising up in the fractionating column become richer in more volatile component. By the time the vapours reach to the top of the fractionating column, these are rich in the more volatile component. Fractionating columns are available in various sizes and designs as shown in Fig. A fractionating column provides many surfaces for heat exchange between the ascending vapours and the descending condensed liquid. Some of the condensing liquid in the fractionating column obtains heat from the ascending vapours and revaporises. The vapours thus become richer in low boiling component. The vapours of low boiling component ascend to the top of the column. On reaching the top, the vapours become pure in low boiling component and pass through the condenser and the pure liquid is collected in a receiver. After a series of successive distillations, the remaining liquid in the distillation flask gets enriched in high boiling component. Each successive condensation and vaporisation unit in the fractionating column is called a theoretical plate. Commercially, columns with hundreds of plates are available. One of the technological applications of fractional distillation is to separate different fractions of crude oil in petroleum industry.

Ex. Mixture of methanol (B.P. 65°C) and acetone (B.P. 56°C) can be separated using fractional distillation method.

Distillation under reduced pressure: (Vacuum distillation)

This method is used to purify liquids having very high boiling points and those, which decompose at or below their boiling points. Such liquids are made to boil at a temperature lower than their normal boiling points by reducing the pressure on their surface. A liquid boils at a temperature at which its vapour pressure is equal to the external pressure. The pressure is reduced with the help of a water pump or vacuum pump. Glycerol can be separated from spent-lye in soap industry by using this technique.

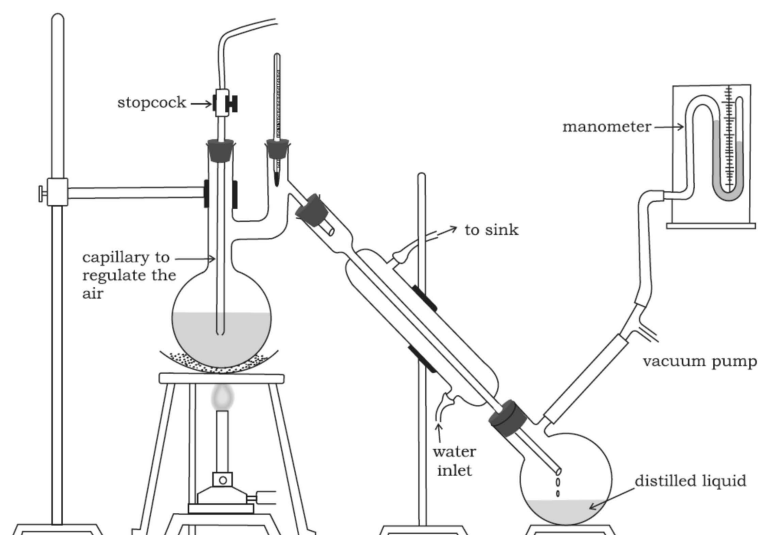


Fig. Distillation under reduced pressure. A liquid boils at a temperature below its vapour pressure by reducing the pressure.

Steam Distillation:

This technique is applied to separate substances which are steam volatile and are immiscible with water. In steam distillation, steam from a steam generator is passed through a heated flask containing the liquid to be distilled. The mixture of steam and the volatile organic compound is condensed and collected. The compound is later separated from water using a separating funnel. In steam distillation, the liquid boils when the sum of vapour pressures due to the organic liquid (p_1) and that due to water (p_2) becomes equal to the atmospheric pressure (p), i.e. $p = p_1 + p_2$. Since p_1 is lower than p_2 the organic liquid vaporises at lower temperature than its boiling point. Thus, if one of the substances in the mixture is water and the other, a water insoluble substance, then the mixture will boil close to but below, 373K. A mixture of water and the substance is obtained which can be separated by using a separating funnel. Aniline is separated by this technique from aniline – water mixture.

Ex. Ortho nitro phenol and para nitro phenol can be separated using steam distillation method.

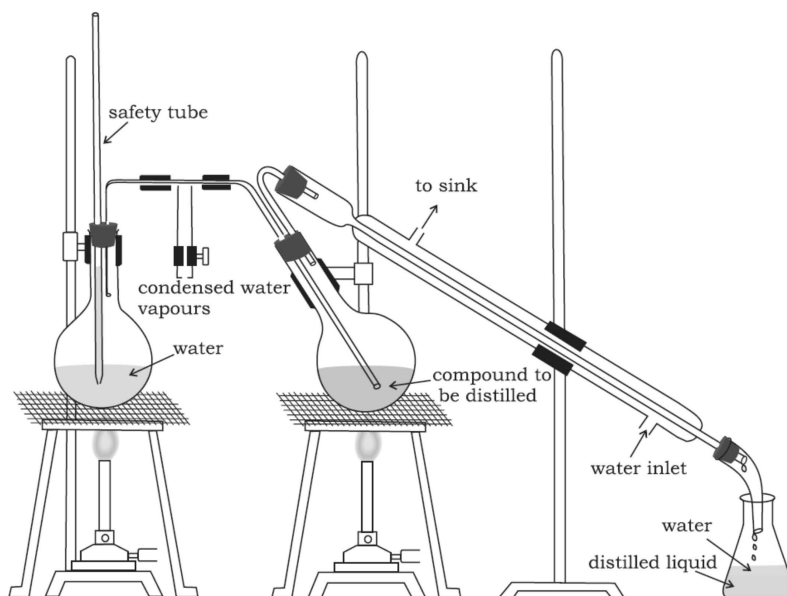


Fig. Steam distillation. Steam volatile component volatilizes, the vapours condense in the condenser and the liquid collects in conical flask.

Differential Extraction :

When an organic compound is present in an aqueous medium, it is separated by shaking it with an organic solvent in which it is more soluble than in water. The organic solvent and the aqueous solution should be immiscible with each other so that they form two distinct layers which can be separated by separatory funnel. The organic solvent is later removed by distillation or by evaporation to get back the compound. Differential extraction is carried out in a separatory funnel. If the organic compound is less soluble in the organic solvent, a very large quantity of solvent would be required to extract even a very small quantity of the compound. The technique of continuous extraction is employed in such cases. In this technique same solvent is repeatedly used for extraction of the compound.

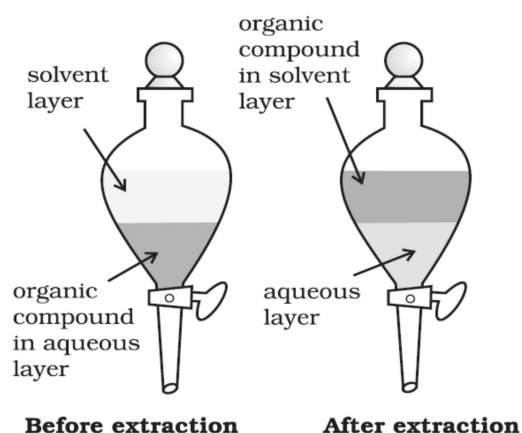


Fig. Differential extraction. Extraction of compound takes place based on difference in solubility.

Chromatography :

The name chromatography is based on the Greek word chroma, for colour since the method was first used for the separation of coloured substances found in plants. In this technique, the mixture of substances is applied on to a stationary phase, which may be a solid or a liquid. A pure solvent, a mixture of solvents, or a gas is allowed to move slowly over the stationary phase. The components of the mixture get gradually separated from one another. The moving phase is called the mobile phase. Based on the principle involved, chromatography is classified into different categories. Two of these are:

- (a) Adsorption chromatography, and
- (b) Partition chromatography.

(a) Adsorption Chromatography:

Adsorption chromatography is based on the fact that different compounds are adsorbed on an adsorbent to different degrees. Commonly used adsorbents are silica gel and alumina. When a mobile phase is allowed to move over a stationary phase (adsorbent), the components of the mixture move by varying distances over the stationary phase. Following are two main types of chromatographic techniques based on the principle of differential adsorption.

- (i) Column chromatography, and
- (ii) Thin layer chromatography.

(i) Column Chromatography:

Column chromatography involves separation of a mixture over a column of adsorbent (stationary phase) packed in a glass tube. The column is fitted with a stopcock at its lower end.

The mixture adsorbed on adsorbent is placed on the top of the adsorbent column packed in a glass tube. An appropriate eluant which is a liquid or a mixture of liquids is allowed to flow down the column slowly. Depending upon the degree to which the compounds are adsorbed, complete separation takes place. The most readily adsorbed substances are retained near the top and others come down to various distances in the column.

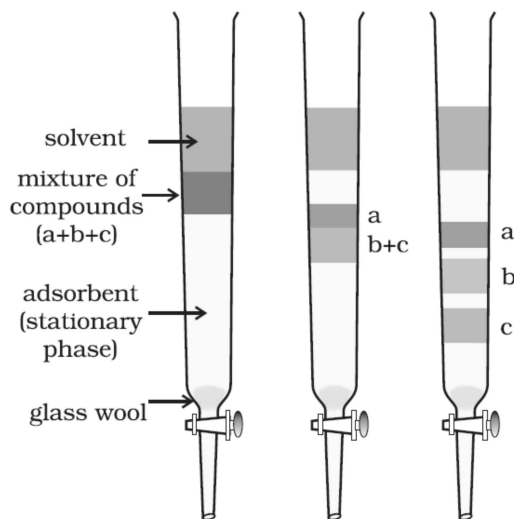


Fig. Column chromatography. Different stages of separation of components of a mixture.

(ii) Thin Layer Chromatography:

Thin layer chromatography (TLC) is another type of adsorption chromatography, which involves separation of substances of a mixture over a thin layer of an adsorbent coated on glass plate. A thin layer (about 0.2mm thick) of an adsorbent (silica gel or alumina) is spread over a glass plate of suitable size. The plate is known as thin layer chromatography plate or chromaplate. The solution of the mixture to be separated is applied as a small spot about 2 cm above one end of the TLC plate. The glass plate is then placed in a closed jar containing the eluant. As the eluant rises up the plate, the components of the mixture move up along with the eluant to different distances depending on their degree of adsorption and separation takes place. The relative adsorption of each component of the mixture is expressed in terms of its retardation factor i.e. R_f value

$$R_f = \frac{\text{Distance moved by the substance from base line (x)}}{\text{Distance moved by the solvent from base line (y)}}$$

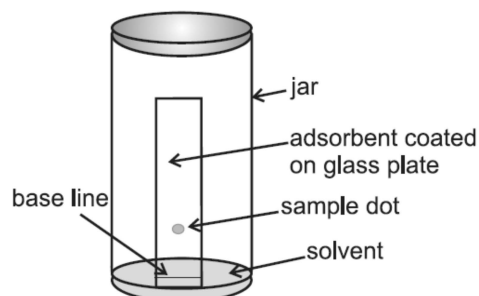


Fig. (a) Thin layer chromatography. Chromatogram being developed

The spots of coloured compounds are visible on TLC plate due to their original colour. The spots of colourless compounds, which are invisible to the eye but fluoresce in ultraviolet light, can be detected by putting the plate under ultraviolet light. Another detection technique is to place the plate in a covered jar containing a few crystals of iodine. Spots of compounds, which adsorb iodine, will show up as brown spots. Sometimes an appropriate reagent may also be sprayed on the plate. For example, amino acids may be detected by spraying the plate with ninhydrin solution.

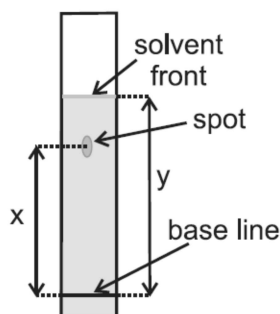


Fig. (b) Developed chromatogram.

Partition Chromatography:

Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases. Paper chromatography is a type of partition chromatography. In paper chromatography, a special quality paper known as chromatography paper is used. Chromatography paper contains water trapped in it, which acts as the stationary phase. A strip of chromatography paper spotted at the base with the solution of the mixture is suspended in a suitable solvent or a mixture of solvents. This solvent acts as the mobile phase. The solvent rises up the paper by capillary action and flows over the spot. The paper selectively retains different components according to their differing partition in the two phases. The paper strip so developed is known as a chromatogram. The spots of the separated coloured compounds are visible at different heights from the position of initial spot on the chromatogram. The spots of the separated colourless compounds may be observed either under ultraviolet light or by the use of an appropriate spray reagent as discussed under thin layer chromatography.

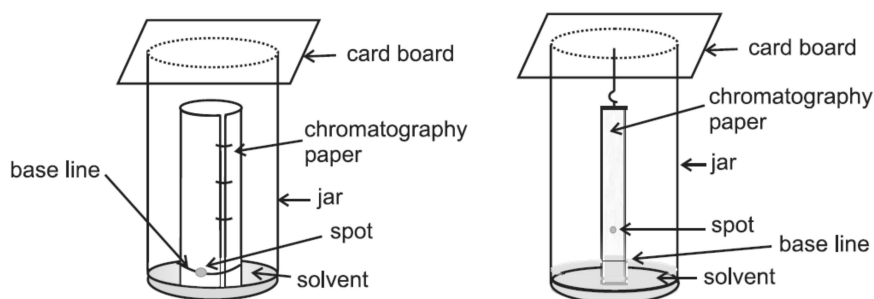


Fig. Paper chromatography. Chromatography paper in two different shapes.

Exercise 1.1

1. Benzoic acid is soluble in hot water but sparingly soluble in cold water. An impure sample of benzoic acid can be purified by:
 - (1) distillation
 - (2) steam distillation
 - (3) crystallisation
 - (4) solvent extraction
2. Naphthalene is a highly volatile solid. It can be separated from non-volatile impurities by:
 - (1) crystallisation
 - (2) sublimation
 - (3) fractional crystallisation
 - (4) distillation
3. From a mixture of methanol (b.p. 65°C) and acetone (b.p. 56°C) the components can be separated by:
 - (1) simple distillation
 - (2) vacuum distillation
 - (3) steam distillation
 - (4) fractional distillation
4. Which of the following compounds can be purified by steam distillation?
 - (1) Benzamide
 - (2) Benzoic acid
 - (3) Aniline
 - (4) Cinnamic acid
5. Which of the following pairs of compounds can be separated from their mixture by steam distillation?
 - (1) o-Nitrophenol and p-nitrophenol
 - (2) Camphor and NaCl
 - (3) NaCl and KCl
 - (4) All the above
6. In steam distillation, the steam-volatile compound and water distil together at:
 - (1) The normal boiling point of water
 - (2) The normal boiling point of the compound
 - (3) A temperature below the normal boiling point of the compound
 - (4) A temperature below the normal boiling points of both the compound and water
7. If two volatile liquids have almost identical boiling points, they are best separated from a mixture by:
 - (1) simple distillation
 - (2) fractional distillation
 - (3) column chromatography
 - (4) paper chromatography
8. The separation of a mixture of organic compounds into individual components by column chromatography is based on the differences in their:
 - (1) solubilities
 - (2) adsorptivities
 - (3) absorptivities
 - (4) partition coefficients
9. A mixture of naphthalene and sodium chloride can be separated into pure compounds by:
 - (1) crystallisation
 - (2) sublimation
 - (3) distillation
 - (4) vacuum distillation
10. The most commonly used adsorbent in column chromatography is:
 - (1) sand
 - (2) Calcium carbonate
 - (3) Charcoal
 - (4) silica gel
11. Method which is based on the difference in the solubilities of the compound-
 - (1) Crystallisation
 - (2) Distillation
 - (3) Sublimation
 - (4) All

12. Different fractions of crude oil are separated by
- (1) Simple distillation
 - (2) Fractional distillation
 - (3) Steam distillation
 - (4) Vacuum distillation

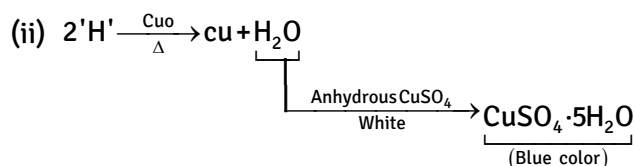
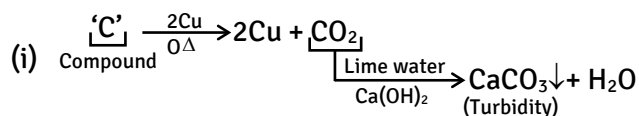
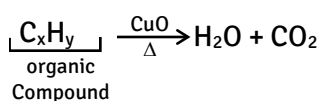
13. Which process is used for purification of liquids having high B.P. and decomposition below their B.P.
- (1) Distillation
 - (2) Steam distillation
 - (3) Distillation under reduced pressure
 - (4) Differential extraction

14. Which purification technique is used for high boiling organic compound (decomposes near its boiling point)?
- (1) Simple distillation
 - (2) Steam distillation
 - (3) Fractional distillation
 - (4) Reduced pressure distillation

Qualitative Analysis of Organic Compounds

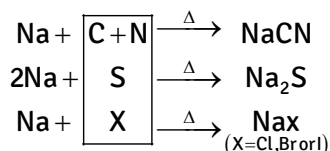
(A) Detection of carbon and hydrogen → By Liebig's method

Carbon and hydrogen are detected by heating the compound with copper (II) oxide. carbon present in the compound is oxidised to carbon dioxide and Hydrogen to water



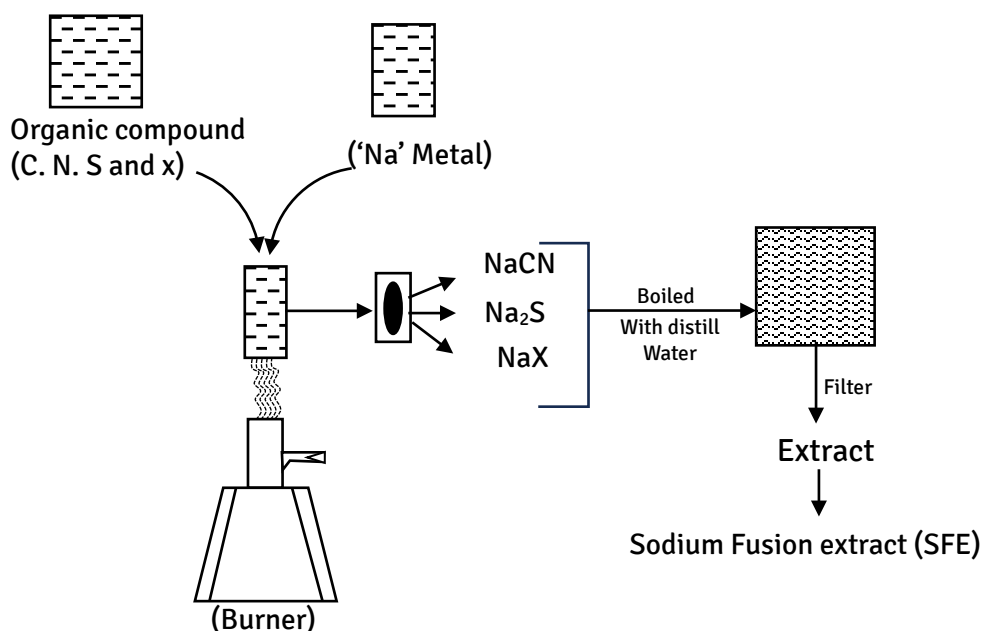
(B) Detection of other Elements:

- P, S, N and X Present in an organic compound are detected by "Lassaigne's Test"
- The elements present in the compound are converted from covalent form into the ionic form by fusing the compound with sodium metal.



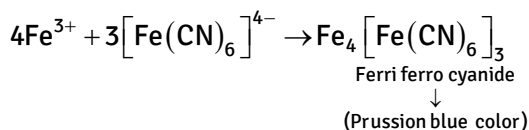
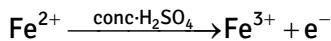
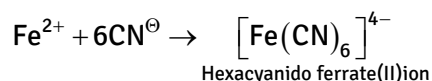
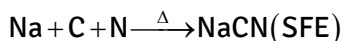
↓
Part of organic compound

Process →



(1) Test For Nitrogen →

The sodium fusion extract is boiled with Iron (II) sulphate and then acidified with concentrated sulphuric acid. The formation of prussian blue color confirm the Presence of Nitrogen.



Note: Lassaigne Test of Nitrogen is not given if

(i) carbon is absent in the given compound

Ex. $\text{NH}_2 - \text{NH}_2, \text{NH}_4\text{OH}$ etc.
They don't form NaCN

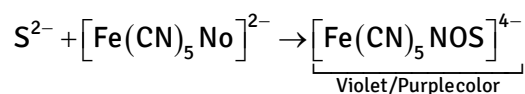
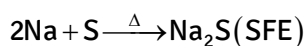
(ii) Diazonium compound

Ex. Alkyl diazonium or Aryl diazonium compound $\xrightarrow[\text{N}_2 \uparrow]{\Delta} \text{NaCN}$ is not formed

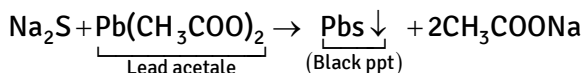
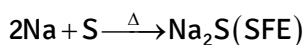
(R-N₂⁺Cl⁻) (Ar-N₂⁺Cl⁻)

(2) Test For sulphur →

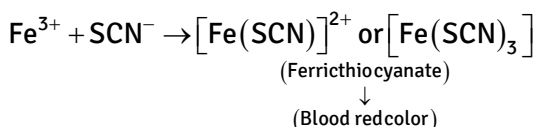
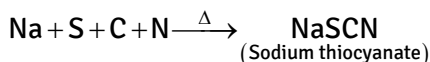
(i) On treating sodium fusion extract with sodium nitro prusside, appearance of a violet/purple color further indicates the presence of sulphur



- (ii) The sodium fusion extract is acidified with acetic acid and Lead acetate is added to it. A black Precipitate of Lead sulphide indicates the presence of Sulphur



Note: (i) if 's' and 'N' both are present in the compound

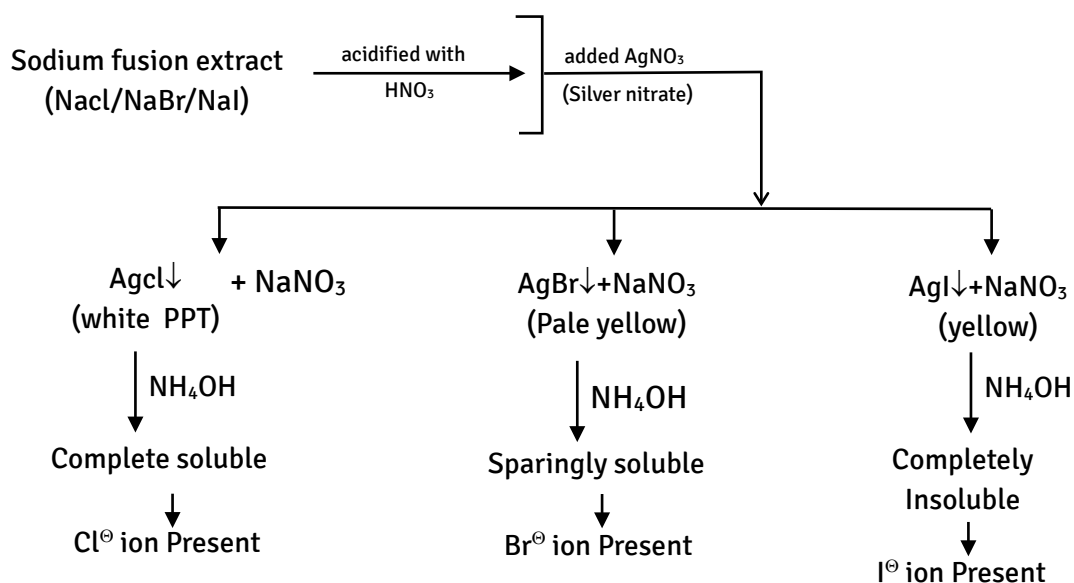


- Prussian blue color is not observed since there are no free cyanide ions (CN^-)

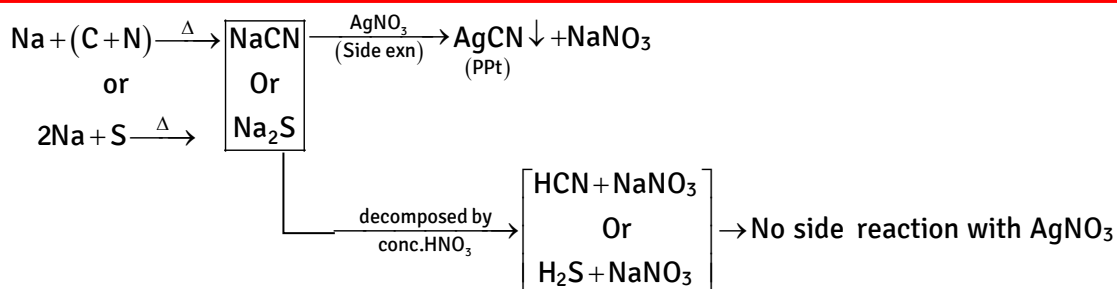
Note: (ii) if sodium fusion is carried out with excess of Sodium, the thiocyanate decomposes to yield cyanide and sulphide ion. These ions give their usual tests



(3) Test for Halogens →

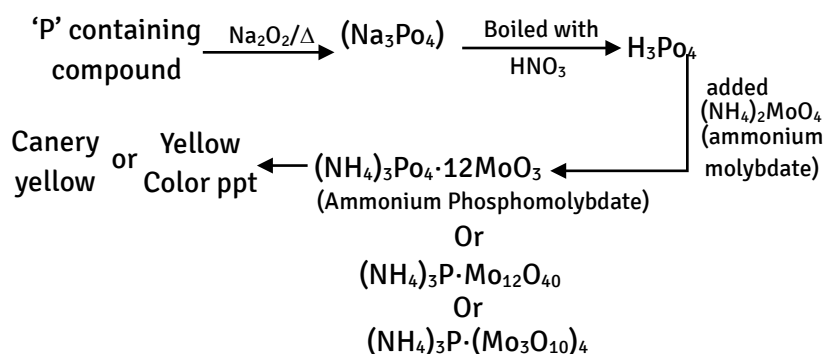


Note: sodium fusion extract is acidified with conc. HNO_3 before adding AgNO_3 because if 'N' or 's' is also Present in the compound then formed NaCN or Na_2S during Lassaigne test will interfere with silver nitrate Test for halogen. therefore to decompose NaCN or Na_2S , sodium fusion extract is acidified with conc. HNO_3



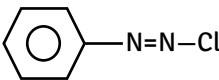
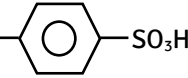
(4) Test For Phosphorus:

- The compound is heated with an oxidizing agent (sodium peroxide). The phosphorus present in the compound is oxidized to phosphate. The solution is boiled with nitric acid and then treated with ammonium molybdate. A yellow colored ppt indicate the presence of phosphorus



Exercise 1.2

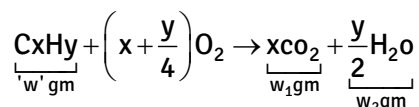
- When a nitrogenous organic compound is fused with metallic sodium, its nitrogen is converted to :
 (1) sodium cyanate
 (2) sodium nitrate
 (3) sodium amide
 (4) sodium cyanide
- The sodium fusion extract of an organic compound on acidification with dilute HCl and subsequent treatment with FeCl_3 develops a blood-red colour. The organic compound contains.
 (1) bromine
 (2) nitrogen
 (3) sulphur
 (4) both nitrogen and sulphur
- The sodium fusion extract of an organic compound on treatment with sodium nitroprusside solution develops violet colour. So, the compound contains.
 (1) nitrogen
 (2) sulphur
 (3) iodine
 (4) phosphorus
- The sodium fusion extract of an organic compound containing both nitrogen and sulphur when acidified with HCl and then treated with FeCl_3 solution develops a :
 (1) blue colour
 (2) green colour
 (3) red colour
 (4) yellow colour
- Lassaigne's Test for the detection of nitrogen will fail in the case of-
 (1) NH_2CONH_2
 (2) $\text{NH}_2\text{—CO—NH—NH}_2 \cdot \text{HCl}$
 (3) $\text{NH}_2\text{—NH}_2 \cdot \text{HCl}$
 (4) $\text{C}_6\text{H}_5\text{NH—NH}_2 \cdot 2\text{HCl}$

6. The Prussian blue color obtained during the test of nitrogen by lassaigine's test is due to the formation of-
- (1) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 - (2) $\text{Na}_3[\text{Fe}(\text{CN})_6]$
 - (3) $\text{Fe}(\text{CN})_3$
 - (4) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
7. In lassaigine's extract of an organic compound both nitrogen and sulphur are present, which given blood red color with Fe^{3+} due to the formation of-
- (1) NaSCN
 - (2) $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$
 - (3) $[\text{Fe}(\text{SCN})]^{2+}$
 - (4) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot \text{H}_2\text{O}$
8. Nitric acid added to sodium extract before adding silver nitrate for testing halogen because
- (1) Nitric acid reduces sulphide
 - (2) Nitric acid decomposes " NaCN " and " Na_2S "
 - (3) Nitric acid oxidises the organic compound
 - (4) Nitric acid acts as a dehydrating agent
9. Which one of the following set of elements can be detected using sodium fusion extract?
- (1) Sulfur, Nitrogen, phosphorous, halogen
 - (2) Phosphorous, oxygen, nitrogen, halogen
 - (3) Nitrogen, phosphorous, carbon, sulphur
 - (4) Halogen, nitrogen, oxygen, sulfur
10. Black precipitate in the detection of sulphur with lead acetate and acetic acid is due to the formation of
- (1) Pb_2S
 - (2) PbS
 - (3) PbS_2
 - (4) PbSO_4
11. Which of the following will give positive lassaigine's test for nitrogen
- (1) NH_2OH
 - (2) NH_2-NH_2
 - (3) KCN
 - (4) 
12. Which of the following will give blood red color in lassaigine's test for nitrogen
- (1) PhNH_2
 - (2) PhNO_2
 - (3) 
 - (4) PhSO_3H
13. The violet color that develops on addition of nitro prusside solution to sodium fusion extract of a sulphur containing organic compound is due to the formation of
- (1) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
 - (2) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
 - (3) $\text{Na}_3[\text{Fe}(\text{SCN})_6]$
 - (4) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{S}]$
14. The sodium fusion extract of an organic compound on acidification with acetic acid followed by addition of lead acetate solution produced a black ppt. So the compound contains-
- (1) Nitrogen
 - (2) phosphorous
 - (3) Sulphur
 - (4) Bromine
15. The sodium fusion extract of an organic compound is boiled with nitric acid and then treated with silver nitrate solution. A white precipitate is formed which dissolves in aqueous Ammonia solution. The organic compound contain
- (1) Sulphur
 - (2) Chlorine
 - (3) Iodine
 - (4) Bromine
16. The formula of sodium nitro prusside is
- (1) $\text{Na}_4[\text{Fe}(\text{CN})_6]$
 - (2) $\text{Na}_3[\text{Fe}(\text{CN})_5\text{NO}]$
 - (3) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
 - (4) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$

Quantitative analysis for organic compound

(1) % Calculation For carbon & hydrogen → by Liebig method

- Both carbon and hydrogen are estimated in one experiment. A known mass of an organic compound is burnt in the presence of excess of oxygen and copper (II) oxide. carbon and hydrogen in the compound are oxidised to CO_2 and H_2O respectively



Let mass of organic compound is 'w' gm

$$44 \text{ gm CO}_2 \rightarrow 12 \text{ gm 'c'}$$

$$1 \text{ gm CO}_2 \rightarrow \frac{12}{44} \text{ gm 'c'}$$

$$w_1 \text{ gm CO}_2 \rightarrow \left(\frac{12}{44} \times w_1\right) \text{ gm 'c'}$$

$$\boxed{\% \text{ of 'C'} = \frac{12}{44} \times \frac{w_1}{w} \times 100}$$

Similarly: $18 \text{ gm H}_2\text{O} \rightarrow 2 \text{ gm 'H'}$

$$1 \text{ gm H}_2\text{O} \rightarrow \frac{2}{18} \text{ gm 'H'}$$

$$w_2 \text{ gm H}_2\text{O} \rightarrow \left(\frac{2}{18} \times w_2\right) \text{ gm 'H'}$$

$$\boxed{\% \text{ of 'H'} = \frac{2}{18} \times \frac{w_2}{w} \times 100}$$

Ex. (1) 0.3960 gm of an organic compound on combustion gives 0.792 gm CO_2 and 0.324 gm of H_2O . Calculate the % of C and H

Ans. 9.09%

Sol. $\% \text{ of C} = \frac{12}{44} \times \frac{\text{Wt. of CO}_2}{\text{Wt. of compound}} \times 100$

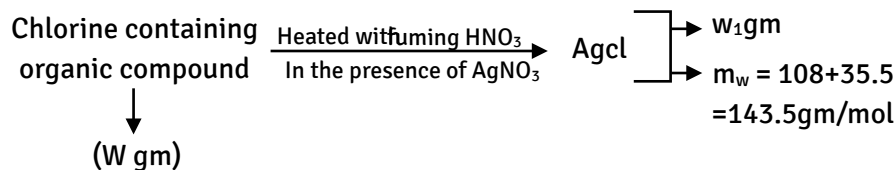
$$= \frac{12}{44} \times \frac{0.792}{0.396} \times 100 = 54.55\%$$

$$\% \text{ of H} = \frac{2}{18} \times \frac{\text{Wt. of H}_2\text{O}}{\text{Wt. of compound}} \times 100$$

$$= \frac{2}{18} \times \frac{0.324}{0.396} \times 100 = 9.09\%$$

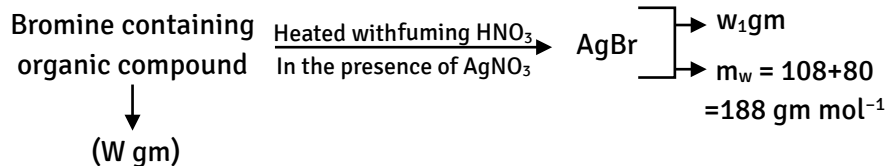
(2) % Calculation for Halogen → by Carius method

Case (a):- for 'Cl'



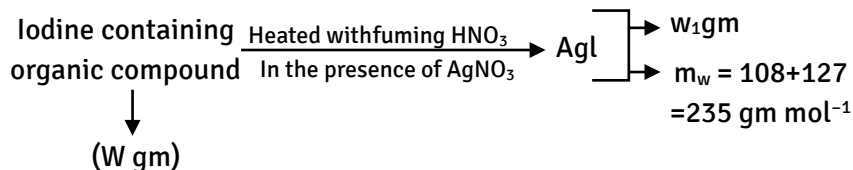
$$\% \text{ of Cl} = \frac{35.5}{143.5} \times \frac{w_1}{w} \times 100$$

Case (b):- for 'Br'



$$\% \text{ of Br} = \frac{80}{188} \times \frac{w_1}{w} \times 100$$

Case (c):- for 'I'



$$\% \text{ of I} = \frac{127}{235} \times \frac{w_1}{w} \times 100$$

Ex. (2) In carius method of estimation of halogen, 0.15g of an organic compound gave 0.12g of AgBr. Find out the percentage of bromine in the compound.

Ans. 34.04%

Sol. Molar mass of AgBr = 108 + 80 = 188 g mol⁻¹

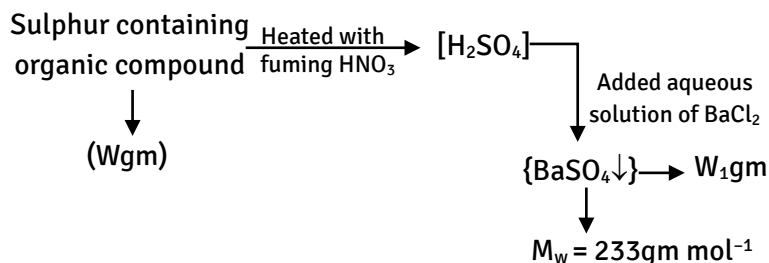
188 g AgBr contains 80g bromine.

0.12g AgBr contains 80 g bromine.

0.12g AgBr contains $\frac{80 \times 0.12}{188}$ g bromine.

Percentage of bromine = $\frac{80 \times 0.12 \times 100}{188 \times 0.15} = 34.04\%$

(3) % Calculation for Sulphur → by Carius method.



$$\% \text{ of 'S'} = \frac{32}{233} \times \frac{w_1}{w} \times 100$$

Ex. (3) In Sulphur estimation, 0.157g of an organic compound gave 0.4813 g of barium sulphate. What is the percentage of Sulphur in the compound ?

Ans. 42.10%

Sol. Molecular mass of $\text{BaSO}_4 = 137 + 32 + 64 = 233\text{g}$

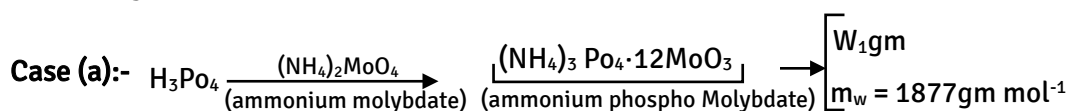
233g BaSO_4 contains $\frac{32 \times 0.4813}{233}$ g Sulphur

$$\text{percentage of Sulphur} = \frac{32 \times 0.4813 \times 100}{233 \times 0.157} = 42.10\%$$

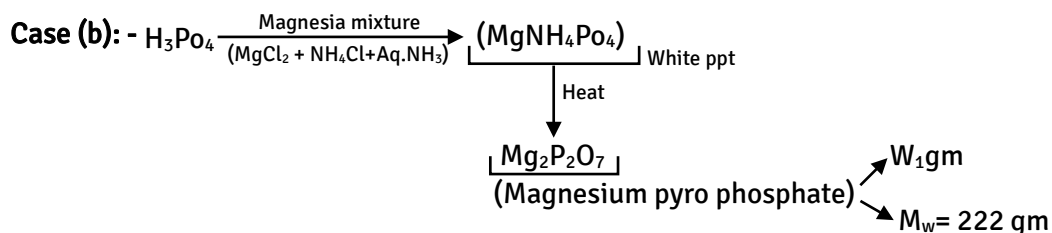
(4) % Calculation for Phosphorus →

Phosphorus containing organic compound $\xrightarrow[\text{fuming HNO}_3]{\text{Heated with}}$ $[\text{H}_3\text{PO}_4]$
(Phosphoric acid)

(Wgm)



$$\% \text{ of 'P'} = \frac{31}{1877} \times \frac{W_1}{W} \times 100$$



$$\% \text{ of P} = \frac{62}{222} \times \frac{W_1}{W} \times 100$$

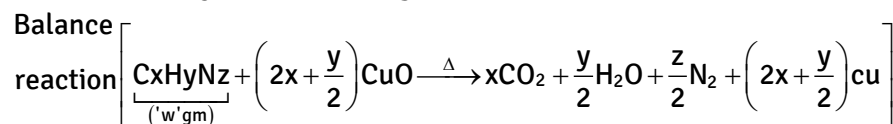
(5) % Calculation for Nitrogen

There are two methods for estimation of nitrogen

(i) Dumas method (ii) Kjeldahl's method

(i) Dumas method :

when nitrogen containing organic compound is heated with copper (II) oxide in an atmosphere of carbondioxide, yields free nitrogen in addition to carbondioxide and water.



if traces amount of nitrogen oxides formed then they are reduced to N_2 by Passing the gaseous mixture over heated copper gauze.

- Produced gaseous mixture is collected over aqueous Solution of KOH which absorbs CO_2
 $\{2\text{KOH} + \text{CO}_2 \rightarrow \text{K}_2\text{CO}_3 + \text{H}_2\text{O}\}$

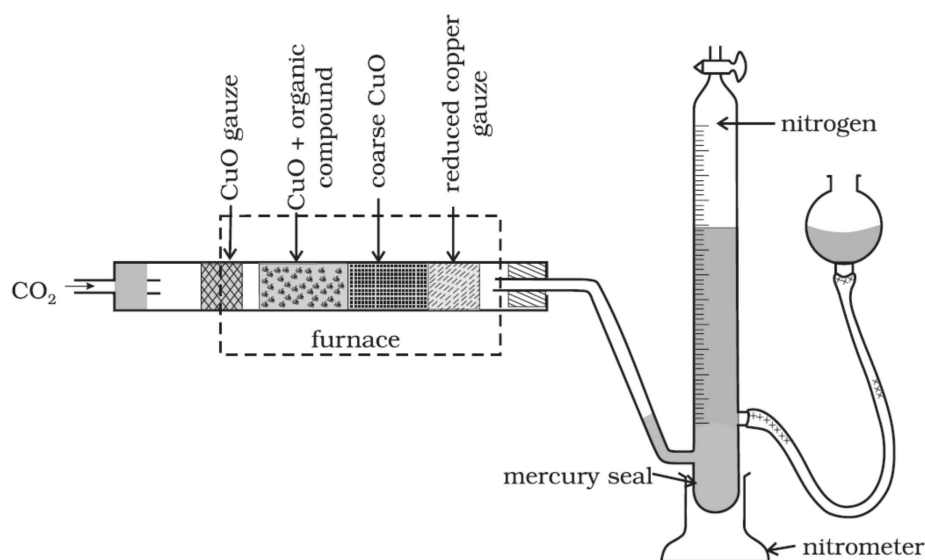


Fig. Dumas method. The organic compound yields nitrogen gas on heating it with copper (II) oxide in the presence of CO_2 gas. The mixture of gases is collected over potassium hydroxide solution in which CO_2 is absorbed and volume of nitrogen gas is determined.

From ideal gas equation

$$PV = nRT$$

$$PV \propto T$$

$$\frac{(P_1 - P)V_1}{T_1} = \frac{(P_2 - P)V_2}{T_2} \rightarrow \text{at STP}$$

P_1 = Pressure of moist gas

P = Aqueous tension

T_1 = Temp.(K) at which N_2 gas is collected

V_1 = Volume of N_2 gas (m^3)

$P_2 = 1 \text{ atm or } 760 \text{ mm of Hg}$

$T_2 = 273 \text{ K}$

V_2 = Volume of N_2 gas collected at STP

At STP; 22.4L or 22400 ml $\rightarrow$ 1 mole N_2 or 28 gm N_2

$$1 \text{ ml} \rightarrow \left(\frac{28}{22400} \right) \text{ gm } \text{N}_2$$

$$(V_2)_{\text{ml}} \rightarrow \left(\frac{28}{22400} \times V_2 \right) \text{ gm } \text{N}_2$$

$$\% \text{ of 'N'} = \frac{28}{22400} \times \frac{V_{2(\text{ml})}}{w} \times 100$$

Ex. (4) 0.25gm of an organic compound NTP gives 31 ml of N_2 gas by Duma's method. Find out of % of N.

Ans. 15.5%

Sol.

$$\begin{aligned} \% \text{ of N} &= \frac{28}{22400} \times \frac{V}{W} \times 100 \\ &= \frac{28}{22400} \times \frac{31}{0.25} \times 100 = 15.5\% \end{aligned}$$

Ex. (5) 31.7 ml of moist N_2 was obtained from 0.2033 gm of an organic compound in Duma's method at $14^\circ C$ and 758 mm pressure. If aq. tension at $14^\circ C = 14$ mm then calculate of % of nitrogen in the compound

Ans. 18.15%

Sol. $P_1 = \text{Pressure of } N_2 - \text{Aq. tension} = 758 - 14 = 744 \text{ mm}$

$$P_2 = 760 \text{ mm}$$

$$T_2 = 273 \text{ K}$$

$$T_1 = 273 + 14$$

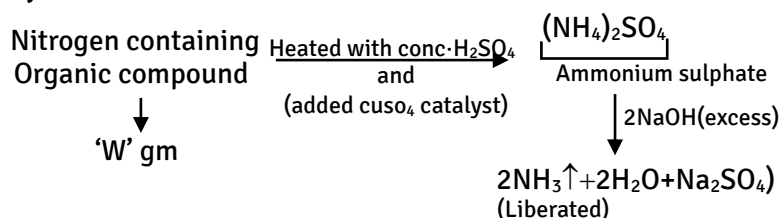
$$T_1 = 287 \text{ K}$$

$$V_2 = \frac{P_1 V_1}{T_1} \times \frac{273}{760} = \frac{744 \times 31.7 \times 273}{287 \times 760} = 29.51$$

$$\%N = \frac{28}{22400} \times \frac{V}{W} \times 100$$

$$= \frac{28}{22400} \times \frac{29.51}{0.2033} \times 100 = 18.15\%$$

(ii) Kjeldahl's method →



- Liberated ammonia is absorbed in excess of standard solution of sulphuric acid
- The amount of ammonia Produced is determined by estimating the amount of H_2SO_4 consumed in the reaction it is done by estimating unreacted sulphuric acid Left after the absorption of ammonia by titrating it with standard alkali solution (NaOH).
- The difference between the initial amount of acid taken and that Left after the reaction gives the amount of acid reacted with ammonia.

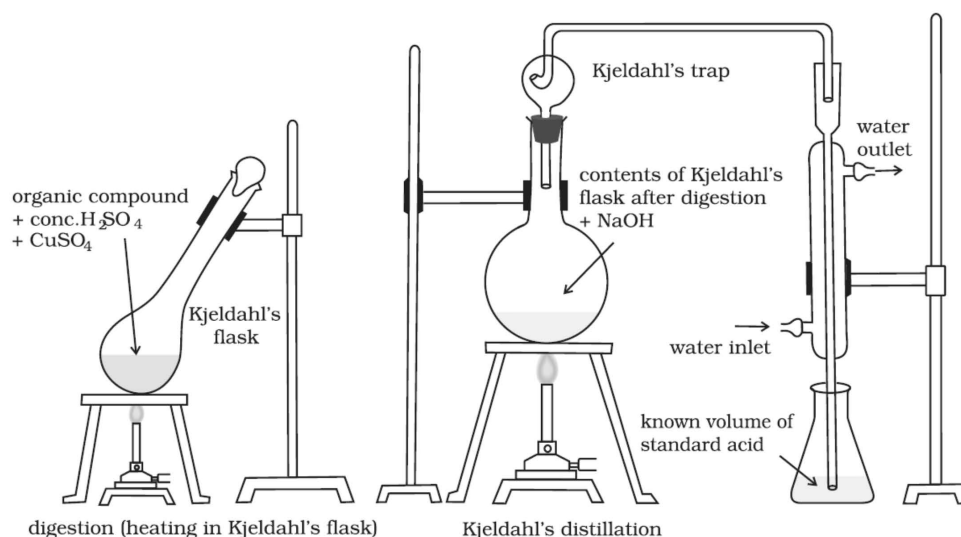


Fig. Kjeldahl method. Nitrogen-containing compound is treated with concentrated H_2SO_4 to get ammonium sulphate which liberates ammonia on treating with NaOH; ammonia is absorbed in known volume of standard acid.

From low of equivalent:gm equivalent of H_2SO_4 = gm equivalent of NH_3

$$N = \frac{\text{gm eq. of solute } (\text{H}_2\text{SO}_4) \times 1000}{\text{Volume of solution (ml)}} \quad \left| \quad \begin{aligned} \text{gm eq.} &= \frac{W}{E} \\ \text{gm eq. of } \text{H}_2\text{SO}_4 &= \frac{N \times \text{Vml}}{1000} \end{aligned} \right. \quad \text{gm eq. of } \text{NH}_3 = \frac{\text{Weight of } \text{NH}_3}{17/1}$$

$$\frac{N \times \text{Vml}}{1000} = \frac{\text{weight of } \text{NH}_3}{17/1}$$

$$\text{Weight of } \text{NH}_3 = \left(\frac{N \times (\text{Vml}) \times 17}{1000} \right) \text{ gm}$$

$$17 \text{ gm } \text{NH}_3 \longrightarrow 14 \text{ gm 'N'}$$

$$1 \text{ gm} \longrightarrow \frac{14}{17} \text{ gm 'N'}$$

$$\left(\frac{N \times \text{Vml} \times 17}{1000} \right) \text{ gm } \text{NH}_3 \longrightarrow \left(\frac{14}{17} \times \frac{N \times \text{Vml} \times 17}{1000} \right) \text{ gm 'N'}$$

$$\Rightarrow \left(\frac{14}{1000} \times N \times \text{Vml} \right) \text{ gm 'N'}$$

$$\% \text{ of 'N'} = \frac{14}{1000} \times \frac{N \times \text{Vml}}{w} \times 100$$

$$\boxed{\% \text{ of 'N'} = \frac{1.4 \times N \times \text{Vml}}{W \text{ gm}}}$$

W → weight of organic compound

 $N \times \text{Vml}$ = milli eq. of H_2SO_4 consumed in the reaction to neutralized ammonia.= milli eq. of H_2SO_4 taken – milli eq. of H_2SO_4 remaining.= milli eq. of H_2SO_4 taken – milli eq. of NaOH .**Note:-** This method is not applicable in the following Compound.(I) Nitro comp ($-\text{NO}_2$) (II) Azo compound ($-\text{N}=\text{N}-$)

(III) Nitrogen in the ring

Ex.

(Pyridine)

(reason:- 'N' of these compound does not change to ammonium sulphate under these condition)

Ex. (6) 30 ml 0.25 N H_2SO_4 is used in neutralizing NH_3 obtained from 0.75 gm of an organic compound in Kjeldahl's method. Find out % of N in the compound.**Ans.** 14%

$$\text{Sol. } N\% = \frac{1.4}{W} \times NV$$

$$= \frac{14}{0.75} \times 30 \times 0.25 = 14\%$$

Ex. (7) 0.35 gm of an organic compound was analysed by Kjeldahl's method. Ammonia evolved was absorbed in

100 ml $\frac{N}{10}$ H_2SO_4 . Unused acid required 30 ml $\frac{N}{10}$ NaOH for neutralization calculate % of N

Ans. 28%

Sol. $W = 0.35 \text{ gm NH}_3$

$$100 \text{ ml } \frac{N}{10} \text{ H}_2\text{SO}_4 (10 \text{ ml H}_2\text{SO}_4)$$

↓
30 ml $\frac{N}{10}$ NaOH (3 ml N NaOH)

Remaining acid 3 ml $\text{N H}_2\text{SO}_4$

Consumed acid (10 ml N – 3ml N) = 7 ml N H₂SO₄

$$\% \text{ of N} = \frac{1.4}{W} \times NV$$

$$= \frac{1.4}{0.35} \times 1 \times 7 = 28\%$$

Exercise 1.3

1. Carbon and hydrogen in an organic compound are estimated by :
(1) Dumas method
(2) Carius method
(3) Liebig's method
(4) Kjeldahl's method
 2. In Dumas method of estimation of nitrogen, the gas collected in the nitrometer is :
(1) N_2 (2) NH_3
(3) NO_2 (4) N_2O
 3. Kjeldahl's method is used for the estimation of :
(1) Sulphur (2) Nitrogen
(3) Halogen (4) Phosphorus
 4. In Dumas method of estimation of nitrogen, an organic compound containing nitrogen is heated with :
(1) Oxygen (2) Carbon dioxide
(3) Copper (4) Copper oxide
 5. In Kjeldahl's method, nitrogen present in a compound is estimated by converting it finally into :
(1) NH_3 (2) $(NH_4)_2SO_4$
(3) N_2 (4) NH_2-NH_2
 6. Kjeldahl's method of estimation of nitrogen is not applicable to :
(1) Amides
(2) Amines
(3) Diazo compounds
(4) Amino acids
 7. 0.20 g of an organic compound on combustion gave 0.50 g of CO_2 . The percentage of carbon in the compound is
(1) 48.25 (2) 68.18
(3) 58.28 (4) 78.81
 8. In the estimation of nitrogen by Dumas method, 0.112 g of an organic compound gave 22.4 mL of nitrogen gas at stp. The percentage of nitrogen in the compound is :
(1) 20.2 (2) 47.5
(3) 25.0 (4) 15.5
 9. Halogens present in organic compounds are estimated by :
(1) Carius method
(2) Dumas method
(3) Liebig's method
(4) Kjeldahl's method

10. In the Carius method, halogens present in organic compounds are estimated in the form of :
(1) Barium halide
(2) Silver halide
(3) Hydrogen halide
(4) Gaseous halogen
11. Phosphorus present in an organic compound is generally estimated gravimetrically in the form of :
(1) Magnesium pyrophosphate
(2) Ammonium phosphomolybdate
(3) Either of the above
(4) None of the above
12. 0.32 g of an organic compound containing sulphur was heated with fuming nitric acid to convert sulphur into sulphuric acid. The latter was then precipitated as BaSO_4 by adding excess of BaCl_2 solution. 0.233 g of BaSO_4 was obtained. The percentage of sulphur in the compound is approximately :
(1) 15.0 (2) 10.0
(3) 30.5 (4) 20.8

Exercise 2

1. Match Column I with Column II

Column I		Column II	
(i)	Dumas method	(a)	AgNO ₃
(ii)	Kjeldahl's method	(b)	Silica gel
(iii)	Carius Method	(c)	Nitrogen Gas
(iv)	chromatography	(d)	Free radicals
		(e)	Ammonium sulphate

(1) i → c; ii → e; iii → a; iv → b

(2) i → d; ii → a; iii → c; iv → e

(3) i → b; ii → c; iii → a; iv → d

(4) i → e; ii → b; iii → d; iv → c

2. The principle involved in paper chromatography is :

- (1) Adsorption (2) Partition
(3) Solubility (4) Volatility

3. In carius method of estimation of halogen 0.15 g of an organic compound gave 0.12g of AgBr. Find out the percentage of bromine in the compound.

- (1) 23.03% (2) 34.04%
(3) 42.10% (4) 45.05%

4. In Sulphur estimation, 0.157g of an organic compound gave 0.4813g of barium sulphate. What is the percentage of sulphur in the compound ?

- (1) 23.03% (2) 34.04%
(3) 42.10% (4) 48.05%

5. In kjeldahl's method nitrogen present is estimated as:

- (1) N₂ (2) NO₂
(3) NH₄SO₄ (4) NH₃

6. An organic compound give 0.220 gm of CO₂ and 0.126gm of H₂O on complete combustion if the % of carbon is 24 then % hydrogen is-

- (1) 76 (2) 8.6
(3) 5.6 (4) 50

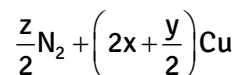
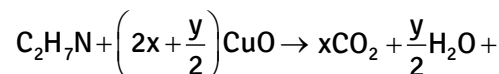
7. When 0.01 mol of an organic compound containing 60% carbon was burnt completely. 4.4g of CO₂ was produced. The molar mass of comp. is----- g mol⁻¹.

- (1) 300 (2) 250
(3) 200 (4) 150

8. A sample of 0.125 g of an organic compound when analyzed by duma's method yields 22.78 ml of nitrogen gas collected over KOH solution at 280K and 759 mm Hg. The % of nitrogen in the given organic compound is.....? (Consider aqueous tension zero)

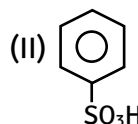
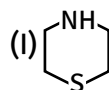
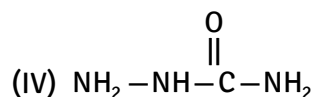
- (1) 36 (2) 22
(3) 12 (4) 48

9. The transformation occurring in duma's method is given below:-



The value of 'y' is_

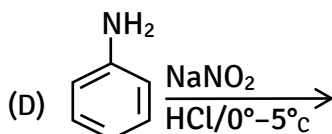
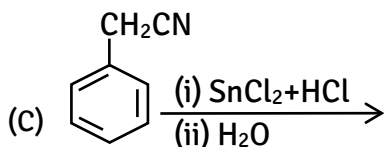
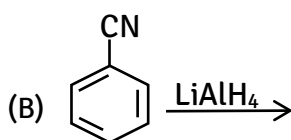
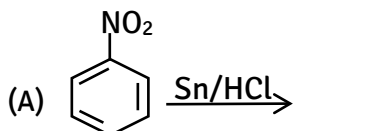
- (1) 14 (2) 7
(3) 7/2 (4) None of these

10. Which of the following compounds will give Prussian blue color when it is converted into lassaing's extract and FeSO₄ is added followed by FeCl₃ (Fe³⁺ ion)(III) NH₂OH

- (1) I and IV (2) IV only
(3) I, III and IV (4) I, II, III and IV

11. The number of moles of CuO, that will be utilised in Dumas method for estimation of nitrogen in a sample of 57.5gm of $C_7H_{17}N$ is-
- (1) 11.25 (2) 18.6
(3) $\frac{45}{2}$ (4) 30.5

12. The kjeldahl method of nitrogen estimation fails in which of the following reaction major product?



- (1) (C) and (D)
(2) (A) and (D)
(3) (A), (C) and (D)
(4) (B) and (C)

13. In the recrystallisation of solid organic compounds, the coloured impurities are removed by boiling the solution of the compound with :
- (1) Activated charcoal
(2) oxalic acid
(3) magnesium sulphate
(4) calcium chloride

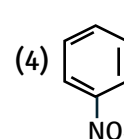
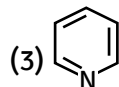
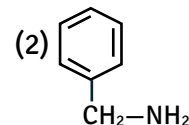
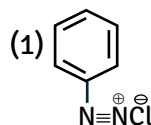
14. The best and latest technique for isolation purification and separation of organic compounds is
- (1) Crystallisation
(2) Distillation
(3) Sublimation
(4) Chromatography

15. The fragrance of flower is due to the presence of some steam volatile organic compound called essential oils. These are generally insoluble in water at room temperature but are miscible with water vapour in the vapour phase. A suitable method for the extraction of these oils from the flowers is :
- (1) Distillation
(2) Crystallisation
(3) Distillation under reduced pressure
(4) steam distillation

16. If a liquid compound decomposes at its boiling point, which method (s) can you choose for its purification it is known that the compound is stable at low pressure, steam volatile and insoluble in water
- (1) Steam distillation
(2) fractional distillation
(3) chromatography
(4) Crystallisation

17. A sample of 0.50 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.25 M H_2SO_4 . The residual acid required 30 ml of 0.5 M solution of NaOH for neutralization. Find % composition of nitrogen in the compound.
- (1) 56% (2) 28%
(3) 36% (4) 63%

18. Nitrogen can be estimated by kjeldahl's method for which of the following compound?



19. In carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of
- (1) HNO_3 (2) $AgNO_3$
(3) $CuSO_4$ (4) $FeSO_4$

20. Which of the following is a false statement?
- (1) Carius tube is used in the estimation of sulphur in an organic compound
 - (2) Carius method is used for the estimation of nitrogen in an organic compound
 - (3) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $\text{Mg}_2\text{P}_2\text{O}_7$ by adding magnesia mixture
 - (4) Kjeldahl's method is used for the estimation of nitrogen in an organic compound
21. The compound formed upon subjecting an aliphatic amine to Lassaigne's test is
- (1) NaNH_2
 - (2) NaNO_2
 - (3) NaCN
 - (4) NaN_3
22. Which of the following compound is formed when an organic compound containing both nitrogen and sulphur is fused with sodium (avoiding excess)
- (1) Sodium sulphide
 - (2) sodium cyanide
 - (3) sodium thiocyanate
 - (4) A mixture of (1) & (2)
23. An organic compound is fused with sodium peroxide and the fused mass is extracted with water. The extract is heated with nitric acid and ammonium molybdate. A yellow precipitate is obtained. The compound contains
- (1) Sulphur
 - (2) Chlorine
 - (3) nitrogen
 - (4) Phosphorus
24. An organic compound with 51.6% sulphur is heated in a Carius tube. The amount of this compound which forms 18.74 gm of barium sulphate is-
- (1) 15.9 gm
 - (2) 11.8 gm
 - (3) 18.6 gm
 - (4) 4.99 gm
25. Distillation under reduced pressure is useful for?
- (1) Crude oil
 - (2) Glycerol
 - (3) Benzene
 - (4) Naphthalene
26. The most suitable method of separation of 1 : 1 mixture of ortho and para nitrophenols is
- (1) Chromatography
 - (2) Crystallisation
 - (3) Steam distillation
 - (4) Sublimation
27. The distillation technique most suited for separating glycerol from spent lye in the soap industry is-
- (1) Distillation under reduced pressure
 - (2) Simple distillation
 - (3) Fractional distillation
 - (4) Steam distillation

Exercise 3

1. **Assertion (A)** If one of the substances in the mixture is water and the other a water insoluble substance, then the mixture will boil close to but higher than 373 K.
Reason (R) Aniline is separated by using steam distillation technique from aniline-water mixture.
 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

2. **Assertion (A)** A liquid boils at a temperature at which its vapour pressure is lower than the external pressure.
Reason (R) Glycerol can be separated from spent-lye in soap industry by using the technique of distillation under reduced pressure.
 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

3. **Assertion (A)** In steam distillation, the liquid boils when the sum of vapour pressures due to the organic liquid (p_1) and that due to water (p_2) becomes equal to the atmospheric pressure (p), i.e. $p = p_1 + p_2$
Reason (R) p_1 is lower than p_2 the organic liquid vaporises at lower temperature than its boiling point.
 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

4. **Assertion (A)** Carbon and hydrogen are detected by heating the compound with copper (II) oxide.
Reason (R) Carbon and hydrogen present in the compound are oxidised to carbon dioxide and water respectively.
 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

5. **Assertion (A)** Phosphorus present in the organic compound is oxidised to phosphoric acid. It is precipitated as ammonium phosphomolybdate by adding ammonium hydroxide
Reason (R) Phosphoric acid may be precipitated as MgNH_4PO_4 by adding magnesia mixture which on ignition yields $\text{Mg}_2\text{P}_2\text{O}_7$.
 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

6. **Assertion (A)** In Dumas method, the nitrogen containing organic compound, when heated with copper oxide in an atmosphere of carbon dioxide yields free nitrogen in addition to carbon dioxide and water.
Reason (R) The correct equation representing this method is

$$\text{C}_x\text{H}_y\text{N}_z + (2x + y/2)\text{CuO} \rightarrow x\text{CO}_2 + y/2\text{H}_2\text{O} + z/2\text{N}_2 + (2x + y/2)\text{Cu}$$

 (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

7. **Assertion (A)** Kjeldahl's method is not applicable to compounds containing nitrogen in the form of nitro and azo groups and nitrogen present in the ring (e.g. pyridine).

Reason (R) Nitrogen of nitro and azo group containing compounds does not change to ammonium sulphate by Kjeldahl's method.

- (1) Both A and R are correct ; R is the correct explanation of A.
 (2) Both A and R are correct ; R is not the correct explanation of A.
 (3) A is correct ; R is incorrect.
 (4) R is correct ; A is incorrect.

8.

Mixture		Purification technique	
(A)	Chloroform & aniline	(P)	Distillation
(B)	Glycerol & spent lye	(Q)	Chromatography
(C)	Aniline & water	(R)	Steam distillation
(D)	Amino acid	(S)	Distillation under reduced pressure

Correct match

- (1) A-P, B-Q, C-R, D-S
 (2) A-P, B-S, C-R, D-Q
 (3) A-P, B-R, C-S, D-Q
 (4) A-R, B-S, C-P, D-Q

9. Match list I with list II

	List-I (Mixture)		List -II (Separation Technique)
(A)	$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	(i)	Steam distillation
(B)	$\text{C}_6\text{H}_{14} + \text{C}_5\text{H}_{12}$	(ii)	Differential extraction
(C)	$\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2\text{O}$	(iii)	Distillation (Simple)
(D)	Organic compound in water	(iv)	Fractional distillation

- (1) A-(iv), B-(i), C-(iii), D-(ii)
 (2) A-(iii), B-(iv), C-(i), D-(ii)
 (3) A-(ii), B-(i), C-(iii), D-(iv)
 (4) A-(iii), B-(i), C-(iv), D-(ii)

10. Match list I with list II

	List-I Element detected		List -II Reagent used/ Product formed
(A)	Nitrogen	(i)	$\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
(B)	Sulphur	(ii)	AgNO_3
(C)	Phosphorous	(iii)	$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
(D)	Halogen	(iv)	$(\text{NH}_4)_2\text{MoO}_4$

- (1) A-(ii), B-(iv), C-(i), D-(iii)
 (2) A-(iv), B-(ii), C-(i), D-(iii)
 (3) A-(ii), B-(i), C-(iv), D-(iii)
 (4) A-(iii), B-(i), C-(iv), D-(ii)

Exercise 4 (Previous Year's Questions)

1. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralized 10 mL of 1 M H_2SO_4 . The percentage of nitrogen in the soil is : **[AIPMT-2014]**

(1) 35.33 (2) 43.33
(3) 37.33 (4) 45.33

2. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is - **[AIPMT-2015]**

(1) 16.76 (2) 15.76
(3) 17.36 (4) 18.20

3. Paper chromatography is an example of: **[NEET 2020]**

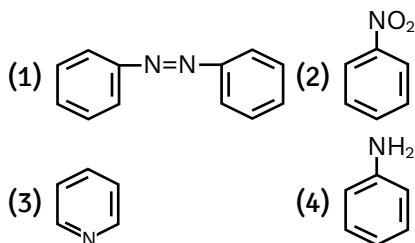
(1) Column chromatography
(2) Adsorption chromatography
(3) Partition chromatography
(4) Thin layer chromatography

4. A liquid compound (x) can be purified by steam distillation only if it is:

[NEET(UG) 2020 (Covid-19)]

(1) Steam volatile, immiscible with water
(2) Not steam volatile, miscible with water
(3) Steam volatile, miscible with water
(4) Not steam volatile, immiscible with water

5. The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds - **[AIPMT-2022]**



6. In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of - **[NEET-2023]**

(1) NaSCN (2) $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$
(3) $[\text{Fe}(\text{SCN})]^{2+}$ (4) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$

7. On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

[NEET-2024]

(1) Distillation (2) chromatography
(3) Crystallization (4) Sublimation

8. A steam volatile organic compound which is immiscible with water has a boiling point of 250°C . During steam distillation, a mixture of this organic compound and water will boil.

[RE-NEET-2024]

(1) above 100°C but below 250°C .
(2) above 250°C .
(3) at 250°C .
(4) close to but below 100°C .

9. Which one of the following reactions does NOT belong to "Lassaignes' test"?

(1) $\text{Na} + \text{X} \xrightarrow{\Delta} \text{NaX}$
(2) $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$
(3) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \text{NaCN}$
(4) $2\text{Na} + \text{S} \xrightarrow{\Delta} \text{Na}_2\text{S}$

10. Match List I with List II

	List- I (Mixture)		List- II (Method of Separation)
A.	$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	I.	Distillation under reduced pressure
B.	Crude oil in petroleum industry	II.	Steam distillation
C.	Glycerol from spent-lye	III.	Fractional distillation
D.	Aniline-water	IV.	Simple distillation

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II (2) A-III, B-IV, C-II, D-I
(3) A-IV, B-III, C-I, D-II (4) A-IV, B-III, C-II, D-I

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	3	2	4	3	1	4	2	2	2	4	1	2	3	4	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Ans.	4	4	2	3	3	1	3	2	1	2	3	3	1	3	2	3	

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	3	1	2	4	1	3	2	3	1	2	3	2	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	2	3	4	3	3	2	2	2	1	1	1	4	4	1	2	2	2	2
Que.	21	22	23	24	25	26	27													
Ans.	3	3	4	4	2	3	1													

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	4	4	1	1	4	1	1	2	2	4	

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	3	1	3	1	4	3	4	4	2	3	

